MA255

Adv. Multi-Variable Calculus

UNITED STATES MILITARY ACADEMY
WEST POINT.

MA255 ADVANCED MULTI-VARIABLE CALCLUS

ABET eCOURSEBOOK AY2021-2026

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SECTION I

COURSE ASSESSMENTS

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB A AY2021 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-MATH 3 JUNE 2021

MEMORANDUM FOR LTC Starling, Associate Program Director, Advanced Math Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

SUBJECT: Course End Report for MA255, AY21-2

1. **Background.** The goal of the Core Advanced Mathematics Program (AMP) is to produce confident students who possess the knowledge, skills, and interest necessary for future studies of science, technology, engineering, and mathematics (STEM). However, we also provide exposure to higher level mathematics for students who do not become STEM majors. A list of detailed course objectives is found in the Course Memo. MA255 used the following required resources:

a. *Text*: Calculus Early Transcendentals, 9th Edition, James Stewart (2021), chapters 12 through 16.

b. *Software*: Mathematica and Microsoft Office (MS Teams, Word, PowerPoint).

c. *Online*: WebAssign (*www.webassign.net*) and Blackboard *(usma.blackboard.com*)

2. **Enrollment and Validation.** The course consisted of 21 sections scheduled during A, B, E, and K hours. This is a change from previous years.

a. *Enrollment*: 286 Cadets and two (2) cadet candidates enrolled during FY21-2.

b. *Sectioning*: 21 Sections, 11 instructors. Four sections were designated as advanced or "honors" sections. The students are selected based on grades achieved in MA153. We recommend continuing this practice because it allows instructors to create a tailored, challenging environment for the brightest students in the course. Fr. Costa, COL Goethals, and Dr. Beshaj taught these advanced sections.

c. *USMAPS*: We maintain our relationship with the Math Department at USMAPS and offered the enrollment of cadet candidates who showed the propensity to do well in the Program. Two (2) cadet candidates completed MA255 during this academic year. Grades and performance for the cadet candidates are not included in the remainder of the report.

d. *Validation*: We offered validation exams for Multivariable Calculus (MA205 validation credit) and Vector Calculus (MA255 validation credit) to students enrolled in MA153 during the fall AY 21-1.

i. *Multivariable Calculus*: Seven (7) students successfully validated MA205 out of the 14 who participated. (This is an increase from six (6) out of 17 participants in AY20-2.)

ii. *Vector Calculus*: Four (4) students successfully validated MA255 out of the 12 who participated. (This is an increase from zero (0) validations out of ten (10) participants in AY20-2.)

d. *Off-ramp cadets*: In addition to the cadets who have validated MA205 and MA255 above, we also had eighteen (18) cadets exit the AMP for various reasons. Most of these cadets did not perform well in MA153 and were offered the opportunity to enter MA103 (if exiting during MA153), MA104 (and vacate the MA104 waiver), or MA205 (and keep the MA104 waiver). The grades of the four cadets who entered MA103 in AY21-2, were two A's, one B, and one C. The grades of the seven cadets who entered MA104 in AY21-2, were two A's, one B+, three B's, and one B-. The grades for the four cadets who entered MA205 in AY21-2, were one C+, one D, and two F's. There were also three cadets who did not take a math class in AY21-2, one of which will take MA103 and two who will take MA205 in AY22-1.

3. **Administration:** The course consisted of four (4) major blocks of instruction: Block I – Vectors and the Geometry of Space; Block II – Multivariable Differentiation; Block III – Multivariable Integration; and Block IV – Vector Calculus.

a. *Course Summary*. There are a total of 63 required attendances consisting of 56 Lessons and six (6) Problem Solving Labs, further broken down as follows:

56 Lessons:

Lesson based on course text:	41
Technology Labs	4
WPRs	4
Course Project	7

6 Problems Solving Labs:

WPR Review Days	4
TEE Review Day	2

b. *Course Evaluation Plan*. There were 3,000 possible course-wide points available. The point distribution is summarized in Table 1.

Category	Event	Pts	PCT	
Individual Exams	TEE	900	30.00%	70.00%
	WPR1	300	10.00%	
	WPR2	300	10.00%	

	WPR3	300	10.00%	
	WPR4	300	10.00%	
Group Work / Modeling	Project	250	8.33%	8.33%
Coding	Tech Lab 1	50	1.67%	6.67%
	Tech Lab 2	50	1.67%	
	Tech Lab 3	50	1.67%	
	Tech Lab 4	50	1.67%	
HW	WebAssign HW 1	50	1.67%	6.67%
	WebAssign HW 2	50	1.67%	
	WebAssign HW 3	50	1.67%	
	WebAssign HW 4	50	1.67%	
Instructor	Instructor Points	250	8.33%	8.33%
	Total	3000	100.00%	

Table 1: Summary of MA255 Grading Scheme.

i. The use of textbooks or computers was not authorized during any examination (WPRs or TEE). The primary implementation of technology, *Mathematica,* was through technology labs and the course project. The course project was the culminating exercise used to challenge students in the areas of problem solving, critical thinking, and verbal and written communication.

ii. *Term End Exam (TEE).* Cadets were given a comprehensive TEE at the end of the semester apart from students who achieved greater than a 93% in the course prior to the TEE. The TEE was made optional for students who demonstrated mastery (≥ 93% course average) of course concepts. This practice was implemented for the first time in MA153 AY20-1 and continued for this course. This was the first year this option was used in MA255. The TEE consisted of nine (9) questions and covered representative problems and applications from all blocks of instruction. Students had 2.5 hours to complete this final examination. The TEE accounted for 30% of the final grade.

iii. *Written Partial Review (WPR)*. Cadets were assessed with a WPR at the end of each block of instruction. Cadets could use one (1) page of handwritten notes and their academy- issued TI-36x calculator during examinations. The WPRs included a mix of computational questions, problem solving questions and concept check questions. WPRs accounted for 40% of the final grade.

iv. *Course Project*. The course project was used to exercise interdisciplinary applications, collaboration, mathematical modeling, and written and verbal communication skills. Students were introduced to the project at the beginning of Block 4 and chose a project from a set of five (5) interdisciplinary topics (civil engineering, operations research, small arms dynamics, Army's future vertical lift, and sabermetrics). The due outs for the project were a project proposal, in-progress review, a 20 minute in-class presentation, and a written report. The project accounted for 8.33% of the final grade.

v. *Technology Labs*. Tech Labs were published at the beginning of each block, and instructors incorporated the required Mathematica commands/instruction throughout each lesson leading up to the designated Tech Lab working day at the end of each block. The Tech Labs were design to supplement each lesson. The Tech Lab prompt identified the exact sections of the textbook to which each question corresponded. The overarching intent of the Tech Labs was to provide a tool for students to visualize and help compute the concepts covered in class. Tech Labs accounted for 6.67% of the final grade.

vi. *WebAssign Homework*. Cadets completed WebAssign homework for each lesson in the course. The new edition of the textbook (9th edition) implemented several new features available through WebAssign and were adopted for this course. In addition to traditional questions, we implemented *Just in Time* questions, *Master It* questions, *Explore It* questions, and *lecture videos* covering the examples from the textbook. The majority of the *Master It* questions were optional with no credit due, but they served as scaffolding for the questions worth points. WebAssign accounted for 6.67% of the final grade.

vii. *Instructor Points*. The course had a total of 250 instructor points. Each instructor created their own assessments tailored to their students and teaching. In general, instructor points consisted of quizzes, problem sets, and reviews. Instructor points accounted for 8.33% of the final grade.

c. *Faculty Development.* All 11 instructors were introduced to the course through a three-day Faculty Development Workshop (FDW) held before and during re-orgy week. In addition, instructors met on a weekly basis for Program Director (PD) meetings to discuss course administration, instructional techniques, teaching tips, available resources, and upcoming lessons, assessments, and requirements. By rotating which instructors are responsible for each of the WPRs, Tech Labs, and project prompts, all instructors had a role in developing the course content. Instructors also contributed to the weekly PD meetings by offering instructional and mathematical insights.

d. *Course Guest Lectures*. No guest lectures were provided due to COVID-19 restrictions and a shortened semester.

4. **Course Statistics and Final Grades:** The average scores for each course-wide graded event and the final grade distribution are shown below. *Note:* No cutline adjustments were made to the A+ grades because of the implementation of the optional TEE for students who had above a 93% prior to the TEE.

Event	Average
WPR 1	86%
WPR 2	91.2%
WPR 3	89.1%
WPR 4	83.3%
Project	93.5%
Tech Labs	93.8%
WebAssign	93.1%
TEE	88.47%

Grade	AY21-01
A+	44 (15.38%)
A	133 (46.5%)
A-	33 (11.54%)
B+	37 (12.94%)
B	23 (8.04%)
B-	5 (1.75%)
C+	7 (2.45%)
C	2 (0.69%)
C-	1 (0.35%)
D	0
F/NC	1 (0.35%)
Total	286

5. **Evaluation of Student Learning Outcomes:** MA255 builds skills to support the four Academic Program Goals (APGs) pertinent to the core mathematics sequence. Each APG is paired with one What Graduates Can Do (WGCD) statement. Table 2 below shows our ten (10) course objectives described in the Course Memo, mapped to the WGCD statements and APGs. We collected supporting information about each of the APGs through a combination of student course evaluations surveys, faculty input, and performance on select course assessment tools (course project, technology labs, WPRs, etc.). We conducted a short mid-course survey and end of course survey to measure student progress along these various goals.

While the course supports all six (6) primary core mathematics goals, the evaluation for the course focuses primarily on "Building a Body of Knowledge" and "Build Competent and Confident Problem Solvers". These goals are described in this report primarily by addressing WGCD statements 5.1 and 2.4.

APG Category	WGCDs	Supporting Course Objectives
Communication: Graduates communicate effectively with all audiences.	**1.5:** Use sound logic and relevant evidence to make convincing arguments.	**III.** Students build critical thinking skills to solve problems through learning complex mathematical concepts in multivariable and vector calculus. **V.** Students improve the quality of both their written and verbal communication of mathematics

Critical Thinking and Creativity: Graduates think critically and creatively.	**2.4:** Reason qualitatively and quantitatively.	**IV.** Students can design, solve, and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.
Lifelong Learning: Graduates demonstrate the capability and desire to pursue progressive and continues intellectual development.	**3.1:** Demonstrate the willingness to and ability to learn independently.	**I.** Students learn good scholarly habits for progressive intellectual development.
Science Technology, Engineering, and Mathematics (STEM): Graduates apply science, technology, engineering, and mathematics concepts and processes to solve complex problems.	**5.1:** Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.	**II.** Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines. **VI.** Students can use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. **VII.** Students can use vectors and vector functions to model motion and surfaces in three-dimensional space. **VIII.** Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. **IX.** Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. **X.** Students understand and apply the concepts of vector calculus.

Table 2: Course Objective Mapping to APGs.

a. WGCD 5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.

i. Summary: MA255 increased cadets' body of knowledge by introducing a widespread set of topics. This increase is evident in student performance evaluated by major graded event grades, a comparison of TEE results across the years, and student self-reported confidence in solving multivariable calculus problems.

ii. Core Mathematics Goal: Build a Body of Knowledge: The learning outcomes that support this goal are (VII) Students can use vectors and vector functions to model motion and surfaces in three-dimensional space. (VIII) Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. (IX) Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. (X) Students understand and apply the concepts of vector calculus.

iii. Course Objective VII - X (Multivariable and Vector Calculus). We give a comprehensive treatment of vector geometry, multivariable calculus (differentiation and integration), and vector calculus (line and surface integrals, Fundamental Theorem of Line Integrals, Green's Theorem, Stokes' Theorem, and the Divergence Theorem). This is extremely challenging material to learn, understand, and master. We measure our effectiveness in this task primarily through graded assessments. A combination of WPR, WebAssign, and the Term End Exam (TEE) grades provides the best estimate of effectiveness in this task. Major graded event results are presented in Section 4: Course Statistics and Final Grades.

The TEE period was reduced to 2.5-hours for this year, so the standard course TEE was modified by choosing 9 out of the original 12 questions. The questions used remained comprehensive, challenging, and essentially unchanged from the TEEs issued for 7-10 years prior to AY20-2. This provides a consistent barometer year-to-year of understanding. Table 3 below gives a comparison of TEE performance from AY18-2 to AY21-2.

		Average		
Term	**Number of Cadets**	**TEE Points**	**Grade**	**Note:**
AY 21-2	134	796.23	88.47%	9 Questions 2.5 hr Exam Students with >93% pre-TEE grade optional
AY 20-2	276	810.11	90.01%	Full Remote Exam
AY 19-2	279	835.59	92.84%	12 Questions 3.5 hr Exam
AY 18-2	250	823.86	91.54%	12 Questions 3.5 hr Exam

Table 3: TEE Grade Average for AY18-2 to AY21-2

Performance in the TEE was remarkably similar to previous iterations. The question with the largest spread in deficiencies was lines, tangent vectors, and planes. The same discrepancy is seen in previous iterations of the course, this could be caused by the fact that the material is covered during the first block of instruction. The course average on the TEE appears slightly lower this semester but we attribute this at least partially to the fact that over 150 students who had grades above 93% did not participate in the TEE. It also appears evident that the more challenging WPR 4 causes students to place more emphasis on preparation for the TEE.

iv. Student Reported Confidence: Mid-course and Course end Surveys. Our students demonstrated an increased confidence in all aspects of the modeling process including Solve. Figure 1 below displays the self-reported confidence level for the various surveys presented in MA153 and MA255. Columns S4 and S5 highlight student confidence in Solving at the MA255 midcourse survey and end of course survey, respectively. There is a clear increase in “very confident” (response 4) responses.

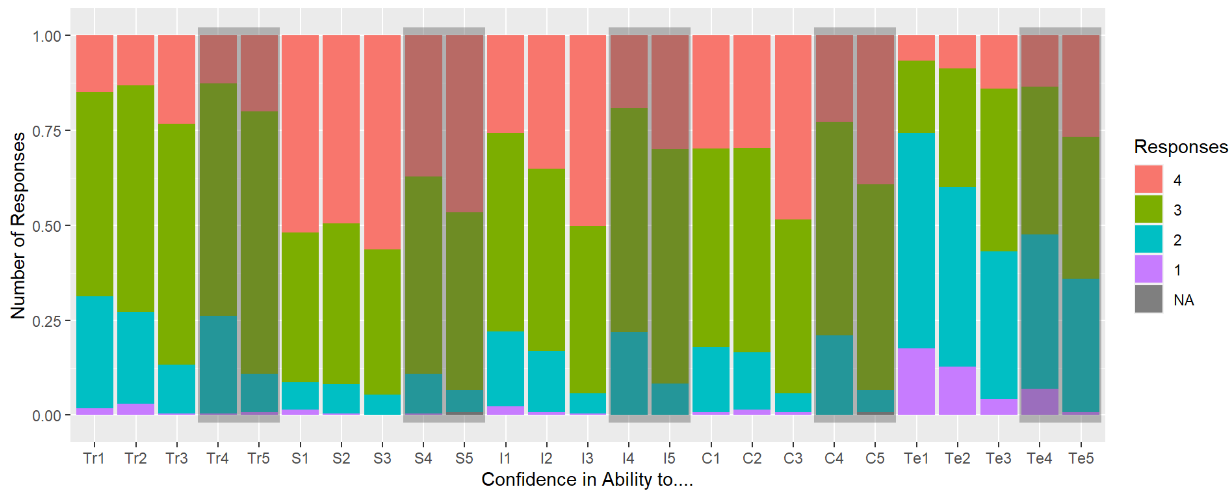

Figure 1: Student Survey Response for Confidence Questions

b. WGCD 2.4: Reason qualitatively and quantitatively.

i. Summary: MA255 increased cadets’ confidence and competence in problem solving. This increase is evident in student performance evaluated by the course end project as well as student self-reported confidence in transforming, solving, interpreting, communicating, and using technology to find and present solutions.

ii. Core Mathematics Goal: Build Competent and Confident Problem Solvers. The course learning outcome that best supports this objective is (IV) Students can design, solve, and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.

iii. Course Objective I - VI: (Transform, Solve, and Interpret). The course applies students’ understanding of the USMA Modeling Process learned in MA153 the previous semester to complete a course-end project. The MA255 Project is a group-based (2- or 3-person), out-of-class exercise that demonstrates various interdisciplinary applications of multivariable calculus. Requirements for the project include the completion of a written report and an oral presentation. Students chose a topic from a pool of five (5) scenarios that was of greatest interest to them. Project topics included: civil engineering, small arms dynamics, future vertical lift, operations research, and sabermetrics. The objective for allowing students a choice was to increase interest in the project, promote independent learning of the topic, and vary project presentations by each group.

Cadets performed exceptionally well in the project. The average grade for the course project was 93.5%.

iv. Student Reported Confidence: Mid-course and Course end Surveys. As seen in Figure 1 above, students reported higher accounts of "very confident" for every question posed, including transform, solve, and interpret. We attribute this to the fact that the survey was presented after the completion of the course project. Their work on this assessment allowed them the opportunity to communicate their research, methodology, solution, and interpretation of different ill posed problems. They did so exceptionally well as evident in the average grade and this allowed them the opportunity to exercise these learning objectives.

c. Overview of other Core Mathematics Goals:

i. Develop Habits of Mind. The course makes a continuous effort to promote self-study and learning of mathematics outside of the classroom. This semester, we included the video lessons and *master it* questions on WebAssign as an optional resource to aid students in their understanding of the material. We saw in the course surveys a significant number of responses stating that this was a "Very Helpful" resource. This, along with the strong number of responses in favor of WebAssign and low use of Additional Instruction, indicates our population is comprised of strong independent learners. Unfortunately, the use of the textbook was ranked lower than other resources. See Figure 2.

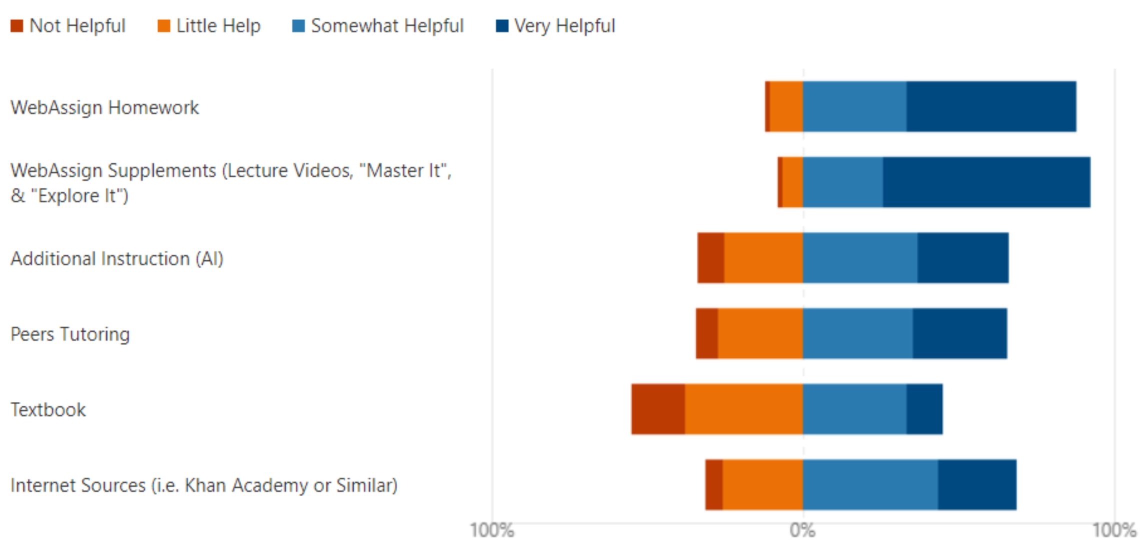

Figure 2: Student Survey Response for Helpful Resources Question

ii. Interdisciplinary Problem Solving. The course presents a range of topics that are presented in a variety of applications including motion in space, optimization, and physics applications like work and flux. However, the highlighting event used to promote interdisciplinary problem solving is the course-end project. Students were given a set of five different project topics to choose from, all from different disciplines and applications of course material. Students performed exceptionally well and presented, in general, quality reports and presentations. The course project is described in Section 3b and in Section 5b as a course evaluation tool.

iii. Apply Technology. We use Mathematica as a tool to solve and visualize the complex concepts in this class. Each of the four (4) Tech Labs was created to highlight and support the material covered in each section from the textbook. Instructors were encouraged to include Mathematica examples in every lesson and students were encouraged to solve each section's portion of the Tech Lab as the block of instruction progressed. Students were allotted a Tech Lab day at the end of each block to get any necessary help with their Mathematica code or concepts, and to allow students to finalize their Tech Lab work. The material covered in the course is well suited for Mathematica, due to the ease of creating plots and the symbolic solution options available. The course surveys demonstrate that some students remain apprehensive of learning and using Mathematica, but in general their skills with technology improved over the semester. More importantly, their individual confidence over the semester also improved. This is evident in Figure 1 above.

iv. Communicate Effectively. There were minimal opportunities at the course level for students to exercise their communication during the first three blocks of instruction. This, along with the classroom capacity limitations posed by COVID restrictions, diminished instructors' abilities to use board problems and emphasize communication skills. Students' confidence levels were lower than expected after the midcourse survey as seen in Figure 1 above. This is something that need to be considered for future iterations of the course. The course-end project, however, proved to be a successful tool to exercise our students' written and verbal communication. The course-end survey showed a significant increase in student confidence as evident in Figure 1 above.

6. Location of Course Documents:
https://usarmywestpoint.sharepoint.com/:f:/r/sites/math.teaching/Shared%20Documents/Teaching/Program%20Director%20Information/Course%20End%20Reports/AY21?csf=1&web=1&e=c2ZZ11

Course syllabus & schedule:	\ 01 Course Guide & Schedule
WPRs & TEE:	\ 02 Exams
Course Project & TechLabs:	\ 03 Course Assignments
Final Grades & OML:	\ 04 Final Grades & OML
Course Surveys:	\ 05 Course Surveys

Final OML \ 06 Final Grades

IVAN D. BERMUDEZ
MAJ, LG
Course Director

DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
West Point, NY 10996-1786

MADN-MATH 16 December 2020

MEMORANDUM FOR RECORD

SUBJECT: Student Memorandum for MA255 - Advanced Multivariable Calculus

1. **Purpose**. This memorandum outlines guidance for students in MA255 AY21-2

2. **Requirements.**

   a. Text. *Multivariable Calculus (Loose-leaf)*, 9th Edition, James Stewart, Copyright 2016.

   b. Software. *Mathematica* – Wolfram Research, Inc. and Microsoft Office (*Word*, *Excel*, *PowerPoint, MS Teams*)

   c. Online. *WebAssign/Cengage* – www.webassign.net, *Blackboard* usma.blackboard.com

3. **Attendance.**

   a. MA255 is a 4.5 credit hour course. It does not follow the Day 1 / Day 2 schedule of a 3.0 credit hour course. The course schedule can be found on the MA255 Blackboard site under Course Resources. https://usma.blackboard.com/

   b. The first two weeks of MA255 AY21-2 will be presented fully remote, over MS Teams or Blackboard Collaborate. After the first two weeks, the course will transition to the hybrid-rotate synchronous form; meaning that for each lesson, half of the class will attend in person while the other half will attend over MS Teams/or Blackboard Collaborate. This allows the maximum amount of in-classroom time possible while maintaining all prescribed COVID-19 safety measures. Each section will be assigned to one of two "chalks" and each chalk will alternate in-person attendance as described in the course schedule.

   c. Attendance at Written Partial Reviews (WPRs) is mandatory. Except in emergencies, cadets will not make any appointments that could preclude their attendance for a WPR. Any cadet missing a WPR will be required to take a make-up or make-ahead exam.

4. **Course Learning Outcomes.** This course is structured to develop the six core mathematics program goals with focus on 10 Student Leaning Outcomes.

a. **Build a body of knowledge: (1)** Students can use vectors and vector functions to model motion and surfaces in three dimensional space. **(2)** Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. **(3)** Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. **(4)** Students understand and apply the concepts of vector calculus.

b. **Communicate Effectively: (5)** Students improve the quality of both their written and verbal communication of mathematics**.** Effective communication is an essential characteristic of great leaders. You, as a future leader, must be able to clearly articulate your thoughts. You will get many opportunities to improve your communications through in-class board presentations, course homework assignments, technology labs, and the course project.

c. **Apply Technology: (6)** Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. You will become competent in using a computer algebra system (Mathematica) through in-class learning and self-paced technology labs. You will leverage technology to assist in your daily learning as well as in completing the course project. Ultimately, you will understand the connection between the mathematical theory and its implementation using technology.

d. **Build Competent and Confident Problems Solvers: (7)** Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus. The problem solving will be integrated in class problems, technology labs and the course project. You will hone your problem-solving skills by formulating and solving challenging mathematical problems.

e. **Provide Experience in Interdisciplinary Problem Solving: (8)** Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines. **(9)** Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems. There will be multiple opportunities in class and during the course lectures for you to see real world problems from a variety of disciplines. You will see how mathematics is used to solve these problems.

f. **Develop Habits of Mind: (10)** Students learn good scholarly habits for progressive intellectual development. You will learn habits in this course that will benefit you throughout your academic career. You will develop an intellectual curiosity in general, and for mathematics, by doing challenging but interesting mathematical problems. You will improve your work ethic and critical thinking through learning and doing mathematics. You will develop skills to be independent learners through the conduct of the course. These habits of mind are essential traits of leaders and scholarly students.

5. **Lesson Assignments.** The Course Schedule and Course Gide serve as the road map for the course. These documents are both available in the MA255 Blackboard site under the Course Resources tab.

6. **Course Summary.** There are a total of **63** required attendances consisting of **56 Lessons** and **6 Problem Solving Labs,** which are further broken down as follows:

56 Lessons:

Lesson based on course text:	41
Technology Labs	4
WPRs	4
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6 Problems Solving Labs:

WPR Review Days	4
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7. **Course Evaluation Plan**

Category	Event	Pts	PCT	
Individual Exams	TEE	900	30.00%	70.00%
	WPR1	300	10.00%	
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HW	WebAssign HW 1	50	1.67%	6.67%
	WebAssign HW 2	50	1.67%	
	WebAssign HW 3	50	1.67%	
	WebAssign HW 4	50	1.67%	
Instructor	Instructor Points	250	8.33%	8.33%
	Total	3000	100.00%	

8. **Grading**

a. Final course grades will follow the Department of Mathematical Sciences' guidelines:

Letter Grade	% Range	Quality Points	Subjective Interpretation
A+	97 ≤ Mark ≤ 100	4.33	Beyond expectations of course
A	93 ≤ Mark < 97	4	Dominates the material
A-	90 ≤ Mark < 93	3.67	Mastery
B+	87 ≤ Mark < 90	3.33	Excellent performance
B	83 ≤ Mark < 87	3	Good understanding
B-	80 ≤ Mark < 83	2.67	Proficient; Aptitude for the subject
C+	77 ≤ Mark < 80	2.33	Can build upon this foundation
C	73 ≤ Mark < 77	2	Passing; Proficient now (short range)
C-	70 ≤ Mark < 73	1.67	Short-range understanding
D	65 ≤ Mark < 70	1	Marginal performance with some elementary understanding
F	Mark < 65	0	Failing; Definitely failed to demonstrate understanding

b. Cadets who score less than 50% on the TEE, regardless of their final course average, or whose final course average is less than 65% may fail the course.

c. Instructors may award up to 30 bonus points at their discretion. Bonus points will be added directly to the total number of points you earn in the course.

9. **Documentation**. All cadet work in MA255 will be in accordance with the procedures found in the Dean's *Documentation of Academic Work* (DAW), *June 2018*. All required out-of-class work constitutes homework as defined in the DAW. The course standard for MA255 is MLA. Refer to the DAW for further information. Proper documentation is required for all graded exercises assigned by your instructor and all out-of-class course-wide assignments outlined in this memorandum.

10. **Out-of-Class Assessments.**

a. **Daily Homework.** Daily web-based homework problems will be assigned via WebAssign. After you submit each assignment, you will receive immediate feedback, to include your grade. Additionally, all daily WebAssign problem sets require students to complete an online acknowledgement statement. Failure to complete the documentation steps outlined here will result in a score of zero for the given assignment. Lastly, MA255 uses the following *WebAssign* assignment extension policy: students may ask for up to two *automatic* extensions, each three days in duration, for a deduction of 10% per period. The point deduction only applies to problems not completed prior to the assignment deadline. A *WebAssign* automatic extension is an assignment extension for which instructor permission is *not* required. Students may use automatic extensions for a 14 day period after the exercise is assigned, or through the end of the given block of instruction, whichever is earlier.

b. **Technology Labs.** There is a technology lab associated with each block of instruction in MA255 which involve solving mathematical problems with a computer using the *Mathematica*. Technology labs will be made available at the beginning of the block and are due before the end of the block. Your instructor will be available during a dedicated tech lab period to provide required assistance, such as help with computer syntax/functions involved in the assignments.

c. **Project.** The MA255 Project is a group-based (2- or 3-person), out-of-class, graded exercise that will demonstrate various interdisciplinary applications of multi-variable calculus. The project will require completion of a *written report* and an *oral presentation*.

11. **In-class Assessments.**

a. **Written Partial Reviews.** WPRs assess your understanding of lesson objectives in the course. Each WPR is cumulative (meaning any content covered up to the point may be assessed) but focuses on content from the corresponding block. Attendance at WPRs is mandatory as described in Section 3 Attendance.

b. **Term End Exam.** The TEE is mandatory and cumulative for the entire course.

12. **Classroom Rules**. All cadets are expected to maintain proper military bearing and appearance (both remote and in-classroom) in accordance with appropriate regulations.

13. **Chain of Command**. The Department of Mathematical Sciences has an open-door policy. Your initial attempt to address problems or concerns – including respect in the classroom, unprofessional behavior, sexual assault, harassment – should go through the chain of command:

a. Your instructor.

b. MA255 Course Director: MAJ Bermudez, Ivan, Office: TH255B, DSN: 938-5905.

c. Advanced Mathematics Associate Program Director: LTC Starling, James, Office TH255A, DSN: 938-4544.

d. Core Mathematics Program Director: COL Lindquist, Joseph, Office TH225, DSN: 938-2807.

e. Head of the Department: COL Hartley, Tina, Office TH238, DSN: 938-5285.

However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain to include the Head of the Department. The entire math department faculty is here to help you.

JAMES K. STARLING
LTC, FA (FA49)
Associate Program Director

AYT21-2		MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SAT/SUN
Block 1: Vectors and Geometry of Space	1	18-Jan-21 N/A No Class RSOI 8 MLK	19-Jan-21 1-1 Lesson 1 Course Intro	20-Jan-21 1-2 Lesson 2 3D Coordinate Systems (12.1)	21-Jan-21 2-1 Lesson 3 Vectors (12.2)	22-Jan-21 1-3 Lesson 4 Dot Product/Cross Product (12.3 / 12.4)	A/D
	2	25-Jan-21 2-2 Lesson 5 Equations of Lines and Planes (12.5)	26-Jan-21 1-4 Lesson 6 Cylinders and Quadric Surfaces (12.6)	27-Jan-21 1-5 No Class Drop	28-Jan-21 2-3 Lesson 7 Vector Functions & Space Curves (13.1)	29-Jan-21 1-6 Lesson 8 Derivatives & Integrals of Vector Functions (13.2)	A/C
	3	1-Feb-21 2-4 First Hybrid Day Lesson 9 Arc Length & Curvature (13.3)	2-Feb-21 1-7 Lesson 10 Velocity & Acceleration (13.4)	3-Feb-21 N/A Study Day Majors Open House	4-Feb-21 1-8 Lesson 11 Tech Lab 1	5-Feb-21 2-5 PSL 01 WPR Review Due: Tech Lab 1	A/C
Block 2: Multivariable Differentiation	4	8-Feb-21 1-9 D Lesson 12 WPR 1	9-Feb-21 2-6 No Class Drop	10-Feb-21 1-10 Lesson 13 Functions of Several Variables (14.1)	11-Feb-21 2-7 Lesson 14 Limits & Continuity (14.2)	12-Feb-21 1-11 Lesson 15 Partial Derivatives (14.3)	B 500th Night
	5	15-Feb-21 N/A No Class President's Day	16-Feb-21 2-8 Lesson 16 Tangent Planes & Linear Approximations (14.4)	17-Feb-21 1-12 D Lesson 17 Chain Rule (14.5)	18-Feb-21 2-9 Lesson 18 Directional Derivatives & Gradient Vector (14.6)	19-Feb-21 1-13 Lesson 19 Max & Min Values I (14.7)	A/D YWW
	6	22-Feb-21 2-10 Lesson 20 Max & Min Values II (14.7) Lagrange Multipliers (14.8)	23-Feb-21 1-14 D Lesson 21 Lagrange Multipliers II (14.8)	24-Feb-21 N/A Study Day	25-Feb-21 1-15 D Lesson 22 Tech Lab 2	26-Feb-21 2-11 PSL 02 WPR 2 Review Due: Tech Lab 2	A/C
Block 3: Multivariable Integration	7	1-Mar-21 1-16 D Lesson 23 WPR 2	2-Mar-21 2-12 No Class Drop	3-Mar-21 1-17 Lesson 24 Double Integrals over Rectangles I (15.1)	4-Mar-21 2-13 Lesson 25 Double Int. over Rectangles II & General Regions I (15.1 / 15.2)	5-Mar-21 1-18 Lesson 26 Double Integrals over General Regions II (15.2)	B Plebe Weekend
	8	8-Mar-21 N/A No Class USMA	9-Mar-21 2-14 Lesson 27 Double Integrals in Polar Coordinates (15.3)	10-Mar-21 N/A Study Day	11-Mar-21 1-19 D Lesson 28 Applications of Double Integrals (15.4)	12-Mar-21 2-15 Lesson 29 Triple Integrals (15.6)	A/D 100th Night
	9	15-Mar-21 1-20 Lesson 30 Triple Integrals in Cylindrical Coordinates (15.7)	16-Mar-21 2-16 Lesson 31 Triple Integrals in Spherical Coordinates (15.8)	17-Mar-21 1-21 No Class Drop	18-Mar-21 2-17 Lesson 32 Triple Integrals in Spherical Coordinates (15.8)	19-Mar-21 1-22D PSL 03 WPR 3 Review	A/C
	10	22-Mar-21 2-18 Lesson 33 WPR 3	23-Mar-21 1-23 Lesson 34 Tech Lab 3	24-Mar-21 N/A Study Day	25-Mar-21 1-24D Lesson 35 Project Introduction Due: TL 3	26-Mar-21 2-19 Lesson 36 Vector Fields (16.1)	A/C SH Validation
Block 4: Vector Calculus	11	29-Mar-21 1-25 Lesson 37 Line Integrals I (16.2)	30-Mar-21 N/A No Class Honorable Living Day	31-Mar-21 2-20 Lesson 38 Line Integrals II (16.2)	1-Apr-21 1-26D Lesson 39 Fundamental Theorem of Line Integrals (16.3)	2-Apr-21 2-21 Lesson 40 Green's Theorem I (16.4)	B
	12	5-Apr-21 N/A No Class USMA	6-Apr-21 1-27 Lesson 41 Green's Theorem II (16.4)	7-Apr-21 N/A Study Day	8-Apr-21 1-28 Lesson 42 Curl & Divergence (16.5)	9-Apr-21 2-22 Lesson 43 Parametric Surfaces I and their Areas (16.6)	A/D
	13	12-Apr-21 1-29 Lesson 44 Parametric Surfaces II and their Areas (16.6)	13-Apr-21 2-23 Lesson 45 Surface Integrals I (16.7)	14-Apr-21 1-30 Lesson 46 Surface Integrals II (16.7)	15-Apr-21 2-24 Lesson 47 Project IPR	16-Apr-21 Modified 1-31 Lesson 48 Project IPR	A/C Sandhurst
	14	19-Apr-21 2-25 Lesson 49 Stokes' Theorem I (16.8)	20-Apr-21 1-32 Lesson 50 Stokes' Theorem II (16.8)	21-Apr-21 N/A Study Day	22-Apr-21 2-26 Lesson 51 Divergence Theorem (16.9)	23-Apr-21 1-33 Lesson 52 Tech Lab 4	A/D
Project and TEE	15	26-Apr-21 2-27 PSL 04 WPR 4 Review Due: Tech Lab 4	27-Apr-21 N/A Study Day WPR 4	28-Apr-21 1-34 No Class Drop	29-Apr-21 N/A No Class USMA Project's Day	30-Apr-21 2-28 Lesson 53 Project Work Day	A/C
	16	3-May-21 1-35 Lesson 54 Project Presentations	4-May-21 2-29 Lesson 55 Project Presentations	5-May-21 1-36D Lesson 56 Project Presentations	6-May-21 N/A Study Day	7-May-21 1-37 No Class Drop	A/D
	17	10-May-21 2-30 TEE Review	11-May-21 1-38D TEE Review	12-May-21 N/A No Class TEE Week	13-May-21 N/A No Class TEE Week	14-May-21 N/A No Class TEE Week	15-May-21 No Class TEE Week

KEY			
	No Class	Problem Solving Lab	WPR
	Project	Tech Lab	Due Date or Mod Sched.

TAB B AY2022 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-MATH 31 MAY 2022

MEMORANDUM FOR LTC Starling, Associate Program Director, Advanced Math Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

SUBJECT: Course End Report for MA255, AY22-2

1. **Background.** The goal of the Core Advanced Mathematics Program (AMP) is to produce confident students who possess the knowledge, skills, and interest necessary for future studies of science, technology, engineering, and mathematics (STEM). However, we also provide exposure to higher level mathematics for students who do not become STEM majors. A list of detailed course objectives is found in the Course Memo. MA255 used the following required resources:

a. *Text*: Calculus Early Transcendentals, 9th Edition, James Stewart (2021), chapters 11 through 16.

b. *Software*: Mathematica and Microsoft Office (MS Teams, Word, PowerPoint).

c. *Online*: WebAssign (*www.webassign.net*) and Blackboard *(usma.blackboard.com*)

2. **Enrollment and Validation.** The course consisted of 21 sections scheduled during A, B, C, D, and E hours. This is a change from previous years where math courses were almost exclusively taught during the morning with very few sections in the afternoon.

a. *Enrollment*: 316 Cadets and one (1) cadet candidate enrolled during FY22-2.

b. *Sectioning*: 21 Sections, 8 dedicated instructors. Four sections were designated as advanced or “honors” sections. If scheduling permitted, the 16 students with the highest scores in MA153 were scheduled for Fr. Costa’s MA255 section (the only section he teaches). MAJ Mussmann also taught students who achieved grades in the top 20% of their MA153 course. This allowed the flexibility to schedule the top tier of AMP students in either A, C, or D hours. We recommend continuing this practice because it allows instructors to create a tailored, challenging environment for the brightest students in the course.

c. *USMAPS*: We maintain our relationship with the Math Department at USMAPS and offered the enrollment of cadet candidates who showed the propensity to do well in the Program. One (1) cadet candidate completed MA255 during this academic year. Grades and performance for the cadet candidate are not included in the remainder of the report.

d. *Validation*: We offered validation exams for Multivariable Calculus (MA205 validation credit) and Vector Calculus (MA255 validation credit) to students enrolled in MA153 during the fall AY 22-1.

i. *Multivariable Calculus*: One (1) student successfully validated MA205 out of the 12 who participated. (This is a decrease from seven (7) out of 14 participants in AY21-2.)

ii. *Vector Calculus*: Nine (9) students successfully validated MA255 out of the 12 who participated. (This is an increase from four (4) validations out of twelve (12) participants in AY21-2.)

e. *Off-ramp cadets*: In addition to the cadets who have validated MA205 and MA255 above, we also had six (6) cadets exit the AMP for various reasons. This is down from eighteen (18) in AY21-2. Most of these cadets did not perform well in MA153 and were offered the opportunity to enter MA103 (if exiting during MA153), MA104 (and vacate the MA104 waiver), or MA205 (and keep the MA104 waiver). We believe this reduction in losses was due to the writ taken in the first two weeks of MA153 covering essential skills students would need to be successful in the AMP. The fall semester began with 354 cadets in the program, however, the early assessments provided thirty-two (32) cadets the opportunity to move to MA103 rather than continue in the AMP. Early assessments and the ability to off-ramp should certainly remain in future iterations of the course.

3. **Administration:** The course consisted of four (4) major blocks of instruction: Block I – Vectors and the Geometry of Space; Block II – Infinite Sequences and Series & Multivariable Differentiation; Block III – Multivariable Integration; and Block IV – Vector Calculus. Inclusion of sequences and series is a change from previous years as it was typically introduced in MA153.

a. *Course Summary*. There are a total of 64 required attendances consisting of 56 Lessons and eight (8) Modeling/Project days, further broken down as follows:

56 Lessons:	
Lesson based on course text:	48
Synthesis/Review Days	5
WPRs	3
8 Modeling/Project Days:	
In-Class Project Work	6
Project Presentations	2

b. *Course Evaluation Plan*. There were 3,000 possible course-wide points available. The point distribution is summarized in Table 1.

Category	Event	Pts	PCT	
Individual Exams	TEE	900	30.00%	60.00%
	WPR1	300	10.00%	
	WPR2	300	10.00%	
	WPR3	300	10.00%	
Group Work / Modeling/ Technology	Two-Day Project #1	150	5.00%	20.00%
	Two-Day Project #2	150	5.00%	
	Two-Day Project #3	150	5.00%	
	Two-Day Project #4	150	5.00%	
HW	WebAssign HW 1	75	2.50%	10.00%
	WebAssign HW 2	75	2.50%	
	WebAssign HW 3	75	2.50%	
	WebAssign HW 4	75	2.50%	
Instructor	Instructor Points	300	10.0%	10.00%
	Total	3000	100.00%	100.00%

Table 1: Summary of MA255 Grading Scheme.

i. The use of textbooks or computers was not authorized during any examination (WPRs or TEE). The primary implementation of technology, *Mathematica,* was through the four mini-projects (one per block) discussed later.

ii. *Term End Exam (TEE).* Cadets were given a mandatory comprehensive TEE at the end of the semester. This was a departure from the last several years where students with over a 93% course average were allowed to choose to take the exam. The TEE consisted of fourteen (14) questions and covered representative problems and applications from all blocks of instruction but was slightly weighted toward Block IV since there was not a Block IV WPR. Students had 3.5 hours to complete the final examination which accounts for 30% of the final grade.

iii. *Written Partial Review (WPR)*. Cadets were assessed with a WPR at the end of each block of instruction. Cadets could use one (1) page of handwritten notes and their academy- issued TI-36x calculator during examinations. The WPRs included a mix of computational questions, problem solving questions and concept check questions. WPRs accounted for 30% of the final grade.

iv. *Course Mini Projects*. In a shift from previous years, the course included four (4) small projects to exercise interdisciplinary applications, collaboration, mathematical modeling, and written and verbal communication skills. In previous years, this was accomplished through a single, large, nearly month-long semester project. Instead, students conducted four (4) two-day group projects in the days following the WPR. This work, referred to as “mini-projects”, was to highlight the application of the material to a real-world application while providing two opportunities to produce technical reports (mini-projects 1 and 3) and two opportunities to present their findings orally (mini-projects 2 and 4). The prompts and topics to be investigated intentionally asked students to answer questions that would be prohibitively difficult to do without the use of technology. While the prompts did not require students to use Wolfram's Mathematica, correct solutions would be infeasible without using some form of software. This led the project groups (2-3 students each) to find for themselves the utility of tools like Python, R, Excel, and Mathematica. In Block 1, students wrote a report discussing the application of the ballistic equations to our military's indirect fire systems; after WPR 2, students presented research on infinite series or optimized decision variables for construction of a baseball stadium; at the end of Block 3, they wrote reports outlining their recommendations for excavating a road through a mountain; and prior to the TEE, students optimized flight paths for between Air Force bases while accounting for the effects of the jet stream. Overall, the mini-projects accounted for 20% of the final grade.

v. *WebAssign Homework*. Cadets completed WebAssign homework for each lesson in the course. The new edition of the textbook (9^{th} edition) implemented several new features available through WebAssign which were adopted in AY21-2 and continued for this semester. In addition to traditional questions, we implemented *Just in Time* questions, *Master It* questions, *Explore It* questions, and *lecture videos* covering the examples from the textbook. The majority of the *Master It* questions were optional with no credit due, but they served as scaffolding for the questions worth points. The only change from previous years to this framework was an adjustment to a weekly due date. Students were no longer required to do WebAssign on a nightly basis but were encouraged to manage their own time and weekly workload to get the practice needed to do well in the course. WebAssign accounted for 10% of the final grade.

vi. *Instructor Points*. The course had a total of 250 instructor points. Each instructor created their own assessments tailored to their students and teaching. In general, instructor points consisted of quizzes, problem sets, and reviews. Instructor points accounted for 10% of the final grade.

c. *Faculty Development.* All 8 instructors were introduced to the course through two Faculty Development Workshop (FDW) sessions held before and during re-orgy day. Typically, the Academy schedules a "re-orgy" week, but this was truncated to a single day for this semester. In addition, instructors met on a weekly basis for Program Director (PD) meetings to discuss course administration, instructional techniques, teaching tips, available resources, and upcoming lessons, assessments, and requirements. These meetings were critical for ensuring instructors were prepared not only for their daily lessons, but for the projects following each exam. The PD meetings provided the time for the instructors who created the project to brief their drafts and final product. With this, every instructor knew how best to help their cadets discover the utility of the mathematical skills covered in the block. By rotating which instructors are responsible for each of the WPRs and project prompts, all instructors had a role in developing the course content. Instructors also contributed to the weekly PD meetings by offering instructional and mathematical insights.

4. **Course Statistics and Final Grades:** The average scores for each course-wide graded event and the final grade distribution are shown below.

Event	Average
WPR 1	86.00%
WPR 2	87.50%
WPR 3	87.86%
WebAssign	91.00
Projects	90.75%
TEE	89.12%

Table 1: Major Graded Events

Grade	AY22-02
A+	34 (10.75%)
A	106 (33.50%)
A-	61 (19.30%)
B+	37 (11.71%)
B	42 (13.29%)
B-	18 (5.70%)
C+	10 (3.16%)
C	6 (1.90%)
C-	2 (0.63%)
D	0
F/NC	0
Total	316

Table 2: Frequency of Letter Grades

5. **Evaluation of Student Learning Outcomes:** MA255 builds skills to support the four Academic Program Goals (APGs) pertinent to the core mathematics sequence. Each APG is paired with one What Graduates Can Do (WGCD) statement. Table 2 below shows our ten (10) course objectives described in the Course Memo, mapped to the WGCD statements and APGs. We collected supporting information about each of the APGs through a combination of student course evaluations surveys, faculty input, and performance on select course assessment tools (course project, technology labs, WPRs, etc.). We conducted a short mid-course survey and end of course survey to measure student progress along these various goals.

While the course supports all six (6) primary core mathematics goals, the evaluation for this year's course focuses primarily on "Building a Body of Knowledge" and "Build Competent and Confident Problem Solvers". These goals are described in this report primarily by addressing WGCD statements 5.1 and 2.4.

Academic Program Goals / MA255 Assessment Techniques					
Supported Academic Program Goals (From Educating Army Leaders)	1. Graduates communicate effectively with all audiences	2. Graduates think critically and creatively	3. Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development	5. Graduates apply science, technology, engineering, and math concepts and processes to solve complex problems	7. Graduates integrate and apply knowledge and methodological approaches gained through in-depth study of an academic discipline.

MA255 Course Goals (Align with Core Math Goals)	2. Communicate effectively	3. Apply technology; 4. Build competent and confident problem-solvers; 5. Foster an interdisciplinary perspective	6. Develop Habits of Mind; 4. Build competent and confident problem-solvers	3. Apply technology; 4. Build competent and confident problem-solvers	5. Foster an interdisciplinary perspective
Middle States Accreditation Standards	3. Design and Delivery of Student Learning Experience			5. Educational Effectiveness Assessment	
MA255 Course Learning Objectives	5. Students improve the quality of both their written and verbal communication of mathematics.	4. Students understand and apply the concepts of vector calculus. 7. Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus.	6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. 8. Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines. 10. Students learn good scholarly habits for progressive intellectual development.	2. Understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. 4. Understand and apply the concepts of vector calculus.	1. Use vectors and vector functions to model motion in three-dimensional space. 3. Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. 9. Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.
MA255 Assessment Techniques	Mini-projects (2x Oral presentations, 2x written reports), daily in-class board problems, surveys (2)	1x Modeling problem per WPRs (x3) Mini-projects (x4) with interdisciplinary topics	WebAssign Homework (x37) Mini-project read-ahead, independent group work (8x in-class workdays)	2x oral projects requiring visual aids 2x written projects requiring technology to solve and adequately interpret results	1x Modeling problem per WPRs (x3) Mini-projects (x4) with interdisciplinary topics

Table 3: Course Objective Mapping to APGs.

a. WGCD 5.1 Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.

i. Summary: MA255 increased cadets' body of knowledge by introducing a widespread set of topics. This increase is evident in student performance evaluated by major graded event grades, a comparison of TEE results across the years, and student self-reported confidence in solving multivariable calculus problems.

ii. Core Mathematics Goal: Build a Body of Knowledge: The learning outcomes that support this goal are (VII) Students can use vectors and vector functions to model motion and surfaces in three-dimensional space. (VIII) Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. (IX) Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. (X) Students understand and apply the concepts of vector calculus.

iii. Course Objective VII - X (Multivariable and Vector Calculus). We give a comprehensive treatment of vector geometry, multivariable calculus (differentiation and integration), and vector calculus (line and surface integrals, Fundamental Theorem of Line Integrals, Green's Theorem, Stokes' Theorem, and the Divergence Theorem). This is extremely challenging material to learn, understand, and master. We measure our effectiveness in this task primarily through graded assessments. A combination of WPR, WebAssign, and the Term End Exam (TEE) grades provides the best estimate of effectiveness in this task. Major graded event results are presented in Section 4: Course Statistics and Final Grades.

The TEE period returned to the standard 3.5-hours as in years preceding COVID-19. Because we did not conduct a Block 4 (Vector Calc) WPR, the TEE covered all major topics from Block 4 while still being comprehensive and pulling representative questions from previous TEEs to assess the other three blocks. By using the same questions from the past 7-10 years of TEEs, we can have a consistent, year-over-year metric comparison. Table 3 below gives a comparison of TEE performance from AY18-2 to AY22-2.

Term	**Number of Cadets**	**Avg TEE Points**	**Avg Grade**	**Note:**
AY 22-2	316	802	89.12	13 Questions 3.5hr Exam (Block 4 Heavy)
AY 21-2	134	796.23	88.47%	9 Questions 2.5 hr Exam Students with >93% pre-TEE grade optional
AY 20-2	276	810.11	90.01%	Full Remote Exam
AY 19-2	279	835.59	92.84%	12 Questions 3.5 hr Exam
AY 18-2	250	823.86	91.54%	12 Questions 3.5 hr Exam

Table 2: TEE Grade Average for AY18-2 to AY22-2

Average scores on this year's TEE were about 1% lower on average than in previous years with full-length (3.5hr) exams. It's worth noting, however, that this year's cohort was substantially larger than previous years, AY18 and AY19 only had 12 questions, and previous exams did not test as many Vector Calculus concepts as this year's exam. As is typical, students struggled with questions about lines, tangent vectors, and planes. The same discrepancy is seen in previous iterations of the course and may be due to the time-distance between this material and the TEE. Some intentional interleaving could alleviate this, but care should be taken not to "teach to the test" shortly before the final exam. Unique to this year is the students' apparent difficulty answering a question using the Fundamental Theorem of Line Integrals (FTLI). It's possible that students struggled to see this as an FTLI problem because the question before it could also have been solved using the theorem. In a basic line integral question with a conservative vector field, students could use either a standard line integral or they could apply the FTLI. If they used the FTLI earlier in the exam, there's a chance that in the subsequent question—

which is very difficult to compute without the theorem—students were deterred from using the theorem a second time

iv. Student Reported Confidence: Mid-course and Course end Surveys. Our students demonstrated an increased confidence in all aspects of the modeling process including Solve. Figure 1 below displays the self-reported confidence level for the various surveys presented in MA153 and MA255. Columns Solve3 and Solve4 highlight student confidence in Solving at the MA255 midcourse survey and end of course survey, respectively. While those reporting "very confident" (4) regressed after Block 4, the total number of students reporting a lack of confidence (levels 1 and 2) nearly disappears. This same behavior is noted in the students' communication skills in columns Com3 and Com4. Again, by the end of MA255 and the AMP, the majority of cadets are confident or very confident in their abilities to give written and oral technical reports.

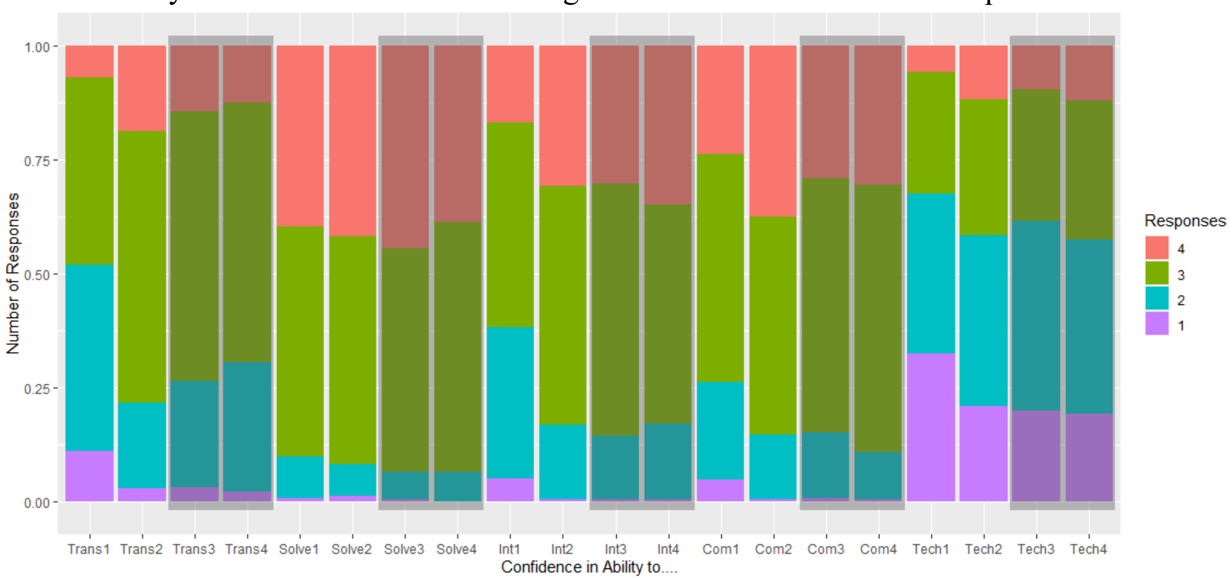

Figure 1: Student Survey Response for Confidence Questions

b. WGCD 2.4: Reason qualitatively and quantitatively.

i. Summary: MA255 increased cadets' confidence and competence in problem solving. This increase is evident in student performance evaluated on the four course mini-projects as well as student self-reported confidence in transforming, solving, interpreting, communicating, and using technology to find and present solutions.

ii. Core Mathematics Goal: Build Competent and Confident Problem Solvers. The course learning outcome that best supports this objective is (IV) Students can design, solve, and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.

iii. Course Objective I - VI: (Transform, Solve, and Interpret). The course applies students' understanding of the USMA Modeling Process learned in MA153 the previous semester to complete four small projects. The MA255 projects described in section 3 of this report provide cadets with several opportunities to apply the mathematical skills taught in each block to a variety of problems one may see in the real world. Cadets performed quite well on these projects and saw an average grade increase of 3% from the first two projects to the last two projects.

iv. Student Reported Confidence: Mid-course and Course end Surveys. As seen in Figure 1 above, students reported higher accounts of "very confident" for both the "solve" and "interpret" categories when compared to MA153. While MA153 is the course in which the AMP focuses on modeling, MA255 is very much a "solution technique" course. Students learn a variety of skills and focus on applying multivariable mechanics accurately. Nevertheless, we used the mini-projects to build in modeling opportunities. It appears from Figure 1 that virtually every student in the AMP is confident in their ability to interpret and communicate mathematical information.

c. Overview of other Core Mathematics Goals:

i. Develop Habits of Mind. The course makes a continuous effort to promote self-study and learning of mathematics outside of the classroom. We continued to use the *master it* questions on WebAssign as an optional resource to aid students in their understanding of the material. Once again, the course survey indicates a significant number of students find this to be a "Very Helpful" resource. This indicator, along with the responses in favor of WebAssign and low use of Additional Instruction, indicates our population is comprised of strong independent learners. Unfortunately, cadets still refrain from using the textbook. Even more concerning is that this year's survey indicates students do not seek out peer tutoring as much as previous years. This may be an outlying instance for this cohort—perhaps due to the social distancing/remote learning this group learned in high school, but it's worth keeping track of how frequently cadets lean on each other for tutoring. See Figure 2 below.

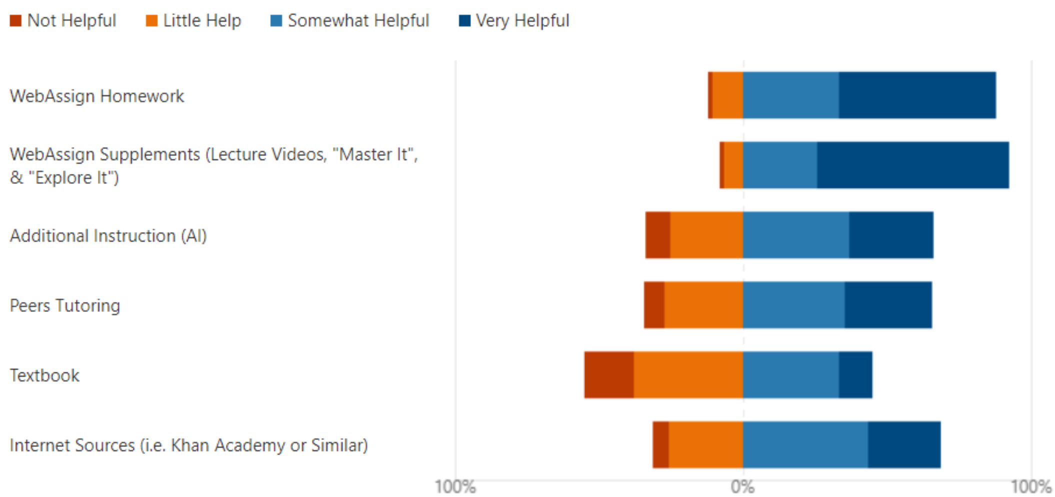

Figure 2: Student Survey Response for Helpful Resources Question

ii. Interdisciplinary Problem Solving. The course presents a range of topics with a variety of applications including motion in space, optimization, and physics applications like work and flux. However, the highlighting events used to promote interdisciplinary problem solving are the course mini-projects. The course projects are described in Section 3 of this report and they serve as opportunities to expose cadets to several topics including civil engineering, optimization, and ballistics.

iii. Communicate Effectively. Each student worked as part of a three-person team to write two technical reports and brief two oral reports to their class. In previous years the course had only one major course project and lamented the lost opportunity to hone cadets' communication skills. The advent of the four two-day projects solved this problem and proved to be effective.

6. Location of Course Documents:
https://usarmywestpoint.sharepoint.com/sites/math.teaching/Shared%20Documents/Forms/AllItems.aspx?newTargetListUrl=%2Fsites%2Fmath%2Eteaching%2FShared%20Documents&viewpath=%2Fsites%2Fmath%2Eteaching%2FShared%20Documents%2FForms%2FAllItems%2Easpx&id=%2Fsites%2Fmath%2Eteaching%2FShared%20Documents%2FCore%20Courses%2Fccma255%2FMA255%2022%2D2%20%28MV%20Calc%29&viewid=a15e8bfd%2D4cbd%2D462d%2D9867%2D0752be80ad62

Course syllabus & schedule:	\ 01 Admin \02 Course Documents
Exams:	\ 02 WPRs
Mini-Projects:	\ 03 Projects
Final Grades & OML:	\ 01 Admin \03 Final Grades
Course Surveys:	\ 01 Admin \05 PD Analysis Surveys

CALEB J. SHEFFIELD
MAJ, FA49/IN
Course Director

DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
West Point, NY 10996-1786

MADN-MATH 3 January 2022

MEMORANDUM FOR RECORD

SUBJECT: Student Memorandum for MA255 - Advanced Multivariable Calculus

1. **Purpose**. This memorandum outlines guidance for students in MA255 AY22-2

2. **Requirements.**

   a. Text. *Multivariable Calculus (Loose-leaf)*, 9th Edition, James Stewart, Copyright 2016.

   b. Software. *Mathematica* – Wolfram Research, Inc. and Microsoft Office (*Word*, *Excel*, *PowerPoint, MS Teams*)

   c. Online. *WebAssign/Cengage* – www.webassign.net, *Blackboard* usma.blackboard.com

3. **Attendance.**

   a. MA255 is a 4.5 credit hour course. It does not follow the Day 1 / Day 2 schedule of a 3.0 credit hour course. The course schedule can be found on the MA255 Blackboard site under Course Resources. https://usma.blackboard.com/

   b. Attendance at Written Partial Reviews (WPRs) is mandatory. Except in emergencies, cadets will not make any appointments that could preclude their attendance for a WPR. Any cadet missing a WPR will be required to take a make-up or make-ahead exam.

4. **Course Learning Outcomes.** This course is structured to develop the six core mathematics program goals with focus on 10 Student Leaning Outcomes.

   a. **Build a body of knowledge: (1)** Students can use vectors and vector functions to model motion and surfaces in three dimensional space. **(2)** Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. **(3)** Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. **(4)** Students understand and apply the concepts of vector calculus.

   b. **Communicate Effectively: (5)** Students improve the quality of both their written and verbal communication of mathematics**.** Effective communication is an essential characteristic of great leaders. You, as a future leader, must be able to clearly articulate your thoughts. You will get many opportunities to improve your communications through in-class board presentations, course homework assignments, technology labs, and the course project.

c. **Apply Technology: (6)** Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. You will become competent in using a computer algebra system (Mathematica) through in-class learning and self-paced technology labs. You will leverage technology to assist in your daily learning as well as in completing the course project. Ultimately, you will understand the connection between the mathematical theory and its implementation using technology.

d. **Build Competent and Confident Problems Solvers: (7)** Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus. The problem solving will be integrated in class problems, technology labs and the course project. You will hone your problem-solving skills by formulating and solving challenging mathematical problems.

e. **Provide Experience in Interdisciplinary Problem Solving: (8)** Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines. **(9)** Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems. There will be multiple opportunities in class and during the course lectures for you to see real world problems from a variety of disciplines. You will see how mathematics is used to solve these problems.

f. **Develop Habits of Mind: (10)** Students learn good scholarly habits for progressive intellectual development. You will learn habits in this course that will benefit you throughout your academic career. You will develop an intellectual curiosity in general, and for mathematics, by doing challenging but interesting mathematical problems. You will improve your work ethic and critical thinking through learning and doing mathematics. You will develop skills to be independent learners through the conduct of the course. These habits of mind are essential traits of leaders and scholarly students.

5. **Lesson Assignments.** The Course Schedule and Course Guide serve as the road map for the course. These documents are both available in the MA255 Blackboard site under the Course Resources tab.

6. **Course Summary.** There are a total of **64** required attendances consisting of **56 Lessons** and **8 Problem Solving Labs** split into two parts each. A more thorough break down follows:

56 Lessons:

Lesson based on course text:	48
Synthesis/Review Days:	5
WPRs:	3

8 Problems Solving Labs:

2-Day Projects: 4

7. **Course Evaluation Plan**

Category	Event	Pts	PCT	
Individual Exams	TEE	900	30.00%	60.00%
	WPR1	300	10.00%	
	WPR2	300	10.00%	
	WPR3	300	10.00%	
Group Work / Modeling/ Technology	Two-Day Project #1	150	5.00%	20.00%
	Two-Day Project #2	150	5.00%	
	Two-Day Project #3	150	5.00%	
	Two-Day Project #4	150	5.00%	
HW	WebAssign HW 1	75	2.50%	10.00%
	WebAssign HW 2	75	2.50%	
	WebAssign HW 3	75	2.50%	
	WebAssign HW 4	75	2.50%	
Instructor	Instructor Points	300	10.0%	10.00%
	Total	3000	100.00%	100.00%

8. **Grading**

a. Final course grades will follow the Department of Mathematical Sciences' guidelines:

Letter Grade	% Range	Quality Points	Subjective Interpretation
A+	97 ≤ Mark ≤ 100	4.33	Beyond expectations of course
A	93 ≤ Mark < 97	4	Dominates the material
A-	90 ≤ Mark < 93	3.67	Mastery
B+	87 ≤ Mark < 90	3.33	Excellent performance
B	83 ≤ Mark < 87	3	Good understanding
B-	80 ≤ Mark < 83	2.67	Proficient; Aptitude for the subject
C+	77 ≤ Mark < 80	2.33	Can build upon this foundation
C	73 ≤ Mark < 77	2	Passing; Proficient now (short range)
C-	70 ≤ Mark < 73	1.67	Short-range understanding
D	65 ≤ Mark < 70	1	Marginal performance with some elementary understanding
F	Mark < 65	0	Failing; Definitely failed to demonstrate understanding

b. Cadets who score less than 50% on the TEE, regardless of their final course average, or whose final course average is less than 65% may fail the course.

c. Instructors may award up to 30 bonus points at their discretion. Bonus points will be added directly to the total number of points you earn in the course.

9. **Documentation**. All cadet work in MA255 will be in accordance with the procedures found in the Dean's *Documentation of Academic Work* (DAW), *June 2017*. All required out-of-class work constitutes homework as defined in the DAW. The course standard for MA255 is MLA. Refer to the DAW for further information. Proper documentation is required for all graded exercises assigned by your instructor and all out-of-class course-wide assignments outlined in this memorandum.

10. **Out-of-Class Assessments.**

a. **Daily Homework.** Daily web-based homework problems will be assigned via WebAssign. After you submit each assignment, you will receive immediate feedback, to include your grade. Additionally, all daily WebAssign problem sets require students to complete an online acknowledgement statement. Failure to complete the documentation steps outlined here will result in a score of zero for the given assignment. Lastly, MA255 uses the following *WebAssign* assignment extension policy: students may ask for up to two *automatic* extensions, each three days in duration, for a deduction of 10% per period. The point deduction only applies to problems not completed prior to the assignment deadline. A *WebAssign* automatic extension is an assignment extension for which instructor permission is *not* required. Students may use automatic extensions for a 14 day period after the exercise is assigned, or through the end of the given block of instruction, whichever is earlier.

b. **Two-Day Projects.** At the end of each block, students will be assigned a project that synthesized the material covered in the block. These will be group-based (2- or 3-person) and can use out-of-class time to complete the work. The projects will include the effective use of technology and will each use a different mode of communicating the team's results. The purpose is to reinforce the material and demonstrate various interdisciplinary applications of multi-variable calculus.

11. **In-class Assessments.**

a. **Written Partial Reviews.** WPRs assess your understanding of lesson objectives in the course. Each WPR is cumulative (meaning any content covered up to the point may be assessed) but focuses on content from the corresponding block. Attendance at WPRs is mandatory as described in Section 3 Attendance.

b. **Term End Exam.** The TEE is mandatory and cumulative for the entire course.

12. **Classroom Rules**. All cadets are expected to maintain proper military bearing and appearance (both remote and in-classroom) in accordance with appropriate regulations.

13. **Chain of Command**. The Department of Mathematical Sciences has an open-door policy. Your initial attempt to address problems or concerns – including respect in the classroom, unprofessional behavior, sexual assault, harassment – should go through the chain of command:

a. Your instructor.

b. MA255 Course Director: MAJ Sheffield, Caleb, Office: TH255C, DSN: 938-4018.

c. Advanced Mathematics Associate Program Director: LTC Starling, James, Office TH255A, DSN: 938-4544.

d. Core Mathematics Program Director: COL Lindquist, Joseph, Office TH225, DSN: 938-2807.

e. Head of the Department: COL Hartley, Tina, Office TH238, DSN: 938-5285.

However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain to include the Head of the Department. The entire math department faculty is here to help you.

JAMES K. STARLING
LTC, FA (FA49)
Associate Program Director

MA255: Advanced Multivariable Calculus - AY22-2 (Spring 2022) **Printed on: 2/2/2022**

Block	AY 22-2	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SAT/SUN
Block 1: Vectors and Geometry of Space	1	3-Jan-22 N/A No Class REORGY DAY	4-Jan-22 1-1 Lesson 1 Course Intro	5-Jan-22 2-1 Lesson 2 3D Coordinate Systems (12.1)	6-Jan-22 1-2 Lesson 3 Intro to Vectors I (12.2)	7-Jan-22 2-2 Lesson 4 Intro to Vectors II (12.2)	A/C A/C
	2	10-Jan-22 1-3 Lesson 5 Dot Product / Cross Product (12.3 / 12.4)	11-Jan-22 2-3 Lesson 6 Equations of Lines and Planes (12.5)	12-Jan-22 N/A No Class Study Day	13-Jan-22 1-4 Lesson 7 Cylinders and Quadric Surfaces (12.6)	14-Jan-22 2-4 Lesson 8 Vector Functions & Space Curves (13.1)	B B RMC
	3	17-Jan-22 N/A No Class MLK Day	18-Jan-22 1-5 Lesson 9 Derivatives & Integrals of Vec. Funcs (13.2)	19-Jan-22 2-5 Lesson 10 Arc Length, Curvature, and TNB Frame (13.3)	20-Jan-22 1-6 Lesson 11 Arc Length, Curvature, and TNB Frame (13.3 / 13.4)	21-Jan-22 2-6 Lesson 12 Velocity & Acceleration (13.4)	B 500th Night
	4	24-Jan-22 1-7 Lesson 13 Synthesis / Board Problems	25-Jan-22 2-7 Lesson 14 WPR 1 Mini Proj. 1 Released	26-Jan-22 N/A No Class Study Day	27-Jan-22 1-8 PSL 1A Mini Project 1 Ballistics	28-Jan-22 2-8 PSL 1B Mini Project 1 Ballistics	A/D
Block 2: Multivariable Differentiation	5	31-Jan-22 1-9 Lesson 15 Sequences / Series (11.1 / 11.2) Due: Mini Proj. 1 Write-Up	1-Feb-22 2-9 Lesson 16 Alt. Series/Ratio/Root Test (11.5 / 11.6)	2-Feb-22 1-10 Lesson 17 Power / Taylor / Maclaurin (11.8 / 11.10)	3-Feb-22 2-10 Lesson 18 Funcs. of Sev. Vars / Limits & Cont. (14.1 / 14.2)	4-Feb-22 1-11 No Class Drop	B 100th Night
	6	7-Feb-22 1-12 Lesson 19 Limits. & Cont. / Part. Derivs. (14.2 / 14.3)	8-Feb-22 2-11 Lesson 20 Tangent Planes & Linear Approximations (14.4)	9-Feb-22 N/A Study Day	10-Feb-22 2-12 Lesson 21 Chain Rule (14.5)	11-Feb-22 1-13 Lesson 22 Directional Derivatives & Gradient Vector (14.6)	A/D TEST
	7	14-Feb-22 1-14 Lesson 23 Max & Min Values I (14.7)	15-Feb-22 N/A No Class Study Day	16-Feb-22 2-13 Lesson 24 Max & Min Values II (14.7) Lagrange Multipliers (14.8)	17-Feb-22 1-15 Lesson 25 Lagrange Multipliers II (14.8)	18-Feb-22 2-14 Lesson 26 Instructor Day	B YWW
	8	21-Feb-22 N/A No Class Presidents Day	22-Feb-22 1-16 Lesson 27 Synthesis / Board Problems	23-Feb-22 2-15 Lesson 28 WPR 2 Mini Proj. 2 Released	24-Feb-22 1-17 No Class Drop	25-Feb-22 2-16 PSL 2A Mini Project 2 Optimization	A/C
Block 3: Multivariable Integration	9	28-Feb-22 1-18 PSL 2B Mini Project 2 Due: In-Class Presentation	1-Mar-22 2-17 Lesson 29 Double Integrals over Rect. (15.1)	2-Mar-22 N/A No Class Study Day	3-Mar-22 2-18 Lesson 30 Double Int. over Rectangles II & General Regions I (15.1 / 15.2)	4-Mar-22 Modified 1-19 No Class Drop	Plebe Wkn Plebe Weekend
	10	7-Mar-22 N/A No Class Spring Break	8-Mar-22 N/A No Class Spring Break	9-Mar-22 N/A No Class Spring Break	10-Mar-22 N/A No Class Spring Break	11-Mar-22 N/A No Class Spring Break	Spg Brk
	11	14-Mar-22 1-20 Lesson 31 Double Integrals over General Regions II (15.2)	15-Mar-22 2-19 Lesson 32 Double Integrals in Polar Coordinates (15.3)	16-Mar-22 1-21 No Class Drop	17-Mar-22 2-20 Lesson 33 Applications of Double Integrals I (15.4)	18-Mar-22 1-22 Lesson 34 Applications of Double Integrals II (15.4)	A/C
	12	21-Mar-22 2-21 Lesson 35 Triple Integrals (15.6)	22-Mar-22 1-23 Lesson 36 Triple Integrals in Cylindrical Coordinates (15.7)	23-Mar-22 N/A No Class Study Day	24-Mar-22 2-22 Lesson 37 Triple Integrals in Spherical Coordinates (15.8)	25-Mar-22 1-24 Lesson 38 Triple Integrals in Spherical Coordinates (15.8)	A/D
	13	28-Mar-22 1-25 Lesson 39 Synthesis / Board Problems	29-Mar-22 2-23 Lesson 40 WPR 3 Mini Proj. 3 Released	30-Mar-22 1-26 No Class Drop	31-Mar-22 2-24 PSL 3A Mini Project 3 Design Optimization	1-Apr-22 1-27 PSL 3B Mini Project 3 Design Optimization	A/C
Block 4: Vector Calculus	14	4-Apr-22 1-28 Lesson 41 Vector Fields (16.1) Due: Mini Proj. #3 Write-Up	5-Apr-22 2-25 Lesson 42 Line Integrals I (16.2)	6-Apr-22 Modified 1-29 No Class Drop	7-Apr-22 N/A No Class Study Day	8-Apr-22 1-30 Lesson 43 Line Integrals II (16.2)	A/C Sandhurst Validation
	15	11-Apr-22 1-31 Lesson 44 Fundamental Theorem of Line Integrals (16.3)	12-Apr-22 N/A Lesson 45 Green's Theorem I (16.4)	13-Apr-22 1-32 Lesson 46 Green's Theorem II & Curl (16.4 / 16.5)	14-Apr-22 2-27 Lesson 47 Curl & Divergence (16.5)	15-Apr-22 1-33 Lesson 48 Parametric Surfaces I and their Areas (16.6)	B
	16	18-Apr-22 2-28 Lesson 49 Parametric Surfaces II and their Areas (16.6)	19-Apr-22 1-34 Lesson 50 Surface Integrals I (16.7)	20-Apr-22 N/A No Class Study Day	21-Apr-22 2-29 Lesson 51 Surface Integrals II (16.7)	22-Apr-22 1-35 Lesson 52 Stokes' Theorem I (16.8)	A/C
	17	25-Apr-22 1-36 Lesson 53 Stokes' Theorem II (16.8) Mini Proj. #4 Released	26-Apr-22 N/A No Class Study Day	27-Apr-22 1-37 Lesson 54 Divergence Theorem (16.9)	28-Apr-22 N/A No Class USMA Project's Day	29-Apr-22 N/A No Class Sandhurst	30-Apr-22 Sandhurst
TEE Prep	18	2-May-22 1-38 PSL 4A Mini Project 4 Airline / Aero / Decision Support	3-May-22 2-30 PSL 4B Mini Project 4 Due: In-Class Presentation	4-May-22 1-39 Lesson 55 TEE Review (Blocks 1 & 2)	5-May-22 1-40 Lesson 56 TEE Review (Blocks 3 & 4)	6-May-22 N/A No Class Study Day	7-May-22 TEE

KEY			
	No Class	Problem Solving Lab	WPR
	Due Date or Mod Sched.	Synthesis Day / Review	

TAB C AY2023 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996

MADN-MATH 23 JUN 2023

MEMORANDUM THRU

COL Lindquist, Calculus Program Director, Dept. of Mathematical Sciences, USMA, West Point, NY 10996
COL Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: Annual Executive Summary for MA255 – Advanced Multivariable and Vector Calculus, AY23-2

1. **Document Summary.** Section two provides a brief overview of the course. Section three highlights the curriculum changes that were implemented based on feedback from last semester. Section four covers the grading scheme and a breakdown of each of the graded course events. Section five addresses student learning outcomes and an assessment of how well MA255 met both WGCD statements and core math goals. Section six discusses possible future changes for next year. Section seven describes the role of MA255 as a course in the Advanced Math Program (AMP). Section eight includes the methodology for validation of students for MA255 as well as off-ramping. Section nine is a compilation of notes from the Course Director for future Course Directors. Section 10 provides the location of this document and supporting documents on SharePoint.

2. **Background.**

The goal of the Core Advanced Mathematics Program (AMP) is to produce confident, competent students who possess the knowledge, skills, and interest necessary for the future study of science, technology, engineering, and mathematics (STEM). In AY23-2, MA255 began the semester with a student population of **276** plebes split into 21 sections and ended with **273** plebes. The final course average was **89.8%.** The required course textbook was *Calculus Early Transcendentals*, 9th Edition by James Stewart. Homework was assigned through the Webassign application and Canvas was used for course-wide knowledge management.

The course consisted of four major blocks of instruction: Block I – Vectors and the Geometry of Space; Block II – Infinite Sequences and Series & Multivariable Differentiation; Block III – Multivariable Integration; and Block IV – Vector Calculus.

Course-wide graded events consisted of a Written Partial Review (WPR) for each of the first three blocks, a comprehensive Term End Examination (TEE), graded homework assignments for each lesson (online), four Mini-Projects (two written papers and two oral presentations). The use of textbooks or computers were not allowed during any examination (WPR or TEE). Cadets primarily implemented technology (*Mathematica*) through mini-projects. The projects were also the primary exercises used for challenging students in the areas of problem solving, critical thinking, and both written and oral communication.

3. **Curriculum Changes for AY23-2.** This semester, we only designated “honors” sections during the A and B course hours. These sections were assigned to Fr. Costa, who teaches with a focus on mathematical theory. All other sections were randomly assigned. Each Mini-Project was assigned with the intent to provide students a whole week to complete the project, in addition to the assigned class periods. This added time reduced issues arising from conflicting student schedules. Projects one and three culminated in an oral presentation, while projects two and four required a written report. Minor changes were made to the course schedule, including the replacement of the “Instructor Day” previously included in Block II with an additional vector calculus review day in Block IV. Additionally, WPRs were more deliberately structured to assess lesson objectives and key learning outcomes from the course with the use of crosswalks.

4. **Administration.**

a. Overall Grading Scheme. There were 3,000 possible course points for MA255. The point distribution is summarized below:

Category	Event	Pts	PCT	
Individual Exams	TEE	900	30.0%	60.0%
	WPR1	300	10.0%	
	WPR2	300	10.0%	
	WPR3	300	10.0%	
Group Work / Modeling/ Technology	Two-Day Project #1	150	5.0%	20.0%
	Two-Day Project #2	150	5.0%	
	Two-Day Project #3	150	5.0%	
	Two-Day Project #4	150	5.0%	
HW	WebAssign HW 1	75	2.5%	10.0%
	WebAssign HW 2	75	2.5%	
	WebAssign HW 3	75	2.5%	
	WebAssign HW 4	75	2.5%	
Instructor	Instructor Points	300	10.0%	10.0%
Total		3000	100.00%	

Table 1: MA255 Grading Scheme

b. **Term End Examination.** Cadets were given a mandatory comprehensive TEE at the end of the semester. The TEE consisted of thirteen (13) questions and covered representative problems and applications from all blocks of instruction but was slightly weighted toward block IV since there was not a Block IV WPR. We replaced one vector calculus question on the previous year's TEE with a similar problem this year. Students had 3.5 hours to complete the final examination which accounts for 30% of the final grade. Cadets were allowed four pages of handwritten notes and their issued TI-36x calculator.

c. **Written Partial Reviews.** Cadets were assessed with a WPR at the end of Blocks I, II and III. For each WPR, cadets were allowed one page of handwritten notes and their issued calculator. The WPRs included a mix of computational questions, problem solving questions, and concept check questions. WPRs accounted for 30% of the final grade.

d. **Course Mini-Projects.** The course included four small projects to exercise interdisciplinary applications, collaboration, mathematical modeling, and written and verbal communication skills. Students conducted four week-long group projects with two designated class days per project following each WPR. This work, referred to as "mini-projects", highlighted the application of material to real-world applications while providing two opportunities to produce technical reports (mini-projects two and four) and two opportunities to present their findings orally (mini-projects one and three). The prompts and investigated topics intentionally asked students to answer questions that would be prohibitively difficult to do without the use of technology. While the prompts did not require students to use Wolfram's Mathematica, correct solutions would be infeasible without using some form of software. This led the project groups (2-3 students each) to explore the utility of tools like Python, R, Excel, and Mathematica, with most groups using Mathematica. In Block 1, students applied ballistic equations to determine the interception of an inter-continental ballistic missile; after WPR 2, students optimized the location of a logistics node; at the end of Block 3, they outlined their recommendations for excavating a road through a mountain; and prior to the TEE, students optimized flight paths between Air Force bases while accounting for the effects of a jet stream. Overall, the mini-projects accounted for 20% of the final grade.

e. **Webassign Homework**. Cadets completed graded homework assignments (for each lesson) through the online system, WebAssign. In addition to traditional questions, we implemented *Just in Time* questions, *Master It* questions, *Explore It* questions, and *lecture videos* covering the examples from the textbook. The majority of the *Master It* questions were optional with no credit due, but they served as scaffolding for the questions worth points. Altogether, the graded homework accounted for 10% of the final grade.

f. **Instructor Points**. The course had a total of 300 instructor points. These points were assigned for instructor use and were generally allocated to quizzes and other instructor-prepared grading tools. Instructor points accounted for 10% of the final grade.

g. **Faculty Development**.

(1) Over the course of the semester, each instructor observed a senior MA255 instructor, a junior MA255 instructor, and an instructor from a course outside the Advanced Mathematics Program. These observations gave instructors the opportunity to learn new methods of instruction and to provide valuable feedback to peers.

(2) Each instructor was assigned to a different department at USMA (i.e. civil engineering, life sciences, computer science, etc.) to investigate how multivariable and vector calculus are integrated into their curricula and to determine whether some of their course material could be integrated into MA255 to provide better context to our students. Instructors then briefed MA255 course-wide faculty on their findings, and excerpts were collected for future integration.

(3) Eight instructors (including the Program and Course Director) met weekly to discuss course administration, instructional techniques, mathematics pertaining to the course material, and future lessons and requirements. All instructors contributed to the course by developing mini-projects and WPRs as well as offering instructional and mathematical insights. This collaborative effort, along with the flexibility and candor among instructors, contributed significantly to the success of the course.

5. **Student Learning Outcomes and Assessment.**

Our assessment emphasis is on the core math focus areas for AY23, Habits of Mind and Communication. Of these, habits of mind maps to What Graduates Can Do statement 3.1, demonstrate the willingness and ability to learn independently, which is one of the Dean’s AY23 EXSUM guidance focus areas. Each of the two areas is measured on a 1-5 scale (1=Outstanding (A+,A), 2=Very Good (A-,B+), 3=Good (B-,C+), 4=Fair (C, C-, D), 5= Poor (F)). Figure 1 outlines the M255 student outcomes, mapped through core math goals to Academic Program goals.

Academic Program Goals / MA255 Assessment Techniques					
Supported Academic Program Goals (From Educating Army Leaders)	1. Graduates communicate effectively with all audiences	2. Graduates think critically and creatively	3. Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development	5. Graduates apply science, technology, engineering, and math concepts and processes to solve complex problems	7. Graduates integrate and apply knowledge and methodological approaches gained through in-depth study of an academic discipline.
MA255 Course Goals (Align with Core Math Goals)	2. **Communicate effectively**	3. Apply technology; 4. Build competent and confident problem-solvers; 5. Foster an interdisciplinary perspective	6. **Develop Habits of Mind;** 4. Build competent and confident problem-solvers	3. Apply technology; 4. Build competent and confident problem-solvers	5. Foster an interdisciplinary perspective
MA255 Course Learning Objectives	5. Students improve the quality of both their written and verbal communication of mathematics.	4. Students understand and apply the concepts of vector calculus. 7. Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus.	6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. 8. Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines. 10. Students learn good scholarly habits for progressive intellectual development.	2. Understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. 4. Understand and apply the concepts of vector calculus.	1. Use vectors and vector functions to model motion in three-dimensional space. 3. Students understand the concepts of multivariable integral calculus and can apply them to solve problems involving accumulation. 9. Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.
MA255 Assessment Techniques	**Mini-projects (2x presentations, 2x written reports), daily board problems, surveys**	1x Modeling problem per WPRs (x3) Mini-projects (x4) with interdisciplinary topics	**WebAssign Homework (x37)** **Mini-project read-ahead, independent group work (8x in-class workdays)**	2x oral projects requiring visual aids 2x written projects requiring technology to solve and adequately interpret results	1x Modeling problem per WPR (x3) Mini-projects (x4) with interdisciplinary topics

Table 2: Assessment Crosswalk

a. Communication (2). In addition to in-class board work presentations, the course included four mini-projects with a significant emphasis on presentation. For the first and third mini-projects, students were required to work together and give a group oral presentation. For the second and fourth mini-projects, students wrote a technical report directed at a specific target audience specified in the project documentation. For the reasons detailed below, MA255 forced students to practice their communication skills and led to an increase in their self-confidence in this arena.

(1) In the Core Math Book for the Department of Mathematical Sciences, the emphases for effective communication include: 1) actively listening to a presentation, 2) appropriately documenting work, and 3) articulating mathematical ideas orally. For our two oral presentations, all three of these goals were directly assessed. As part of the grading for the projects, the presentations were graded separately from the

mathematical work to determine whether students objectively improved from the first project to the third project. The student presentation average for project one was 89%, and the presentation average for project three was 92%. Instructors also observed a marked improvement in the quality of presentations from the first to second oral presentation.

(2) The Core Math Book also emphasizes that cadets should 'articulate mathematical ideas in writing to include development of a mathematical model, methods used to solve the model, and interpretation of the results.' Cadets were required to follow the USMA mathematical modeling process when writing the technical reports in mini-projects two and four. As with the previous mini-projects, the communication grades for the projects were graded separately from the mathematical scores. The student written communication average for project two was 90%, and the written communication average for project four was 92%.

(3) Students were provided a mid-course survey and final course survey to allow them to self-assess their ability in the key areas of communication referenced in the Core Math Book.

(a) Writing: At the mid-course survey, 57% percent of respondents rated their ability to communicate mathematical ideas in writing as good or very good, with that percentage increasing to 71% by the end-of-course survey. 78% of students stated that MA255 improved their ability.

(b) Oral Communication: At the mid-course survey, 68% percent of respondents rated their ability to communicate mathematical ideas orally as good or very good, with that percentage increasing to 75% by the end-of-course survey. 83% of students stated that MA255 improved their ability.

b. Habits of Mind (2). In the Core Math Book, the emphases for effective communication include: 1) reasoning and critical thinking, 2) creativity, 3) work ethic, 4) thinking interdependently, and 5) life long learning and curiosity. While we made efforts to promote these traits in the classroom, most of the students' development would be gained through out-of-class experience where students were forced to engage with material on their own terms. These skills were most honed with engagement in mini-projects, self-study, and Webassign homework. Although habits of mind are inherently difficult to assess, survey results indicate positive correlations between course structure/requirements and improvement in student ability.

(1) In the four course mini-projects students were forced to work in groups (think interdependently) to solve complex problems stemming from real-world applications (life-long learning). When modeling their problems, students were required to make reasonable assumptions in order to build a feasible model (critical thinking). At the culmination of each project students were given the opportunity to answer open-ended bonus questions that stretched the material into new applications (creativity). Although

55% of students initially reported that the course projects were not beneficial to their learning as a course resource, later in the survey, 83% of students stated that the projects improved their reasoning and critical thinking, while 70% stated that the projects assisted them in thinking interdependently. Despite this attribution to self-development, the most common recommendations to improve the course were to either remove the mini-projects or to make them easier/remove coding from them. As is common in learning, my hypothesis would be that the difficulty of mini-projects likely caused students to dislike them, despite the fact that cadets knew they were attaining real benefits from them.

(2) In MA255, students are required to complete Webassign homework on each lesson learned in the classroom. By practicing problems out of class, students are forced to engage their work ethic and, at times, think interdependently with their peers. This homework also teaches students life-long learning skills that will assist them with future learning. In the end-of-course survey, 85% of students reported that WebAssign was helpful to their learning, while 93% of students reported that working with their peers assisted their learning. By forcing these students to engage the material out of class, we ensured that students ‘knew what they did not know’ and could effectively engage theory learned in class. 52% of students also reported that MA255 improved their work ethic, while 53% reported an increase in life-long learning and curiosity.

6. **Possible Future Changes.** We acknowledge that making all the following changes in the next semester is not feasible, but we leave our thoughts here as a record.

a. **Move Sequences/Series to Block I.** The sequences/series block of instruction is currently located at the beginning of Block II: Multivariable Differentiation. The Stewart textbook introduces sequences and series prior to the material from Block I: Vectors and Geometry of Space. In order to minimize disruption to the flow of the course, we recommend reorganizing the course structure to reflect the order of instruction in the textbook. Since we will not be returning to sequences or series for the duration of MA255, this change will avoid unnecessary disruption.

b. **Include Writs for Sequences/Series and Block IV.** Sequences and series are currently evaluated as a part of WPR II, and Block IV: Vector Calculus material is currently assessed during the TEE. In order to provide students with a check on learning and to emphasize the importance of this separate material, we recommend that one writ be included for sequences and series and one writ be included for the first half of vector calculus.

c. **Replace Curvature and TNB Frame with an additional Sequences/Series Lesson**. The lesson on curvature and the TNB Frame does not tie to any future lessons in the course, and it consists of material that would more effectively be taught in conjunction with the application of these concepts to an engineering course. Additionally, we currently complete all of our sequences/series lesson objectives in a three-course span. The recommendation is to add one day to our sequences/schedule lessons so that we can more effectively engage the course material.

d. **Redesign Mini-Project 3 or consolidate with Mini-Project 4.** Block III: Multivariable Integration contains some of the most important material for students in the course, but incorporation into a project is limited, because the primary assessment would be setting up an integral. The current course project does not adequately nest this material with other course material. Since Block IV: Vector Calculus material integrates double and triple integrals, we could effectively evaluate their ability to use Block III material in a composite Block III/IV mini-project.

7. **MA255 in the Advanced Math Program.** MA255 is the second course of a two-semester advanced mathematics sequence for selected cadets who have validated single variable calculus and demonstrated strength in mathematical sciences. The course aims to provide a foundation for the continued study of mathematics, sciences, and engineering through the advanced coverage of topics in multivariable calculus, to include: a detailed study of vectors and geometry of space, vector functions, partial derivatives, multiple integrals, and vector calculus along with a brief introduction to infinite sequences and series. Students who successfully pass MA255 typically go on to take MA256, Advanced Statistics, or MA365, Advanced Engineering Mathematics.

8. **Cadets Off-Ramping from MA255.** This year, there were 20 students who completed MA153 but did not proceed to MA255.

a. **MA205/MA255 Validation**. In contrast to previous years, students were not able to validate MA205 and MA255 separately. Since both courses now include both Multivarable Calculus and Vector Calculus with similar course outcomes, there was only one exam offered to validate the course material. This exam consisted of two parts, a Multivariable Calculus Exam, which 17 of 29 test-takers passed, and a Vector Calculus Exam, which 16 of 29 test-takers passed. The 16 students who passed both exams were validated from MA255. Of these students, four were required to attend a Vector Calculus retraining session to revisit key concepts and ensure that they left the advanced mathematics program meeting the intent of the course.

b. **Off-ramping to MA103/MA104/MA205**. When students enter the Advanced Mathematics Program, they have demonstrated that they have the requisite skills and knowledge that is an output of MA104. Additionally, during MA153, students are required to conduct single variable integration and differentiation on almost a daily basis. Because of this, we strongly recommend that students are not allowed to move to MA104. There is also a minimal difference in learning outcomes between MA205 and MA255 so we likely do not recommend off-ramping to MA205. There were four total students who off-ramped this semester. One student failed MA153 and was required to take MA103. The other three students off-ramped to MA104/MA205.

9. **Notes from the Course Director to future Course Directors.**

Start planning early for the validation exam by coordinating approval through the Program Director and Associate Program Director. The exam needs to be scheduled on top of other course requirements, and it will also need to be advertised early to the students to ensure that they have time to prepare. It is also a good idea to make the course schedule laydown well before instructors are assigned so that you can disseminate to your new instructors and give them time to prepare lesson material. Furthermore, it's a good idea to get a head start on collaborating with your fellow CDs (MA104, MA206) to lock in classrooms for your semester. Once that is done, you can reach out to AARS (POC: Mr. Mark Walters) to request your desired classrooms. Using this strategy, we were able to lock in TH327 – 336 without any issues. Make sure that you cc your Assistant Course Director on all of these communications so that they will be able to plan on the correct timelines next year. The most important document that went into early course planning was the 'Teaching Plan' document that is included on the CCMA. This document includes a roster with phone numbers, templates to coordinate sectioning between instructors, a coverage plan, and major graded event planning considerations. I would recommend that you assign instructors who have previously taught the course as the primaries for mini-projects and WPRs. This will minimize rework on your part due to scopes inconsistent with course expectations. Lastly, although it is a big task, I would recommend that you systematically go through the course guide to validate lesson objectives, ensuring that the associated Webassign assignments accurately assess those lesson objectives. The MA255 Course/Instructor Guide is a living, breathing document that was extremely beneficial to faculty development efforts throughout the semester.

10. **Location of Documents.**

https://usarmywestpoint.sharepoint.com/sites/math.teaching then: Documents \ Teaching \ Program Director Information \ Course End Reports \ AY23 \ MA255

a. Course Syllabus & Schedule:	\ Course Guide & Schedule
b. WPRs & TEE:	\ Major Exams
c. Mini-Projects:	\ Mini-Projects
d. Final Grades & OML:	\ Final Grades & OML

11. Point of contact for this memorandum is MAJ Jeremy Reynolds at jeremyjreynolds@gmail.com or the undersigned at jonathan.paynter@westpoint.edu.

JONATHAN L. PAYNTER
LTC, AR/FA49
Associate Program Director

DEPARTMENT OF MATHEMATICAL SCIENCES
UNITED STATES MILITARY ACADEMY
WEST POINT, NY 10996-1786

MADN-MATH 9 January 2023

MEMORANDUM FOR Cadets, Faculty, and Staff in the Advanced Mathematics Program, West Point, New York, 10996

SUBJECT: Student Memorandum for MA255 - Advanced Multivariable Calculus

1. **Purpose**. This memorandum outlines guidance for students in MA255: Advanced Multivariable Calculus for Academic Year (AY) 23-2.

2. **Requirements.**

a. Text. *Multivariable Calculus (Loose-leaf)*, 9th Edition, James Stewart, Copyright 2016.

b. Software. *Mathematica* – Wolfram Research, Inc. and Microsoft Office (*Word*, *Excel*, *PowerPoint, MS Teams*)

c. Online. *WebAssign/Canvas* – www.webassign.net, https://canvas.instructure.com

3. **Attendance.**

a. MA255 is a 4.5 credit hour course. It does not follow the Day 1 / Day 2 schedule of a 3.0 credit hour course. The course schedule is found on the MA255 Canvas site under Course Resources. https://canvas.instructure.com/

b. Attendance at Written Partial Reviews (WPRs) is mandatory. Except in emergencies, cadets will not make any appointments that could preclude their attendance for a WPR. Any cadet missing a WPR will be required to take a make-up or make-ahead exam.

4. **Course Learning Outcomes.** This course is structured to develop the six core mathematics program goals with a focus on 10 Student Learning Outcomes.

a. **Build a body of knowledge: (1)** Students can use vectors and vector functions to model motion and surfaces in three-dimensional space. **(2)** Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. **(3)** Students understand the concepts of multivariable integral calculus and can apply it to solve problems involving accumulation. **(4)** Students understand and apply the concepts of vector calculus.

b. **Communicate Effectively: (5)** Students improve the quality of both their written and verbal communication of mathematics**.** Effective communication is an essential characteristic of great leaders. You, as a future leader, must be able to clearly articulate your thoughts. You will get many opportunities to improve your communications through in-class board presentations, course homework assignments, technology labs, and the course projects.

c. **Apply Technology: (6)** Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. You will become competent in using a computer algebra system (Mathematica) through in-class learning and self-paced technology labs. You will leverage technology to assist in your daily learning as well as in completing the course projects. Ultimately, you will understand the connection between the mathematical theory and its implementation using technology.

d. **Build Competent and Confident Problems Solvers: (7)** Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus. The problem solving will be integrated in class problems, technology labs and the course project. You will hone your problem-solving skills by formulating and solving challenging mathematical problems.

e. **Provide Experience in Interdisciplinary Problem Solving: (8)** Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines. **(9)** Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems. There will be multiple opportunities in class and during the course lectures for you to see real world problems from a variety of disciplines. You will see how mathematics is used to solve these problems.

f. **Develop Habits of Mind: (10)** Students learn good scholarly habits for progressive intellectual development. You will learn habits in this course that will benefit you throughout your academic career. You will develop an intellectual curiosity in general, and for mathematics, by doing challenging but interesting mathematical problems. You will improve your work ethic and critical thinking through learning and doing mathematics. You will develop skills to be independent learners through the conduct of the course. These habits of mind are essential traits of leaders and scholarly students.

5. **Lesson Assignments.** The Course Schedule and Course Guide serve as the road map for the course. These documents are both available on the homepage of the MA255 course website.

6. **Course Summary.** There are a total of **64** required attendances consisting of **56 Lessons** and **8 Problem Solving Labs** split into two parts each.

7. **Course Evaluation Plan**

Category	Event	Pts	PCT	
Individual Exams	TEE	900	30.0%	60%
	WPR1	300	10.0%	
	WPR2	300	10.0%	
	WPR3	300	10.0%	

Group Work / Modeling/ Technology	Two-Day Project #1	150	5.0%	20%
	Two-Day Project #2	150	5.0%	
	Two-Day Project #3	150	5.0%	
	Two-Day Project #4	150	5.0%	
HW	WebAssign HW 1	75	2.5%	10%
	WebAssign HW 2	75	2.5%	
	WebAssign HW 3	75	2.5%	
	WebAssign HW 4	75	2.5%	
Instructor	Instructor Points	300	10.0%	10%
	Total	3000	100.0%	100%

8. **Grading**

a. Final course grades will follow the Department of Mathematical Sciences' guidelines:

Letter Grade	% Range	Quality Points	Subjective Interpretation
A+	97≤ Mark ≤100	4.33	Beyond expectations of course
A	93 ≤ Mark < 97	4	Dominates the material
A-	90 ≤ Mark < 93	3.67	Mastery
B+	87 ≤ Mark < 90	3.33	Excellent performance
B	83 ≤ Mark < 87	3	Good understanding

B-	80 ≤ Mark < 83	2.67	Proficient; Aptitude for the subject
C+	77 ≤ Mark < 80	2.33	Can build upon this foundation
C	73 ≤ Mark < 77	2	Passing; Proficient now (short range)
C-	70 ≤ Mark < 73	1.67	Short-range understanding
D	65 ≤ Mark < 70	1	Marginal performance with some elementary understanding
F	Mark < 65	0	Failing; Definitely failed to demonstrate understanding

b. Cadets who score less than 50% on the TEE, regardless of their final course average, or whose final course average is less than 65% may fail the course.

c. Instructors may award up to 30 bonus points at their discretion. Bonus points will be added directly to the total number of points you earn in the course.

9. **Documentation**. All cadet work in MA255 will be in accordance with the procedures found in the Dean's *Documentation of Academic Work* (DAW), *June 2017*. All required out-of-class work constitutes homework as defined in the DAW. The course standard for MA255 is MLA. Refer to the DAW for further information. Proper documentation is required for all graded exercises assigned by your instructor and all out-of-class course-wide assignments outlined in this memorandum.

10. **Out-of-Class Assessments.**

a. **Daily Homework.** Daily web-based homework problems will be assigned via WebAssign. After you submit each assignment, you will receive immediate feedback, to include your grade. Additionally, all daily WebAssign problem sets require students to complete an online acknowledgement statement. Failure to complete the documentation steps outlined here will result in a score of zero for the given assignment. Lastly, MA255 uses the following *WebAssign* assignment extension policy: students may ask for up to two *automatic* extensions, each three days in duration, for a deduction of 10% per period. The point deduction only applies to problems not completed prior to the assignment deadline. A *WebAssign* automatic extension is an assignment extension for which instructor permission is *not* required. Students may use automatic extensions for a 14 day period after the exercise is assigned, or through the end of the given block of instruction, whichever is earlier.

b. **Two-Day Projects.** At the end of each block, students will be assigned a project that synthesizes the material covered in the block. These will be group-based (2- or 3-person) and can use out-of-class time to complete the work. The projects will include the effective use of technology and will each use a different mode of communicating the team's results. The purpose is to reinforce the material and demonstrate various interdisciplinary applications of multi-variable calculus.

11. **In-class Assessments.**

a. **Written Partial Reviews.** WPRs assess your understanding of lesson objectives in the course. Each WPR is cumulative (meaning any content covered up to the point may be assessed) but focuses on content from the corresponding block. Attendance at WPRs is mandatory as described in Section 3 Attendance.

b. **Term End Exam.** The TEE is mandatory and cumulative for the entire course.

12. **Classroom Rules**. Cadets will conform to the highest standards of ethical behavior and military bearing in the performance of their academic duties. Cadets will treat every member of the faculty with the courtesy appropriate to commissioned officers. Smoking, using smokeless tobacco, chewing gum, and consuming food is always prohibited in classrooms, laboratories, lecture halls, and auditoriums. Drinks are allowed in a Dean-approved container, defined as a container capable of completely sealing to prevent accidental spillage. The uniform for classes will be the duty uniform, with exceptions made for injury and special class functions that occur during the class day. Cadets will leave outer garments and bags in the hallway. The uniform for additional instruction (AI) is the duty uniform.

13. **Chain of Command**. The Department of Mathematical Sciences has an open-door policy. Your initial attempt to address problems or concerns – including respect in the classroom, unprofessional behavior, sexual assault, harassment – should go through your instructor. However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain to include the Head of the Department. The entire math department faculty is here to help you.

a. Your instructor.

b. MA255 Course Director: MAJ Reynolds, Thayer 255C, x7618.

c. Associate Program Director: LTC Paynter, Thayer 255A, x4544.

d. Calculus Program Director: COL Lindquist, Thayer 225, x7436.

e. Head of the Department of Mathematical Sciences: COL Hartley, Thayer 238, x5870.

JONATHAN L. PAYNTER
LTC, AR(49)
AMP Associate Program Director

MA255: Advanced Multivariable Calculus - AY23-2 Spring 2023 **Printed on: 3/2/2023**

AY 23-2		MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY
Block 1: Vectors and Geometry of Space	1	9-Jan-23 N/A No Class REORGY	10-Jan-23 2-1 Lesson 1 Course Intro	11-Jan-23 1-1 Lessson 2 3D Coordinate Systems (12.1)	12-Jan-23 2-2 Lesson 3 Intro to Vectors I (12.2)	13-Jan-23 1-2 Lesson 4 Intro to Vectors II (12.2)
	2	16-Jan-23 N/A No Class MLK Day	17-Jan-23 2-3 Lesson 5 Dot Product / Cross Product (12.3 / 12.4)	18-Jan-23 1-3 Lesson 6 Equations of Lines and Planes (12.5)	19-Jan-23 2-4 Lesson 7 Cylinders & Quadric Surfaces (12.6)	20-Jan-23 1-4 Lesson 8 Vector Functions & Space Curves (13.1)
	3	23-Jan-23 1-5 Lesson 9 Derivatives & Integrals of Vect. Fcns (13.2)	24-Jan-23 2-5 Lesson 10 Arc Length, Curvature, and TNB Frame (13.3)	25-Jan-23 N/A No Class Study Day	26-Jan-23 1-6 Lesson 11 Arc Length, Curvature, and TNB Frame (13.3 / 13.4)	27-Jan-23 2-6 Lesson 12 Velocity & Acceleration (13.4)
	4	30-Jan-23 1-7 Lesson 13 Synthesis / Board Problems	31-Jan-23 2-7 Lesson 14 WPR 1 Mini-Project 1 Released	1-Feb-23 1-8 No Class Course Drop	2-Feb-23 2-8 PSL 1A Project Work	3-Feb-23 1-9 Lesson 15 Sequence / Series (11.1 / 11.2)
Block 2: Multivariable Differentiation	5	6-Feb-23 1-10 PSL 1B Presentations Due: In-Class Presentation	7-Feb-23 2-9 Lesson 16 Alt. Series / Ratio / Root Test (11.5 / 11.6)	8-Feb-23 N/A No Class Study Day	9-Feb-23 1-11 Lesson 17 Power / Taylor / Maclaurin (11.8 / 11.10)	10-Feb-23 2-10 Lesson 18 Funcs. Of Sev. Vars / Limits of Cont. (14.1 / 14.2)
	6	13-Feb-23 1-12 Lesson 19 Limts & Cont. / Part. Derivs. (14.2 / 14,3)	14-Feb-23 2-11 Lesson 20 Tangent Planes & Linear Approximations (14.4)	15-Feb-23 N/A No Class Study Day	16-Feb-23 1-13 Lesson 21 Chain Rule (14.5)	17-Feb-23 2-12 Lesson 22 Directional Derivatives & Gradient Vector (14.6)
	7	20-Feb-23 N/A No Class Presidents Day	21-Feb-23 2-13 Lesson 23 Max & Min Values I (14.7)	22-Feb-23 1-14 Lesson 24 Max & Min Values II (14.7) Lagrange Multipliers I (14.8)	23-Feb-23 2-14 Lesson 25 Lagrange Multipliers II (14.8)	24-Feb-23 1-15 Lesson 26 Synthesis / Board Problems
	8	27-Feb-23 1-16 Lesson 27 WPR 2 Mini-Project 2 Released	28-Feb-23 2-15 No Class Course Drop	1-Mar-23 1-17 PSL 2A Project Work	2-Mar-23 2-16 No Class Study Day	3-Mar-23 Modified 1-18 PSL 2B Project Work
	9	6-Mar-23 N/A No Class Spring Break	7-Mar-23 N/A No Class Spring Break	8-Mar-23 N/A No Class Spring Break	9-Mar-23 N/A No Class Spring Break	10-Mar-23 N/A No Class Spring Break
Block 3: Multivariable Integration	10	13-Mar-23 1-19 Lesson 28 Double Integrals over Rect. I (15.1)	14-Mar-23 N/A No Class Course Drop	15-Mar-23 1-20 Lesson 29 Dbl Int. over Rect. II & Gen Reg I Due: Mini Proj. 2 Write-Up	16-Mar-23 2-17 Lesson 30 Double Integrals over General Regions II (15.2)	17-Mar-23 1-21 Lesson 31 Double Integrals in Polar Coordinates (15.3)
	11	20-Mar-23 1-22 Lesson 32 Applications of Double Integrals I (15.4)	21-Mar-23 2-18 Lesson 33 Applications of Double Integrals II (15.4)	22-Mar-23 N/A No Class Study Day	23-Mar-23 1-23 Lesson 34 Triple Integrals (15.6)	24-Mar-23 2-19 Lesson 35 Triple Integrals in Cylindrical Coordinates (15.7)
	12	27-Mar-23 1-24 Lesson 36 Triple Integrals in Spherical Coordinates I (15.8)	28-Mar-23 2-20 Lesson 37 Triple Integrals in Spherical Coordinates II (15.8)	29-Mar-23 1-25 Lesson 38 Synthesis / Board Problems	30-Mar-23 2-21 Lesson 39 WPR 3 Mini-Project 3 Released	31-Mar-23 1-26 No Class Course Drop
	13	3-Apr-23 1-27 PSL 3A Project Work	4-Apr-23 2-22 Lesson 40 Vector Fields (16.1)	5-Apr-23 N/A No Class Study Day	6-Apr-23 1-28 PSL 3B Presentations Due: In-Class Presentation	7-Apr-23 2-23 Lesson 41 Line Integrals I (16.2)
Block 4: Vector Calculus	14	10-Apr-23 1-29 Lesson 42 Line Integrals II (16.2)	11-Apr-23 2-24 Lesson 43 Fundamental Theorem of Line Integrals (16.3)	12-Apr-23 1-30 No Class Course Drop	13-Apr-23 2-25 Lesson 44 Green's Theorem I (16.4)	14-Apr-23 1-31 Lesson 45 Green's Theorem II & Curl (16.4 / 16.5)
	15	17-Apr-23 1-32 Lesson 46 Curl & Divergence (16.5)	18-Apr-23 2-26 Lesson 47 Parametric Surfaces I and their Areas (16.6)	19-Apr-23 N/A No Class Study Day	20-Apr-23 1-33 Lesson 48 Parametric Surfaces II and their Areas (16.6)	21-Apr-23 2-27 Lesson 49 Surface Integrals I (16.7)
	16	24-Apr-23 1-34 Lesson 50 Surface Integrals II (16.7)	25-Apr-23 2-28 Lesson 51 Vector Calculus Float / Review Day	26-Apr-23 1-35 Lesson 52 Stokes' Theorem I (16.8)	27-Apr-23 2-29 Lesson 53 Stokes' Theorem II (16.8) Mini Proj. 4 Released	28-Apr-23 N/A No Class Sandhurst
	17	1-May-23 1-36 Lesson 54 Divergence Theorem (16.9)	2-May-23 2-30 PSL 4A Project Work	3-May-23 N/A No Class Study Day	4-May-23 N/A No Class Projects Day	5-May-23 1-37 No Class Course Drop
TEE Prep	18	8-May-23 1-38 PSL 4B Project Work	9-May-23 N/A No Class Study Day	10-May-23 1-39 Lesson 55 TEE Review (Blocks 1 & 2) Due: Mini Proj. 4 Write-Up	11-May-23 1-40 Lesson 56 TEE Review (Blocks 3 & 4)	12-May-23 N/A No Class Study Day

KEY					
KEY	No Class	Review	Course-wide Graded Event	Projects	Due Date or Modified Sched.

TAB D AY2024 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MTH 24 May 2024

MEMORANDUM THRU

COL Joseph Lindquist, Director, Calculus Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: MA255 Advanced Multivariable Calculus Executive Summary (AY24)

1. **Purpose:** The purpose of this memorandum is to assess the MA255 curriculum.

2. **Assessment:** The Advanced Math Program (AMP) has a three year assessment strategy. This year's assessment examined Student Learning Objectives (SLO) listed below in Table 1 with our findings for each assessed area. The full list of SLOs can be found in Enclosure 1.

MA255 SLOs	Strength of Attainment	Strength of Evidence	Overall Evaluation
SLO 7	B	B	B
SLO 8	A	B	B
SLO 9	A	B	B
SLO 10	B	B	B

Table 1: MA255 AY24-2 SLO assessment evaluations.

To read Table 1, we note that A is defined as exceeding the standard, B meets the standard, C meets the standard with some weakness, D fails the standard (but can be addressed in the short-term), and F catastrophically fails the standard (requires significant remediation).

3. **Assessment Discussion:** This section presents an overview of assessment findings for AY24. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report located in Enclosure 4.

a. Status of significant changes: the only significant change was the addition of an additional Written Partial Review (WPR) to cover the final block of content (Vector Calculus) with a slight reallocation of class time to accommodate it.

b. Strengths in the course: the major strengths of the course lie in its mathematical maturation of high-performing students and its focus on getting those students a high volume of experience performing the mathematics while providing an opportunity to attempt real-world applications. SLOs 8 and 9 are assessed as above the standard because of exceptionally high performance on the corresponding exam and final exam questions. The course did an overall good job of covering the content for SLOs 7 and 8.

c. Areas for improvement in the course: finding another way to validate student understanding, including a chance for additional retraining, could strengthen student performance in SLO 10. Performance as measured in WPR 4 and the Term End Examination (TEE) indicate that many students still struggle with the applications of these concepts.

d. Recommended changes to be implemented in AY25: continue to use and refine the use of GradeScope to facilitate and potentially customize additional student follow-on assignments or quizzes. Maintain WPR 4 and continue to refine “Block Projects” to give students a chance to learn hands-on.

e. Proposed changes to the SLOs: None.

4. The point of contact for this memorandum is MAJ Devon Zillmer, the AY24 MA255 Course Director, at 910-728-2051 or devon.p.zillmer.mil@army.mil, or LTC Jonathan Paynter at jonathan.paynter@westpoint.edu.

Encl

JONATHAN L. PAYNTER
LTC, AR/FA49
Associate Program Director

1 Full set of SLOs

Below are the Student Learning Objectives (SLOs) for MA255 Advanced Multivariable Calculus.

1. Students learn good scholarly habits for progressive intellectual development.
2. Students build critical thinking skills to solve problems through learning complex mathematical concepts in multivariable and vector calculus.
3. Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines.
4. Students can design, solve, and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.
5. Students improve the quality of both their written and verbal communication of mathematics.
6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems.
7. Students can use vectors and vector functions to model motion and surfaces in three dimensional space.
8. Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems.
9. Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation.
10. Students understand and apply the concepts of vector calculus.

Anticipated changes to SLOs: None.

2 Mapping of Academic Program Goals to SLOs

***Supported Academic Program Goals** (From Educating Army Leaders)*	*1. Graduates communicate effectively with all audiences.*	*2. Graduates think critically and creatively.*	*3. Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development.*	*5. Graduates apply science, technology, engineering, and math concepts and processes to solve complex problems.*
***Supported What Graduates Can Do** Tasks*	1.5: Use sound logic and relevant evidence to make convincing arguments.	2.4: Reason both quantitatively and qualitatively.	3.1 : Demonstrate the willingness and ability to learn independently.	5.1: Apply mathematics, science, and computing to model devices, systems, processes, or behaviors. 5.3 : Collect and analyze data in support of decision making.
Mathematics Program Goals	2. Communicate effectively	1. Acquire a body of knowledge 3. Apply technology	6. Develop habits of mind	4. Build competent and confident problem-solvers 5. Provide experience in interdisciplinary problem solving
MA255 Student Learning Objectives	4. Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems. 5. Students improve the quality of both their written and verbal communication of mathematics.	2. Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus. 6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems.	1. Students learn good scholarly habits for progressive intellectual development. 3. Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines.	7. Students can use vectors and vector functions to model motion and surfaces in three dimensional space. 8. Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. 9. Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation. 10. Students understand and apply the concepts of vector calculus.
MA255 Assessment Techniques	Mini-projects (2x presentations, 2x written reports), daily board problems, surveys	1x Modeling problem per WPRs (x4) Mini-projects (x4) with interdisciplinary topics	WebAssign Homework (x37) Mini-project read-ahead, independent group work (8x in-class workdays)	4x exams Term End Examination 4x Mini-Projects exploring applications

Figure 1: Assessment Crosswalk

3 Cyclical schedule for assesments

Academic Year	*Assessment Focus*		
	WGCD	**MPG**	**SLO**
2024	5.1, 5.3	4, 8	7, 8, 9, 10
2025	2.4, 3.1	1, 6	1, 2, 3
2026	1.5, 2.4	2, 3	4, 5, 6
20(3n)	5.1, 5.3	4, 8	7, 8, 9, 10
20(3n+1)	2.4, 3.1	1, 6	1, 2, 3
20(3n+2)	1.5, 2.4	2, 3	4, 5, 6

Figure 2: Assessment Schedule

4 Detailed Course Design and Analysis Report

Associate Program Director (APD): LTC Jon Paynter
Course Director (CD): MAJ Devon Zillmer
Assistant Course Director (ACD): MAJ Joseph Maxwell

This document provides an introduction and overview for MA255 (Advanced Multivariable and Vector Calculus) for Academic Year 24-2. We first introduce the academic field and general environment in Section 4.1, then provide a background and brief introduction to this particular course in Section 4.2, and move into an in-depth discussion of the course design in Section 4.3. Section 4.4 discusses our assessment based on the criteria laid out in Section 4.3, and we discuss any conclusions or future potential work in Section 4.5.

4.1 Introduction

In this section we will open with the broader academic situation in which the course is situated, including guidance from the Mathematical Association of America (MAA), highlight of comparable courses at other academic institutions, and then briefly discuss general use of technology in the classroom.

4.1.1 MAA Guidelines

As seen in [12], there is not a uniform structure from MAA to indicate what mathematical content belongs in which particular Calculus course. While there are some trends among particular universities, and the MAA acknowledges a wealth of textbooks, there is not a specific delineated progression of topics.

4.1.2 Comparable University Courses

To provide a groundwork of other universities' advanced calculus courses, we consider the curriculum for United States Naval Academy (USNA) [2], United States Air Force Academy (USAFA) [1], Pomona College [10], and Ithaca College [9]. The first two colleges are the primary competitors of the United States Military Academy (USMA), the other two have comparable acceptance rates (Pomona College) or student body sizes (Ithaca). In comparing their classes equivalent to MA255, similarities and differences are worth noting.

Two major similarities: common textbook, and comparable course structure. The Stewart *Calculus with Early Transcendentals* is universally used, and not only as a reference, but also as a guide: all of the reviewed syllabi follow in near lock-step. Accordingly, much (but not all) of the course content is introduced with roughly the same progression. Most of the "Calculus III" content also generally covers through Stokes' Theorem in Vector Calculus.

Four major differences of note between USMA and these four other universities. First, MA255 is the only course that places deliberate emphasis on modeling and projects;

the others have course work and tests, but do not have comparable projects or time dedicated to modeling. Second, none of the other schools place a comparable course (usually called Calc III) in their core mathematics program; most use it like a service courses or as a part of their major in mathematics. Third, we are the only course that places relatively heavy emphasis on leveraging technology (Mathematica (MMA), as opposed to the MAA's recommendation of Python with NumPy/SciPi). Finally, none of the other single courses span the breadth of topics we cover: sequences and series, optimization, projects/modeling, and still following Stewart's framework from vectors through the Divergence Theorem.

4.1.3 Calculus as a Core Course

Discussed in several places from the MAA perspective (such as [12]), but mentioned also on some university cites (like at Ithaca College in [9]), there is an ongoing discussion about how to make Calculus more accessible to the average student. While we have a tradition at USMA of requiring all Cadets to take Calculus (and until recently, multivariate Calculus), the increasing requirement of all students in four year colleges to pass an introductory Calculus course means that accessibility of material and breadth of application frequently are an increasing part of the curriculum. Given MA255's role as part of the core curriculum within the AMP, we fit well into this discussion, and our projects and focus on modeling are a potential inputs to the conversation.

4.1.4 Course Technology Use

AMP current leverages MMA as the primary technology tool in the classroom. It is primarily used as a "superpowered graphing calculator," to verify solutions, and to help students build intuition for behavior.

The decision to use MMA and not Python is deliberate. The current CD and ACD both invested time in attempting to transition the course software to use Python, but ultimate chose not to do so for several reasons. First, there are things MMA can do that Python cannot. Two examples are plotting implicit functions (e.g., surfaces) and symbolic manipulation (in the same traditionally done in mathematics, such as "solving for x"). Second, even if we restructured the course to avoid these pitfalls, we would still have a considerable startup cost to give students more accessible graphing and solving tools. In practice, this means we would have to give them a script that translates a lot of computer science realities into the traditional math form; one example is ploting $f(x, y)$ from -3 to 3 on both axes requires generating a mesh of points, declaring several baseline settings for the plot, and then the student can plot, as opposed to MMA's straightforward "Plot" command. Finally, the course does not prevent students from using a variety of resources in modeling or problem-solving contexts, and we encourage the use of built-in help functions, peers and online resources, and even the use of generative Artificial Intelligence (AI) such as ChatGPT or WolframAlpha.

4.1.5 Previous Course Feedback

MA255 is teaching content that is very well established, travelling a route that is well trod; there is seldom course After Action Review (AAR) feedback indicating large changes are needed, with one exception. The only feedback specifically recommended but not enacted for the last four years (since 2020 it has been mentioned in course end reports, see [6] and [5]) is to reverse the sequence of courses between MA153 and MA255 (i.e., return to the sequence of classes presented to students in 2015 and previously, teaching Multivariate Calculus in the autumn and Modeling with Ordinary Differential Equations in the spring).

4.2 Background

In this section, we discuss the required content for MA255 from the Redbook, its intended audience, how it differs from MA205, and who its primary customers are. We then discuss the course SLOs and time allocation, both from credit-hour perspective and instructor perspective.

4.2.1 Redbook

The Redbook course description (from [11]) is seen below. We do not recommend any changes.

> This is the second course of a two-semester advanced mathematics sequence for selected cadets who have validated single variable calculus and demonstrated strength in the mathematical sciences. It is designed to provide a foundation for the continued study of mathematics, sciences, and engineering. This course consists of an advanced coverage of topics in multivariable calculus. Topics may include a study of infinite sequences and series, vectors and the geometry of space, vector functions, partial derivatives, multiple integrals, and vector calculus. An understanding of course material is enhanced through the use of a computer algebra system.

4.2.2 Audience

MA255 is the second course in the two-course advanced sequence to fulfill the general mathematical requirements for students at the United States military Academy. AMP rigorously screens incoming plebes and provides an in-person exam during the first week of classes to validate that all students enrolled in AMP have a solid Calculus background (up through familiarity with basic integration techniques, including at a minimum substitution and integration by parts). Thus these students have a solid mathematical background and often have good study skills and habits. The purpose of the course, as described by the previous several course directors, is to “produce confident, competent students who possess the knowledge, skills, and interest necessary for the future study of Science, Technology, Engineering, and Mathematics (STEM)” (from [7]).

This is true insofar as it goes, but it should be noted that this is a core math course, addressed to advanced students who go on to major in a variety of academic majors, including non-STEM fields. This distinguishes the audience of MA255 from MA205.

4.2.3 Comparison with MA205

MA205 is a comparable course, but three distinct differences stand out. First, MA255 covers more content than MA205, including several lessons teaching from sequences and series through applications of Taylor Series (Stewart Chapter 10). Second, the focus of the audience is different, as MA205 is only required for STEM degrees (i.e., a service course), while MA255 satisfies the Academy’s Calculus graduation requirement in the core curriculum (i.e., a core course). Finally, and most significantly, the student population is different: as discussed previously, the AMP student population is more mathematically mature, allowing for a better grasp of and more in-depth exploration of the topics, including more challenging projects. For these reasons, despite the similarities, we do not recommend leaving MA255 in the AMP and not combining course leadership with MA205.

4.2.4 Courses for which MA255 is a Prerequisite

From the customer perspective, however, MA255 is analogous to an advanced version of MA205 (as it covers comparable content), and so satisfies the MA205 prerequisite for other department's courses (per [11], as seen in [4]). As of July 2023, 38 courses require either MA255 or MA205, listed by department as seen in Table 2.

Department	**Courses**
Chemistry and Life Science	3
Civil and Mechanical Engineering	7
Electrical Engineering and Computer Science	4
Geography and Environmental Engineering	1
Mathematical Sciences	19
Physics and Nuclear Engineering	3
Social Sciences	1

Table 2: Number of courses offered by a department which require MA255 or MA205 as a prerequisite.

4.2.5 Course Mathematical Goals

This course has ten major SLOs in the Core Math Book (see [8]), including six general mathematical goals (e.g., learn good scholarly habits, build critical thinking skills, improve quality of written and verbal habits) and four course-focused goals (use vectors and vector functions to model motion and surfaces, understand and apply: multivariable differential calculus, multivariable integral calculus, and vector calculus). These goals shape the way lessons are built and how each major graded event is constructed.

A **possible change** may be to add or modify a SLO about broadening students intellectual perspectives to include thinking about infinity, including assorted applications of the calculus (limits at infinity), sequences and series (culminating in Taylor Series), and their utility in modern computational approximation. Of note, the only two courses to do this are MA255 (situating Taylor Series in a "normal calculus" context) and MA364 (discussing them in rapid-fire succession with other engineering topics), but *not* in MA104 (a more traditional "normal calculus" context for series).

4.2.6 Credit Hour Justification

As defined in the Redbook in [11], each credit hour should accompany 40 hours of planned time. The usual planning rule is for two hours out of class for each hour in class. MA255 is a 4.5 credit hour course, as justified in Table 3.

Lesson Types	Planned Time	Total
Baseline 40 Lessons	3	120
MA255: 46 Lessons	3	138
4 WPRs	3	12
8 Project Days	3	24
7 Problem Solving Labs	1	7
Total: 65 lessons		**181**

Table 3: Lesson count with recommended out of class time, and total credit hour justification.

4.2.7 Instructor Requirements

Table 4 highlights the approximate allocation of time during an average week for an instructor in MA255. For seasoned instructors, the decrease in course preparation required generally corresponds to much higher time spent building course products and helping new instructors master course material.

Activity	Hours	Total
In-Class Instruction (3 sections, 4 lessons/week)	12	12
Weekly PD Meeting	1	13
Course Taskings (Product generation, validation, etc)	2	15
Department (Coffee Call, Broadening, MTP, etc)	1	16
Grading (average estimate)	4	20
Lesson Preparation (3 hours per lesson)	12	32
Administrative Tasks (daily “eaches”, Q&A before/after class, emails, etc)	5	37
Instructor Choice (admin. tasks, AI, research, intramurals)	3	40

Table 4: Instructor average time allocation, per 40 hour week.

Figure 3 provides an example of how instructors are assigned tasks to assist in generating course products. The instructors are listed along the left, whereas the products required by block are listed across the top. In the body of the matrix, P stands for the Primary writer, A stands for the Alternate or Assistant writer supporting the primary, V indicates an assignment to Validate and time a project, T assigns a (usually first-time teaching) instructor to Time how long a test took them to complete. As can been seen by inspecting the product, most instructors (aside from the CD and APD) are responsible for generating or contributing substantially to several major course assignments.

As of: 22 NOV 23	Block 1		Block 2		Block 3		Block 4			Tasks
	MP1	WPR1	MP2	WPR2	MP3	WPR3	MP4	WPR4	TEE	
LTC Paynter (APD)			P							1
MAJ Zillmer (CD)		P			P					2
MAJ Maxwell (ACD)	P	A				A			P	4
Fr. Costa				P				P		2
Dr. Groves						P	A			2
MAJ Oletti				A			P	A		3
CPT Bates		T	V		V			T		4
CPT Colgan	V			T		T	V			4
Brief to Leadership:	09 Jan 24	16 Jan 24	30 Jan 24	13 Feb 24	27 Feb 24	05 Mar 24	19 Mar 24	16 Apr 24	23 Apr 24	
Brief to Instructors:	16 Jan 24	23 Jan 24	06 Feb 24	20 Feb 24	05 Mar 24	12 Mar 24	02 Apr 24	23 Apr 24	30 Apr 24	
Assignment Date:	22 Jan 24	26 Jan 24	13 Feb 24	21 Feb 24	19 Mar 24	13 Mar 24	09 Apr 24	29 Apr 24	07 May 24	

Figure 3: Product generation matrix for AY24-2.

4.3 Course Design

4.3.1 SLO Assessment Focus

Per Enclosure 3, the focus for assessments in academic year 2024 is SLOs 7-10, each of which deals with the use of course concepts in class and their application: vectors and surfaces in three-dimensional space, multivariable differentiation, multivariable integration, and vector calculus. To assess these SLOs, we will consider the scores on WPRs 1-4 (mapping to SLOs 7-10, respectively) and the student performance on the TEE questions which relate to the specific block's content.

The last time the comparable SLOs were assessed (AY2021-2, see [6]) the assessment was performed through the framework of assessing Mathematics Program Goal (MPG)s, and was focused on Body of Knowledge and forming Confident and Competent Problem Solvers (CCPS). When it was last assessed, the assessment technique was largely qualitative and intuitive. The assessment for Body of Knowledge was "Very Good," based on overall high course grades, including 52% of the course not taking a TEE (due COVID, there was a variety of department and Academy guidance directing fewer in-class events). The assessment of CCPS was "Good," and was based on course projects score and course survey feedback. In both cases, the assessments were largely qualitative and intuitive, without clearl metrics.

4.3.2 Lesson Modification

There are only two small changes we want to make to the MA255 schedule from AY23-2 to this year (AY24-2): an additional day on Sequences and Series and a Block 4 WPR. The Sequences and Series additional day is to provide instructors and students the time and space to allow students to better understand the concepts and applications of Taylor Series. The Block 4 WPR will provide students an additional formal opportunity to perform the key objectives for the Vector Calculus block of instruction, under WPR conditions, and receive feedback on their work prior to the TEE. Both of these slight changes will be assessed by comparing the distributions of scores between this and the previous year's scores on the relevant TEE questions.

4.3.3 Instructor Preparation and Course Guide

We have two additional efforts of emphasis for MA255 in the instructor preparation and resources: deliberate Faculty Development Workshop (FDW)-style pre-semester classes to onboard new instructors, and an improved Course Guide. The FDW will include two deliberate thirty-minute sessions during the final two weeks of the semester, running the three new faculty to the Jedi Math Program through several key lessons in the Vector Calculus block of instruction.

The course guide overhaul includes a deliberate restructure of the Course Guide (both student- and instructor-versions). Changes include paring down Lesson Objectives to clearly indicate the focus for a given course and to codify the "level" of a given objective on Bloom's Taxonomy (see [3] for a more in-depth discussion). Both of these are internally-focused measures, but to try to make a quantitative assessment of these measures, we will do two things: conduct sentiment analysis on the text provided for the AAR and ask for subjective feedback from the instructors on a variety of dimensions across the last few years, and we will try to use course survey question feedback on the utility of the course guide. While these are not ideal metrics to use for such a small course, these measures should allow us to provide a general assessment of if the changes were valuable.

4.4 Assessment

In this section we will discuss the three major planned assessment topics: (1) planned course SLOs; (2) course lesson changes, including an additional day on sequences and series and an additional WPR for vector calculus; (3) updates to the pre-course instructor preparation process and modifications to the course guide.

In summary, the methods used to instruct students were largely acceptable, but the evidence that the course modifications were helpful are inconclusive.

4.4.1 SLOs

As mentioned in Section 4.3, the primary means by which we will assess SLOs 7-10 will be by considering the scores on the WPRs and the relevant TEE question scores (e.g., SLO 7 against Block 1, SLO 8 against Block 2, and so forth). See Figure 4 for a visual overview of the relevant student scores.

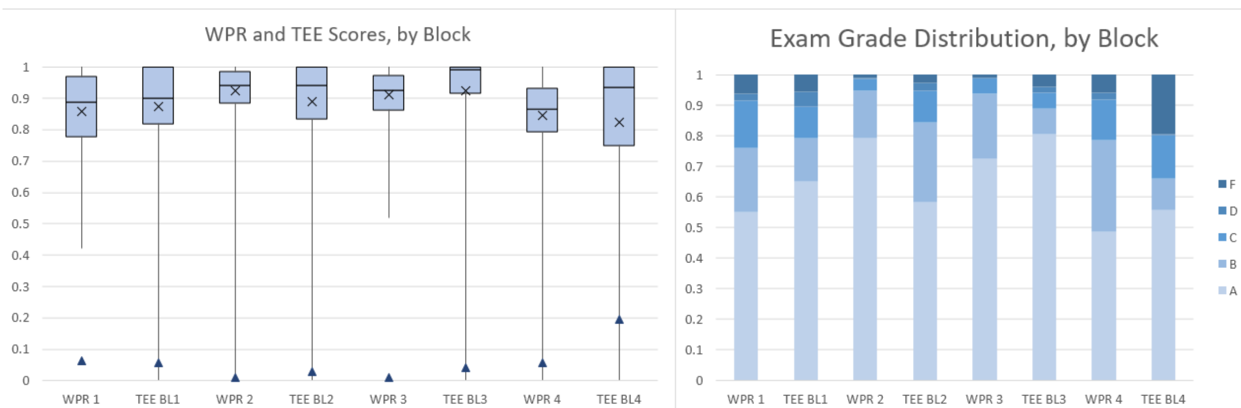

Figure 4: At left, WPR and TEE scores by block. The Box and Whisker diagram indicates the minimum and all quartile scores, the “X” indicates the course mean, and the Triangle indicates the proportion of the students that earned an “F” on that portion. At right, the distribution of grades by block, where the lightest colours earned an “A.”

We assess SLOs 7 and 10 as meeting the standard for the course, with the average scores solidly in the “B” range, failure rate between 5-20%, and less than 80% of the course scoring above 80% of the course points. These data are a pretty clear indication of student performance, and largely corroborate instructor experience on the most challenging blocks of the course.

We assess SLOs 8 and 9 as exceeding the standard for the course, with average scores in the “A-” range (see Table 5), failure rate below 5%, and 85% or greater of the course earning an “A” or “B” in the course. This is also correlated with instructor observations of the material students tend to quickly understand.

Assessment	**Block 1**	**Block 2**	**Block 3**	**Block 4**
WPR Avg	85.92%	92.27%	91.07%	84.54%
TEE Question Avg	87.42%	89.03%	92.23%	82.40%

Table 5: Comparison of the average unsuccessful validation exam score to the average WPR and TEE scores.

4.4.2 Lesson Modification

As described in Section 4.3, there was one additional lesson reallocated to Sequences and Series, and an additional WPR was added to test students on block 4. To assess the efficacy of these measures, we will compare student performance on the exam questions on the TEE this year, as compared to the previous year.

One statistical note: while the score data are non-normal, the sample sizes are so large that traditional nonparametric confidence interval tests (such as the Sign Test or Wilcoxon Sign-Rank Test) become tedious to calculate (and don't get much additional power). As such, we will assume that around the mean we can use traditional confidence interval testing (i.e., assume a t-distribution of scores around the mean). Also of note is that the very large sample sizes ($n_{23} = 273$ and $n_{24} = 311$ students each) also significantly restrict the 95% confidence interval.

The mean scores on the Sequences and Series TEE questions were 93.99% in AY23 and 91.40% in AY24, resulting in a difference of means of 2.59%. Considering a 95% confidence interval, we get the intervals $[92.80, 95.17]$ and $[89.93, 92.86]$. This means that, while the average scores decreased with an additional day of content, we cannot say the difference is statistically significant at the $p = .05$ level.

The mean score on the collective Vector Calculus section was 84.93% in AY23 and 92.40% in AY24, resulting in a difference of means of 2.54%. Again using a 95% confidence interval, we have the intervals $[83.98, 85.89]$ and $[81.27, 83.52]$. This means that at the $p = .05$ level we would say the average scores decreased by a statistically significant amount. With that said, because of the large number of observations (5 questions across 273 or 311 students, respectively), we are much more inclined to find statistical significance. If there were half as many observations, we would see overlap in our confidence intervals.

4.4.3 Instructor Preparation and Course Guide Updates

To assess the deliberate additional instructor preparation and the modifications to the course guide, we intended to conduct sentiment analysis on course AAR comments (instructor and student). However, as of the publishing of this document, there is no way to retrieve the data from the course-end survey to do this analysis. As such, there is no quantitative way to assess student or instructor feedback on the course guide, aside from anecdotal feedback from instructors.

4.5 Conclusions, Future Work

In this section we will discuss the efficacy of the current course structure and design, the assessed effect of any of the changes made from the previous year, and discuss recommendations for future course directors.

As discussed in Section 4.4.1 and visualized in Figure 4, the students generally do very well in the course. The course average this year was 88.82% (as compared to 89.75% and 89.74% in AY23 and AY22, respectively), with 95% of the students earning an “A” or “B” in the course. However, there were two students in the course that failed – more than any course in the AMP since the curriculum was re-ordered in AY16. This could be caused by many things, but two potential confounding factors were the larger program size (312 versus 273) and the “off-ramp” of 20 students from MA153 to MA255 in AY23 (see [7]), whereas only 2 cadets from MA153 in AY24 were not brought into MA255 (they were separated from the academy before the close of the year). Both of these changes, in addition to the other usual class-to-class variance, make identifying specific issues challenging. That aside, as indicated previously, SLOs 7-10 can be assessed as having met the standard.

Regarding the modifications to the course, the data are not clear. As spelled out in Section 4.4.2, there is not sufficient evidence to demonstrate statistically significant changes in course performance on sequences and series, and weak evidence that the students performed worse on the vector calculus TEE questions than they did in the previous year. Anecdotal feedback from instructors, however, has generally been of the opinion that both changes were good. The additional day on sequences and series meant cadets were able to demonstrate during work at the boards they could find and assess the radius of convergence of series. Many also felt that the final WPR was the right call and forced many students to study where they otherwise might not have.

In conclusion, the recommendations for the future course director are three-fold. First, to retain the basic structure of the course, including the Mini- or Block Projects, as the general student performance has been excellent. Second, to retain the major course changes, as they are at least anecdotally supported by instructors (as born out in the course AAR). Third, as mentioned at the closing of Section 4.1, consider petitioning to reorder the AMP to introduce cadets to sequences and series and multivariable concepts prior to Modeling with Ordinary Differential Equations so as to help students have more firm mathematical foundations in that course.

POC for this document are MAJ Devon Zillmer, the AY24 MA255 Course Director, at 910-728-2051 or devon.p.zillmer.mil@army.mil, or LTC Jonathan Paynter at jonathan.paynter@west

5 List of Acronyms

AAR After Action Review

AI Artificial Intelligence

AMP Advanced Math Program

CCPS Confident and Competent Problem Solvers

FDW Faculty Development Workshop

MPG Mathematics Program Goal

MAA Mathematical Association of America

MMA Mathematica

STEM Science, Technology, Engineering, and Mathematics

SLO Student Learning Objectives

TEE Term End Examination

USAFA United States Air Force Academy

USMA United States Military Academy

USNA United States Naval Academy

WPR Written Partial Review

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DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
WEST POINT, NEW YORK 10996

MADN-MTH 3 January 2024

MEMORANDUM FOR Cadets enrolled in MA255

SUBJECT: Student Memorandum for MA255, AY24 - Advanced Multivariable Calculus

1. **Purpose**. This memorandum provides an overview of MA255: Advanced Multivariable Calculus for Cadets enrolled in Academic Year (AY) 23-2.

2. **Requirements**.

a. Text. *Calculus: Early Transcendentals*, 9th Edition, James Stewart, Copyright 2016.

b. Software. *Wolfram Mathematica*, V13 or greater; and *Microsoft Office*, including *Word, Excel, PowerPoint, Teams*.

c. Online. *Canvas*, for all course documents, and *WebAssign* for automated scoring and feedback of daily homework.

3. **Attendance**.

a. MA255 is a 4.5 credit hour course. It does not follow the Day 1 / Day 2 schedule of a 3.0 credit hour course. The course schedule is found on the MA255 Canvas site under Course Resources.

b. Attendance at Written Partial Reviews (WPRs) is mandatory. Except in emergencies, cadets will not make any appointments that could preclude their attendance for a WPR. Any cadet missing a WPR will be required to take a make-up or make-ahead exam.

4. **Course Learning Outcomes**. This course is structured to develop the six core mathematics program goals with a focus on 10 Student Learning Outcomes.

a. **Build a body of knowledge**: **(7)** Students can use vectors and vector functions to model motion and surfaces in three-dimensional space. **(8)** Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems. **(9)** Students understand the concepts of multivariable integral calculus and can apply it to solve problems involving accumulation. **(10)** Students understand and apply the concepts of vector calculus.

b. **Communicate Effectively**: **(5)** Students improve the quality of both their written and verbal communication of mathematics. Effective communication is an essential characteristic of great leaders. You, as a future leader, must be able to clearly articulate your thoughts. You will get many opportunities to improve your communications through in-class board presentations, course homework assignments, technology labs, and the course projects.

c. **Apply Technology**: **(6)** Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems. You will become competent in using a computer algebra system (Mathematica) through in-class learning and self-paced technology labs. You will leverage technology to assist in your daily learning as well as in completing the course projects. Ultimately, you will understand the connection between the mathematical theory and its implementation using technology.

d. **Build Competent and Confident Problems Solvers**: **(2)** Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus. The problem solving will be integrated in class problems, technology labs and the course project. You will hone your problem-solving skills by formulating and solving challenging mathematical problems.

e. **Provide Experience in Interdisciplinary Problem Solving**: **(3)** Students develop a knowledge, curiosity and appreciation for the application of mathematics in different disciplines. **(4)** Students can design, solve and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems. There will be multiple opportunities in class and during the course lectures for you to see real world problems from a variety of disciplines. You will see how mathematics is used to solve these problems.

f. **Develop Habits of Mind**: **(1)** Students learn good scholarly habits for progressive intellectual development. You will learn habits in this course that will benefit you throughout your academic career. You will develop an intellectual curiosity in general, and for mathematics, by doing challenging but interesting mathematical problems. You will improve your work ethic and critical thinking through learning and doing mathematics. You will develop skills to be independent learners through the conduct of the course. These habits of mind are essential traits of leaders and scholarly students.

5. **Lesson Assignments**. The Course Schedule and Course Guide serve as the road map for the course. These documents are both available from the MA255 Canvas page.

6. **Course Summary**. There are a total of **65** required attendances consisting of 46 regular lessons, 8 project lessons, 7 problem-solving labs, and 4 exams.

7. **Course Evaluation Plan**. See Table 1 for the distribution of course points.

8. **Grading**.

a. Final course grades will follow the Advanced Math Program modification to the Department of Mathematical Sciences' guidelines, per Table 2.

Category	Event	Points	Per Cent	Overall
Individual Exams	Term End Exam	900	30%	63%
	WPR 1	250	8.3%	
	WPR 2	250	8.3%	
	WPR 3	250	8.3%	
	WPR 4	250	8.3%	
Group Work (Modeling)	Mini-Project 1	125	4.2%	17%
	Mini-Project 2	125	4.2%	
	Mini-Project 3	125	4.2%	
	Mini-Project 4	125	4.2%	
Homework	WebAssign Block 1	75	2.5%	10%
	WA Block 2	75	2.5%	
	WA Block 3	75	2.5%	
	WA Block 4	75	2.5%	
Instructor	Instructor Points	300	10%	10%
	Total:	3000	100%	

Table 1: Point values by Assignment.

Grade	Per Cent Range	QPA	Word Picture
A+	$95 \leq$ Mark ≤ 100	4.33	Beyond expectations of course
A	$90 \leq$ Mark < 95	4.0	Dominates the material
A-	$87 \leq$ Mark < 90	3.67	Mastery
B+	$83 \leq$ Mark < 87	3.33	Excellent performance
B	$80 \leq$ Mark < 83	3.0	Good understanding
B-	$77 \leq$ Mark < 80	2.67	Proficient, with aptitude for the subject
C+	$73 \leq$ Mark < 77	2.33	Can build upon this foundation
C	$70 \leq$ Mark < 73	2.0	Passing; proficient now
C-	$67 \leq$ Mark < 70	1.67	Short-range understanding
D	$64 \leq$ Mark < 67	1.0	Marginal performance with elementary understanding
F	Mark < 64	0	Failing; definitely failed to demonstrate understanding

Table 2: Grade Scale by percent score out of course points.

b. Cadets who score less than 50% on the TEE, regardless of their final course average, or whose final course average is less than 65%, may fail the course.

c. Instructors may award up to 30 bonus points at their discretion. Bonus points will be added directly to the total number of points you earn in the course.

9. **Documentation**. All cadet work in MA255 will be in accordance with the procedures found in the Dean's *Documentation and Acknowledgment of Academic Work* (DAAW), June 2023. All required out-of-class work constitutes homework as defined in the DAAW.

The course standard for MA255 is MLA. Refer to the DAAW for further information. Proper documentation is required for all graded exercises assigned by your instructor and all out-of-class course-wide assignments outlined in this memorandum.

10. **Out-of-Class Assignments**.

a. **Daily Homework**. Daily web-based homework problems will be assigned via WebAssign. To help build good habits, the assignment for a given lesson in Block 1 will be due prior to the start of the next lesson; for Block 2 and beyond, assignments will be due weekly. After you submit each assignment, you will receive immediate feedback, to include your grade. Additionally, all daily WebAssign problem sets require students to complete an online acknowledgement statement. After the first assignment, completion of the previous assignment is required in order to begin the next assignment. Finally, MA255 uses the following WebAssign assignment extension policy: students may ask for up to two *automatic* extensions, each three days in duration, for a deduction of 10% per period. The point deduction only applies to problems not completed prior to the assignment deadline. A WebAssign automatic extension is an assignment extension for which instructor permission is *not* required. Students may use automatic extensions for a 14 day period after the exercise is assigned, or through the end of the given block of instruction, whichever is earlier.

b. **Miniature Projects**. During each block, students will be assigned a project that synthesizes the material covered in the block. These will be group-based (2- or 3-person) and are estimated to require approximately 3 hours per cadet outside of class for each day in class. The projects will require the effective use of technology and will alternate delivery media (presentations or reports). The purpose of the projects is to reinforce course content and demonstrate various interdisciplinary applications of multivariable calculus while helping develop cadets' communications skills.

11. **In-Class Assignments**

a. **Written Partial Reviews**. WPRs assess your understanding of lesson objectives in the course. Each WPR is cumulative (meaning any content covered up to the point may be assessed) but focuses on content from the corresponding block. Attendance at WPRs is mandatory as described in the Attendence paragraph above.

b. **Term End Exam**. The TEE is mandatory and cumulative for the entire course.

12. **Classroom Rules**. Cadets will conform to the highest standards of ethical behavior and military bearing in the performance of their academic duties. Cadets will treat every member of the faculty with the courtesy appropriate to commissioned officers. Smoking, using smokeless tobacco, chewing gum, and consuming food is always prohibited in classrooms, laboratories, lecture halls, and auditoriums. Drinks are allowed. The uniform for classes will be the duty uniform, with exceptions made for injury and special class functions that occur during the class day. Cadets will leave outer garments and bags in the hallway. The uniform for additional instruction (AI) is the duty uniform.

13. **Chain of Command**. The Department of Mathematical Sciences has an open-door policy. Your initial attempt to address problems or concerns – including respect in the classroom, unprofessional behavior, sexual assault, harassment – should go through your instructor. However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain to include the Head of the Department. The entire math department faculty is here to help you.

a. Your instructor.

b. MA255 Course Director: MAJ Zillmer, Thayer 255B, x6308.

c. Associate Program Director: LTC Paynter, Thayer 255A, x4544.

d. Calculus Program Director: COL Lindquist, Thayer 225, x7436.

e. Head of the Department of Mathematical Sciences: COL Scioletti, Thayer 238, x5870.

14. Point of contact for this memorandum is MAJ Devon Zillmer at 845-938-6308 or devon.zillmer@westpoint.edu.

JONATHAN L. PAYNTER
LTC, AR(49)
AMP Associate Program Director

MA255 AY 24-2

Advanced Multivariable and Vector Calculus

Spring 2024 Schedule

	WK	Monday	Tuesday	Wednesday	Thursday	Friday	Weekend
Block 1: Vectors and Geometry of Space	1	08 Jan 24 LSN 1 Course Introduction	09 Jan 24 LSN 2 3D Coordinate Systems (12.1)	10 Jan 24 LSN 3 Vectors in 3D I (12.2)	11 Jan 24 LSN 4 Vectors in 3D II (12.2)	12 Jan 24 Course Drop	13 Jan 24 | 14 Jan 24 B
	2	15 Jan 24 Martin Luther King Day	16 Jan 24 LSN 5 Dot Product (12.3) Cross Product (12.4)	17 Jan 24 LSN 6 Equations of Lines and Planes (12.5)	D 18 Jan 24 LSN 7 Cylinders and Quadric Surfaces (12.6)	19 Jan 24 Course Drop	20 Jan 24 | 21 Jan 24 B 500th Night
	3	22 Jan 24 LSN 8 Vector Functions and Space Curves (13.1) **MP1 Intro**	23 Jan 24 LSN 9 MP1 Working Day 1	D 24 Jan 24 Course Drop	25 Jan 24 LSN 10 Derivatives and Integrals of Vector Functions (13.2)	26 Jan 24 LSN 11 Arc Length and Curvature (13.3)	27 Jan 24 | 28 Jan 24 B
	4	29 Jan 24 LSN 12 Velocity and Acceleration (13.4)	30 Jan 24 LSN 13 MP1 Working Day 2	D 31 Jan 24 Course Drop **MP1 Due**	01 Feb 24 LSN 14 PSL 1 Synthesis / Board Problems	02 Feb 24 LSN 15 WPR 1	03 Feb 24 | 04 Feb 24 B
Block 2: Optimization and Taylor Series	5	D 05 Feb 24 LSN 16 Functions of Several Variables, Partial Derivatives (14.1-3)	06 Feb 24 LSN 17 Tangent Planes and Linear Approximations (14.4)	07 Feb 24 LSN 18 The Chain Rule (14.5)	08 Feb 24 LSN 19 Directional Derivatives and the Gradient Vector (14.6)	09 Feb 24 Course Drop	10 Feb 24 | 11 Feb 24 B 100th Night
	6	12 Feb 24 LSN 20 Maximum and Minimum Values (14.7)	13 Feb 24 LSN 21 Lagrange Multipliers I (14.8) **MP2 Intro**	D 14 Feb 24 LSN 22 MP2 Working Day 1	15 Feb 24 LSN 23 Lagrange Multipliers II (14.8)	16 Feb 24 Course Drop	17 Feb 24 | 18 Feb 24 B
	7	19 Feb 24 Presidents Day	20 Feb 24 LSN 24 Sequences and Series (11.1-2)	21 Feb 24 LSN 25 Alternating Series, Ratio and Root Tests (11.5-6)	D 22 Feb 24 LSN 26 MP2 Working Day 2 **MP2 Due**	23 Feb 24 LSN 27 Power Series (11.8)	24 Feb 24 | 25 Feb 24 B Yearling Winter Wkd
	8	26 Feb 24 LSN 28 Taylor and Maclaurin Series (11.10)	27 Feb 24 LSN 29 PSL 2 Synthesis / Board Problems	28 Feb 24 LSN 30 WPR 2	29 Feb 24 Course Drop	01 Mar 24 LSN 31 Double Integrals over Rectangles (15.1)	02 Mar 24 | 03 Mar 24 A By-Stander
Block 3: Triple Integrals	9	04 Mar 24 LSN 32 Double Integrals over General Regions I (15.2)	05 Mar 24 LSN 33 Double Integrals over General Regions II (15.2)	06 Mar 24 Course Drop	07 Mar 24 LSN 34 Double Integrals in Polar Coordinates (15.3)	08 Mar 24 LSN 35 Applications of Double Integrals I (15.4)	09 Mar 24 | 10 Mar 24 B
	10	11 Mar 24 LSN 36 Applications of Double Integrals II (15.4)	12 Mar 24 LSN 37 Triple Integrals (15.6) **MP3 Intro**	D 13 Mar 24 LSN 38 Triple Integrals in Cylindrical Coordinates (15.7)	14 Mar 24 Course Drop	15 Mar 24 LSN 39 Triple Integrals in Spherical Coordinates I (15.8)	16 Mar 24 | 17 Mar 24 A 1CL ACFT
	11	18 Mar 24 LSN 40 Triple Integrals in Spherical Coordinates II (15.8)	19 Mar 24 LSN 41 PSL 3 Synthesis / Board Problems	20 Mar 24 LSN 42 WPR 3	21 Mar 24 Course Drop	22 Mar 24 LSN 43 MP3 Working Day 1	23 Mar 24 | 24 Mar 24 PPW Spring Break
	12	25 Mar 24 Spring Break	26 Mar 24 Spring Break	27 Mar 24 Spring Break	28 Mar 24 Spring Break	29 Mar 24 Spring Break	30 Mar 24 | 31 Mar 24 B Spring Break
Block 4: Vector Calculus	13	01 Apr 24 Corps Returns	02 Apr 24 LSN 44 MP3 Working Day 2	03 Apr 24 LSN 45 Vector Fields (16.1)	D 04 Apr 24 LSN 46 Line Integrals I (16.2) **MP3 Due**	05 Apr 24 LSN 47 Line Integrals II (16.2)	06 Apr 24 | 07 Apr 24 A Sandhurst Validation
	14	08 Apr 24 LSN 48 The Fundamental Theorem for Line Integrals (16.3)	09 Apr 24 LSN 49 Green's Theorem I (16.4)	D 10 Apr 24 Course Drop	11 Apr 24 LSN 50 Green's Theorem II (16.4) **MP4 Intro**	D 12 Apr 24 LSN 51 Curl & Divergence (16.5)	13 Apr 24 | 14 Apr 24 A ACFT
	15	D 15 Apr 24 LSN 52 PSL 4 Line Integral PSL	16 Apr 24 LSN 53 Parametric Surfaces and Their Areas I (16.6)	17 Apr 24 Course Drop	18 Apr 24 LSN 54 Parametric Surfaces and Their Areas II (16.6)	19 Apr 24 LSN 55 Surface Integrals I (16.7)	20 Apr 24 | 21 Apr 24 B
	16	22 Apr 24 LSN 56 Surface integrals II (16.7)	23 Apr 24 LSN 57 Stokes' Theorem I (16.8)	24 Apr 24 LSN 58 Stokes' Theorem II (16.8)	25 Apr 24 LSN 59 Divergence Theorem (16.9)	26 Apr 24 Sandhurst	27 Apr 24 | 28 Apr 24 A Sandhurst
	17	29 Apr 24 LSN 60 PSL 5 Surface Integral PSL	30 Apr 24 LSN 61 WPR 4	D 01 May 24 LSN 62 MP4 Working Day 1	02 May 24 Projects Day	03 May 24 Course Drop	04 May 24 | 05 May 24 A
	18	06 May 24 Course Drop	07 May 24 LSN 63 MP4 Working Day 2 **MP4 Due**	08 May 24 Course Drop	09 May 24 LSN 64 PSL 6 Course Synthesis	10 May 24 LSN 65 PSL 7 Course Synthesis	11 May 24 | 12 May 24 B Reading Day
TEE	19	13 May 24 TEEs	14 May 24 TEEs	15 May 24 TEEs	16 May 24 TEEs	17 May 24 TEEs	18 May 24 | 19 May 24 TEE

D Dean's Day   M Modified Day   (12.1) <- Stewart's *Calculus,* Chapter 12, Section 1

Tech Lab | MAD / Project | WPR | Problem Solving Lab | Holiday / Course Drop | Day 2 (blue)

Version Date: 12/27/2023

TAB E AY2025 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
DEPARTMENT OF MATHEMATICAL SCIENCES
WEST POINT, NEW YORK 10996

MADN-MTH 5 June 2025

MEMORANDUM THRU

Dr. Rachelle DeCoste, Director, Mathematical Sciences Program, Department of Mathematical Sciences, USMA, West Point, NY 10996

COL Michael Scioletti, Professor and Head, Department of Mathematical Sciences, USMA, West Point, NY 10996

FOR Office of the Dean, AARS, United States Military Academy, West Point, NY 10996

SUBJECT: Executive Summary for MA255, AY25-2 - Advanced Multi-Variable Calculus

1. **Purpose:** The purpose of this memorandum is to detail the methodology for results of the MA255 curriculum assessment.

2. **Assessment:** The Advanced Math Program (AMP) has a three year assessment strategy. This year's assessment examined each Student Learning Outcome (SLO) listed below in Table 1 with our findings for each assessed area. Detailed results of the findings can be found in Enclosure 4 - Detailed Course Design and Assessment Report, Section 4. The full list of SLOs can be found in Enclosure 2.

MA255 SLOs	Strength of Attainment 1	Strength of Attainment 2	Strength of Evidence	Overall Evaluation
SLO 1	F	B	A	C
SLO 2	A	B	A	B
SLO 3	C	B	B	B

Table 1: MA255 AY25-2 SLO assessment evaluations.

To read table 1, letter grades in Strength of Attainment are assigned as "A" for surpassing instructor expectations, "B" for approximately meeting instructor expectations, "C" for meeting instructor expectations with some weaknesses, "D" for not meeting the standard (but can be addressed in the short-term), and "F" for significantly failing to meet the standard.

a. Strength of Attainment will be measured differently for each SLO. See Enclosure 2 for full descriptions of AY25-2 SLO strength of attainment methodology.

b. Strength of Evidence

(1) A strength of evidence score of "A" is assigned if the course director has multiple samples of either quantitative or qualitative data ***generated by the Cadets*** (e.g. a pre- and post-event survey).

(2) "B" will be assigned for single data points of qualitative data generated by the Cadets **or** multiple subjective analyses from the instructors.

(3) "C" will be assigned for single subjective analyses from the instructors.

3. **Assessment Discussion:** This section presents an overview of assessment findings for AY25. A detailed discussion of direct and indirect indicators can be found in the Course Design and Analysis Report located in Enclosure 3.

a. Assessment Summary with Abbreviated SLOs:

(1) SLO 1 - Students learn good scholarly habits for progressive intellectual development. Strength of Attainment 1 essentially assesses if Cadets do at least 80% of their homework on time. It is assessed at an F because Cadets largely do not do their homework *on time* (a critical part of this method of assessment). Strength of Attainment 2 assess Cadet confidence in their study habits based on survey responses. It is assessed at a B because there is no significant change from their first semester, though we have data to suggest that their confidence in their habits has increased since they arrived to USMA. Over 86% of responses indicated an increase in the confidence in their study habits.

Tables 8 and 9 in Enclosure 4.1.1 give us insight into how and when Cadets do their homework. Only 42.3% of Cadets achieve over 80% on at least 80% of their **on time** homework assignments. Conversely, as final grades approach, Cadets recognize the benefit of homework exclusively as a means to earning course points and the number of Cadets who achieve more than 80% on at least 80% of the assignments (Table 9) jumps to 74.1%. Recommendations for closing this gap between Cadet-perceived utility and instructor-driven incentive follow in paragraph 3. e. below.

(2) SLO 2 - Students build critical thinking skills to solve problems through learning complex mathematical concepts in multivariable and vector calculus. Strength of Attainment 1 assesses Cadet performance on Block Projects and is assessed at an A because they have an average of 92.95% across the four Block Projects. Strength of Attainment 2 assesses Cadet performance on 'modeling' questions on Written Partial Review (WPR)s and the Term End Examination (TEE). This is assessed at a B because the Cadets have a weighted average of 85.22% across the modeling questions.

(3) SLO 3 - Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines. Strength of Attainment 1 assesses the consistency and utility of instructors bringing problems from other departments directly into the classroom. This is assessed at a C because of the limited number of concrete problems that were imported into the classroom. Instructors instead mostly brought in concepts. Strength of Attainment 2 assesses Cadet confidence in their ability to apply math in other disciplines based on survey responses.

This is assessed at a B because there is no significant difference from AY25-1, but again, we have strong data to suggest that their confidence has increased since they arrived to USMA. Over 94% of responses indicated an increase in their confidence in this realm.

b. Status of significant changes:

(1) We have changed the sequence of Block 1 (Vectors and Geometry of Space) to present a block of instruction on Stewart Ch. 13.4 (“Velocity and Acceleration”) prior to the first Block Project (BP) and Ch. 13.3 (“Arc Length and Curvature”) to more fully enable the Cadets to work independently on BP1. This new sequence makes Ch. 13.3 the final lecture that presents new material before WPR1. Course leadership does not assess any significant difference in risk to Cadet performance between Ch. 13.3 or Ch 13.4 being the final lecture before WPR1.

(2) BPs 1, 3, and 4 are now due after their respective WPRs. This gives the Cadets additional time to work on the BPs, which have traditionally been a relatively large intellectual hurdle for the Cadets (particularly BP1). More importantly, it deconflicts the Cadets’ priorities and enables them to focus on the foundational analytical skills required for a WPR before fully focusing on the associated BP, which is intended to be a summative assessment through which the Cadets demonstrate their mastery of the material. See Fig. 1 for a calendar of the course.

(3) A supplemental document to Stewart 10.1 (Curves Defined by Parametric Equations) is now a deliberate effort in LSN 07. Previously, this lesson focused on only Cylinders and Quadric Surfaces (Stewart 12.6), and has historically been an easier lesson. The addition of deliberate instruction on 2D parametrizations is intended to more rigorously expose Cadets to the concept of parametrizations before there is deliberate instruction on parametrizations of surfaces in $\mathbb{R}^3$.

c. Strengths in the course: The major strengths of the course lie in its mathematical maturation of high-performing students and its focus on getting those students a high volume of experience performing the mathematics while providing an opportunity to attempt real-world applications.

d. Areas for improvement in the course: Creating time and formalizing the process of supplemental teaching/reteaching and reassessing material is a shortcoming for the course. The course assesses a given student’s ability to study sufficiently well, but we do not have a powerful metric into the students’ commitment to additional scholarly habits in SLO 1 (e.g., reviewing material on which they did not perform well or if they continue any of the forced nightly study habits that the course requires).

e. Recommended changes to be implemented in AY26:

(1) Implement stricter penalties for late turn-in on daily WebAssign assignments. Analysis of SLO 1 reveals that Cadets will not do homework unless they see the direct benefit to their overall grade. This stands in opposition to the actual intent of reinforcing and supplementing the material from the daily lecture. Significantly increasing the penalty on daily homework supports the intent of the assignments by more directly motivating the behavior that the instructors expect. Recommend modest (10% - 25%) penalties for late homework on the weekly assignments.

These are 'review-style' assignments and serve a skill-building purpose beyond their contemporaneous use and should not be as heavily penalized. I fear that too heavy a penalty will make Cadets see homework as 'not worth it' if they miss the original due date and I firmly belive that 'late practice is better than no practice.' Further recommend to **not** change the structure of either the daily or weekly assignments.

(2) Split Block Projects into a writing/presentation grade and a mathematics grade. The instructor team convention of targeting an A- average across an instructors' sections overly homogenizes the grade to not give Cadets enough feedback on their shortcomings. Splitting into a math grade that assesses the *technique* and a writing/presentation grade that assess their *communication skills* more fully gives Cadets feedback on how they can improve and supports analysis of SLOs related to communication for the course. Additionally, recommend that the Block Projects themselves be increasingly developed in a way that concretely and simply illustrates an application of the technique at hand and then goes into "exploration" of more advanced techniques. This has qualitatively shown to increase Cadet engagement for high-performers and is expected to simultaneously lower the barrier for average performers to earn (approximately) a B on their project without having to intentionally 'curve' the projects around an A-.

(3) Maintain the split between conceptual and technical problems on the WPRs while slightly increasing the weight of the conceptual questions. WPRs in AY25-2 did a good job of isolating Cadets' ability to understand concepts and to apply core techniques from the course. This is the correct direction for the major assessments in the course, but more weight should be given to the conceptual topics. This year, approximately 15% of an exam was devoted to concepts. That percentage should likely be 30% - 40%.

f. Proposed changes to the SLOs:

(1) The primary objective for SLO rewrites throughout the AY25-2 is to reduce the SLO list to only those items that are directly and quantifiably assessable in line with the curriculum committee's guidance to MA205. This will most likely move the current SLOs 1, 2, and 3 into a category of "Course Goals" (working name) as opposed to SLOs.

(2) Recommend removal of the term "complex" from SLO 2 to avoid confusion with complex algebra or geometries. Replace with any of "challenging/difficult/complicated" to preserve original feeling and structure. Revised version as "Students build critical thinking skills to solve problems through learning challenging mathematical concepts in multivariable and vector calculus."

(3) In accordance with department guidance to update SLOs to include ethical reasoning, recommend that SLO 4 be re-written to say "Students can objectively design, appropriately solve, and dispassionately interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems." Recommend this change to address the need for students to more fully separate themselves from discouraged research practices that may lead to selective presentation of data to support a hypothesis (e.g. "p-hacking") or retroactively adjusting a research question to fit results. Text color for emphasis only.

(4) To more fully capture the scope of what MA255 students are asked to do in their Block Projects, recommend SLO 6 be rewritten as “Students use technology to illustrate, solve, and communicate the solution to applied calculus problems.” The change in verbiage from “visualize” to “illustrate” captures the fact that the students are asked to explain their graphics. Additionally, “Illustrate” carries a connotation of purpose for use by others, vice “visualizations” carrying a self-use connotation. The objective of communicating the illustrations is core to the reports and presentations in the Block Projects, which are the primary use case of technology in the course. Further recommend removing the usage of “complex” IAW with proposed change (1) above.

4. The point of contact for this memorandum is MAJ Joseph Maxwell, the AY25 MA255 Course Director, at 256-468-7272 or joseph.b.maxwell.mil@army.mil, or LTC Karoline Hood at karoline.hood@westpoint.edu.

KAROLINE M. HOOD
LTC, EN(47)
AMP Associate Program Director

4 Encls
1. Course Calendar AY25-2
2. Full Set of AY25-2 SLOs
3. Mapping of Academic Program Goals to SLOs
4. Detailed Course Design and Analysis Report

1. Course Calendar AY25-2

MA255 AY 25-2
Advanced Multivariable and Vector Calculus
Spring 2025 Schedule

Block	WK	Monday	Tuesday	Wednesday	Thursday	Friday	Weekend
Block 1: Vectors and Geometry of Space	1	06 Jan 25 Reorgy Day	07 Jan 25 LSN 1 Course Intro **Day 1 & Day 2**	08 Jan 25 LSN 2 3D Coordinate Systems (12.1)	09 Jan 25 LSN 3 Vectors I (12.2)	10 Jan 25 LSN 4 Vectors II (12.2)	11 Jan 25 | 12 Jan 25 B 500th Night
	2	13 Jan 25 LSN 5 Dot Product (12.3) Cross Product (12.4)	14 Jan 25 LSN 6 Equations of Lines and Planes I (12.5)	15 Jan 25 Course Drop1	16 Jan 25 LSN 7 Equations of Lines and Planes II (12.5)	17 Jan 25 Course Drop2	18 Jan 25 | 19 Jan 25 B
	3	20 Jan 25 Martin Luther King Day	21 Jan 25 LSN 8 Quadric Surface (12.6) 2D Parametrizations (10.1)	22 Jan 25 LSN 9 Vector Functions and Space Curves (13.1)	D 23 Jan 25 LSN 10 **Der. & Ints. of Vector Fxns (13.2)** **Velocity and Acceleration (13.4)** **BP1 Intro**	24 Jan 25 LSN 11 BP1 Working Day 1	25 Jan 25 | 26 Jan 25 B Yearling Winter Wkd
	4	27 Jan 25 LSN 12 Arc Length and Curvature (13.3)	28 Jan 25 Course Drop3	D 29 Jan 25 LSN 13 PSL 1 Synthesis / Board Problems	30 Jan 25 LSN 14 WPR 1	P 31 Jan 25 Course Drop4	01 Feb 25 | 02 Feb 25 A
Block 2: Optimization and Taylor Series	5	03 Feb 25 LSN 15 BP1 Presentation Day **BP1 Due**	04 Feb 25 LSN 16 Tangent Planes and Linear Approximations (14.4)	05 Feb 25 LSN 17 The Chain Rule (14.5)	06 Feb 25 LSN 18 Directional Derivatives and the Gradient Vector (14.6)	D 07 Feb 25 Course Drop1	08 Feb 25 | 09 Feb 25 B 100th Night
	6	10 Feb 25 LSN 19 Maximum and Minimum Values (14.7) **BP2 Intro**	11 Feb 25 LSN 20 BP2 Working Day 1	12 Feb 25 LSN 21 Lagrange Multipliers I (14.8)	P 13 Feb 25 LSN 22 Lagrange Multipliers II (14.8)	14 Feb 25 Course Drop2	15 Feb 25 | 16 Feb 25 B
	7	17 Feb 25 Presidents Day	18 Feb 25 LSN 23 BP2 Working Day 2	19 Feb 25 LSN 24 Sequences and Series (11.1-2)	D 20 Feb 25 LSN 25 Alternating Series and Ratio Roots Tests (11.5-6)	21 Feb 25 LSN 26 Power Series (11.8) **BP2 Due**	22 Feb 25 | 23 Feb 25 B
	8	D 24 Feb 25 LSN 27 Taylor and Maclaurin Series (11.10)	25 Feb 25 LSN 28 PSL 2 Synthesis / Board Problems	26 Feb 25 LSN 29 WPR 2	27 Feb 25 Course Drop3	28 Feb 25 LSN 30 Double Integrals over Rectangles (15.1)	01 Mar 25 | 02 Mar 25 A By-Stander?
Block 3: Triple Integrals	9	D 03 Mar 25 LSN 31 Double Integrals over General Regions I (15.2)	04 Mar 25 LSN 32 Double Integrals over General Regions II (15.2)	05 Mar 25 Course Drop1	06 Mar 25 LSN 33 Double Integrals in Polar Coordinates I (15.3)	P 07 Mar 25 LSN 34 Double Integrals in Polar Coordinates II (15.3)	08 Mar 25 | 09 Mar 25 A 1CL ACFT?
	10	10 Mar 25 LSN 35 Applications of Double Integrals (15.4)	11 Mar 25 LSN 36 Triple Integrals (15.6) **BP3 Intro**	D 12 Mar 25 Course Drop2	13 Mar 25 LSN 37 BP3 Working Day 1	14 Mar 25 LSN 38 **Triple Integrals in Cylindrical** Coordinates (15.7)	15 Mar 25 | 16 Mar 25 PPW Spring Break
	11	17 Mar 25 Spring Break	18 Mar 25 Spring Break	19 Mar 25 Spring Break	20 Mar 25 Spring Break	21 Mar 25 Spring Break	22 Mar 25 | 23 Mar 25 B Spring Break
	12	24 Mar 25 LSN 39 Triple Integrals in Spherical Coordinates I (15.8)	25 Mar 25 LSN 40 Triple Integrals in Spherical Coordinates II (15.8)	26 Mar 25 LSN 41 PSL 3 Synthesis / Board Problems	27 Mar 25 LSN 42 WPR 3	P 28 Mar 25 Course Drop3	29 Mar 25 | 30 Mar 25 A
Block 4: Vector Calculus	13	D 31 Mar 25 LSN 43 Vector Fields (16.1)	01 Apr 25 LSN 44 Line Integrals I (16.2)	02 Apr 25 LSN 45 Line Integrals II (16.2)	03 Apr 25 LSN 46 The Fundamental Theorem for Line Integrals (16.3)	04 Apr 25 LSN 47 BP3 Presentation Day **BP3 Due**	05 Apr 25 | 06 Apr 25 A Sandhurst Validation
	14	07 Apr 25 LSN 48 Green's Theorem I (16.4)	08 Apr 25 LSN 49 Green's Theorem II & Curl (16.4) **BP4 Intro**	09 Apr 25 Course Drop1	10 Apr 25 LSN 50 PSL 4 Line Integral PSL	P 11 Apr 25 LSN 51 Curl & Divergence (16.5)	12 Apr 25 | 13 Apr 25 B
	15	14 Apr 25 LSN 52 Parametric Surfaces and Their Areas I (16.6)	15 Apr 25 LSN 53 Parametric Surfaces and Their Areas II (16.6)	D 16 Apr 25 Course Drop2	17 Apr 25 LSN 54 Surface Integrals I (16.7)	18 Apr 25 LSN 55 Surface integrals II (16.7)	19 Apr 25 | 20 Apr 25 B
	16	21 Apr 25 No Class Day	D 22 Apr 25 LSN 56 Stokes' Theorem I (16.8)	23 Apr 25 LSN 57 Stokes' Theorem II (16.8)	24 Apr 25 Projects Day	25 Apr 25 Course Drop3	26 Apr 25 | 27 Apr 25 A
	17	28 Apr 25 LSN 58 Divergence Theorem (16.9)	29 Apr 25 LSN 59 BP4 Working Day 1	D 30 Apr 25 LSN 60 PSL 5 Surface Integral PSL	01 May 25 LSN 61 WPR 4	02 May 25 Sandhurst	03 May 25 | 04 May 25 A Sandhurst
	18	05 May 25 LSN 62 BP4 Working Day 2	06 May 25 Course Drop4	D 07 May 25 LSN 63 PSL 6 Course Synthesis **BP4 Due**	08 May 25 LSN 64 PSL 7 Course Synthesis	09 May 25 Course Drop5	10 May 25 | 11 May 25 Reading Day
TEE	19	12 May 25 TEEs	13 May 25 TEEs	14 May 25 TEEs	15 May 25 TEEs	16 May 25 TEEs	17 May 25 | 18 May 25 TEE (17th)

D Dean's Day   M Modified Day   (12.1) <- Stewart Calculus Chapter 12, Section 1

Tech Lab	MAD / Project	WPR	Problem Solving Lab	Holiday / Course Drop	Day 2 (blue)

Version Date:

Figure 1: MA255 AY25-2 Course Calendar

2. Full set of AY25-2 SLOs

Below are the SLOs for MA255 Advanced Multivariable Calculus.

1. Students learn good scholarly habits for progressive intellectual development.

   AY25 Assessment 1: Percent of Cadets who achieve and 80% score on at least 80% of **on time** homework assignments.

Range	Grade
[80%, 100%]	A
[70%, 80%)	B
[65%, 70%)	C
[0%, 65%)	F

Table 2: MA255 AY25-2 SLO 1 assessment 1 methodology

   AY25 Assessment 2: End of course survey question as compared to end of course survey question from MA153 - "I am more confident in my ability to apply math in different academic disciplines than I was when I arrived to USMA." Choices will be "Strongly Disagree", "Slightly Disagree", "Slightly Agree", and "Strongly Agree". The CD team will test a difference of means between the two semesters' responses according to Table 3 below.

Result	Grade
Statistically Significant Increase	A
No Significant Difference	B
Statistically Significant Decrease	C

Table 3: MA255 AY25-2 SLO 1 assessment 2 methodology

2. Students build critical thinking skills to solve problems through learning complex mathematical concepts in multivariable and vector calculus.

   AY25 Assessment 1: Block project scores. This will be assessed at each BP according to MA255's letter scoring criteria with no plus or minus modifiers.

   AY25 Assessment 2: Modeling questions on WPRs/TEE. This will be assessed at each WPR according to MA255's letter scoring criteria with no plus or minus modifiers.

3. Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines.

   AY25 Assessment 1: Qualitative instructor assessment based on exposure through Faculty Development initiative to provide problems from other departments that demonstrate applications.

   AY25 Assessment 2: Compare MA255 end of course question to MA153 - "I am confident in my ability to apply math in different academic disciplines." Choices will be "Very Unconfident", "Slightly Unconfident", "Slightly Confident", and "Very Confident". A difference of means will again be used according to Table 3 above.

4. Students can design, solve, and interpret the results of mathematical models using multi-variable calculus to address complex and interdisciplinary problems.

   AY26-2 overall evaluation: TBD

5. Students improve the quality of both their written and verbal communication of mathematics.

   AY26-2 overall evaluation: TBD

6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems.

   AY26-2 overall evaluation: TBD

7. Students can use vectors and vector functions to model motion and surfaces in three dimensional space.

   AY24-2 overall evaluation: B

8. Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems.

   AY24-2 overall evaluation: B

9. Students understand the concepts of multivariable integral calculus and can apply it solve problems involving accumulation.

   AY24-2 overall evaluation: B

10. Students understand and apply the concepts of vector calculus.

    AY24-2 overall evaluation: B

Anticipated changes to SLOs: Changes proposed in 3.e above.

3. Mapping of Academic Program Goals to SLOs

	Academic Program Goals / MA255 Assessment Techniques				
Supported Academic Program Goals *(From Educating Army Leaders)*	*1. Graduates communicate effectively with all audiences*	*2. Graduates think critically and creatively*	*3. Graduates demonstrate the capability and desire to pursue progressive and continued intellectual development*	*4. Graduates recognize ethical issues and apply ethical perspectives and concepts in decision making.*	*5. Graduates apply science, technology, engineering, and math concepts and processes to solve complex problems*
***Supported What Graduates Can Do** Tasks*	1.2: Speak and write using standard American English.	2.4: Reason both quantitatively and qualitatively.	3.1: Demonstrate the willingness and ability to learn independently.	4.4: Apply ethical perspectives and concepts in solving complex problems, including those found in military settings.	5.1: Apply mathematics, science, and computing to model devices, systems, processes, or behaviors.
	1.3: Effectively convey meaningful information to diverse audiences using appropriate forms and media.				
	1.5: Use sound logic and relevant evidence to make convincing arguments.				5.3: Collect and analyze data in support of decision making.
Mathemtatics Program Goals	G4: Communicate mathematics effectively.	G5: Demonstrate a body of knowledge. G3: Apply technology.	G1: Develop habits of mind.	G1: Develop habits of mind.	G2: Demonstrate problem solving skills. G6: Consider problems from an interdisciplinary perspective.
MA255 Student Learning Objectives	4. Students can design, solve, and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.	2. Students build critical thinking skills to solve problems through learning complex mathematical concepts with multivariable and vector calculus	1. Students learn good scholarly habits for progressive intellectual development.	4. Students can design, solve, and interpret the results of mathematical models using multivariable calculus to address complex and interdisciplinary problems.	7. Students can use vectors and vector functions to model motion and surfaces in three dimensional space.
	5. Students improve the quality of both their written and verbal communication of mathematics.	6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems.	3. Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines.		8. Students understand the concepts of multivariable differential calculus and can apply it to solve unconstrained and constrained optimization problems.
	6. Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex, applied calculus problems.				9. Students understand the concepts of multivariable integral calculus and can apply it to solve problems involving accumulation.
					10. Students understand and apply the concepts of vector calculus.
MA255 Assessment Techniques	Block Projects (2x presentations, 2x written reports), daily board problems, surveys	1x Modeling problem per WPR (x4) Block Projects (x4) with interdisciplinary topics	WebAssign Homework (x45) Block Project read-ahead, independent group work (8x in-class workdays)	Block Projects (2x presentations, 2x written reports) - ethical component of each BP	4x WPRs Term End Exam 4x Block Projects exploring applications

Figure 2: Assessment Crosswalk

Academic Year	***Assessment Focus***		
	WGCD	**MPG**	**SLO**
2024	5.1, 5.3	2, 6	7, 8, 9, 10
2025	2.4, 3.1	1, 3, 5	1, 2, 3
2026	1.2, 1.3, 1.5, 4.4	1, 3, 4	4, 5, 6
20(3n)	Repeat cycle		
20(3n+1)			
20(3n+2)			

Figure 3: Assessment Schedule

Detailed Course Design and Analysis Report

Associate Program Director (APD): LTC Karoline Hood
Course Director (CD): MAJ Joseph Maxwell
Assistant Course Director (ACD): MAJ Daniel Colgan

This document provides an introduction and overview for MA255 (Advanced Multivariable and Vector Calculus) for Academic Year 25-2. We first introduce the academic field and general environment in Section 1, then provide a background and brief introduction to this particular course in Section 2, and move into an in-depth discussion of the course design in Section 3. Section 4 discusses our assessment based on the criteria laid out in Section 3, and we discuss any conclusions or future potential work in Section **??**.

1. Introduction

In this section we will open with the broader academic situation in which the course is situated, including guidance from the Mathematical Association of America (MAA), highlight of comparable courses at other academic institutions, and then briefly discuss general use of technology in the classroom.

1.1. MAA Guidelines

As seen in [11], there is not a uniform structure from MAA to indicate what mathematical content belongs in which particular Calculus course. While there are some trends among particular universities, and the MAA acknowledges a wealth of textbooks, there is not a specific delineated progression of topics.

1.2. Comparable University Courses

To provide a groundwork of other universities' advanced calculus courses, we consider the curriculum for United States Naval Academy (USNA) [2], United States Air Force Academy (USAFA) [1], Pomona College [9], and Ithaca College [8]. The first two colleges are the primary competitors of the United States Military Academy (USMA), the other two have comparable acceptance rates (Pomona College) or student body sizes (Ithaca). In comparing their classes equivalent to MA255, similarities and differences are worth noting.

Two major similarities: common textbook, and comparable course structure. The Stewart *Calculus with Early Transcendentals* is universally used, and not only as a reference, but also as a guide: all of the reviewed syllabi follow in near lock-step. Accordingly, much (but not all) of the course content is introduced with roughly the same progression. Most of the "Calculus III" content also generally covers through Stokes' Theorem in Vector Calculus.

Four major differences of note between USMA and these four other universities. First, MA255 is the only course that places deliberate emphasis on modeling and projects. The others have course work and tests, but do not have comparable projects or time dedicated to modeling. Second, none of the other schools place a comparable course (usually called Calc III) in their core mathematics program. Most use it like a service courses or as a part of their major in mathematics. Third, we are the only course that places relatively heavy emphasis on leveraging technology (Mathematica (MMA), as opposed to the MAA's recommendation of Python with NumPy/SciPi).

Finally, none of the other single courses span the breadth of topics we cover. MA255 includes sequences and series, optimization, and projects/modeling, while still following Stewart's framework from vectors through the Divergence Theorem.

1.3. Calculus as a Core Course

Discussed in several places from the MAA perspective (such as [11]), but mentioned also on some university cites (like at Ithaca College in [8]), there is an ongoing discussion about how to make Calculus more accessible to the average student. While we have a tradition at USMA of requiring all Cadets to take Calculus (and until recently, multivariate Calculus), the increasing requirement of all students in four year colleges to pass an introductory Calculus course means that accessibility of material and breadth of application frequently are an increasing part of the curriculum. Given MA255's role as part of the core curriculum within the AMP, we fit well into this discussion, and our projects and focus on modeling are a potential inputs to the conversation.

1.4. Course Technology Use

AMP current leverages MMA as the primary technology tool in the classroom. It is primarily used as a "superpowered graphing calculator" to verify solutions and to help students build intuition for behavior.

The decision to use MMA and not Python is deliberate. The current and previous CD both invested time in attempting to transition the course software to use Python, but ultimate chose not to do so for several reasons. First, there are things MMA can do that Python cannot. Two examples are plotting implicit functions (e.g., surfaces) and symbolic manipulation (e.g. "solving for x"). Second, even if we restructured the course to avoid these pitfalls, we would still have a considerable startup cost to give students more accessible graphing and solving tools. In practice, this means we would have to give them a script that translates a lot of computer science realities into the traditional math form. One example is plotting $f(x, y)$ from -3 to 3 on both axes requires generating a mesh of points and declaring several baseline settings for the plot for the student to be able to plot, as opposed to MMA's straightforward "Plot" command. Finally, the course does not prevent students from using a variety of resources in modeling or problem-solving contexts, and we encourage the use of built-in help functions, peers and online resources, and even the use of generative Artificial Intelligence (AI) such as ChatGPT or WolframAlpha.

1.5. Previous Course Feedback

MA255 is teaching content that is very well established, travelling a route that is well trod. There are seldom course After Action Review (AAR) comments indicating large changes are needed, with one exception. The only feedback specifically recommended but not enacted for the last four years (since 2020 it has been mentioned in course end reports, see [5] and [4]) is to reverse the sequence of courses between MA153 and MA255 (i.e., return to the sequence of classes presented to students in 2015 and previously, teaching Multivariate Calculus in the autumn and Modeling with Ordinary Differential Equations in the spring).

This restructuring would return USMA Cadets to a more traditional calculus sequence wherein they would complete "Calc 3" and then move into differential equations in their second semester at USMA. This would likely require the calculus refresher week be moved from MA153 into MA255 and thus the sequences and series "mini-block" be moved into MA153 if it were to remain in the

4th Class Advanced Math Sequence. Moving sequences and series into MA153 may enable series solutions to differential equations, which is a commonly noted omission from the typical ordinary differential equations material.

2. Background

In this section, we discuss the required content for MA255 from the Redbook, its intended audience, how it differs from MA204/MA205, and who its primary customers are. We then discuss the course SLOs and time allocation, both from credit-hour perspective and instructor perspective.

2.1. Redbook

The Redbook course description (from [10]) is seen below. We do not recommend any changes.

> This is the second course of a two-semester advanced mathematics sequence for selected cadets who have validated single variable calculus and demonstrated strength in the mathematical sciences. It is designed to provide a foundation for the continued study of mathematics, sciences, and engineering. This course consists of an advanced coverage of topics in multivariable calculus. Topics may include a study of infinite sequences and series, vectors and the geometry of space, vector functions, partial derivatives, multiple integrals, and vector calculus. An understanding of course material is enhanced through the use of a computer algebra system.

2.2. Audience

MA255 is the second course in the two-course advanced sequence to fulfill the general mathematical requirements for students at the United States military Academy. AMP rigorously screens incoming plebes and provides an in-person exam during the first week of classes to validate that all students enrolled in AMP have a solid calculus background (up through familiarity with basic integration techniques, including at a minimum u-substitution and integration by parts). Thus these students have a solid mathematical background and often have good study skills and habits. The purpose of the course, as described by the previous several course directors, is to "produce confident, competent students who possess the knowledge, skills, and interest necessary for the future study of Science, Technology, Engineering, and Mathematics (STEM)" (from [6]).

This is true insofar as it goes, but it should be noted that this is a core math course, addressed to advanced students who go on to major in a variety of academic majors, including non-STEM fields. This distinguishes the audience of MA255 from MA204 and MA205.

2.3. Comparison with MA204

MA204 is a comparable course (less-so than MA205 below), but several differences distinguish the courses. First, MA204 is not available to Cadets who have passed MA153 as it is intended for those Cadets who demonstrate only rudimentary calculus skills and thus do not validate calculus. They are instead placed into a "hybrid" accelerated calculus/abbreviated multivariable calculus course, MA204. MA204's abbreviated schedule omits many topics in sequences and series and some depth in applications of multiple integrals and vector calculus. Additionally, MA204 does not engage the students with project-based learning (i.e. Block Projects) as MA255 does.

2.4. Comparison with MA205

MA205 is a comparable course, but three distinct differences stand out. First, MA255 covers more content than MA205, including several lessons teaching from sequences and series through applications of Taylor Series (Stewart Chapter 11) and BPs with written and/or oral reports from the Cadets. Second, the focus of the audience is different, as MA205 is only required for STEM degrees (i.e., a service course) and is thus only a 4.0 credit hour course. MA255 satisfies the Academy's Calculus graduation requirement in the core curriculum (i.e., a core course) and is a 4.5 credit hour course. Finally, and most significantly, the student population is different: as discussed previously, the AMP student population is more mathematically mature, allowing for a better grasp of and more in-depth exploration of the topics, including more challenging projects. For these reasons, despite the similarities, we recommend leaving MA255 in the AMP and not combining course leadership with MA205.

2.5. Courses for which MA255 is a Prerequisite

From the customer perspective, however, MA255 is analogous to an advanced version of MA205 (as it covers comparable content), and so satisfies the MA205 prerequisite for other department's courses (per [10], as seen in [3]). As of July 2023, 38 courses require either MA255 or MA205, listed by department as seen in Table 4.

Department	Courses
Chemistry and Life Science	3
Civil and Mechanical Engineering	7
Electrical Engineering and Computer Science	4
Geography and Environmental Engineering	1
Mathematical Sciences	19
Physics and Nuclear Engineering	3
Social Sciences	1

Table 4: Number of courses offered by a department which require MA255 or MA205 as a prerequisite.

2.6. Course Mathematical Goals

This course has ten major SLOs in the Core Math Book (see [7]), including six general mathematical goals (e.g., learn good scholarly habits, build critical thinking skills, improve quality of written and verbal habits) and four course-focused goals (use vectors and vector functions to model motion and surfaces, understand and apply: multivariable differential calculus, multivariable integral calculus, and vector calculus). These goals shape the way lessons are built and how each major graded event is constructed.

2.7. Credit Hour Justification

As defined in the Redbook in [10], each credit hour should accompany 40 hours of planned time. The usual planning rule is for two hours out of class for each hour in class. MA255 is a 4.5 credit hour course which authorizes 180 hours between in-class contact and expected out-of-class work. These hours are justified in Table 5.

Lesson Types	Planned Time	Total
MA255: 46 Lessons	3*	136
4 WPRs	1	4
8 Project Days	4	32
7 Problem Solving Labs	1	7
Total: 65 lessons		**179**

Table 5: Lesson count with recommended out of class time, and total credit hour justification.

* In line one, it should be noted that the first lesson is an intro lesson only and required no out-of-class work. The true calculation for lessons is 45 "full" lessons at three hours each and one intro day which requires zero out-of-class work (45x3 + 1 intro day hour = 136).

2.8. Instructor Requirements

Table 6 highlights the approximate allocation of time during an average week for an instructor in MA255. For seasoned instructors, the decrease in course preparation required generally corresponds to much higher time spent building course products and helping new instructors master course material.

Activity	Hours	Sum
In-Class Instruction (3 sections, 4 lessons/week)	12	12
Weekly PD Meeting	1	13
Course Taskings (Product generation, validation, etc.)	2	15
Department (Coffee Call, Broadening, MTP, etc)	2	17
Grading (average estimate)	4	21
Lesson Preparation (2 hours per lesson)	8	29
Administrative Tasks (daily "eaches", Q&A before/after class, emails, etc)	5	34
Instructor Choice (admin. tasks, AI, research, intramurals)	6	40

Table 6: Instructor average time allocation, per 40 hour week.

Figure 4 provides an example of how instructors are assigned tasks to assist in generating course products. The instructors are listed along the left, whereas the products required by block are listed across the top. In the body of the matrix, P stands for the Primary writer, A stands for the Alternate or Assistant writer supporting the primary, V indicates an assignment to Validate and time a project, T assigns a (usually first-time teaching) instructor to Time how long a test took them to complete. As can been seen by inspecting the product, most instructors (aside from the CD and APD) are responsible for generating or contributing substantially to several major course assignments.

Product Generation Matrix											
As of: 19 NOV 24	AY25-1 Validation	Block 1		Block 2		Block 3		Block 4			
		BP1	WPR1	BP2	WPR2	BP3	WPR3	BP4	WPR4	TEE	Tasks
LTC Hood (APD)					P						1
MAJ Maxwell (CD)	P		P			P					3
MAJ Colgan (ACD)	A	P	A		A		A			P	6
Fr. Costa									P		1
Dr. Dorta				V			P				2
MAJ Bates				P				V	A		3
MAJ Rutherford			T			V			T		3
MAJ Borger		V			T		T	P		A	5
Brief to Leadership:	30 Oct 24	17 Dec 24	14 Jan 25	28 Jan 25	11 Feb 25	25 Feb 25	04 Mar 25	25 Mar 25	15 Apr 25	23 Apr 24	
Brief to Instructors:	22 Oct 24	14 Jan 25	21 Jan 25	04 Feb 25	18 Feb 25	04 Mar 25	11 Mar 25	01 Apr 25	22 Apr 25	30 Apr 24	
Assignment Date:	01 Nov 24	22 Jan 24	29 Jan 25	10 Feb 25	26 Feb 25	11 Mar 25	27 Mar 25	08 Apr 25	01 May 25	12 May 25	
Due Date:	-	03 Feb 25	-	21 Feb 25	-	04 Apr 25	-	07 May 25	-	-	

P: Primary A: Alternate T: Timer V: Verifier

Figure 4: Product generation matrix for AY25-2.

3. Course Design

3.1. SLO Assessment Focus

Per Enclosure 4, the focus for assessments in academic year 2025 is SLOs 1-3, each of which deals with the habits of mind of the Cadets and their ability to further their personal journey through higher education. These SLOs have not been quantitatively assessed in recent documents and thus have little meaningful comparison to make against previous years of MA255.

The last time the comparable SLOs were assessed (AY2022-2, see [4]) the assessment was performed qualitatively by instructors against student interest in the BPs - “Anecdotally, the [Block Projects] increased interest in interdisciplinary problems, promoted independent learning, and developed mathematical modeling and communication skills.” Additionally, critical thinking skills were assessed on BPs using similar intuition as above. In both cases, the assessments were largely qualitative and intuitive, without clear metrics.

3.2. Lesson Modification

There were three changes to the lesson structure for AY25-2 from AY24-2. First, we truncated the focus of LSN 7 - Cylinders and Quadric Surfaces (12.6) to allow for a ”half-lesson” on parametrizations. This ”half-lesson” leveraged Stewart’s section 10.1 and instructor insights to produce a supplementary PDF as the primary lesson material. The utility of the lesson in parametrizations is highly subjective, but generally positive from the instructors that have taught MA255 recently.

Second, we rearranged the lessons associated with Stewart 13.3 and 13.4 to more closely follow the common instructor teaching techniques and better enable the Block Project associated with Block 1. Specifically, we placed Velocity and Acceleration (13.4) as LSN 10 in front of Arc Length and Curvature (13.3) as LSN 11. Instructors could flow more naturally from LSN 09

- Derivatives and Integrals of Vector Functions (13.2) into the kinematics discussions. This also enabled a Block Project 1 intro earlier in the block than otherwise possible. In this AY, that afforded the Cadets an extra weekend to work on their Block Project.

Finally, we made a more deliberate effort to ”wrap” Block Projects around the WPRs and place working days before the WPR with the due date after the WPR. In previous AYs, there seemed to be a convergence of deadlines around the end of the blocks that inhibited Cadets' ability to meaningfully focus on the Block Project **and** the WPR. Middle-to-poor performers would regularly feel pressure to choose one event to put time into. ”Wrapping” noticeably (albeit subjectively) reduced the stress in that Cadet time conflict. It also enabled an instructor focus on using the BPs to *explore deeper into the concepts* rather than attempting to use the BPs as another tool to *teach* the concepts.

3.3. Instructor Preparation and Course Guide

In AY25-2, we maintained the mini-Faculty Development Workshop (FDW) from AY24-2, but changed it slightly to have new instructors teach selected ”lighter” lessons from earlier in the semester rather than immediately jumping into vector calculus material from Block 4. The **CD!** (**CD!**) team selected these lessons to cover the main threads of information through the course from each block. The lessons were LSN 12 - Arc Length, LSN 20 - Max and Min Values, LSN 33 - Double Integrals Over General Regions, and LSN 50 - Green's Theorem.

The only major update to the Course Guide was the addition of LSN 07's emphasis on parametrization. All other changes were administrative/typographical in nature.

Course end feedback indicates that very few Cadets use the Course Guide. Without deliberate effort by the CD team to structure the course in a way that **requires** use of the course guide, **it should primarily be used as a tool for the instructor team to focus their teaching priorities and methodologies**.

4. SLO Assessments

In this section we will discuss the planned course SLOs for the semester, their evaluation methods and criteria, and the results of those evaluations. Each SLO will be measured using two methods of assessment. In summary, SLO analysis revealed valuable insights into how instructors can better use and motivate homework assignments and reinforcing Cadet sentiment for the developmental direction of Block Projects.

4.1. SLO 1: Students learn good scholarly habits for progressive intellectual development.

4.1.1. SLO Assessment 1.1

The first method of assessment for SLO 1 is the percent of Cadets who achieve an 80% score on at least 80% of **on time** homework assignments. Evaluation will be based on 7 below.

Percentage of Cadets	Evaluation
[80%, 100%]	A
[70%, 80%)	B
[65%, 70%)	C
[0%, 65%)	F

Table 7: MA255 AY25-2 SLO 1 assessment 1 methodology

For method 1, we assess ourselves at an F grade for the criteria established ahead of the semester. Only 42.3% of Cadets achieved over 80% of the points on 80% or more of their **on-time** homework assignments. The data suggests that Cadets do not view homework as an incremental method of learning and reinforcing material from the recent lectures. This analysis has brought about valuable distinctions in how Cadets use homework because it created a distinction between **if** Cadets do their homework versus **when** Cadets do their homework. The analysis shows that Cadets will do their homework, but only so far as they see it valuable to their overall grade and not for the immediate supplemental learning from lectures.

We believe the conclusion above is representative of Cadet sentiments based on a few reasons. Several anecdotes from instructors and students indicate that they waited until TEE week to complete their homework and only did so because of the point values assigned towards homework. This often also required instructor prompting that they were leaving (in some cases) upwards of 250 points unattempted. We reinforce these anecdotes by viewing the data in tables 8 and 9 below. Table 8 shows the number and percent of Cadets who achieved an on-time homework score of above 80%. Table 9 shows the number and percent of Cadets who achieved a score of 80% or higher on their homework at any time in the course. The data in Table 8, especially when juxtaposed against Table 9, reinforces the conclusions above.

Threshold	CDT Count	CDT Percent
$\geq$ 80%	149	42.3%
$\geq$ 70%	194	55.1%
$\geq$ 65%	208	59.1%
Below 65%	144	40.9%

Table 8: On-Time Homework Analysis

Threshold	CDT Count	CDT Percent
$\geq$ 80%	261	74.1%
$\geq$ 70%	295	83.8%
$\geq$ 65%	299	84.9%
Below 65%	53	15.1%

Table 9: Anytime Homework Analysis

4.1.2. SLO Assessment 1.2

The second method for assessment will use End of Course Survey responses from AY25-2 compared to AY25-1 for the question: “Rate the effectiveness of your current study habits.” Choices will be “Highly Ineffective”, “Slightly Ineffective”, “Slightly Effective”, and “Highly Effective”. The CD team will test a difference of means between the two semesters’ responses according to Table 10 below.

Result	Grade
Statistically Significant Increase	A
No Significant Difference	B
Statistically Significant Decrease	C

Table 10: MA255 AY25-2 SLO 1 assessment 2 methodology

For method 2, we should assess ourselves at a B grade because we find that there is not statistically significant difference between the end of course assessment for MA153 and MA255. The responses indicate that this most likely means that any habits of mind that develop in the Cadets happen early enough to be captured by the end of course survey in the first semester. It may be worth an initial survey that asks Cadets to rank the effectiveness of their study habits at entry to USMA for another snapshot, but not critical.

We do still have data to support that we should suspect an improvement in the Cadets’ habits of mind, even if that improvement is not visible between semesters. In MA153 AY25-1, 70% of respondents indicated a positive change in their study habits. In MA255 AY25-2, 66% of respondents indicated a positive change in their study habits. Nearly 25% in each semester reported ”significantly better” study habits than when they arrive to USMA.

A Mann-Whitney U Test indicated that the populations did not have statistically significant different response populations. The null hypothesis is that there is not a significant difference in the populations’ responses and the resulting p-value for the test was 0.754, so we should accept the null hypothesis at any reasonable level of significance.

4.1.3. SLO 1 Assessment Overall

With the F and the B evaluations from assessment methods 1 and 2, respectively, **I evaluate this SLO at a C level**. Further, I recommend no cause for alarm despite the potentially concerning evaluation from assessment method 1. I strongly believe that this pattern of homework use by Cadets is consistent with previous semesters. The strict utilitarianism by Cadets raises an interesting opportunity for future CD teams in the Jedi program.

Recommend that instructor teams strongly consider increasing the penalty for nightly homework assignments to 50% or greater (up from 10%). Nightly homework assignments in MA255 are hand-selected to be supplementary to the daily lesson and have built-in guiderails for the Cadets to continue learning. Given that these daily assignments reinforce the daily lessons, their utility diminishes significantly as time passes and the aggressive late penalties may be justified.

Weekly homework assignments are a mix of all problem types throughout the week and test a Cadet's ability to identify the correct strategy in addition to executing it. Given that these have utility beyond the week that they are assigned, it would be reasonable to leave the late penalty at something in the range of 10%-25%.

4.2. SLO 2: Students build critical thinking skills to solve problems through learning complex mathematical concepts in multivariable and vector calculus.

4.2.1. SLO Assessment 2.1

The first method for assessing SLO 2 will use Cadets' Block Project scores. We will record the average BP score for each Block Project without plus/minus modifiers to determine the ability of the course to achieve SLO 2. Table 11 below shows the summary of results. Overall, the BP averages and their "average of averages" are both at an A level.

	Raw	**Percent**
BP1 Avg	116.24	92.99%
BP2 Avg	114.91	91.93%
BP3 Avg	119.10	95.28%
BP4 Avg	114.63	91.70%
Avg of Avgs	-	92.98%

Table 11: Block Project Raw Score Averages and Average Percentages

4.2.2. SLO Assessment 2.2

The second method for assessing SLO 2 will use Cadets' modeling questions scores on WPRs and the TEE. This will be assessed at each WPR according to MA255's letter scoring criteria with no plus or minus modifiers. The assessment criteria evaluates Cadets' ability to solve critical thinking problems in testing scenarios at a B level. See Table 12 for full data.

	Raw	Percent
WPR1 Q5 Avg	46.04	76.73%
WPR2 Q1 Avg	35.30	88.26%
WPR3 Q3 Avg	47.31	94.61%
TEE Q7 Avg	50.31	83.84%
Weighted Avg	-	85.22%

Table 12: Block Project Raw Score Averages and Average Percentages

4.2.3. SLO 2 Assessment Overall

Given the two evaluation methods, **I evaluate this SLO at a B level**. In hindsight, this SLO is challenging to evaluate in the context of the way the course has been run for several years. By convention, the instructor team targets an A- average for their sections when grading Block Projects. This effort exists to control for discrepancies in how much assistance individual instructors may give to teams and to contribute to the grade 'softening' that is afforded to Cadets that opt into the advanced math track. It is worth mentioning that neither of these conventions is ever transmitted to the Cadets. Block Projects (method of assessment 1) are almost certainly the best way to assess this SLO, but the instructor team faces challenges in meaningfully assessing this SLO in the context of timed WPRs.

The initial description of evaluating “modeling“ questions on WPRs was not fully developed. My current belief is that a “true” modeling question should be one that encourages or requires Cadets to make assumptions and has some degree of open-endedness or ambiguity in the response. MA255, as a course, is not well suited to asking these types of questions in timed scenarios. I suspect it is possible, but would require significant consideration by future CD teams. My personal opinion is that for timed exams, the course would be better served by having more comprehensive conceptual questions integrated into the exam than attempted to shoehorn “true” modeling questions into the MA255 curriculum. Overall, **I recommend against using method of assessment 2** for this or other SLOs in the future.

4.3. SLO 3: Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines.

4.3.1. SLO 3 Assessment 3.1

The first method for assessing SLO 3 is using a qualitative instructor assessment based on exposure through Faculty Development initiative to provide problems from other departments that demonstrate applications. The semester's faculty development initiative was an effort to pull problems directly from other courses at USMA that leveraged the techniques in MA255. Examples are kinematics from the Physicas and Nuclear Engineering Department for tying in with LSN 10's Velocity and Acceleration or using Lagrange Multipliers in an economics problem from the Social Sciences Department to tie in with LSN 22 and 23's Lagrange Multipliers.

Subjectively, I assess this at a C level of proficiency. Faculty members drifted from the original intent of the faculty development program and did not provide concrete problems for other instructors to use. Instead, instructors began presenting the associated concepts and applications of techniques.

4.3.2. SLO 3 Assessment 3.2

The second method of assess SLO 3 is to compare the MA255 end of course survey question to MA153's similar question: "I am more confident in my ability to apply math in different academic disciplines than I was when I arrived to USMA." Choices will be "Strongly Disagree", "Slightly Disagree", "Slightly Agree", and "Strongly Agree". We will again use the Mann-Whitney U Test to see if the populations differ in their responses.

The p-value for the test is $\tilde{0}.773$ and supports the conclusion that we should accept the null hypothesis at any reasonable confidence level. Thus, we conclude there is no statistical significance in the Cadet's confidence between semesters and assess ourselves at a B in this category.

For transparency, it is worth noting that there is some survey error in this assessment. In the MA153 survey, there was a fifth response available that was simply "Agree," and could have been interpreted by respondents as a middling response (less than "Slightly Agree") or as something between "Slightly Agree" and "Strongly Agree." For consistency between the assessments, I omitted the results from MA153's survey that responded as simply "Agree" and reduce the effective number of respondents from 87 to 68.

4.3.3. SLO 3 Assessment Overall

Overall, **I evaluate this SLO at a B**. Straying from the intent of the faculty development initiative associated with method of assessment 1 damaged the overall evaluation. Mission drift failure is the responsibility of the Course Director and I accept that failing in the initiative as my own. I do believe that the initiative is workable and valuable if the instructor team adheres to the principles of the initiative in providing problems that can be worked in class instead of discussion points.

For method of assessment 2, I recommend maintaining the question's current wording that considers the Cadet's entire time at USMA and not just the semester. If MA153 and MA255 are truly to be consdiered a two-semester sequence, then we should evaluate their holistic development across the semesters when possible and SLOs 1-3 for MA255 are well-suited to that end.

The analysis may not reveal significant differences within a semester, but almost 94% of respondents in MA153 and in MA255 indicated a positive sentiment in their ability to apply math in different disciplines compared to when they arrived to USMA. There were not comparable questions on the AY24-2 End of Course Survey to compare against.

List of Acronyms

AAR	After Action Review
AI	Artificial Intelligence
AMP	Advanced Math Program
CCPS	Confident and Competent Problem Solvers
FDW	Faculty Development Workshop
MPG	Mathematics Program Goal
MAA	Mathematical Association of America
MMA	Mathematica
STEM	Science, Technology, Engineering, and Mathematics
SLO	Student Learning Outcome
TEE	Term End Examination
USAFA	United States Air Force Academy
USMA	United States Military Academy
USNA	United States Naval Academy
WPR	Written Partial Review
BP	Block Project
AMP	Advanced Math Program

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DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
DEPARTMENT OF MATHEMATICAL SCIENCES
WEST POINT, NEW YORK 10996

MADN-MTH 12 December 2024

MEMORANDUM FOR Cadets enrolled in MA255

SUBJECT: Student Memorandum for MA255, AY25-2 - Advanced Multi-Variable Calculus

1. **Purpose**. This memorandum provides an overview of MA255: Advanced Multi-Variable Calculus for Cadets enrolled in Academic Year (AY) 25-2.

2. **Requirements**.

a. Text. *Calculus: Early Transcendentals*, 9th Edition, James Stewart, Copyright 2016.

b. Software. *Wolfram Mathematica*, V14 or greater; and *Microsoft Office Suite*, including *Outlook, Word, Excel, PowerPoint, and Teams*.

c. Online. *Canvas*, for all course documents, and *WebAssign* for automated scoring and feedback of daily homework.

3. **Attendance**.

a. MA255 is a 4.5 credit hour course. It does not follow the Day 1 / Day 2 schedule of a 3.0 credit hour course and Cadets will have class every academic day that is not designated as a course drop on the course schedule. The course schedule can be found on the MA255 Canvas site under “Administrative Documents” on the Modules page.

b. Attendance at Written Partial Reviews (WPRs) is mandatory. Except in emergencies, Cadets will not make any appointments that could preclude their attendance for a WPR. Any Cadet missing a WPR will be required to take a make-up or make-ahead exam within a time frame determined by their instructor and/or course leadership.

c. Course drops may move on the calendar due to weather considerations. Students will be notified of any course drop movements and the subsequent impacts via Canvas.

4. **Mathematics Program Goals**. This course is structured to develop the six core mathematics program goals through a focus on 10 Student Learning Outcomes (SLO). Each program goal and its supporting outcomes are listed below

a. **Students display effective habits of mind in their intellectual process.**

(1) **SLO 1:** Students learn good scholarly habits for progressive intellectual development.

(2) **SLO 3:** Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines.

(3) **SLO 4:** Students can design, solve, and interpret the results of mathematical models using multi-variable calculus to address complex and interdisciplinary problems.

b. **Student demonstrate problem solving skills.**

(1) **SLO 7:** Students can use vectors and vector functions to model motion and surfaces in three dimensional space.

(2) **SLO 8:** Students understand the concepts of multi-variable differential calculus and can apply it to solve unconstrained and constrained optimization problems.

(3) **SLO 9:** Students understand the concepts of multi-variable integral calculus and can apply it to solve problems involving accumulation.

(4) **SLO 10:** Students understand and apply the concepts of vector calculus.

c. **When appropriate, students apply technology to solve mathematical problems.**

(1) **SLO 2:** Students build critical thinking skills to solve problems through learning complex mathematical concepts with multi-variable and vector calculus.

(2) **SLO 6:** Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex applied calculus problems.

d. **Students demonstrate an ability to communicate mathematics effectively.**

(1) **SLO 4:** See SLO in part (a.) above.

(2) **SLO 5:** Students improve the quality of both their written and verbal communication of mathematics.

e. **Students demonstrate knowledge of mathematics.** See SLOs for program goal in part (c.) above.

f. **Students demonstrate an ability to consider problems from the perspective of multiple disciplines including mathematics.** See SLOs for program goal in part (b.) above.

5. **Course Summary**. The Course Schedule and Course Guide serve as the guiding documents for the course. These documents are both available on the MA255 Canvas page under “Administrative Documents” on the Modules page. There are a total of **65** required attendances consisting of 46 regular lessons, 8 project lessons, 7 problem-solving labs, and 4 exams.

6. **Course Evaluation Plan**. See Table 1 below for the distribution of course points.

7. **Grading**.

a. Final course grades will follow the Advanced Math Program modification to the Department of Mathematical Sciences’ guidelines, per Table 2.

b. Cadets who score less than 50% on the TEE, regardless of their final course average, or whose final course average is less than 65%, may fail the course.

c. Instructors may award up to 30 bonus points at their discretion. Bonus points will be added directly to the total number of points you earn in the course.

Category	Event	Points	Percent	Overall
Individual Exams	Term End Exam (TEE)	900	30%	63%
	WPR 1	250	8.3%	
	WPR 2	250	8.3%	
	WPR 3	250	8.3%	
	WPR 4	250	8.3%	
Group Work (Modeling)	Block Project 1	125	4.2%	17%
	Block Project 2	125	4.2%	
	Block Project 3	125	4.2%	
	Block Project 4	125	4.2%	
Homework	WebAssign (WA) Block 1	72	2.4%	10%
	WA Block 2	87	2.9%	
	WA Block 3	60	2.0%	
	WA Block 4	81	2.7%	
Instructor	Instructor Points	300	10%	10%
	Total:	3000	100%	

Table 1: Point values by Assignment.

Grade	Per Cent Range	QPA	Word Picture
A+	$95 \leq$ Mark ≤ 100	4.33	Beyond expectations of course
A	$90 \leq$ Mark < 95	4.0	Dominates the material
A-	$87 \leq$ Mark < 90	3.67	Mastery
B+	$83 \leq$ Mark < 87	3.33	Excellent performance
B	$80 \leq$ Mark < 83	3.0	Good understanding
B-	$77 \leq$ Mark < 80	2.67	Proficient, with aptitude for the subject
C+	$73 \leq$ Mark < 77	2.33	Can build upon this foundation
C	$70 \leq$ Mark < 73	2.0	Passing; proficient now
C-	$67 \leq$ Mark < 70	1.67	Short-range understanding
D	$64 \leq$ Mark < 67	1.0	Marginal performance with elementary understanding
F	Mark < 64	0	Failing; definitely failed to demonstrate understanding

Table 2: Grade Scale by percent score out of course points.

8. **Documentation**. All Cadet work in MA255 will be in accordance with the procedures found in the Dean's *Documentation and Acknowledgment of Academic Work* (DAAW), June 2024. All required out-of-class work constitutes homework as defined in the DAAW. The MA255 standard for citation is the Modern Language Association (MLA) format. Refer to the DAAW for further information, guidance, and examples. Proper documentation is required for all out-of-class assignments outlined in this memorandum and any assigned by the instructor.

9. **Out-of-Class Assignments**.

a. **Daily Homework**. Daily and weekly web-based homework problems will be assigned via WebAssign. To help build good habits, the daily assignment for a given lesson will be due the night after the in-class lesson to allow adequate time for Cadets in morning and afternoon sections to complete the assignment. The weekly assignment will be due the night prior to the start of the next week of class. Attempting WebAssign homework is ideal to facilitate productive in-class discussions. After submitting each question in the assignment, Cadets will receive immediate feedback, to include their grade. In accordance with the DAAW, each daily WebAssign problem set requires students to complete an online acknowledgment statement. **After the first assignment, completion of the previous cover sheet assignment is required in order to begin the next assignment**.

MA255 uses the following WebAssign assignment extension policy: *automatic* extensions can be requested. An automatic extension is an assignment extension for which instructor permission is *not* required. Each extension is 72 hours from the time of the request and carries for a deduction of 10% of the unscored points on the assignment. For example, if 6 of 8 points have been earned, an extension will make the new max score 7.8 out of 8, a 0.2 point reduction from the 2 points remaining).

b. **Block Projects**. During each block, students will be assigned a project that synthesizes the material covered in the block. These will be group-based (2- or 3-person) and are estimated to require approximately 6 hours outside of class per Cadet. The projects will require the effective use of technology and will alternate delivery medium (in-class presentations or written reports). The purpose of the projects is to reinforce course content and demonstrate various interdisciplinary applications of multi-variable calculus while helping develop Cadets' communications skills.

10. **In-Class Assessments**

a. **Written Partial Reviews**. WPRs are summative assessments that evaluate the lesson objectives in the course. Each WPR is cumulative (meaning any content covered up to the point may be assessed) but focuses on content from the corresponding block. Attendance at WPRs is mandatory as described in the Attendance paragraph above.

b. **Term End Exam**. The TEE is a mandatory summative assessment and is cumulative for the entire course.

c. **Instructor Assessments**. Most instructors will implement some sort of formative assessment as part of their 300-point instructor point plan. These assessments typically give the instructor an idea of what the Cadets' understanding of material at a point in time in order to develop reinforcement learning plans. Cadets should use these assessments to understand their own strengths and weaknesses in the course and allocate study time accordingly.

11. **Classroom Rules**. Cadets will conform to the highest standards of ethical behavior and military bearing in the performance of their academic duties. Cadets will treat every member of the faculty with the courtesy appropriate to commissioned officers. Smoking, using smokeless tobacco, chewing gum, and consuming food is always prohibited in classrooms, laboratories, lecture halls, and auditoriums. Drinks are allowed. The uniform for classes will be the duty uniform, with exceptions made for injury and special class functions that occur during the class day. Cadets will leave outer garments and bags in the hallway. The uniform for additional instruction

(AI) is the duty uniform.

12. **Chain of Command**. The Department of Mathematical Sciences has an open-door policy. Your initial attempt to address problems or concerns – including respect in the classroom, unprofessional behavior, sexual assault, harassment – should go through your instructor. However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain to include the Head of the Department. The entire math department faculty is here to help you. Your MA255 chain of command is as follows:

a. Your instructor.

b. MA255 Course Director: MAJ Maxwell, Thayer 255B.

c. Advanced Math Program (AMP) Associate Program Director: LTC Hood, Thayer 255A.

d. Mathematical Sciences Program Director: Dr. DeCoste, Thayer 251.

e. Head of the Department of Mathematical Sciences: COL Scioletti, Thayer 238.

13. Point of contact for this memorandum is MAJ Joseph Maxwell at joseph.maxwell@westpoint.edu.

KAROLINE M. HOOD
LTC, EN(47)
AMP Associate Program Director

MA255 AY 25-2

Advanced Multivariable and Vector Calculus

Spring 2025 Schedule

Block	WK	Monday	Tuesday	Wednesday	Thursday	Friday	Weekend
Block 1: Vectors and Geometry of Space	1	06 Jan 25 Reorgy Day	07 Jan 25 LSN 1 Course Intro	08 Jan 25 LSN 2 3D Coordinate Systems (12.1)	09 Jan 25 Pres. Carter Memoriam	10 Jan 25 LSN 3 Vectors I (12.2)	11 Jan 25 / 12 Jan 25 B 500th Night
	2	13 Jan 25 LSN 4 Vectors II (12.2)	14 Jan 25 LSN 5 Dot Product (12.3) Cross Product (12.4)	15 Jan 25 LSN 6 Equations of Lines and Planes (12.5)	16 Jan 25 LSN 7 Quadric Surface (12.6) 2D Parametrizations (10.1)	17 Jan 25 Course Drop 1.1	18 Jan 25 / 19 Jan 25 B
	3	20 Jan 25 Martin Luther King Day	21 Jan 25 LSN 8 Vector Functions and Space Curves (13.1)	22 Jan 25 LSN 9 Derivatives & Integrals of Vector Functions (13.2)	D 23 Jan 25 LSN 10 Velocity and Acceleration (13.4) **BP1 Intro** Plebe Majors Open House	24 Jan 25 LSN 11 BP1 Working Day 1	25 Jan 25 / 26 Jan 25 B Yearling Winter Wkd
	4	27 Jan 25 LSN 12 Arc Length and Curvature (13.3)	28 Jan 25 LSN 13 PSL 1 Synthesis / Board Problems	D 29 Jan 25 LSN 14 WPR 1	30 Jan 25 Course Drop 1.2	D 31 Jan 25 LSN 15 Functions of Several Variables, Partial Derivatives (14.1-3)	01 Feb 25 / 02 Feb 25 A
Block 2: Optimization and Taylor Series	5	03 Feb 25 LSN 16 BP1 Presentation Day **BP1 Due**	04 Feb 25 LSN 17 Tangent Planes and Linear Approximations (14.4)	05 Feb 25 LSN 18 The Chain Rule (14.5)	06 Feb 25 LSN 19 Directional Derivatives and the Gradient Vector (14.6)	P 07 Feb 25 Course Drop 2.1	08 Feb 25 / 09 Feb 25 B 100th Night
	6	10 Feb 25 LSN 20 Maximum and Minimum Values (14.7) **BP2 Intro**	11 Feb 25 LSN 21 BP2 Working Day 1	12 Feb 25 LSN 22 Lagrange Multipliers I (14.8)	P 13 Feb 25 LSN 23 Lagrange Multipliers II (14.8)	14 Feb 25 Course Drop 2.2	15 Feb 25 / 16 Feb 25 B
	7	17 Feb 25 Presidents Day	18 Feb 25 LSN 24 BP2 Working Day 2	19 Feb 25 LSN 25 Sequences and Series (11.1-2)	D 20 Feb 25 LSN 26 Alternating Series and Ratio Roots Tests (11.5-6)	21 Feb 25 LSN 27 Power Series (11.8) **BP2 Due**	22 Feb 25 / 23 Feb 25 B
	8	D 24 Feb 25 LSN 28 Taylor and Maclaurin Series (11.10)	25 Feb 25 LSN 29 PSL 2 Synthesis / Board Problems	26 Feb 25 LSN 30 WPR 2	27 Feb 25 Course Drop 2.3	28 Feb 25 LSN 31 Double Integrals over Rectangles (15.1)	01 Mar 25 / 02 Mar 25 A By-Stander (Tentative)
Block 3: Triple Integrals	9	D 03 Mar 25 LSN 32 Double Integrals over General Regions I (15.2)	04 Mar 25 LSN 33 Double Integrals over General Regions II (15.2)	05 Mar 25 LSN 34 Double Integrals in Polar Coordinates (15.3)	06 Mar 25 LSN 35 Applications of Double Integrals I (15.4)	07 Mar 25 Course Drop 3.1	08 Mar 25 / 09 Mar 25 A 1CL ACFT?
	10	10 Mar 25 LSN 36 Applications of Double Integrals II (15.4)	11 Mar 25 LSN 37 Triple Integrals (15.6) **BP3 Intro**	D 12 Mar 25 Course Drop 3.2	13 Mar 25 LSN 38 Triple Integrals in Cylindrical Coordinates (15.7)	14 Mar 25 LSN 39 BP3 Working Day	15 Mar 25 / 16 Mar 25 PPW Spring Break
	11	17 Mar 25 Spring Break	18 Mar 25 Spring Break	19 Mar 25 Spring Break	20 Mar 25 Spring Break	21 Mar 25 Spring Break	22 Mar 25 / 23 Mar 25 B Spring Break
	12	24 Mar 25 LSN 40 Triple Integrals in Spherical Coordinates I (15.8)	25 Mar 25 LSN 41 Triple Integrals in Spherical Coordinates II (15.8)	26 Mar 25 LSN 42 PSL 3 Synthesis / Board Problems	27 Mar 25 LSN 43 WPR 3	P 28 Mar 25 Course Drop 3.3	29 Mar 25 / 30 Mar 25 A
Block 4: Vector Calculus	13	D 31 Mar 25 LSN 44 Vector Fields (16.1)	01 Apr 25 LSN 45 Line Integrals I (16.2)	02 Apr 25 LSN 46 Line Integrals II (16.2)	03 Apr 25 LSN 47 The Fundamental Theorem for Line Integrals (16.3)	P 04 Apr 25 LSN 48 BP3 Presentation Day **BP3 Due**	05 Apr 25 / 06 Apr 25 A Sandhurst Validation
	14	07 Apr 25 LSN 49 Green's Theorem I (16.4)	08 Apr 25 LSN 50 Green's Theorem II & Curl (16.4)	09 Apr 25 Course Drop 4.1	10 Apr 25 LSN 51 Curl & Divergence (16.5)	11 Apr 25 LSN 52 PSL 4 Line Integral PSL	12 Apr 25 / 13 Apr 25 B
	15	14 Apr 25 LSN 53 Parametric Surfaces and Their Areas I (16.6)	15 Apr 25 LSN 54 Parametric Surfaces and Their Areas II (16.6) **BP4 Intro**	D 16 Apr 25 LSN 55 Surface Integrals I (16.7)	17 Apr 25 LSN 56 Surface Integrals II (16.7)	P 18 Apr 25 Course Drop 4.2	19 Apr 25 / 20 Apr 25 B
	16	21 Apr 25 No Class Day	D 22 Apr 25 LSN 57 Stokes' Theorem I (16.8)	23 Apr 25 LSN 58 Stokes' Theorem II (16.8)	24 Apr 25 Projects Day	D 25 Apr 25 Course Drop 4.3	26 Apr 25 / 27 Apr 25 A
	17	28 Apr 25 LSN 59 Divergence Theorem (16.9)	29 Apr 25 LSN 60 BP4 Working Day 1	30 Apr 25 LSN 61 PSL 5 Surface Integral PSL	01 May 25 LSN 62 WPR 4	02 May 25 Sandhurst	03 May 25 / 04 May 25 A Sandhurst
	18	05 May 25 LSN 63 BP4 Working Day 2	06 May 25 Course Drop 4.4	D 07 May 25 LSN 64 PSL 6 Course Synthesis **BP4 Due**	08 May 25 LSN 65 PSL 7 Course Synthesis	09 May 25 Course Drop 4.5	10 May 25 / 11 May 25 Reading Day
TEE	19	12 May 25 TEEs	13 May 25 TEEs	14 May 25 TEEs	15 May 25 TEEs	16 May 25 TEEs	17 May 25 / 18 May 25 TEEs (both days)

D Dean's Day | M Modified Day | P Prime Day | (12.1) <- Stewart Calculus Chapter 12, Section 1

Tech Lab | MAD / Project | WPR | Problem Solving Lab | Holiday / Course Drop | Day 2 (blue)

Version Date:

TAB F AY2026 COURSE ASSESSMENT

DEPARTMENT OF THE ARMY
UNITED STATES MILITARY ACADEMY
DEPARTMENT OF MATHEMATICAL SCIENCES
WEST POINT, NEW YORK 10996

MADN-MTH 19 December 2025

MEMORANDUM FOR Cadets Enrolled in MA255

SUBJECT: Student Memorandum for MA255, AY26-2 - Advanced Multi-Variable Calculus

1. **Purpose**. This memorandum provides an overview of MA255: Advanced Multi-Variable Calculus for Cadets enrolled in Academic Year (AY) 26-2.

2. **Course Overview**. The Course Schedule and Course Guide serve as the guiding documents for the course. These documents are both available on the MA255 Canvas page under "Administrative Documents" on the Modules page with "Quick Links" on the home page. There are a total of **65** required attendances consisting of 46 regular lessons, 8 project lessons, 7 problem-solving labs, and 4 exams.

a. **Attendance**. MA255 is a 4.5 credit hour course. It does not follow the Day 1 / Day 2 schedule of a 3.0 credit hour course and Cadets will have class every academic day that is not designated as a course drop on the course schedule. The course schedule can be found on the MA255 Canvas site under "Administrative Documents" on the Modules page. Attendance at Written Partial Reviews (WPRs) is mandatory. Except in emergencies, Cadets will not make any appointments that could preclude their attendance for a WPR. Any Cadet missing a WPR will be required to take a make-up or make-ahead exam within a time frame determined by their instructor and/or course leadership. Course drops may move on the calendar due to weather considerations. Students will be notified of any course drop movements and the subsequent impacts via Canvas.

b. **Course Assessments**. See Table 1 below for the distribution of course points. Instructors may award up to 30 bonus points at their discretion. Bonus points will be added directly to the total number of points earned in the course. Refer to "Out of Class Assignments" for an in-depth description of types of assessments.

c. **Grading**. Final course grades will follow the Advanced Math Program modification to the Department of Mathematical Sciences' guidelines, per Table 2. Cadets who score less than 50% on the TEE, regardless of their final course average, or whose final course average is less than 65%, may fail the course.

d. **Documentation**. All Cadet work in MA255 will be in accordance with the procedures found in the Dean's *Documentation and Acknowledgment of Academic Work* (DAAW), June 2025. All required out-of-class work constitutes homework as defined in the DAAW. The MA255 standard for citation is the Modern Language Association (MLA) format. Refer to the DAAW for further information, guidance, and examples. Proper documentation is required for all out-of-class assignments outlined in this memorandum and any assigned by the instructor.

Category	Event	Points	Percent	Overall
Individual Exams	Term End Exam (TEE)	900	30%	60%
	WPR 1	225	7.5%	
	WPR 2	225	7.5%	
	WPR 3	225	7.5%	
	WPR 4	225	7.5%	
Group Work (Capstone)	Block Capstone 1	150	5%	20%
	Block Capstone 2	150	5%	
	Block Capstone 3	150	5%	
	Block Capstone 4	150	5%	
Homework	Weekly Problem Set (10x)	300	10%	10%
Instructor	Instructor Points	300	10%	10%
	Total:	3000	100%	

Table 1: Point values by Assignment.

Grade	Per Cent Range	QPA	Holistic Description
A+	$95 \leq$ Mark ≤ 100	4.33	Beyond expectations of course
A	$90 \leq$ Mark < 95	4.0	Dominates the material
A-	$87 \leq$ Mark < 90	3.67	Mastery
B+	$83 \leq$ Mark < 87	3.33	Excellent performance
B	$80 \leq$ Mark < 83	3.0	Good understanding
B-	$77 \leq$ Mark < 80	2.67	Proficient, with aptitude for the subject
C+	$73 \leq$ Mark < 77	2.33	Can build upon this foundation
C	$70 \leq$ Mark < 73	2.0	Passing; proficient now
C-	$67 \leq$ Mark < 70	1.67	Short-range understanding
D	$64 \leq$ Mark < 67	1.0	Marginal performance with elementary understanding
F	Mark < 64	0	Failing; definitely failed to demonstrate understanding

Table 2: Grade Scale by percent score out of course points.

e. **Guidance on Generative AI**. The use of Generative AI is generally encouraged to assist in conceptual understanding or debugging practices. Learning is enhanced by asking Generative AI to explain a difficult concept for how concepts are connected. Using Generative AI to quickly jump to a solutions degrades a Cadets ability to critically think and reason through difficult problems. Performance on summative assessments highlight over reliance and potential lack of conceptual understanding caused by improper use of Generative AI. Use of Generative AI must be appropriately documented in accordance with the DAAW.

3. **Out-of-Class Assignments**.

a. **Daily Practice and Weekly Homework Sets**. Daily and weekly web-based homework problems will be assigned via WebAssign. To help build good habits, the daily assignment for a given lesson will be due the night after the in-class lesson to allow adequate time for Cadets in morning and afternoon sections to complete the assignment. The daily assignments are completely voluntary and exist to reinforce concepts taught during in-class lessons. The weekly assignment will be due Sunday nights on assigned weeks. Attempting WebAssign homework is ideal to facilitate productive in-class discussions. After submitting each question in the assignment, Cadets will receive immediate feedback, to include their grade. In accordance with the DAAW, each weekly WebAssign problem set requires students to complete an online acknowledgment statement. **After the first assignment, completion of the previous cover sheet assignment is required in order to begin the next assignment**. There is no cover sheet requirement for ungraded daily assignments

MA255 uses the following WebAssign assignment extension policy: *automatic* extensions can be requested. An automatic extension is an assignment extension for which instructor permission is *not* required. Each extension is 72 hours from the time of the request and carries for a deduction of 10% of the unscored points on the assignment. For example, if 6 of 8 points have been earned, an extension will make the new max score 7.8 out of 8, a 0.2 point reduction from the 2 points remaining). All weekly assignments for a block will be locked two full calendar weeks after the administering of the respective WPR.

b. **Block Capstones**. During each block, students execute a capstone that assesses personal understanding of relevant concepts and synthesizes the material covered in the block in a collaborative setting. These capstones will be group-based (2- or 3-person) and are estimated to require approximately 6+ hours outside of class per Cadet and be administered in three parts. Cadets will complete a challenging check on understanding prior to the first working day. At the start of the first working day, Instructors will assess and provide feedback on the check on understanding while Cadets complete a hands on, visual reinforcement exercise. Upon receiving feedback and completing the activity, Cadets will begin the open-ended exploration of block concepts. The capstone will require the effective use of technology and will alternate delivery medium (in-class presentations or written reports). The purpose of the capstone is to reinforce course content and demonstrate various interdisciplinary applications of multi-variable calculus while helping develop Cadets' communications skills.

4. **In-Class Assessments**

a. **Written Partial Reviews**. WPRs are summative assessments that evaluate the lesson objectives in the course. Each WPR is cumulative (meaning any content covered up to the point may be assessed) but focus on content from the corresponding block. Attendance at WPRs is mandatory as described in the Attendance paragraph above.

b. **Term End Exam**. The TEE is a mandatory summative assessment and is cumulative for the entire course.

c. **Instructor Assessments**. Most instructors will implement some sort of formative assessment as part of their 300-point instructor point plan. These assessments typically give the instructor an idea of what the Cadets' understanding of material at a point in time in order to develop reinforcement learning plans. Cadets should use these assessments to understand their own strengths and weaknesses in the course and allocate study time accordingly.

5. **Classroom Rules**. Cadets will conform to the highest standards of ethical behavior and military bearing in the performance of their academic duties. Cadets will treat every member of the faculty with the courtesy appropriate to commissioned officers. Smoking, using smokeless tobacco, chewing gum, and consuming food is always prohibited in classrooms, laboratories, lecture halls, and auditoriums. Drinks are allowed. The uniform for classes will be the duty uniform, with exceptions made for injury and special class functions that occur during the class day. Cadets will leave outer garments and bags in the hallway. The uniform for additional instruction (AI) is the duty uniform.

6. **Mathematics Program Goals**. This course is structured to develop the six core mathematics program goals through a focus on 10 Student Learning Outcomes (SLO). Each program goal and its supporting outcomes are listed below

a. Students display effective habits of mind in their intellectual process.

(1) **SLO 1:** Students learn good scholarly habits for progressive intellectual development.

(2) **SLO 3:** Students develop a knowledge, curiosity, and appreciation for the application of mathematics in different disciplines.

(3) **SLO 4:** Students can design, solve, and interpret the results of mathematical models using multi-variable calculus to address complex and interdisciplinary problems.

b. Student demonstrate problem solving skills.

(1) **SLO 7:** Students can use vectors and vector functions to model motion and surfaces in three dimensional space.

(2) **SLO 8:** Students understand the concepts of multi-variable differential calculus and can apply it to solve unconstrained and constrained optimization problems.

(3) **SLO 9:** Students understand the concepts of multi-variable integral calculus and can apply it to solve problems involving accumulation.

(4) **SLO 10:** Students understand and apply the concepts of vector calculus.

c. When appropriate, students apply technology to solve mathematical problems.

(1) **SLO 2:** Students build critical thinking skills to solve problems through learning complex mathematical concepts with multi-variable and vector calculus.

(2) **SLO 6:** Students use technology appropriately to visualize complex geometries, facilitate their understanding of concepts, and solve complex applied calculus problems.

d. Students demonstrate an ability to communicate mathematics effectively.

(1) **SLO 4:** See SLO in part (a.) above.

(2) **SLO 5:** Students improve the quality of both their written and verbal communication of mathematics.

e. Students demonstrate knowledge of mathematics. See SLOs for program goal in part (c.) above.

f. Students demonstrate an ability to consider problems from the perspective of multiple disciplines including mathematics. See SLOs for program goal in part (b.) above.

7. **Course References and Digital Requirements**.

a. Text. *Calculus: Early Transcendentals*, 9th Edition, James Stewart, Copyright 2016.

b. Software. *Wolfram Mathematica*, V14 or greater; and *Microsoft Office Suite*, including but limited to *Outlook, Word, Excel, PowerPoint, and Teams*.

c. Online. *Canvas*, primary learning management system for all course content, and *WebAssign* for automated scoring and feedback of out-of-class homework.

8. **Chain of Command**. The Department of Mathematical Sciences has an open-door policy. Your initial attempt to address problems or concerns – including respect in the classroom, unprofessional behavior, sexual assault, harassment – should go through your instructor. However, if you are uncomfortable addressing an issue with your instructor, please do not hesitate to approach any individual in the chain to include the Head of the Department. The entire math department faculty is here to help you. Your MA255 chain of command is as follows:

a. Your instructor.

b. MA255 Course Director: MAJ Colgan, Thayer 255B.

c. Advanced Math Program (AMP) Associate Program Director: LTC Hood, Thayer 255A.

d. Mathematical Sciences Program Director: Dr. DeCoste, Thayer 251.

e. Head of the Department of Mathematical Sciences: COL Scioletti, Thayer 238.

9. Point of contact for this memorandum is MAJ Daniel Colgan at daniel.colgan@westpoint.edu.

KAROLINE M. HOOD
LTC, EN(47)
AMP Associate Program Director

MA255 AY 26-2

Advanced Multivariable and Vector Calculus

Spring 2026 Schedule

Block	WK	Monday	Tuesday	Wednesday	Thursday	Friday	Weekend
Block 1: Vectors and Geometry of Space	1	05 Jan 26 Reorgy Day	P 06 Jan 26 Course Drop 1.1	07 Jan 26 LSN 1 Course Intro	08 Jan 26 LSN 2 3D Coordinate Systems (12.1)	D 09 Jan 26 LSN 3 Vectors I (12.2)	10 Jan 26 / 11 Jan 26 B 500th Night
	2	12 Jan 26 LSN 4 Vectors II (12.2)	13 Jan 26 LSN 5 Dot Product (12.3) Cross Product (12.4)	14 Jan 26 LSN 6 Equations of Lines and Planes (12.5)	15 Jan 26 LSN 7 Quadric Surface (12.6) 2D Parametrizations (10.1)	P 16 Jan 26 Course Drop 1.2	17 Jan 26 / 18 Jan 26 B **Problem Set**
	3	19 Jan 26 Martin Luther King Day	20 Jan 26 LSN 8 Vector Functions and Space Curves (13.1)	21 Jan 26 LSN 9 Derivatives & Integrals of Vector Functions (13.2)	D 22 Jan 26 LSN 10 Velocity and Acceleration (13.4) Plebe Majors Open House	23 Jan 26 LSN 11 Arc Length and Curvature (13.3) **BC1 Intro / CoU**	24 Jan 26 / 25 Jan 26 B Yearling Winter Wkd **Problem Set**
	4	26 Jan 26 LSN 12 BC1 Activity Day Velocity & Acceleration	27 Jan 26 LSN 13 PSL 1 Synthesis / Board Problems	D 28 Jan 26 LSN 14 WPR 1	29 Jan 26 Course Drop 1.3	30 Jan 26 LSN 15 BC1 Working Day	31 Jan 26 / 01 Feb 26 A SAMI **BC1 Due**
Block 2: Optimization and Taylor Series	5	02 Feb 26 LSN 16 Functions of Several Variables, Partial Derivatives (14.1-3)	03 Feb 26 LSN 17 Tangent Planes and Linear Approximations (14.4)	D 04 Feb 26 LSN 18 The Chain Rule (14.5)	05 Feb 26 LSN 19 Directional Derivatives and the Gradient Vector (14.6)	P 06 Feb 26 Course Drop 2.1	07 Feb 26 / 08 Feb 26 B 100th Night
	6	09 Feb 26 LSN 20 Maximum and Minimum Values (14.7)	10 Feb 26 LSN 21 Lagrange Multipliers I (14.8)	11 Feb 26 LSN 22 Lagrange Multipliers II (14.8)	12 Feb 26 LSN 23 Sequences and Series (11.1-2) **BC2 Intro / CoU**	D 13 Feb 26 Course Drop 2.2	14 Feb 26 / 15 Feb 26 B **Problem Set**
	7	16 Feb 26 Presidents Day	17 Feb 26 LSN 24 BC2 Activity Day Lagrange & Gradient	D 18 Feb 26 LSN 25 Alternating Series and Ratio Roots Tests (11.5-6)	19 Feb 26 LSN 26 Power Series (11.8)	20 Feb 26 LSN 27 Taylor and Maclaurin Series (11.10)	21 Feb 26 / 22 Feb 26 B **Problem Set**
	8	23 Feb 26 LSN 28 BC2 Working Day	24 Feb 26 LSN 29 PSL 2 Synthesis / Board Problems	25 Feb 26 LSN 30 WPR 2	26 Feb 26 Course Drop 2.3	P 27 Feb 26 LSN 31 Double Integrals over Rectangles (15.1)	28 Feb 26 / 01 Mar 26 A **BC2 Due**
Block 3: Triple Integrals	9	D 02 Mar 26 LSN 32 Double Integrals over General Regions I (15.2)	03 Mar 26 LSN 33 Double Integrals over General Regions II (15.2)	04 Mar 26 LSN 34 Double Integrals in Polar Coordinates (15.3)	05 Mar 26 LSN 35 Applications of Double Integrals I (15.4)	06 Mar 26 Course Drop 3.1	07 Mar 26 / 08 Mar 26 B **Problem Set**
	10	09 Mar 26 LSN 36 Applications of Double Integrals II (15.4)	10 Mar 26 LSN 37 Triple Integrals (15.6)	11 Mar 26 LSN 38 Triple Integrals in Cylindrical Coordinates (15.7)	12 Mar 26 LSN 39 Triple Integrals in Spherical Coordinates I (15.8) **BC3 Intro / CoU**	P 13 Mar 26 Course Drop 3.2	14 Mar 26 / 15 Mar 26 B **Problem Set**
	11	16 Mar 26 LSN 40 Triple Integrals in Spherical Coordinates II (15.8)	17 Mar 26 LSN 41 BC3 Activity Day Regions of Integrations	D 18 Mar 26 LSN 42 PSL 3 Synthesis / Board Problems	19 Mar 26 LSN 43 WPR 3	20 Mar 26 Course Drop 3.3	21 Mar 26 / 22 Mar 26 B
	12	23 Mar 26 LSN 44 Vector Fields (16.1)	24 Mar 26 LSN 45 Line Integrals I (16.2)	D 25 Mar 26 LSN 46 Line Integrals II (16.2)	26 Mar 26 LSN 47 The Fundamental Theorem for Line Integrals (16.3)	27 Mar 26 LSN 48 BC3 Presentation Day	28 Mar 26 / 29 Mar 26 Plebe Parent Weekend
Block 4: Vector Calculus	13	30 Mar 26 Spring Break	31 Mar 26 Spring Break	01 Apr 26 Spring Break	02 Apr 26 Spring Break	03 Apr 26 Spring Break	04 Apr 26 / 05 Apr 26 Spring Break **Problem Set**
	14	06 Apr 26 Easter No Class Day	07 Apr 26 LSN 49 Green's Theorem I (16.4)	08 Apr 26 LSN 50 Green's Theorem II & Curl (16.4)	09 Apr 26 LSN 51 Curl & Divergence (16.5)	10 Apr 26 LSN 52 PSL 4 Line Integral PSL	11 Apr 26 / 12 Apr 26 A Sandhurst Validation **Problem Set**
	15	13 Apr 26 LSN 53 Parametric Surfaces and Their Areas I (16.6)	14 Apr 26 LSN 54 Parametric Surfaces and Their Areas II (16.6)	D 15 Apr 26 LSN 55 Surface Integrals I (16.7)	16 Apr 26 LSN 56 Surface Integrals II (16.7)	17 Apr 26 Course Drop 4.1	18 Apr 26 / 19 Apr 26 B **Problem Set**
	16	20 Apr 26 LSN 57 Stokes' Theorem I (16.8)	21 Apr 26 LSN 58 Stokes' Theorem II (16.8)	D 22 Apr 26 Course Drop 4.2	23 Apr 26 Projects Day	24 Apr 26 LSN 59 Divergence Theorem (16.9) **BC4 Intro / CoU**	25 Apr 26 / 26 Apr 26 A **Problem Set**
	17	27 Apr 26 LSN 60 BC4 Working Day	D 28 Apr 26 LSN 61 PSL 5 Surface Integral PSL	29 Apr 26 LSN 62 WPR 4	30 Apr 26 Course Drop 4.3	01 May 26 Sandhurst	02 May 26 / 03 May 26 A Sandhurst
	18	04 May 26 LSN 63 BC4 Presentation Day	05 May 26 Course Drop 4.4	D 06 May 26 LSN 64 PSL 6 Course Synthesis **BC4 Due**	07 May 26 LSN 65 PSL 7 Course Synthesis	08 May 26 Course Drop 4.5	09 May 26 / 10 May 26 B
TEE	19	11 May 26 Reading Day	12 May 26 TEEs	13 May 26 TEEs	14 May 26 TEEs	15 May 26 TEEs	16 May 26 / 17 May 26 TEEs

D Dean's Day   M Modified Day   P Prime Day   (12.1) <- Stewart Calculus Chapter 12, Section 1

Tech Lab	MAD / Project	WPR	Problem Solving Lab	Holiday / Course Drop	Day 2 (blue)

Version Date: 11/26/2025

SECTION II

EXAMINATIONS

TAB G WPR #1

WPR 1 - MA255 **Section** ____________

1. This WPR is worth **225 points**. There are 9 pages to this exam. You have 55 minutes to complete it. All work on this WPR is subject to grading unless marked through or clearly marked to denote "do not grade" and must be your own.
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____________	____________	____________	____________
Cadet ID	Printed Name	Signature	Time/Date Signed

Question:	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Total**
Points:	**40**	**40**	**30**	**30**	**40**	**45**	**0**	**225**

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1. (a) (20 points) Find a vector equation for a line segment that starts at P and ends at Q. Clearly state the domain restriction.

$$P = (-2, -8, 3) \qquad Q = (3, 2, -7)$$

(b) (20 points) Find the point where the line in part (a) intersects the following plane:

$$2x - y + 3z = 8$$

.

2. (40 points) Find the scalar equation of the plane that contains the following point and the locations defined by position vectors. Fully simplify your answer.

$$A = (1, 1, 5) \qquad \vec{r}_B = \langle 2, -1, 3 \rangle \qquad \vec{r}_C = -\hat{i} + 2\hat{j} + 8\hat{k}$$

3. (30 points) Find a vector equation of the line created by the intersection of the following two planes.

$$3x + y - 5z = 0 \qquad x + 2y + z = -4$$

.

4. (30 points) A CNC tool head is milling 6062 aluminum in Mahan Hall for a research project. The tool head follows the space curve (in millimeters) to cut the aluminum.

$$\vec{r}(t) = \langle 2t^2 + 8, \frac{3}{2}t^2, 6t^2 - 4 \rangle$$

If the tool head starts at $t = 0$ and ends at $P = (16, 6, 20)$, how far did the tool head travel over the course of the milling job?

5. Monitoring the skies, a tracking radar picks up an incoming attack drone. The targeting computer outputs the projected sky track of the incoming drone, $\vec{r}_1$, and immediately launches an interceptor drone modeled by $\vec{r}_2$ in meters.

$$\vec{r}_1(t) = \langle t+1, 2t+1, 3t+1 \rangle \qquad \vec{r}_2(t) = \langle 2t+1, 2t^2+3, 6t+1 \rangle$$

   (a) (30 points) Where do the trajectories intersect?

   (b) (10 points) Does the interceptor drone successfully interdict the incoming attack drone? If it is not, determine how many seconds elapse between when the two drones arrive at the intersection point.

6. (45 points) In an alternate reality where there is only war, artillery is raining down on a group of Space Marines on a hill top. Unfortunately for our heroes, a round lands in front of the Lieutenant at $\vec{r}_{initial}$, sending him flying through the air for 4 seconds before plummeting back to the surface at $\vec{r}_{final}$. A sensory array notes the planet's gravity is $4\ m/s^2$ and locations are in meters relative to the location of the sensory array.

$$\vec{r}_{initial} = \langle 3, 2, 6 \rangle \qquad \vec{r}_{final} = \langle 11, 14, -2 \rangle$$

If the Space Marine can withstand a ground impact speed of $11\ m/s$, does the Lieutenant survive?

7. **EXTRA CREDIT: (10 points)** In your own words, explain the meaning of curvature. What does the numerical value of κ denote? A properly labeled graphical diagram should accompany your explanation.

Instructor Solution

WPR 1 - MA255 **Section** ANSWER KEY

1. This WPR is worth **225 points**. There are 9 pages to this exam. You have 55 minutes to complete it. All work on this WPR is subject to grading unless marked through or clearly marked to denote "do not grade" and must be your own.
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Cadet ID	Printed Name	Signature	Time/Date Signed
	INSTRUCTOR		

Question:	1	2	3	4	5	6	7	Total
Points:	40	40	30	30	40	45	0	225

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1. (a) (20 points) Find a vector equation for a line segment that starts at P and ends at Q. Clearly state the domain restriction.

$$P = (-2, -8, 3) \qquad Q = (3, 2, -7)$$

Displacement: $\vec{PQ} = \langle 5, 10, -10 \rangle$

Equation: $\vec{r}(t) = \langle -2, -8, 3 \rangle + t \langle 5, 10, -10 \rangle$

or

$\vec{r}(t) = \langle 5t - 2, 10t - 8, 3 - 10t \rangle$

where

$0 \le t \le 1$

(b) (20 points) Find the point where the line in part (a) intersects the following plane:

$$2x - y + 3z = 8$$

Parametric: $x = 5t - 2$ $\quad y = 10t - 8$ $\quad z = 3 - 10t$

Solve for t: $2(5t-2) - (10t - 8) + 3(3 - 10t) = 8$

$-30\,t = -5$

$t = 1/6$

Point: $r(1/6) = \langle -7/6, -19/3, 4/3 \rangle$

2. (40 points) Find the linear equation of the plane that contains the following three points.

3 Points: $A = (1, 1, 5)$ $\vec{r}_B = \langle 2, -1, 3\rangle$ $\vec{r}_C = -\hat{\imath} + 2\hat{\jmath} + 8\hat{k}$

2 Vectors: $\vec{AB} = \langle 1, -2, -2\rangle$

$\overline{AC} = \langle -2, 1, 3\rangle$

1 Normal: $\vec{AB} \times \vec{AC} = \vec{n} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -2 & -2 \\ -2 & 1 & 3 \end{vmatrix} = \langle -4, 1, -3\rangle$

Scalar Eqn: $-4(x-1) + (y-1) - 3(z-5) = 0$

Linear Eqn: $-4x + y - 3z + 18 = 0$

USE A CONTINUATION SHEET IF NECESSARY.

3. (30 points) Find a vector equation of the line created by the intersection of the following two planes.

$$3x + y - 5z = 0 \qquad x + 2y + z = -4$$

Normals: $\bar{n}_1 = \langle 3, 1, -5 \rangle$ $\bar{n}_2 = \langle 1, 2, 1 \rangle$

Displacement of Line: $\bar{v} = \bar{n}_1 \times \bar{n}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & -5 \\ 1 & 2 & 1 \end{vmatrix} = \langle 11, -8, 5 \rangle$

Point of Int: If $z = 0$, $3x + y = 0$, $x + 2y = -4$ $\therefore$ $x = 4/5$, $y = -12/5$

$\bar{r}_I = \langle 4/5, -12/5, 0 \rangle$.

Equation of a Line: $\bar{r}(t) = \underbrace{\langle 4/5, -12/5, 0 \rangle}_{\text{Various Solutions}} + t \langle 11, -8, 5 \rangle$

4. (30 points) A CNC tool head is milling 6062 aluminum in Mahan Hall for a research project. The tool head follows the space curve (in millimeters) to cut the aluminum.

$$\vec{r}(t) = \langle 2t^2 + 8, \frac{3}{2}t^2, 6t^2 - 4 \rangle$$

If the tool head starts at $t = 0$ and ends at $P = (16, 6, 20)$, how far did the tool head travel over the course of the milling job?

Solve for time: $2t^2+8=16 \quad \therefore \quad t = \pm 2$

Magnitude of $\vec{r}\,'$: $|\vec{r}\,'| = \left((4t)^2+(3t)^2+(12t)^2\right)^{1/2} = \left(169t^2\right)^{1/2} = 13t$

Arclength: $s = \int_0^2 13t \, dt = 26$

5. Monitoring the skies, a tracking radar picks up an incoming attack drone. The targeting computer outputs the projected sky track of the incoming drone, $\vec{r}_1$, and immediately launches an interceptor drone modeled by $\vec{r}_2$ in meters.

$$\vec{r}_1(t) = \langle t+1, 2t+1, 3t+1\rangle \qquad \vec{r}_2(t) = \langle 2t+1, 2t^2+3, 6t+1\rangle$$

(a) (30 points) Where do the trajectories intersect?

Solve for s & t: $t+1 = 2s+1 \Rightarrow t = 2s$

$2t+1 = 2s^2+3 \Rightarrow (s-1)^2 = 0 \quad \therefore s = 1, t = 2$

Validate: (May happen) $3t+1 = 6s+1$

$3(2)+1 = 6(1)+1$

$7 = 7$ ✓

Point: $r_1(2) = \langle 3, 5, 7\rangle$ m

$r_2(1) = \langle 3, 5, 7\rangle$ m $\therefore$ Intersect.

(b) (10 points) Does the interceptor drone successfully interdict the incoming attack drone? If it is not, determine how many seconds elapse between when the two drones arrive at the intersection point.

Conceptual: $t \neq s \;\therefore$ No collision.

$\Delta t = t - s = 2 - 1 = 1$ sec

6. (45 points) In an alternate reality where there is only war, artillery is raining down on a group of Space Marines on a hill top. Unfortunately for our heroes, a round lands in front of the Lieutenant at $\vec{r}_{initial}$, sending him flying through the air for 4 seconds before plummeting back to the surface at $\vec{r}_{final}$. A sensory array notes the planet's gravity is $4\ m/s^2$ and locations are in meters relative to the location of the sensory array.

$$\vec{r}_{initial} = \langle 3, 2, 6\rangle \qquad \vec{r}_{final} = \langle 11, 14, -2\rangle$$

If the Space Marine can withstand a ground impact speed of $11\ m/s$, does the Lieutenant survive?

Acceleration: $\bar{a} = \langle 0, 0, -4\rangle$

Velocity: $\bar{v} = \langle V_x, V_y, V_z - 4t\rangle$

Position: $\bar{r} = \langle V_x t + 3, V_y t + 2, V_z t - 2t^2 + 6\rangle$

System of Eqn: @ $t = 4$ $\quad 4V_x + 3 = 11 \quad \therefore V_x = 2$

$4V_y + 2 = 14 \quad \therefore V_y = 3$

$4V_y - 32 + 6 = -2 \quad \therefore V_z = 6$

Speed: $|\bar{v}(4)| = \left(2^2 + 3^2 + (6 - 4(4))^2\right)^{1/2} = (4 + 9 + 100)^{1/2} = \sqrt{113}$

$|\bar{v}(4)| = \sqrt{113} < 11 \quad \therefore$ Lieutenant Survives.

7. **EXTRA CREDIT: (10 points)** In your own words, explain the meaning of curvature. What does the numerical value of κ denote? A properly labeled graphical diagram should should accompany your explanation.

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB H WPR#2

WPR 2 - MA255 **Section** ____________

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Cadet ID Printed Name Signature Time/Date Signed

Question:	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**Total**
Points:	**40**	**40**	**35**	**30**	**40**	**40**	**0**	**225**

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1. (40 points) Consider the following surface:

$$f(x, y) = \frac{x^2 y}{3} + \frac{y^3}{2}$$

   (a) Determine an equation of the tangent plane to the given surface at the point $P(-3, 2)$.

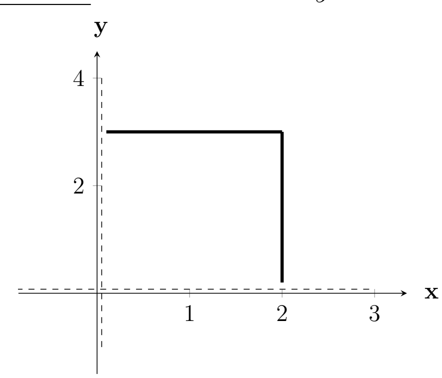
   (b) Use your equation of a tangent plane to approximate the height of the surface at $Q(-2, 1)$ and compute the associated absolute error at Q.

2. (40 points) Consider the following function:

$$f(x, y) = x^2 + 2y^2 - 2x - 4y + 1$$

(a) Find and classify the critical point of the given function. Your final answer should contain the location, value, and classification of the critical point.

(b) What is the absolute/global maximum and minimum of $f(x, y)$ if $0 \leq x \leq 2$ and $0 \leq y \leq 3$? The maximum and minimum **do not occur** where $x = 0$ or $y = 0$.



3. (35 points) Consider the following problems related to Lagrange Multipliers.

   (a) (5 points) Consider a function $f(x, y)$ subject to a constraint $g(x, y) = c$. What is the behavior of the functions at a point along the boundary corresponding to an extreme value which is a solution to the Method of Lagrange Multipliers.

   (b) (30 points) Use Lagrange Multipliers to find the maximum of the following function subject to the given constraint. Put your final answer with the corresponding location in the box below.

$$f(x, y) = 3x + 5y + 2, \qquad x^2 + 2y^2 = 86$$

4. (30 points) Use your knowledge of sequences and series to explain why the following statements are **FALSE**.

   (a) The sequence $a_n = \dfrac{6n^3 + 2n^2}{2n^3 + 3}$ converges to 3, therefore the series $\displaystyle\sum_{n=1}^{\infty} a_n$ also converges.

   (b) $\dfrac{1}{3} + \dfrac{2}{9} + \dfrac{1}{6} + \dfrac{4}{45} + \cdots$ is a geometric series where $a = \dfrac{1}{3}$ and $r = \dfrac{4}{3}$.

   (c) The sum of the geometric series, $\displaystyle\sum_{n=1}^{\infty} \frac{1}{2}(3)^{n-1}$, is $s = \dfrac{1}{8}$.

5. (40 points) Determine of radius of convergence (R) and the interval of convergence (I) for the following power series. **Do not evaluate the endpoints of the interval.** Put your final answer for R and I in the box below.

$$\sum_{n=0}^{\infty}(-1)^n \frac{(x+2)^{5n+3}}{3^{5n+2}}$$

6. (40 points) Consider the following questions on Taylor series.

   (a) Fill in the circle of the correct form of a Taylor series centered at $a = 2$.

   ◯ $c_0 + c_1\,x + c_2\,x^2 + c_3\,x^3 + c_4\,x^4 + \cdots$

   ◯ $c_0 + c_1\left(\dfrac{x}{2}\right) + c_2\left(\dfrac{x}{2}\right)^2 + c_3\left(\dfrac{x}{2}\right)^3 + c_4\left(\dfrac{x}{2}\right)^4 + \cdots$

   ◯ $c_0 + c_1(x-2) + c_2(x+2)^2 + c_3(x-2)^3 + c_4(x+2)^4 + \cdots$

   ◯ $c_0 + c_1(x+2) + c_2(x+2)^2 + c_3(x+2)^3 + c_4(x+2)^4 + \cdots$

   ◯ $c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4 + \cdots$

   (b) Find the Taylor series of the following function centered at $x = 2$. Include only the first **3** non-zero terms. **Write your Taylor series as a function in the box below.**

$$f(x) = \frac{2}{(5-x)^2}$$

7. **EXTRA CREDIT: (10 points)** Find the sum of the following series where the terms are the reciprocals of the positive integers whose only prime factors are 2s and 3s.

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{12} + \cdots$$

Instructor Solution

WPR 2 - MA255 **Section** ____________

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____________ ____________ ____________ ____________

Cadet ID Printed Name Signature Time/Date Signed

Question:	1	2	3	4	5	6	7	**Total**
Points:	40	40	35	30	40	40	0	225

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1. (40 points) Consider the following surface:

$$f(x,y) = \frac{x^2y}{3} + \frac{y^3}{2}$$

(a) Determine an equation of the tangent plane to the given surface at the point $P(-3,2)$.

DERIVATIVES: $\nabla f = \langle \frac{2}{3}xy, \frac{1}{3}x^2 + \frac{3}{2}y^2 \rangle$

$\nabla f(-3,2) = \langle -4, 9 \rangle$ $f(-3,2) = 10$

TANGENT PLANE:

$z = 10 - 4(x+3) + 9(y-2)$

$z = -4x + 9y - 20$

(b) Use your equation of a tangent plane to approximate the height of the surface at $Q(-2,1)$ and compute the associated absolute error at Q.

APPROXIMATE: $z(-2,1) = -4(-2) + 9(1) - 20 = -3$

EXACT: $f(-2,1) = \frac{11}{6}$

ABS ERROR: $\left| \frac{11}{6} - (-3) \right| = \frac{29}{6}$

USE A CONTINUATION SHEET IF NECESSARY.

2. (40 points) Consider the following function:

$$f(x,y) = x^2 + 2y^2 - 2x - 4y + 1$$

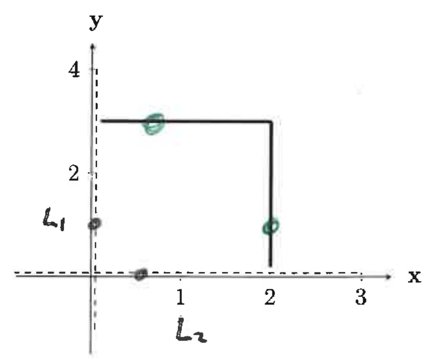
(a) Find and classify the critical point of the given function. Your final answer should contain the location, value, and classification of the critical point.

Critical Point: $\nabla f = \langle 2x-2, 4y-4 \rangle = \vec{0}$ $\therefore f(1,1) = -2$

Second Derivative Test: $D = (2)(4) - 0 = 8$, $f_{xx}, f_{yy} > 0$ $\therefore$ Min

$\boxed{f(1,1) = -2, \text{ Min}}$

(b) What is the absolute/global maximum and minimum of $f(x,y)$ if $0 \le x \le 2$ and $0 \le y \le 3$? The maximum and minimum **do not occur** ~~along the boundaries or endpoints~~ where $x = 0$ or $y = 0$.



Boundaries

FYSA
$L_1: f(0,y) = 2y^2 - 4y + 1 \Rightarrow f' = 4y - 4 = 0$
$f(0,1) = -1$
$L_2: f(x,0) = x^2 - 2x + 1 \Rightarrow f' = 2x - 2 = 0$
$f(1,0) = 0$

$L_3: f(2,y) = 2y^2 - 4y + 1 \Rightarrow f' = 4y - 4 = 0$
$f(2,1) = -1$

$L_4: f(x,3) = x^2 - 2x - 3 \Rightarrow f' = 2x - 2 = 0$
$* \; f(1,3) = -4$

Endpoints

FYSA
$f(0,0) = 1$
$f(2,0) = 1$

FYSA
$f(0,3) = -2$
$* \; f(2,3) = 7$

$\boxed{\begin{array}{l} f(1,3) = -4, \text{ Min} \\ f(2,3) = 7, \text{ Max} \end{array}}$

3. (35 points) Consider the following problems related to Lagrange Multipliers.

   (a) (5 points) Consider a function $f(x, y)$ subject to a constraint $g(x, y) = c$. What is the behavior of the functions at a point along the boundary corresponding to an extreme value which is a solution to the Method of Lagrange Multipliers.

   THE DIRECTION OF THE GRADIENTS ∇f & ∇g ARE PARALLEL.

   (b) (30 points) Use Lagrange Multipliers to find the maximum of the following function subject to the given constraint. Put your final answer with the corresponding location in the box below.

$$f(x, y) = 3x + 5y + 2, \qquad x^2 + 2y^2 = 86$$

Lagrange Multipliers: $\nabla f = \lambda \nabla g$

$\langle 3, 5 \rangle = \lambda \langle 2x, 4y \rangle$

System of Equations:

① $3 = 2x\lambda$

② $5 = 4y\lambda$

③ $x^2 + 2y^2 = 86$

① $\lambda = \frac{3}{2x}$

② $10x = 12y \quad \therefore \quad x = \frac{6}{5}y$

③ $\frac{36+50}{25}y^2 = 86 \quad \therefore \quad y^2 = 25$

$y = 5,\ x = 6$

$y = -5,\ x = -6$

Values

$f(5, 6) = 47$

$f(-5, -6) = -43$

$\boxed{f(5,6) = 47}$

4. (30 points) Use your knowledge of sequences and series to explain why the following statements are **FALSE**.

(a) The sequence $a_n = \dfrac{6n^3 + 2n^2}{2n^3 + 3}$ converges to 3, therefore the series $\sum_{n=1}^{\infty} a_n$ also ~~converges.~~

$\lim_{n\to\infty} a_n \neq 0 \therefore$ Series $\sum_{n=1}^{\infty} a_n$ Diverges.

(b) ~~$\frac{1}{3} + \frac{2}{9} + \frac{1}{6} + \frac{4}{45}$~~ $+ \cdots$ is a geometric series where $a = \frac{1}{3}$ and $r = \frac{4}{3}$.

Geometric Series, $\sum_{n=0}^{\infty} \frac{1}{3}\left(\frac{4}{3}\right)^n = \frac{1}{3} + \frac{4}{9} + \frac{16}{27} + \ldots$

(c) The sum of the geometric series, $\sum_{n=1}^{\infty} \frac{1}{2}(3)^{n-1}$, is ~~$s = \frac{1}{8}$~~.

By Divergence Test $\lim_{n\to\infty} 3^{n-1} \neq 0$, Sum is ∞

5. (40 points) Determine of radius of convergence (R) and the interval of convergence (I) for the following power series. **Do not evaluate the endpoints of the interval.** Put your final answer for R and I in the box below.

$$\sum_{n=0}^{\infty}(-1)^n\frac{(x+2)^{5n+3}}{3^{5n+2}}$$

Ratio Test: $\lim_{n\to\infty}\left|\frac{(-1)^{n+1}(x+2)^{5(n+1)+3}}{3^{5(n+1)+2}}\cdot\frac{3^{5n+2}}{(-1)^n(x+2)^{5n+3}}\right| < 1$

$\lim_{n\to\infty}\left|\frac{(x+2)^5}{3^5}\right| < 1$

$-3^5 < (x+2)^5 < 3^5$

$-3 < x+2 < 3$

$-5 < x < 1$

$R = 3$

$I = (-5, 1)$

$R = 3$

$I = (-5, 1)$

6. (40 points) Consider the following questions on Taylor series.

(a) Fill in the circle of the correct form of a Taylor series centered at $a = 2$.

- ○ $c_0 + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + \cdots$
- ○ $c_0 + c_1\left(\frac{x}{2}\right) + c_2\left(\frac{x}{2}\right)^2 + c_3\left(\frac{x}{2}\right)^3 + c_4\left(\frac{x}{2}\right)^4 + \cdots$
- ○ $c_0 + c_1(x-2) + c_2(x+2)^2 + c_3(x-2)^3 + c_4(x+2)^4 + \cdots$
- ○ $c_0 + c_1(x+2) + c_2(x+2)^2 + c_3(x+2)^3 + c_4(x+2)^4 + \cdots$
- ● $c_0 + c_1(x-2) + c_2(x-2)^2 + c_3(x-2)^3 + c_4(x-2)^4 + \cdots$

(b) Find the Taylor series of the following function centered at $x = 2$. Include only the first **3** non-zero terms. **Write your Taylor series as a function in the box below.**

$$f(x) = \frac{2}{(5-x)^2}$$

		$f^{(n)}(a)$	$C_n(x-a)^n$
0	$2(5-x)^{-2}$	$2/9$	$2/9$
1	$4(5-x)^{-3}$	$4/27$	$4/27\,(x-2)$
2	$12(5-x)^{-4}$	$12/81$	$2/27\,(x-2)^2$

$$\boxed{f(x) = 2/9 + 4/27\,(x-2) + 2/27\,(x-2)^2}$$

USE A CONTINUATION SHEET IF NECESSARY.

7. **EXTRA CREDIT: (10 points)** Find the sum of the following series where the terms are the reciprocals of the positive integers whose only prime factors are 2s and 3s.

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{6} + \frac{1}{8} + \frac{1}{9} + \frac{1}{12} + \cdots$$

(0,0) (1,0) (0,1) (2,0) (1,1) (3,0) (0,2)

$$\cdots = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} \frac{1}{2^n 3^m} = 3$$

UNITED STATES MILITARY ACADEMY
WEST POINT.

TAB I WPR#3

WPR 3 - MA255 **Section** ____________

1. This WPR is worth **225 points**. There are 8 pages to this exam. You have 55 minutes to complete it. All work on this WPR is subject to grading unless marked through or clearly marked to denote "do not grade" and must be your own.

2. Show as much work as necessary to support your answer. A wrong numerical or symbolic solution with no supporting work receives zero credit. Clearly double underline or $\boxed{\text{box}}$ your answer.

3. Use a blank continuation sheet and clearly identify that the problem is continued both on the exam and on the continuation sheet. Use one continuation sheet per problem continued. Be sure to put your name on each continuation sheet.

4. **Approved references**: one $8\frac{1}{2} \times 11$ sheet of paper. The paper must be written in your own handwriting (no photocopying or printing digital notes). You may write on both sides of the paper.

5. You may NOT use any resources that are not on the authorized reference list including, but not limited to: computers, phones, calculators, the internet, your classmates.

6. Cadets are **not** authorized to discuss the content, structure, or any other information about this WPR until this WPR has been released from academic security. Discussion includes all forms of written, electronic, and verbal communication. This WPR will be released from academic security at **1610 on 25 March 2026**.

7. Honor Acknowledgment Statement: sign and date the statement below when you have finished the WPR and are ready to turn it in.

   "I did not use any sources nor did I receive any assistance while completing this WPR. I will not discuss this WPR with anyone until it is released from academic security at **1610 on 25 March 2026**."

______________ ______________ ______________ ______________

Cadet ID Printed Name Signature Time/Date Signed

Question:	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Points:	**45**	**45**	**45**	**45**	**45**	**0**	**225**

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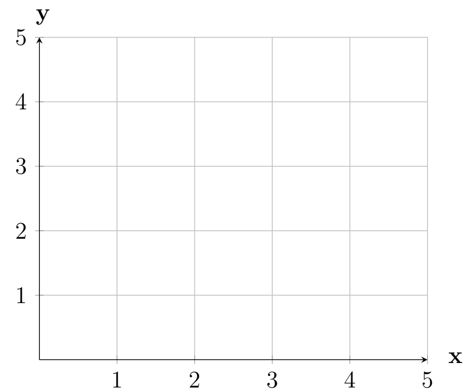
1. (45 points) Let $f(x, y) = 3y + 6$, and D be the triangular region with the vertices $(1, 1)$, $(2, 4)$, and $(4, 4)$.

   (a) Sketch the region D. Label region boundaries with corresponding functions.



   (b) Set up **both sets** of iterated integrals to compute $\iint_D f(x, y)dA$ where $dA = dx\ dy$ and $dA = dy\ dx$. **DO NOT SOLVE.**

   (c) Evaluate one of the iterated integrals in Part b.

2. (45 points) An invasion force of Space Bugs has established a beachhead on a colony planet with a distribution of warriors modeled by the function $\rho(x, y) = 4yx^2$ in millions. Advance scouts have identified a region R that is optimal for an orbital strike:

$$R = \{(x, y)|\ x^2 + y^2 \leq 25, x \geq 0, y \geq 0\},$$

   (a) Find the total number of Space Bugs in the region.

   (b) Concurrently, targeting officers on the orbiting battleship are preparing to launch the first barrage at the planet. Set up the equations to find the optimal target reference point $(\bar{x}, \bar{y})$ through center of mass calculations. The targeting officer assumes 100 million Space Bugs in the region.(***This may not match what you found in Part (a)***). **DO NOT EVALUATE**.

3. (45 points) Set up, but **DO NOT EVALUATE**, a triple integral to determine the volume of the region below the plane $z = 0$ and above the spherical bowl $x^2 + y^2 + z^2 = 9$ in each of the given coordinate systems.

   (a) Rectangular:

   (b) Cylindrical:

   (c) Spherical:

4. (45 points) Set up the following iterated integrals. (**DO NOT EVALUATE**).

   (a) Set up the iterated integral to find the VOLUME of the region bounded by the four planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 4$.

   (b) Set up the iterated integral to find the AREA in the first quadrant of the region contained by the y-axis, the line $y = x$, a circular arc of radius of 1, and a circular arc of radius 3.

   (c) Set up the iterated integral corresponding to the VOLUME of the solid bound by the cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 4$.

5. (45 points) Evaluate the following triple integral.

$$\iiint_E 3\,dV = \int_{-5}^{5}\int_{-\sqrt{25-x^2}}^{\sqrt{25-x^2}}\int_{0}^{5} 3\ dz\ dy\ dx - \int_{-5}^{5}\int_{-\sqrt{25-y^2}}^{\sqrt{25-y^2}}\int_{0}^{\sqrt{25-x^2-y^2}} 3\ dz\ dx\ dy$$

6. **(10 points) BONUS:** Find the optimal targeting point $(\bar{x}, \bar{y})$ for Question 2.

   An invasion force of Space Bugs has established a beachhead on a colony planet with a distribution of warriors modeled by the function $\rho(x, y) = 4yx^2$ in millions. Advance scouts have identified a region R that is optimal for an orbital strike:

$$R = \{(x, y) |\ x^2 + y^2 \leq 25, x \geq 0, y \geq 0\},$$

   Do not use the targeting officer's assumption of 100 million Space Bugs.

Instructor Solution

WPR 3 - MA255 Section INSTRUCTOR

1. This WPR is worth **225 points**. There are 8 pages to this exam. You have 55 minutes to complete it. All work on this WPR is subject to grading unless marked through or clearly marked to denote "do not grade" and must be your own.
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7. Honor Acknowledgment Statement: sign and date the statement below when you have finished the WPR and are ready to turn it in.

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Cadet ID	Printed Name	Signature	Time/Date Signed
	INSTRUCTOR		

Question:	1	2	3	4	5	6	Total
Points:	45	45	45	45	45	0	225

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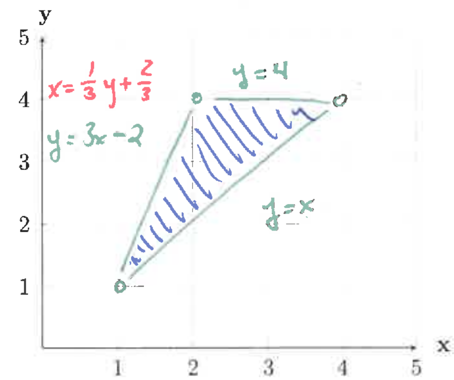
WPR 3 - MA255 Section ______________

1. (45 points) Let $f(x, y) = 3y + 6$, and D be the triangular region with the vertices $(1, 1)$, $(2, 4)$, and $(4, 4)$.

   (a) Sketch the region D. Label region boundaries with corresponding functions.



   (b) Set up **both sets** of iterated integrals to compute $\iint_D f(x, y)dA$ where $dA = dx\,dy$ and $dA = dy\,dx$. **DO NOT SOLVE.**

Vertical: $\int_1^2 \int_x^{3x-2} (3y+6)\,dy\,dx + \int_2^4 \int_x^4 (3y+6)\,dy\,dx$

Horizontal: $\int_1^4 \int_{\frac{1}{3}y+\frac{2}{3}}^{y} (3y+6)\,dx\,dy$

   (c) Evaluate one of the iterated integrals in Part b.

$\int_1^4 \int_{\frac{1}{3}y+\frac{2}{3}}^{y} (3y+6)\,dx\,dy = \int_1^4 \left[3yx + 6x\right]_{\frac{1}{3}y+\frac{2}{3}}^{y} dy = \int_1^4 2y^2 + 2y - 4\,dy$

$= \left[\frac{2}{3}y^3 + y^2 - 4y\right]_1^4 = \frac{128}{3} + 16 - 16 - \frac{2}{3} - 1 + 4 = \frac{135}{3}$

45

2. (45 points) An invasion force of Space Bugs has established a beachhead on a colony planet with a distribution of warriors modeled by the function $\rho(x,y) = 4yx^2$ in millions. Advance scouts have identified a region R that is optimal for an orbital strike:

$$R = \{(x,y) | \cancel{1 \le} x^2 + y^2 \le 25, x \ge 0, y \ge 0\},$$

(a) Find the total number of Space Bugs in the region.

$$4\int_0^{\pi/2}\int_0^5 r^2\cos^2\theta\sin\theta\,dr\,d\theta = \int_0^{\frac{\pi}{2}}\cos^2\theta\sin\theta\,d\theta \int_0^5 r^4\,dr$$

$$\int_0^{\frac{\pi}{2}}\cos^2\theta\sin\theta\,d\theta$$

$u = \cos\theta$ $\quad u(\pi/2) = 0$

$du = -\sin\theta$ $\quad u(0) = 1$

625

$$\int_1^0 -u^2\,du = -\frac{1}{3}u^3\Big|_0^0 = 1/3$$

$= 4(1/3)(625) = \frac{2500}{3}$ millions of warriors.

(b) Concurrently, targeting officers on the orbiting battleship are preparing to launch the first barrage at the planet. Set up the equations to find the optimal target reference point $(\bar{x}, \bar{y})$ through center of mass calculations. The targeting officer assumes 100 million Space Bugs in the region.(***This may not match what you found in Part (a)***). **DO NOT EVALUATE.**

$$\bar{x} = \frac{4}{100}\int_0^{\pi/2}\int_0^5 r^4\cos^3\sin\theta\,dr\,d\theta$$

$$\bar{y} = \frac{4}{100}\int_0^{\pi/2}\int_0^5 r^4\cos^2\theta\sin^2\theta\,dr\,d\theta$$

USE A CONTINUATION SHEET IF NECESSARY.

3. (45 points) Set up, but **DO NOT EVALUATE**, a triple integral to determine the volume of the region below the plane $z = 0$ and above the spherical bowl $x^2 + y^2 + z^2 = 9$ in each of the given coordinate systems.

   (a) Rectangular:

$$\int_{-3}^{3} \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{-\sqrt{9-x^2-y^2}}^{0} dz\,dy\,dx$$

   (b) Cylindrical:

$$\int_{0}^{2\pi} \int_{0}^{3} \int_{-\sqrt{9-r^2}}^{0} r\,dr\,d\theta$$

   (c) Spherical:

$$\int_{0}^{2\pi} \int_{\pi/2}^{\pi} \int_{0}^{3} \rho^2 \sin\phi \, d\rho\, d\phi\, d\theta$$

4. (45 points) Set up the following iterated integrals. (**DO NOT EVALUATE**).

(a) Set up the iterated integral to find the VOLUME of the region bounded by the four planes $x = 0$, $y = 0$, $z = 0$, and $x + y + z = 4$.

$$\int_0^4 \int_0^{4-x} \int_0^{4-x-y} dz\, dy\, dx$$

(b) Set up the iterated integral to find the AREA in the first quadrant of the region contained by the y-axis, the line $y = x$, a circular arc of radius of 1, and a circular arc of radius 3.

$$\int_{\pi/4}^{\pi/2} \int_0^3 r\, dr\, d\theta$$

(c) Set up the iterated integral corresponding to the VOLUME of the solid bound by the cone $z = \sqrt{x^2 + y^2}$ and the plane $z = 4$.

$$\int_0^{2\pi} \int_0^2 \int_r^4 r\, dz\, dr\, d\theta.$$

5. (45 points) Evaluate the following triple integral.

$$\iiint_E 3\,dV = \int_{-5}^{5}\int_{-\sqrt{25-x^2}}^{\sqrt{25-x^2}}\int_0^5 3\,dz\,dy\,dx - \int_{-5}^{5}\int_{-\sqrt{25-y^2}}^{\sqrt{25-y^2}}\int_0^{\sqrt{25-x^2-y^2}} 3\,dz\,dx\,dy$$

Cylindrical

Spherical

$$3\underbrace{\underbrace{\int_0^{2\pi}}_{2\pi}\underbrace{\int_0^{5}}_{\frac{25}{2}}\underbrace{\int_0^{5}}_{5} r\,dz\,dr\,d\theta}_{3(125\pi) = 375\pi}$$

$$-3\underbrace{\underbrace{\int_0^{2\pi}}_{2\pi}\underbrace{\int_0^{\pi/2}}_{1}\underbrace{\int_0^{5}}_{\frac{125}{3}} \rho^2 \sin\theta\, d\rho\, d\phi\, d\theta}_{-3\left(\frac{250\pi}{3}\right) = -250\pi}$$

125π

6. **(10 points) BONUS:** Find the optimal targeting point $(\bar{x}, \bar{y})$ for Question 2.

   An invasion force of Space Bugs has established a beachhead on a colony planet with a distribution of warriors modeled by the function $\rho(x, y) = 4yx^2$ in millions. Advance scouts have identified a region R that is optimal for an orbital strike:

   $$R = \{(x, y) | 1 \le x^2 + y^2 \le 25, x \ge 0, y \ge 0\},$$

   Do not use the targeting officer's assumption of 100 million Space Bugs.

TAB J WPR#4

WPR 4 - MA255 **Section** __________

1. This WPR is worth **225 points**. There are 8 pages to this exam. You have 55 minutes to complete it. All work on this WPR is subject to grading unless marked through or clearly marked to denote "do not grade" and must be your own.

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__________ __________ __________ __________

Cadet ID Printed Name Signature Time/Date Signed

Question:	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
Points:	**30**	**50**	**50**	**50**	**45**	**0**	**225**

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1. (30 points) **Analyze** the following problem prompts to determine the applicable solution techniques and theorems. Every group requires an answer. Use the following bank of techniques and theorems to inform your chosen method(s). **Some questions may have more than one method to solve; list all viable methods. DO NOT ATTEMPT TO SOLVE THE INTEGRALS**.

Line Integral *Fundamental Theorem of Line Integrals* *Surface Integral*
Green's Theorem *Stokes' Theorem* *Divergence Theorem*

(a) Find $\iint_S (x^2y + yx)\, dS$ where S is a portion of a paraboloid above the xy-plane.

◯ Scalar Field	◯ Conservative Field	◯ Open Path
◯ Vector Field	◯ Non-Conservative Field	◯ Closed Path
	◯ Not Applicable	◯ Not Applicable

Method(s) to Solve: ______________________________

(b) Find $\oint_C \langle x^2e^y, x^3y^2\rangle \cdot d\boldsymbol{r}$ where C is composed of two concentric circles (positively oriented).

◯ Scalar Field	◯ Conservative Field	◯ Open Path
◯ Vector Field	◯ Non-Conservative Field	◯ Closed Path
	◯ Not Applicable	◯ Not Applicable

Method(s) to Solve: ______________________________

(c) Find the outward flux of $\boldsymbol{F} = \langle xz, x^2 + y^2, z^2 - y^2\rangle$ across the surface, S, which is a portion of a cone.

◯ Scalar Field	◯ Conservative Field	◯ Open Path
◯ Vector Field	◯ Non-Conservative Field	◯ Closed Path
	◯ Not Applicable	◯ Not Applicable

Method(s) to Solve: ______________________________

2. (50 points) Determine the amount of work done by the wind

$$\mathbf{F} = \langle y^2 + 3x^2y,\ xy + x^3 \rangle$$

while traveling on the route defined by C_1 and C_2:

$$C_1 : \mathbf{r}(t) = \langle t, \quad t^2 \rangle, \quad 0 \leq t \leq 2$$

$$C_2 : \mathbf{r}(t) = \langle -2t + 6, \quad -4t + 12 \rangle, \quad 2 \leq t \leq 3$$

3. (50 points) Find the work along the boundary of the surface described by the paraboloid $z = 25 - x^2 - y^2$ where $z = 0$ subject to the vector field

$$\mathbf{F} = \langle -3xy,\ 3xy,\ 3z \rangle$$

4. (50 points) Find the flux through the region enclosed by the surface $z = 4 - 2x - y$ and the planes $x = 0$, $y = 0$, and $z = 0$, and is positively oriented subject to

$$\mathbf{F} = \langle z^2 + 2x,\ x^2 + z,\ 4z + xy \rangle$$

5. (45 points) Evaluate the following line integral where C is the line $\mathbf{r}(t) = \langle -t^3,\ t\rangle$ and $-1 \leq t \leq 1$.

$$\int_C (2xe^{x^2} - 2y)\ dx + (2y - 2x)\ dy$$

6. **EXTRA CREDIT: (10 points)** Find the potential function f for the conservative vector field $\mathbf{F}(t, w, x, y, z) =$

$$\langle w^2x^3y^4z^5 + 1,\ \ 2twx^3y^4z^5 + 2,\ \ 3tw^2x^2y^4z^5 + 3,\ \ 4tw^2x^3y^3z^5 + 4,\ \ 5tw^2x^3y^4z^4 + 5\rangle$$

Outstanding Student Work

 [Alpha Legion] 29 April 2026

WPR 4 - MA255 Section E27

1. This WPR is worth **225 points**. There are 8 pages to this exam. You have 55 minutes to complete it. All work on this WPR is subject to grading unless marked through or clearly marked to denote "do not grade" and must be your own.

2. Show as much work as necessary to support your answer. A wrong numerical or symbolic solution with no supporting work receives zero credit. Clearly double underline or box your answer.

3. Use a blank continuation sheet and clearly identify that the problem is continued both on the exam and on the continuation sheet. Use one continuation sheet per problem continued. Be sure to put your name on each continuation sheet.

4. **Approved references**: one $8\frac{1}{2} \times 11$ sheet of paper. The paper must be written in your own handwriting (no photocopying or printing digital notes). You may write on both sides of the paper.

5. You may NOT use any resources that are not on the authorized reference list including, but not limited to: computers, phones, calculators, the internet, your classmates.

6. Cadets are **not** authorized to discuss the content, structure, or any other information about this WPR until this WPR has been released from academic security. Discussion includes all forms of written, electronic, and verbal communication. This WPR will be released from academic security at **1610 on 6 May 2026**.

7. Honor Acknowledgment Statement: sign and date the statement below when you have finished the WPR and are ready to turn it in.

   "I did not use any sources nor did I receive any assistance while completing this WPR. I will not discuss this WPR with anyone until it is released from academic security at **1610 on 6 May 2026**."

C4226l663	Chase Lovezoli	[signature]	1500 29APR26
Cadet ID	Printed Name	Signature	Time/Date Signed

Question:	1	2	3	4	5	6	**Total**
Points:	30	50	50	50	45	0	225

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1. (30 points) **Analyze** the following problem prompts to determine the applicable solution techniques and theorems. Every group requires an answer. Use the following bank of techniques and theorems to inform your chosen method(s). **Some questions may have more than one method to solve; list all viable methods. DO NOT ATTEMPT TO SOLVE THE INTEGRALS.**

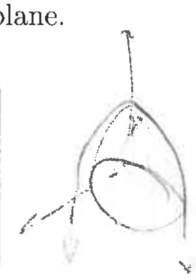
Line Integral ~~*Fundamental Theorem of Line Integrals*~~ *Surface Integral*
Green's Theorem ~~*Stokes' Theorem*~~ ~~*Divergence Theorem*~~

Scalar
3D
Surface

(a) Find $\iint_S (x^2y + yx)\, dS$ where S is a portion of a paraboloid above the xy-plane.

Big

Not div

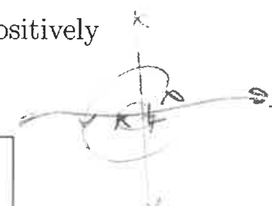


● Scalar Field	○ Conservative Field	○ Open Path
○ Vector Field	○ Non-Conservative Field	○ Closed Path
	● Not Applicable	● Not Applicable

Method(s) to Solve: Surface Integral

Vector
2D
line

(b) Find $\oint_C \langle x^2e^y, x^3y^2\rangle \cdot d\boldsymbol{r}$ where C is composed of two concentric circles (positively oriented).



○ Scalar Field	○ Conservative Field	○ Open Path
● Vector Field	● Non-Conservative Field	● Closed Path
	○ Not Applicable	○ Not Applicable

Method(s) to Solve: Line Integral, Green's Theorem

vectal
3D
Surface

(c) Find the outward flux of $\boldsymbol{F} = \langle xz, x^2 + y^2, z^2 - y^2\rangle$ across the surface, S, which is a portion of a cone.

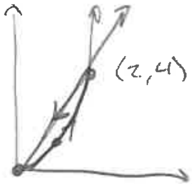
Not div

Flux Sucks
So it doesn't matter

○ Scalar Field	○ Conservative Field	○ Open Path
● Vector Field	● Non-Conservative Field	○ Closed Path
	○ Not Applicable	● Not Applicable

Method(s) to Solve: Surface Integral

USE A CONTINUATION SHEET IF NECESSARY.



2. (50 points) Determine the amount of work done by the wind

$$\mathbf{F} = \langle y^2 + 3x^2y,\ xy + x^3\rangle$$

$xy^2 + x^3y + g(y)$

$\frac{xy^2}{2} + x^3y + g(x)$

while traveling on the route defined by C_1 and C_2:

$$C_1 : \mathbf{r}(t) = \langle t,\ \ t^2\rangle,\ \ 0 \le t \le 2$$

$$C_2 : \mathbf{r}(t) = \langle -2t + 6,\ \ -4t + 12\rangle,\ \ 2 \le t \le 3$$

C_1 $0 \le t \le 2$

$\vec{r}(t) = \langle t, t^2\rangle$

$\vec{F} = \langle y^2 + 3x^2y,\ xy + x^3\rangle$

$= \langle t^4 + 3t^4,\ t^3 + t^3\rangle$

$= \langle 4t^4,\ 2t^3\rangle \cdot \langle 1, 2t\rangle$ $\vec{r}'(t)$

$\int_0^2 4t^4 + 4t^4\,dt \to 8\int_0^2 t^4\,dt$

$8\left[\frac{t^5}{5}\Big|_0^2\right] = \frac{16}{5}(8)$

C_2 $2 \le t \le 3$

$\vec{r}(t) = \langle -2t + 6,\ -4t + 12\rangle$

$\vec{r}'(t) = \langle -2, -4\rangle$

$\vec{F} = \langle y^2 + 3x^2y,\ xy + x^3\rangle$

$\vec{F} = \langle (-4t+12)^2 + 3(-2t+6)^2(-4t+12),\ (-2t+6)(-4t+12) + (-2t+6)^3\rangle$

$(-4t+12)(-4t+12)$ $(-2t+6)(-2t+6)$

$\vec{F} = 16t^2 - 96t + 144 + 3(4t^2 - 24t + 36)(-4t+12),$

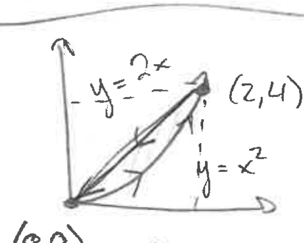
$8t^2 - 48t + 72 + (4t^2 - 24t + 36)(-2t+6)$

$\vec{F} = 16t^2 - 96 + 144 + 3[-16t^3 + 96t^2 - 36(4)t + 48t^2 - 24(12)t + 36(12)$

$\vec{F} = \langle y^2 + 3x^2y,\ xy + x^3\rangle$

$\frac{d}{dy}$ $\frac{d}{dx}$

$= 2y + 3x^2$ $= y + 3x^2$



Pos orient ✓

$\iint_R y + \cancel{3x^2} - (2y + \cancel{3x^2})\,dA$

$= \iint_R -y\,dA$

$\boxed{-}\int_0^2\int_{x^2}^{2x} y\,dy\,dx$

$\frac{y^2}{2}\Big|_{x^2}^{2x}$

$\to \frac{4x^2}{2} - \frac{x^4}{2} = \int_0^2 2x^2 - \frac{1}{2}x^4\,dx$

$\frac{2x^3}{3} - \frac{x^5}{10}\Big|_0^2 =$

$-\left[\frac{(2)^4}{3} - \frac{(2)^5}{10}\right] = \boxed{\frac{(2)^5}{10} - \frac{(2)^4}{3}}$

USE A CONTINUATION SHEET IF NECESSARY.

3. (50 points) Find the work along the boundary of the surface described by the paraboloid $z = 25 - x^2 - y^2$ where $z = 0$ subject to the vector field

$$\mathbf{F} = \langle -3xy,\ 3xy,\ 3z \rangle$$

$\vec{r}(\theta) = \langle 5\cos\theta,\ 5\sin\theta,\ 0 \rangle$

$\vec{r}'(\theta) = \langle -5\sin\theta,\ 5\cos\theta,\ \cancel{0} \rangle$

$\vec{F} = \langle -75\cos\theta\sin\theta,\ 75\cos\theta\sin\theta,\ \cancel{0} \rangle$

$= \int_0^{2\pi} 75(5)\sin^2\theta\cos\theta + 75(5)\sin\theta\cos^2\theta \, d\theta$

$= \boxed{75(5)} \int_0^{2\pi} \sin^2\theta\cos\theta + \cos^2\theta\sin\theta \, d\theta$

$u = \sin\theta$

$du = \cos\theta \, d\theta$

$\int_0^0 u^2 \, du \to \boxed{0}$

$u = \cos\theta$

$-du = +\sin\theta \, d\theta$

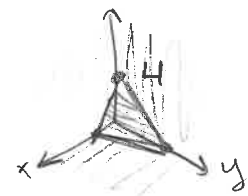
$-\int_1^1 u^2 \, du \quad -\left[\frac{u^3}{3}\Big|_1^1\right] = 0$

$\boxed{= 0}$ Joules

WPR 4 - MA255 Section ____________

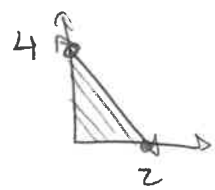
4. (50 points) Find the flux through the region enclosed by the surface $z = 4 - 2x - y$ and the planes $x = 0$, $y = 0$, and $z = 0$, and is positively oriented subject to

$$\mathbf{F} = \langle z^2 + 2x,\ x^2 + z,\ 4z + xy \rangle$$



$\mathrm{Div}(\vec{F}) = 2 + 0 + 4 = 6$

$y = 4 - 2x$



$= \iiint_E 6\,dv = \iiint_0^{4-2x-y} 6\,dz\,dy\,dx$

$6z = \int_0^{4-2x} 24 - 12x - 6y\,dy$

$24y - 12xy - 3y^2 \Big|_0^{4-2x} = 96 - 48x - 12x(4-2x) - 3(16 - 16x + 4x^2)$

$= \int_0^2 \cancel{96}^{\,48} - 48x - \cancel{48x} + 24x^2 - \cancel{48} + \cancel{48x} - 12x^2\,dx$

$\int_0^2 48 - 48x + 12x^2\,dx$

$4(2)(2)(2)$

$48x - 24x^2 + 4x^3 \Big|_0^2$

$4(2)^3$

$4(4)(2)$

$= \cancel{96} - \cancel{24(4)} + 4(16)(4)$

$\boxed{= 32}$

USE A CONTINUATION SHEET IF NECESSARY.

5. (45 points) Evaluate the following line integral where C is the line $\mathbf{r}(t) = \langle -t^3,\ t\rangle$ and $-1 \leq t \leq 1$.

$$\int_C (2xe^{x^2} - 2y)\ dx + (2y - 2x)\ dy$$

$F = \langle 2xe^{x^2} - 2y,\ 2y - 2x \rangle$

$\int dx$ $\quad$ $\int dy$

$e^{x^2} - 2xy + g(y)$ $\quad$ $y^2 - 2xy + g(x)$

$f(x,y) = y^2 - 2xy + e^{x^2}$ $\quad$ $dx = -2y + e^{x^2}(2x)$ ✓

$dy = 2y - 2x$ ✓

Start $t = -1$ $\quad r(-1) = \langle 1, -1 \rangle$

end $t = 1$ $\quad r(1) = \langle -1, 1 \rangle$

$f(-1,1) - f(1,-1)$

$= \left[1 + 2 + e^{(1)}\right] - \left[1 + 2 + e^{(1)}\right]$

$\cancel{1} + \cancel{2} + \cancel{e} - \cancel{1} - \cancel{2} - \cancel{e} = \boxed{0}$

6. **EXTRA CREDIT: (10 points)** Find the potential function f for the conservative vector field $\mathbf{F}(t, w, x, y, z) =$

$$\langle w^2x^3y^4z^5 + 1,\quad 2twx^3y^4z^5 + 2,\quad 3tw^2x^2y^4z^5 + 3,\quad 4tw^2x^3y^3z^5 + 4,\quad 5tw^2x^3y^4z^4 + 5\rangle$$

$w^2x^3y^4z^5t + t + f(w) + f(x) + f(y) + f(z)$

$w^2x^3y^4z^5t + 2w + f(t) + f( ) + f(y) + f(z)$

$$\boxed{f(t,w,x,y,z) = tw^2x^3y^4z^5 + t + 2w + 3x + 4y + 5z}$$

UNITED STATES MILITARY ACADEMY
WEST POINT

TAB K FINAL EXAMINATION

Cadet______________________________ Section ____________

Term End Exam - MA255 **900 Points**

READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING WORK.

1. You have 3.5 hours to complete the exam.
2. Early departure is authorized. Give the exam to your instructor when completed.
3. Approved references: Three $8\frac{1}{2} \times 11$ hand-written note sheets (e.g., no computer-printed or OneNote pages). You may use both sides of the paper. **No calculators are authorized.**
4. Including this cover page, there are 17 pages to this exam.
5. Show as much work as necessary to support your answer. A wrong numerical or symbolic solution with no supporting work receives zero credit. **Even a correct answer without supporting work may receive a deduction.** It is always best to show intermediate steps to illustrate your problem solving process.
6. Clearly identify your final answer. You must double underline your answer.
7. Often a **labeled sketch** will help you see the problem and the grader understand what you are trying to do.
8. Use a continuation sheet and identify that the problem is continued both on the exam and on the continuation sheet. Be sure to put your name on each continuation sheet.

Question:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Total
Points:	100	100	50	60	60	50	60	100	60	80	60	60	60	0	900
Score:															

"I did not use any unauthorized sources nor did I receive any assistance while completing this assessment. I will not discuss any aspects of this assessment with anyone except a MA255 instructor until 20 May 2026."

Sign and date here to show acknowledgment ______________________________

No of Extra Sheets ____________ **Departure time** ____________

Cadet______________________________ **Section** ____________

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Cadet______________________________ **Section** ____________

1. Consider the three points $P_1(3, 1, 2)$, $P_2(5, 3, 3)$, and $P_3(2, 2, 4)$ in a plane.

   (a) (50 points) Find a non-zero vector $\mathbf{n}$ which is orthogonal to the plane containing the three points.

   (b) (50 points) Determine an equation of the plane containing the three points.

Cadet________________________________ **Section** ____________

2. A curve C is traced out by the position function $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$, for time $t \geq 0$.

   (a) (50 points) Find a time, t, such that the tangent line to the curve C is orthogonal to the plane $x + 2y + 3z = 169$.

   (b) (50 points) Find a vector equation or a set of parametric equations for the tangent line to the curve C at time $t = 2$. At what point (x, y, z) does this tangent line intersect the plane $x + 2y + 3z = 169$?

Cadet______________________________ **Section** ____________

3. (50 points) Use the definition of a Taylor series to find the first four nonzero terms of the series for $f(x)$ centered at the value $a = 1$.

$$f(x) = xe^{x-1}$$

Cadet______________________________ **Section** ____________

4. (60 points) Suppose that over a certain region of space the electrical potential E is given by $E(x, y, z) = 7x^2 - 9xy + 3xyz$. Find the rate of change of the potential at $(-3, 2, -10)$ in the direction of the vector $\mathbf{v} = \langle 4, 3, -5 \rangle$.

Cadet________________________ **Section** __________

5. (60 points) Find the maximum and minimum values of $f(x, y) = xy$ subject to the constraint $9x^2 + y^2 = 18$.

6. (50 points) Consider the sum of iterated integrals provided below. **Sketch** the region of integration. **Express** as a single iterated integral.

$$\int_2^4 \int_0^{-y+4} f(x,y)\ dx\ dy + \int_0^2 \int_0^y f(x,y)\ dx\ dy$$

DO NOT EVALUATE.

7. (60 points) You are designing a new swimming pool. The bottom surface of the pool is modeled by:

$$z = g(x, y) = -\frac{x^2}{10} - \frac{y^2}{10} - 1$$

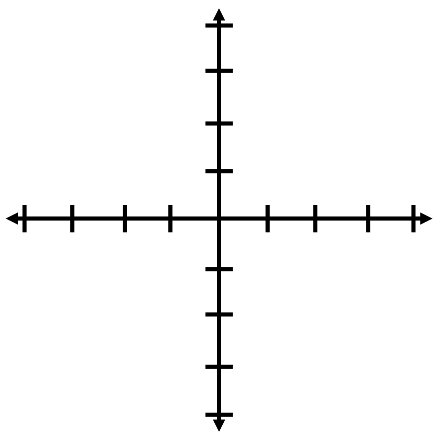
The surface of the pool's water is at $z = 0$. All dimensions are in feet.

You are considering a design for the pool. As viewed from above, the sides of the pool form a half-circle with radius 10 feet, oriented about the origin, in the first two quadrants ($y \geq 0$). How many cubic feet of water could this pool hold if it were full?

Cadet______________________________ **Section** ____________

8. The boundary of a lamina consists of the line $y = -x$, the x-axis, and $x = -3$. The lamina has a density function given by $\rho(x, y) = x^2 + y^2$.

   (a) (10 points) Draw and shade in the region of the lamina; label all lines, axes, and points of intersection.

   (b) (50 points) Find the total mass of the lamina.

Cadet______________________________ **Section** ____________

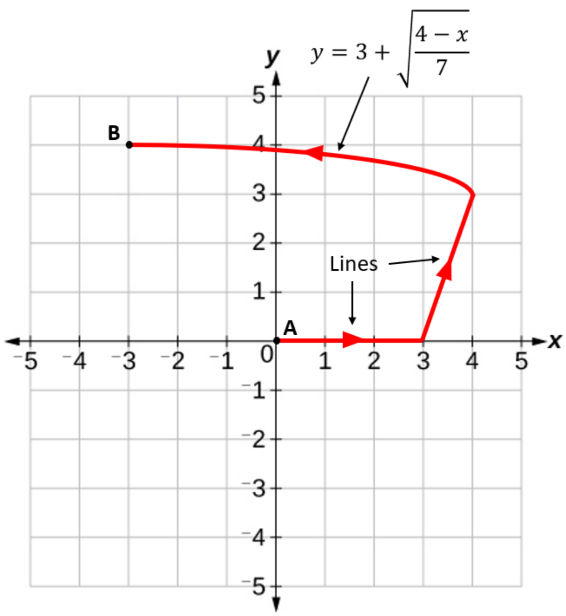
(c) (40 points) This is a continuation of Question 8. Find $\bar{y}$, the y coordinate for the center of mass of the lamina.

9. (60 points) Evaluate the work done by the vector field $\mathbf{F}$ on a particle moving along the path C, where

$$\mathbf{F}(x, y, z) = (xe^y + e^z)\,\mathbf{i} + (\cos y)\,\mathbf{j} + ye^z\,\mathbf{k}$$

and C is the line segment from the origin to the point $(4, 0, 3)$.

10. (80 points) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = (ye^{xy} + 2)\mathbf{i} + (xe^{xy} - 3y^2)\mathbf{j}$ over the curve shown below.



Cadet________________________ **Section** __________

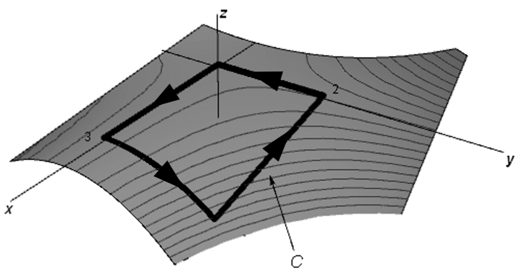
Term End Exam - MA255 **900 Points**

11. (60 points) Evaluate $\oint\limits_C y^3\,dx - x^3\,dy$, where C is the positively-oriented circle of radius 2 centered at the origin.

Cadet______________________________ **Section** ____________

Term End Exam - MA255 **900 Points**

12. (60 points) Consider the vector field $\mathbf{F}(x, y, z) = x^2\,\mathbf{i} + z^2\,\mathbf{j} + 2y\,\mathbf{k}$, and the curve C, where C is the boundary of the part of the surface $z = 1 - xy^2$ over the rectangular region $0 \le x \le 3$, $0 \le y \le 2$, with positive (counterclockwise) orientation when viewed from above. (See the plot below.) Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$.



Cadet______________________________ **Section** ____________

13. (60 points) Use the Divergence Theorem to calculate the flux of $\mathbf{F}$ across S where

$$\mathbf{F}(x, y, z) = ye^z\,\mathbf{i} + 3x^2y\,\mathbf{j} + z^3\,\mathbf{k}$$

and where S is the surface of the solid bounded by the cylinder $x^2 + z^2 = 1$ and the planes $y = -3$ and $y = 1$.

Cadet____________________________ **Section** ____________

Term End Exam - MA255 **900 Points**

14. **EXTRA CREDIT: (10 points)** Why are vectors essential in multivariable calculus, and what new capabilities do they give us?

Instructor Solution

Cadet____________________________ **Section** __________

Term End Exam - MA255 **900 Points**

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Score:															

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Sign and date here to show acknowledgment ________________________

No of Extra Sheets __________ **Departure time** __________

Cadet________________________ **Section** ________

Term End Exam - MA255 **900 Points**

1. Consider the three points $P_1(3,1,2)$, $P_2(5,3,3)$, and $P_3(2,2,4)$ in a plane.

   (a) (50 points) Find a non-zero vector **n** which is orthogonal to the plane containing the three points.

$\vec{A} = \overrightarrow{P_1P_2} = \langle 5-3, 3-1, 3-2 \rangle = \langle 2, 2, 1 \rangle$

$\vec{B} = \overrightarrow{P_1P_3} = \langle 2-3, 2-1, 4-2 \rangle = \langle -1, 1, 2 \rangle$

$$\vec{n} = \vec{A} \times \vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 2 & 1 \\ -1 & 1 & 2 \end{vmatrix} = (4-1)\hat{i} - (4+1)\hat{j} + (2+2)k$$

$$= \underline{\underline{\langle 3, -5, 4 \rangle}}$$

   (b) (50 points) Determine an equation of the plane containing the three points.

$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0$ let $P_1 = P_0 = (3,1,2)$

$\underline{\underline{3(x-3) - 5(y-1) + 4(z-2) = 0}}$

or

$3x - 9 - 5y + 5 + 4z - 8 = 0$

$\underline{\underline{3x - 5y + 4z - 12 = 0}}$

old solution wrote

$3(x-3) - 5(y-1) + 4(z-5) = 0$

or

$3x - 9 - 5y + 5 + 4z - 20 = 0$

$3x - 5y + 4z - 12 = 0$

which is only true if $(z-5)$ was intended to be $(z-2)$.

2. A curve C is traced out by the position function $\mathbf{r}(t) = \langle t, t^2, t^3 \rangle$, for time $t \geq 0$.

   (a) (50 points) Find a time t such that the tangent line to the curve C is orthogonal to the plane $x + 2y + 3z = 169$.

orthogonal to plane, $\vec{n} = \langle 1, 2, 3 \rangle$

tangent to C is $\vec{r}'(t) = \langle 1, 2t, 3t^2 \rangle$

$\vec{r}'(t) = \vec{n}$ when

$$\begin{array}{ccc} \vec{r}'(t) & & \vec{n} \\ 1 & = & 1 \\ 2t & = & 2 \\ 3t^2 & = & 3 \end{array} \Rightarrow t = 1$$

   (b) (50 points) Find a vector equation or a set of parametric equations for the tangent line to the curve C at time $t = 2$. At what point (x, y, z) does this tangent line intersect the plane $x + 2y + 3z = 169$?

let $p(t)$ be the line tangent to C at $t=2$

$p(t) = \vec{r}_0 + t\vec{v}$

$\vec{r}_0 = \vec{r}(2) = \langle 2, 4, 8 \rangle$

$\vec{v} = \vec{r}'(2) = \langle 1, 4, 12 \rangle$

$p(t) = \langle 2, 4, 8 \rangle + t\langle 1, 4, 12 \rangle$

or

$x = 2 + t$

$y = 4 + 4t$

$z = 8 + 12t$

$p(t)$ intersects $x + 2y + 3z = 169$?

$(2+t) + 2(4+4t) + 3(8+12t) = 169$

$2 + t + 8 + 8t + 24 + 36t = 169$

$45t + 34 = 169$

$45t = 135$

$t = 3 \Rightarrow (2+3, 4+4 \cdot 3, 8+12 \cdot 3)$

$(5, 16, 44)$

Cadet____________________________ **Section** __________

Term End Exam - MA255 **900 Points**

3. (50 points) Use the definition of a Taylor series to find the first four nonzero terms of the series for $f(x)$ centered at the value $a = 1$.

$$f(x) = xe^{x-1}$$

$f(x) = xe^{x-1}$ $\quad f(1) = 1e^{1-1} = 1$

$f'(x) = e^{x-1} + xe^{x-1}$ $\quad f'(1) = e^{1-1} + 1e^{1-1} = 2$

$f''(x) = e^{x-1} + e^{x-1} + xe^{x-1}$ $\quad f''(1) = 3$

$f'''(x) = 3e^{x-1} + xe^{x-1}$ $\quad f'''(1) = 4$

$f(x) = 1 + \frac{2}{1!}(x-1) + \frac{3}{2!}(x-1)^2 + \frac{4}{3!}(x-1)^3 + \ldots$

ANS

Cadet____________________ Section ________

Term End Exam - MA255 900 Points

4. (60 points) Suppose that over a certain region of space the electrical potential E is given by $E(x, y, z) = 7x^2 - 9xy + 3xyz$. Find the rate of change of the potential at $(-3, 2, -10)$ in the direction of the vector $\mathbf{v} = \langle 4, 3, -5 \rangle$.

$\nabla E = \langle 14x - 9y + 3yz, -9x + 3xz, 3xy \rangle$

$\nabla E(-3, 2, -10) = \langle 14(-3) - 9(2) + 3(-3)(-10), -9(-3) + 3(-3)(-10), 3(-3)(2) \rangle$

$= \langle -42 - 18 - 60, 27 + 90, -18 \rangle$

$= \langle -120, 117, -18 \rangle$

$\vec{v} = \langle 4, 3, -5 \rangle$ $\quad \vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{5\sqrt{2}} \langle 4, 3, 5 \rangle$

$\|\vec{v}\| = \sqrt{16 + 9 + 25}$

$= \sqrt{50} = 5\sqrt{2}$

$D_{\hat{u}}(E(-3, 2, -10)) = \langle -120, 117, -18 \rangle \cdot \langle 4, 3, -5 \rangle \frac{1}{5\sqrt{2}}$

$\frac{-480 + 351 + 90}{2\sqrt{5}}$ $\quad = \boxed{\frac{-39}{5\sqrt{2}}}$

① gradient ∇E

② Evaluates @ point either before or after

③ $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$

④ $D_u(E) = \nabla E \cdot \vec{u}$

USE A CONTINUATION SHEET IF NECESSARY.

Cadet ______________________ **Section** ________

Term End Exam - MA255 Lagrange **900 Points**

5. (60 points) Find the maximum and minimum values of $f(x, y) = x^2y$ subject to the constraint $x^2 + y^2 = 1$.

$\nabla f = \lambda \nabla g$

① $2xy = 2\lambda x$

② $x^2 = 2\lambda y$

③ $x^2 + y^2 = 1$

$\nabla f = \langle 2xy, x^2 \rangle$

$g(x,y) = x^2 + y^2 - 1$

$\nabla g = \langle 2x, 2y \rangle$

① $x = 0$ or $y = \lambda$

② $2\lambda y = 0$

$\lambda = 0$ or $\cancel{y = 0}$

③ $y = \pm 1$ $\quad 0^2 + 0^2 \neq 0$

② $x^2 = 2y^2$

③ $2y^2 + y^2 = 1$

$3y^2 = 1$

$y = \frac{\pm 1}{\sqrt{3}}$

$x^2 = \frac{2}{3}$

$x = \pm\sqrt{\frac{2}{3}}$

(x, y)	$f(x,y)$	
$(0, 1)$	0	
$(0, -1)$	0	
$(\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{3}})$	$\frac{2}{3\sqrt{3}}$	max
min $(\sqrt{\frac{2}{3}}, -\frac{1}{\sqrt{3}})$	$-\frac{2}{3\sqrt{3}}$	
$(-\sqrt{\frac{2}{3}}, \frac{1}{\sqrt{3}})$	$\frac{2}{3\sqrt{3}}$	max
min $(-\sqrt{\frac{2}{3}}, -\frac{1}{\sqrt{3}})$	$-\frac{2}{3\sqrt{3}}$	

KT

i) $\nabla f = \lambda \nabla g$

ii) solve system

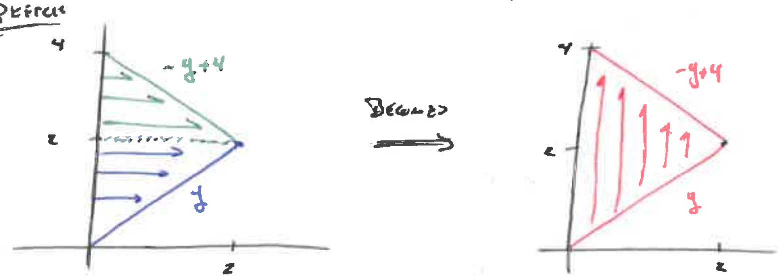
iii) test points

6. (50 points) Consider the sum of iterated integrals provided below. **Sketch** the region of integration and express as a **single** iterated integral (with the order of integration $dA = dy\,dx$).

$$\boxed{\int_2^4 \int_0^{-y+4} f(x,y)\,dx\,dy} + \boxed{\int_0^2 \int_0^y f(x,y)\,dx\,dy}$$

DO NOT EVALUATE.

Sketch



KTS

Plot 1st Integral

Plot 2nd Integral

Integral

$$\int_{x=0}^{x=2} \int_{y=x}^{y=4-x} f(x,y)\,dy\,dx$$

KT

Set up new Integral.

Cadet____________________ Section ________

Term End Exam - MA255 **900 Points**

7. (60 points) You are designing a new swimming pool. The bottom surface of the pool is modeled by:

$$z = g(x,y) = -\frac{x^2}{10} - \frac{y^2}{10} - 1$$

The surface of the pool's water is at $z = 0$. All dimensions are in feet.

You are considering a design for the pool. As viewed from above, the sides of the pool form a half-circle with radius 10 feet, oriented about the origin, in the first two quadrants ($y \geq 0$). How many cubic feet of water could this pool hold if it were full?

Cartesian

$$\int_{-10}^{10}\int_{0}^{(100-x^2)^{1/2}}\int_{-\frac{x^2}{10}-\frac{y^2}{10}-1}^{0} (1)\, dz\, dy\, dx$$

$$= \iint_D -\frac{x^2}{10} - \frac{y^2}{10} - 1 \; dy\, dx$$

$$= \int_{-10}^{10} \frac{x^2}{10}(100-x^2)^{1/2} + \frac{1}{30}(100-x^2)^{3/2} + (100-x^2)^{1/2}$$

= NAH......

Cylindrical.

$$\int_0^{10}\int_0^{\pi}\int_{-\frac{r^2}{10}-1}^{0} (1)\, r\, dz\, dr\, d\theta$$

$$= \pi \int_0^{10} \frac{r^3}{10} + r \; dr$$

$$= \pi \left[\frac{r^4}{40} + \frac{r^2}{2} \right]_0^{10}$$

$$= 300\pi$$

KTs

Interpret Bounds of Integration

Set Up Iterated Integral

Evaluate Integral

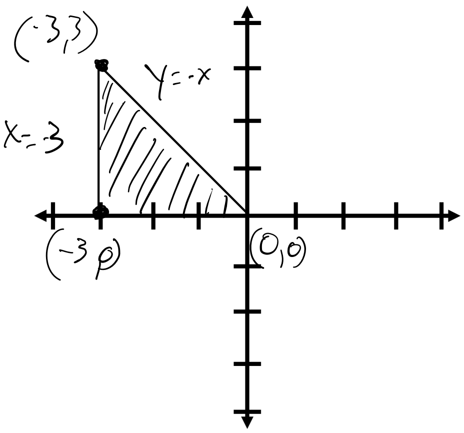
Interpret Result (Ans Cannot Be Negative)

Cadet Answer Key **Section** ______

Term End Exam - MA255 **900 Points**

8. The boundary of a lamina consists of the line $y = -x$, the x-axis, and $x = -3$. The lamina has a density function given by $\rho(x, y) = x^2 + y^2$.

   (a) (10 points) Draw and shade in the region of the lamina; label all lines, axes, and points of intersection.



   (b) (40 points) Find the total mass of the lamina.

$$\int_{-3}^{0}\int_{0}^{-x} x^2+y^2 \, dy \, dx$$

$$\int_{-3}^{0} x^2y + \frac{y^3}{3} \Big|_{y=0}^{y=-x} dx$$

$$\int_{-3}^{0} -x^3 - \frac{x^3}{3} dx = -\frac{4}{3}\int_{-3}^{0} x^3 \, dx$$

$$-\frac{4}{3}\left(\frac{x^4}{4}\right)\Big|_{-3}^{0} \quad 0-\left(-\frac{3^4}{3}\right) = 27$$

Cadet ______________ **Section** ______

Term End Exam - MA255 **900 Points**

(c) (50 points) This is a continuation of Question 8. Find the center of mass of the lamina, $(\bar{x}, \bar{y})$.

$M = 27$

$\bar{x} = \frac{1}{M} \int_{-3}^{0} \int_{0}^{-x} x(x^2+y^2)\,dy\,dx$

$\bar{x} = \frac{1}{27} \int_{-3}^{0} x^3 y + x\frac{y^3}{3} \Big|_{y=0}^{y=-x} dx$

$\bar{x} = \frac{1}{27} \int_{-3}^{0} -x^4 - \frac{x^4}{3}\,dx$

$\bar{x} = -\frac{4}{81} \int_{-3}^{0} x^4\,dx$

$\bar{x} = \frac{-4}{81} \frac{x^5}{5} \Big|_{x=-3}^{x=0}$

$\bar{x} = 0 - \left(\frac{4 \cdot 243}{81 \cdot 5}\right)$

$\bar{x} = -\frac{12}{5}$

$\bar{y} = \frac{1}{M} \int_{-3}^{0} \int_{0}^{-x} y(x^2+y^2)\,dy\,dx$

$\bar{y} = \frac{1}{27} \int_{-3}^{0} \frac{y^2}{2}x^2 + \frac{y^4}{4} \Big|_{y=0}^{y=-x} dx$

$\bar{y} = \frac{1}{27} \int_{-3}^{0} \frac{x^4}{2} + \frac{x^4}{4}\,dx$

$\bar{y} = \frac{1}{36} \int_{-3}^{0} x^4\,dx$

$\bar{y} = \frac{1}{36} \frac{x^5}{5} \Big|_{x=-3}^{x=0}$

$\bar{y} = \frac{1}{36} \frac{243}{5}$

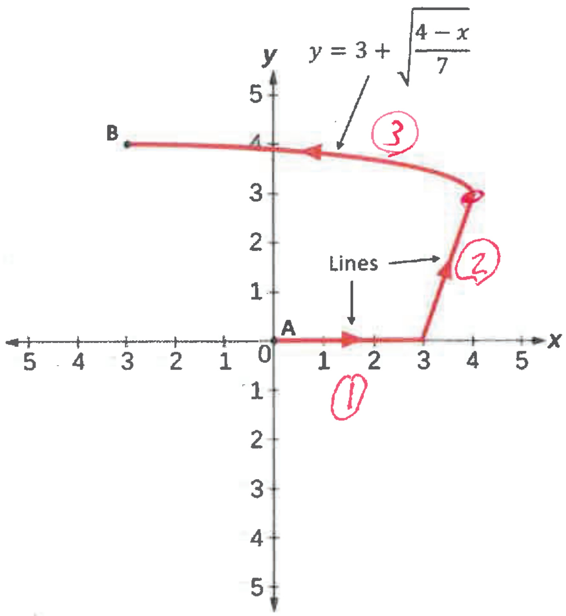
$\bar{y} = \frac{27}{20}$

$\text{COM} = \left(-\frac{12}{5}, \frac{27}{20}\right)$

Cadet KEY Section ______

Term End Exam - MA255 900 Points

10. (80 points) Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for the vector field $\mathbf{F} = (ye^{xy} + 2)\mathbf{i} + (xe^{xy} - 3y^2)\mathbf{j}$ over the curve shown below.



$\Phi = e^{xy} + 2x - y^3$ 35

$\int_{\vec{C}} \vec{F} \cdot d\vec{r} = \Phi(-3,4) - \Phi(0,0)$ 40

$= e^{-12} - 6 - 64 - 1$ 5

$= e^{-12} - 71$ 80

Ans

① $y = 0$, $0 \to x \to 3$ 20

② $y = 3x - 9$, $3 \to x \to 4$ 20

③ $y = 3 + \sqrt{\frac{4-x}{7}}$, $4 \to x \to -3$ 30

TOTAL 10

80

Cadet______________________________ **Section** ____________

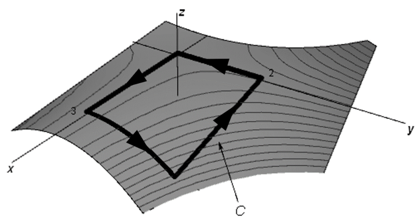
11. (60 points) Evaluate $\oint\limits_C y^3\,dx - x^3\,dy$, where C is the positively-oriented circle of radius 2 centered at the origin.

$$
\begin{aligned}
\int_C \mathbf{F}\cdot dr &= \iint_D Q_x - P_y\,dA \\
&= \iint_D -3x^2 - 3y^2\,dA \\
&= \iint_D -3(x^2+y^2)\,dA \\
&= \int_0^{2\pi}\int_0^2 -3r^2\,r\,dr\,d\theta \\
&= \int_0^{2\pi}\int_0^2 -3r^3\,dr\,d\theta \\
&= \int_0^{2\pi} \frac{-3}{4}r^4\,\Big|_0^2\;d\theta \\
&= \int_0^{2\pi} -12\,d\theta \\
&= -24\pi
\end{aligned}
$$

Cadet______________________________ **Section** __________

Term End Exam - MA255 **900 Points**

12. (60 points) Consider the vector field $\mathbf{F}(x, y, z) = x^2\,\mathbf{i} + z^2\,\mathbf{j} + 2y\,\mathbf{k}$, and the curve C, where C is the boundary of the part of the surface $z = 1 - xy^2$ over the rectangular region $0 \le x \le 3$, $0 \le y \le 2$, with positive (counterclockwise) orientation when viewed from above. (See the plot below.) Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$.



Parameterization :

$$x = x \qquad y = y \qquad z = 1 - xy^2$$

Curl :

$$\begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ dx & dy & dz \\ x^2 & z^2 & 2y \end{vmatrix} = \langle 2 - 2x, -(0-0), 0-0 \rangle = \langle 2 - 2z, 0, 0 \rangle$$

Solution using Special Case Formula $-P\frac{\partial z}{\partial x} - Q\frac{\partial z}{\partial y} + R$

$$\begin{aligned}
\int \vec{F} \cdot dr &= \iint_S \operatorname{curl}(\vec{F}) d\vec{S} \\
&= \iint_D \operatorname{curl}(\vec{F}(\vec{r}(x,y)) \cdot (r_x \times r_y)\, dA \\
&= \iint_D -(2-2z)(-y^2) - 0 + 0\, dA \\
&= \int_0^3 \int_0^2 2y^2 - 2y^2(1 - xy^2)\, dydx \\
&= \int_0^3 \int_0^2 2xy^4\, dydx \\
&= \int_0^3 \frac{2}{5}xy^2|_0^2\, dx \\
&= \int_0^3 \frac{64}{5}x\, dx = \frac{64}{10}x^2|_0^3 = \frac{576}{10} = \frac{288}{5}
\end{aligned}$$

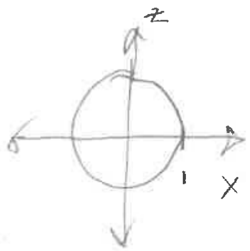
13

DIV THM

$\vec{F}(x,y,z) = ye^{z}\hat{i} + 3x^2y\hat{j} + z^3\hat{k}$

cylinder $x^2+z^2=1$ and $y=-3,\ y=1$

$\nabla \cdot \vec{F} = (0 + 3x^2 + 3z^2) = 3x^2 + 3z^2$



$$\int_0^{2\pi}\int_0^{1}\int_{-3}^{1} (3x^2+3z^2)\, r\, dz\, dr\, d\theta$$

$3x^2+3z^2$
$= 3r^2$

$$\int_0^{2\pi}\int_0^{1}\int_{-3}^{1} (3r^2)\, r\, dz\, dr\, d\theta$$

$$= \int_0^{2\pi}\int_0^{1}\int_{-3}^{1} 3r^3\, dz\, dr\, d\theta$$

$$= \int_0^{2\pi} d\theta \int_0^{1} 3r^3 \int_{-3}^{1} dz$$

$$= \left[\theta\right]_0^{2\pi} \left[\tfrac{3}{4}r^4\right]_0^{1} \left[z\right]_{-3}^{1}$$

$$= [2\pi]\left[\tfrac{3}{4}\right][4] = \underline{6\pi}$$

Cadet______________________________ **Section** ____________

Term End Exam - MA255 **900 Points**

14. **EXTRA CREDIT: (10 points)** Why are vectors essential in multivariable calculus, and what new capabilities do they give us?

SECTION III

ASSIGNMENTS

TAB L KINEMATICS CAPSTONE

KINEMATICS CAPSTONE: SCENARIO

Background

In the modern operating environment, the proliferation of missile technology threatens the safety of the United States and critical infrastructure in our military construct. One of the current threats, most notably seen in the frequent missile tests by North Korea, is inter-continental ballistic missiles (ICBMs). These missiles are capable of traveling thousands of kilometers, and they reach extraordinary altitudes and velocities. Using ICBMs, nations can project military power around the world with both conventional and nuclear munitions.

In order to combat the threat of ICBMs, the United States uses a complex web of sensors and interceptors to detect, characterize, and neutralize incoming threats. The longest ranged interceptor system utilized by the United States is the Ground-based Midcourse Defense (GMD) system. This system relies on ground and space based sensors to detect and track the ICBM while launching an exo-atmospheric interceptor to execute a kinetic kill near the apogee of the incoming missile's path. This is where the incoming ICBM is traveling the slowest and therefore has the highest probability for a successful intercept.

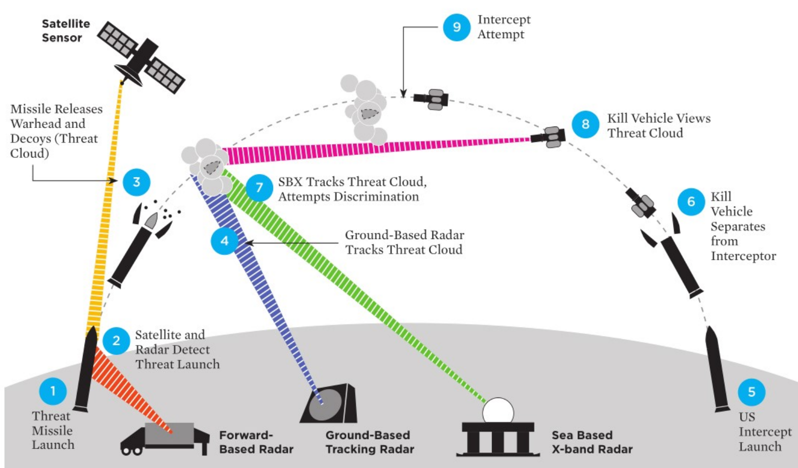


Figure 1: Anatomy of an ICBM Intercept (Arms Control Association, 2019)

This system generally works as follows:

a. Enemy forces launch a missile. (1)

b. Satellite and forward-based radars detect and track the missile in flight as the warhead is released and countermeasures are deployed. (2-4)

c. The Missile Warning Center characterizes the launch, confirms the threat, and issues orders for intercept. (5)

d. Ground and sea based radars continue to track and update the telemetry of interceptor. (7)

e. The interceptor separates from the booster and attempts to execute a kinetic kill of the target warhead. (6, 8, 9)

f. Space based assets monitor the intercept for final kill assessment.

In order to effectively destroy the enemy ICBM, the Missile Warning Center not only needs to be able to predict the future location of the ICBM, but also need to be able to predict the flight path of the interceptor.

ICBMs are initially powered by large booster engine rockets before following an un-powered, parabolic, free-fall trajectory. The flight of an ICBM can can be broken into three phases:

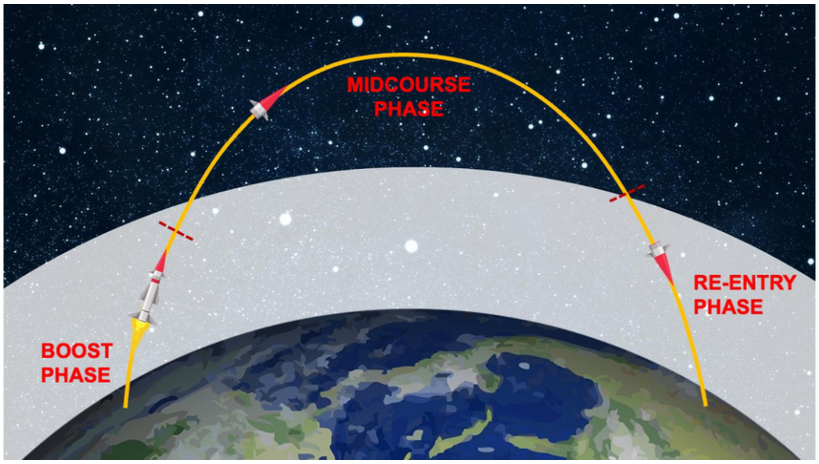


Figure 2: Phases of an ICBM Trajectory (Overview of Sensors, 2022)

1. The Boost Phase: During this phase, the ICBM is accelerated to speeds in excess of Mach 23 (24,000 kilometers per hour) and ends once the booster engines stop firing. After the conclusion of the boosters firing, the trajectory can be calculated and the target of the warhead can be identified.
2. The Midcourse Phase: During this phase, the ICBM begins its un-powered fall back to Earth and is the longest phase of the flight following a predictable flight path. The warhead also detaches from the main boosters and deploys potential countermeasures to obfuscate sensors and defend the warhead from incoming interceptors.
3. The Terminal Phase: During this phase, the warhead reenters the atmosphere and impacts or detonates above the desired target. Upon reentry, the warhead can travel at speeds in excess of 3,200 kilometers per hour and the difficulty of interception increases exponentially. This phase lasts less than a minute.

Due to the complex nature of calculating exo-atmospheric trajectories around a spherical body, for the duration of this project, we will assume all locations are located on the same plane (i.e. the Earth is flat and missiles fly in parabolas above the flat earth). We will also assume that air resistance is negligible. Recall that we have already derived the parametric equations of projectile motion in a vacuum in two dimensions (Stewart §13.4). How would we extend that derivation to three dimensions?

In Part 1 (Individual) of this project, we will study the behavior of an incoming ICBM, given its trajectory at the beginning of the Midcourse Phase. In Part 2 (Group) of this project, we will determined the required trajectory for an interceptor to collide with the ICBM at its apogee. For a challenge, you may incorporate air resistance into your calculations. This leads to a system of second order differential equations whose solution is known as the **ballistic equations**.

UNITED STATES MILITARY ACADEMY
WEST POINT.

KINEMATICS CAPSTONE: CHECK ON UNDERSTANDING

Part 1 - Individual Assignment

Consider that you are the Watch Officer on duty at the Missile Warning Center (MWC) at the Cheyenne Mountain Space Force Station in Colorado. You are tracking an incoming ICBM. To ensure that the United States is protected, the MWC will need to derive the equations of motion based on the initial readings from the space and forward-based ground sensors.

As the Watch Officer, you receive the initial data burst from the early warning sensors at the moment the incoming ICBM achieves burnout and enters the Midcourse phase. The data feed places the ICBM 14,000 kilometers West and 100 kilometers North of your current location in the MWC at an altitude of 1 kilometer. (Hint : How should you establish your coordinate system?) The initial velocity of the warhead as it enters the midcourse phase after separation from the booster is 11 km/s in an eastward direction, 0.5 km/s in a southern direction, and 5.5 km/s in the upward vertical direction.

Submit the answers to the following questions on Canvas no later than **0740 26JAN26** IAW the DAAW. Write your answers in the provided boxes. As applicable, show your work for each problem.

1. What is the acceleration vector of the of the incoming missile in the midcourse phase?

2. What is the velocity vector of the ICBM when you first observe it?

3. What is the position vector of the ICBM when you first observe it? (Use the MWC as the origin.)

4. What is the velocity vector function of the ICBM?

5. What is the position vector function of the ICBM?

6. Where will the ICBM land if it is not intercepted? (in kilometers relative to your location)

7. According to your position vector function, where did the missile take off?

8. What is one source of inaccuracy in using your position vector function to determine where the missile was launched from?

Instructor Solution

Part 1 - Individual Assignment

Consider that you are the Watch Officer on duty at the Missile Warning Center (MWC) at the Cheyenne Mountain Space Force Station in Colorado. You are tracking an incoming ICBM. To ensure that the United States is protected, the MWC will need to derive the equations of motion based on the initial readings from the space and forward-based ground sensors.

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1. What is the acceleration vector of the of the incoming missile in the midcourse phase? $\langle 0, 0, -g\rangle$ where $g = -.00982$
2. What is the velocity vector of the ICBM when you first observe it? $\mathbf{v}_0 =< 11, -.5, 5.5 >$
3. What is the position vector of the ICBM when you first observe it? (Use the MWC as the origin.) $\mathbf{r}_0 =< -14000, 100, 1 >$
4. What is the velocity vector function of the ICBM? Integrate acceleration and apply initial condition. $\mathbf{v}(t) =< 11, -.5, 5.5 - .0098t >$
5. What is the position vector function of the ICBM? Integrate velocity and apply initial condition. $\mathbf{r}(t) =< 11t - 1400, 100 - .5t, -.0049t^2 + 5.5t + 1 >$
6. Where will the ICBM land if it is not intercepted? (in kilometers relative to your location) Set $z = 0$ in position function to find two times that $z = 0$. At $t = 1122.63$ the missile lands at $< -1651.06, -461.315, 0 >$.
7. According to your position vector function, where did the missile take off? Set $z = 0$ in position function to find two times that $z = 0$. At $t = -.181789$ the missile lifts off from $< -14002, 100.091, 0 >$.
8. What is one source of inaccuracy in using your position vector function to determine where the missile was launched from? Missile behavior during the boost phase may not match the trajectory of the unpowered midcourse phase.

Point allocation : Questions 1 and 2 are worth 3 points. All other questions are worth 4 points. Question 8 has many possible correct answers.

KINEMATICS CAPSTONE: GROUP

Part 2 - Group Assignment

Consider that your team is on duty as members of the 49th Missile Defense Battalion in Fort Greeley, Alaska (located 2700 km north and 2747 km east of the Missile Warning Center (MWC). You receive an alert that an incoming ICBM is inbound on the following trajectory:

$$\vec{r}(t) =< 11t - 14000, 100 - .5t, -.0049t^2 + 5.5t + 1 >$$

To increase the likelihood of intercept, you launch an interceptor missile that will collide with the ICBM when the ICBM reaches its apex.

Your intercept missile takes one minute to activate its launch sequence and an additional minute to reach midcourse phase. Your task is the calculate the initial velocity vector that your interceptor missile must have at $t = 120$ and position $(-2747, 2700, 1)$ in order to collide with the ICBM. Use the initial velocity vector that you found to write the vector function for position of your interceptor missile. Confirm that your vector function for position collides with the position function of the ICBM.

Despite your work, your missile did not intercept the ICBM. Confirm that your initial velocity required was more than 7.5 km/s, which is beyond limitations of your interceptor missile. Fortunately, an interceptor fired from a Aegis Ballistic Missile Defense system was able to intercept the ICBM. The Aegis system is a naval defense system. The Navy was able to intercept this missile with an Aegis missile from a cruiser in the Pacific Ocean.

Deliverables

Write a 3 to 5 page report, following the D/Math Writing Template that explains the need for the Aegis Ballistic Missile Defense system. Use the example of this ICBM as the main example in your argument. Your Results section should include an estimated max effective range for the Fort Greely interceptor missile based on the 7.5 km/s limitation. The Methodology section must detail relevant simplifying assumptions. Address these simplifying assumptions in the Discussion section. This section should attempt to interpret your Results in the real, three-dimensional world. At a minimum your discussion section should address the following

1. How would a real world flight path differ from your modeled flight path?
2. What threat are interceptor missiles staged at Fort Greeley best suited for?
3. Where should Aegis enabled Navy cruisers and destroyers be stationed to defend threats from the West?
4. Interpret the impact of two additional assumptions that you made in the Methodology section.

Submit your report on Canvas NLT **01FEB26** IAW the DAAW. You may not use Aritificial Intelligence to draft any portion of your report.

Bonus: Incorporate an air resistance force that is negatively proportional to velocity in your flight calculations. This will require ODEs.

Instructor Solution

Part 2 - Group Assignment

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Deliverables

Write a 3 to 5 page report, following the D/Math Writing Template that explains the need for the Aegis Ballistic Missile Defense system. Use the example of this ICBM as the main example in your argument. Your Results section should include an estimated max effective range for the Fort Greely interceptor missile based on the 7.5 km/s limitation. The Methodology section must detail relevant simplifying assumptions. Address these simplifying assumptions in the Discussion section. This section should attempt to interpret your Results in the real, three-dimensional world. At a minimum your discussion section should address the following

1. How would a real world flight path differ from your modeled flight path?
2. What threat are interceptor missiles staged at Fort Greeley best suited for?
3. Where should Aegis enabled Navy cruisers and destroyers be stationed to defend threats from the West?
4. Interpret the impact of two additional assumptions that you made in the Methodology section.

Submit your report on Canvas NLT **01FEB26** IAW the DAAW. You may not use Aritificial Intelligence to draft any portion of your report.

Bonus: Incorporate an air resistance force that is negatively proportional to velocity in your flight calculations. This will require ODEs.

Solution

Apex of ICBM $\vec{r}(561.315) =< -7826.53, -180.61, 1544.17 >$.
Let $t_0 = 120$ seconds, the moment the interceptor missile inters the mid-course phase.

Derive vector position function for interceptor missile $\vec{p}(t)$.

$$\vec{a}(t) =< 0, 0, -.00982 >$$
$$\vec{v}(t) = -\vec{a}t + C$$

We want $c = \vec{v}_{t_0}$. Reparameterize.

$$\vec{v}(t) = \vec{a}(t - t_0) + c$$
$$\vec{v}(t_0) = 0 + c$$
$$\vec{v}_{t_0} = c$$

$$\vec{v}(t) = \vec{a}(t - t_0) + \vec{v}_{t_0}$$
$$\vec{p}(t) = \frac{1}{2}\vec{a}(t - t_0)^2 + \vec{v}_{t_0}(t - t_0) + \vec{p}(t_0)$$

Solve for $\vec{v}_{t_0}$

$$\vec{v}_{t_0} = \frac{\vec{p}(t) - \vec{p}(t_0) - \frac{1}{2}\vec{a}(t - t_0)^2}{(t - t_0)}$$

Plug in $\vec{p}(561.406) =< -7824.55, -180.703, 1544.37 >$ to obtain

$$\vec{v}_{t_0} =< -11.5077, -6.52744, 5.65964 >$$

Since $|\vec{v}_{t_0}| = 14.3855$ is greater than 7.5, the missile fails to make the intercept.

TAB M OPTIMIZATION CAPSTONE

OPTIMIZATION CAPSTONE: SCENARIO

Background

"Operations Research is a scientific approach to decision making with a focus on how best to design and operate systems, usually under conditions requiring the allocation of scarce resources. However, whether one means the term to be a professional designation, a label for a body of methods, or an approach to problem solving, operations research is today inextricably linked to the direction and management of large systems of people, machines, materials, and money in government, industry, business, and defense." [1]

You are currently serving as a Battalion S4 (Logistics Officer) for 3-66 AR in the First Brigade, First Infantry Division. You are currently on a training exercise at the National Training Center in Fort Irwin, California. Your BN has been tasked to set up a defense in the "central corridor". The current forward line of troops (FLoT) is along the 38 Easting. Able Company, an armor company, is located in the south with a CO HQ at 11S NU 39000 11000. Black Knight Company, also an armor company, is in the North with a CO HQ at 11S NU 39000 17000. Chosen Company, the mechanized infantry company is located with a CO HQ at 11S NU 39000 14000. You have been tasked to pick an optimal location for the Battalion Logistics node, called a Combat Trains Command Post (CTCP). It is currently located at 11S NU 48000 13000.

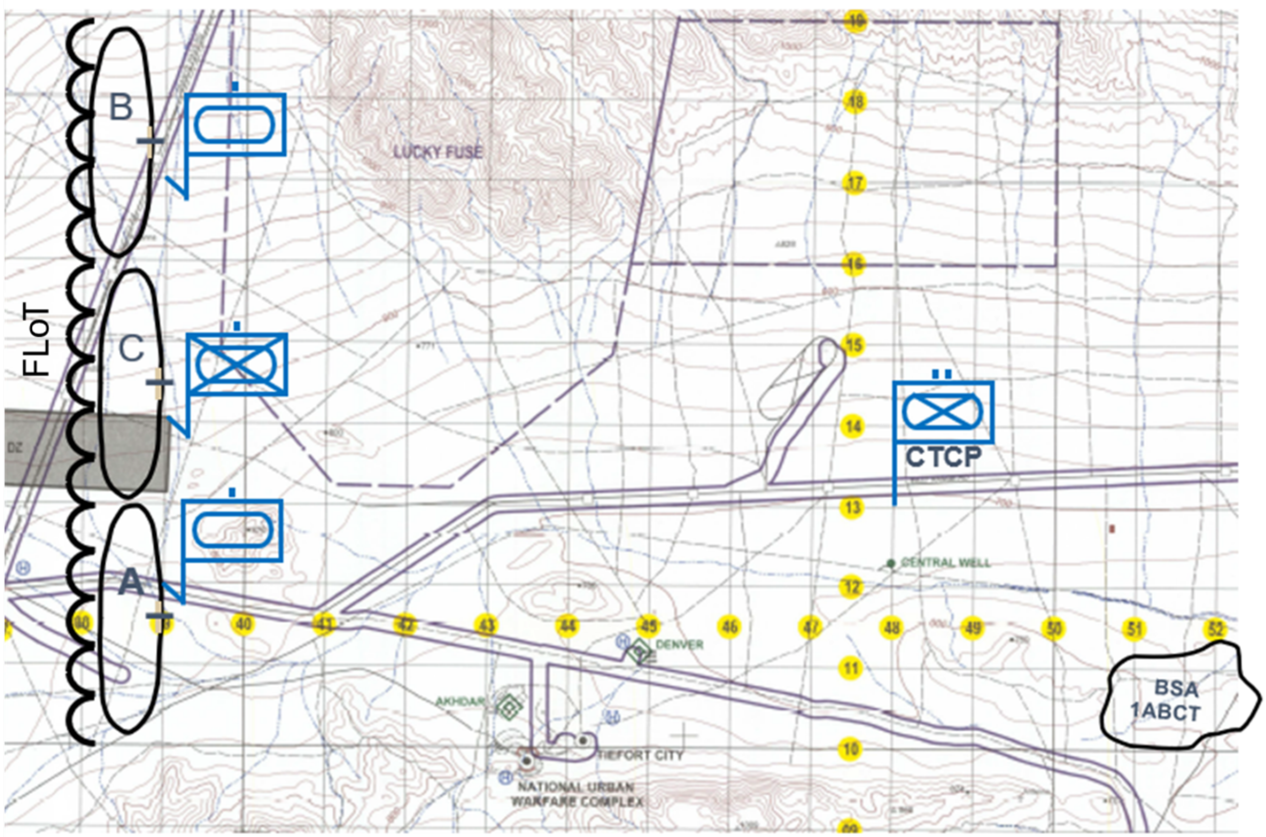

Figure 1: Battalion Operational Graphics

You begin talking to the distribution platoon leader. He lets you know that he takes 5 vehicles to each of the tank companies twice a day and 7 vehicles to the infantry company once a day. The support Soldiers also have to pick up supplies from the Brigade Support Area (BSA) once a day and that requires a convoy of 21 vehicles. The BSA is located at 11S NU 51000 11000. It takes 2 soldiers to operate each vehicle.

You next talk to your BN intelligence officer (S2) who informs you that in the Area of Operations, all travel is restricted to 16 km/hr. She also lets you know that there is a threat from enemy indirect fire that gets bigger as you place the CTCP closer to the front lines. To reduce this risk, the CTCP will need to be camouflaged. Her best approximation is that it takes $C(d) = 18.3 - 1.1d$ hours worth of work to camouflage the CTCP each day to reduce the threat of indirect fire to the CTCP. Where C is soldier hours of labor per day to camouflage the CTCP and d is distance from the CTCP to the FLoT in kilometers. Because the convoys are moving, they are not at risk of being engaged by indirect fire.

Finally, you consult a distribution section leader. He said it took about 24.6 soldier hours yesterday to keep Black Knight Company supplied.

In this project you will find the optimal location for the CTCP with a focus on saving time for the support soldiers. Use all of the skills you have accumulated in MA255 to save time for the operators of the vehicles.

OPTIMIZATION CAPSTONE: CHECK ON UNDERSTANDING

Part 1 - Individual Assignment

Consider that you are the Battalion S4 (Logistics Officer) for 3-66 AR in the 1st Brigade Combat Team, 1st Infantry Division. The current FLOT is along the 38 Easting and you are tasked with selecting a location for the Battalion CTCP. Background facts required for the project are listed in the background document. Refer to the posted scenario for additional details.

Submit the answers to the following questions on Canvas no later than **0740 17FEB26** IAW the DAAW. Write your answers in the provided boxes. As applicable, show your work for each problem. **You are not allowed to use generative AI to assist in completing this assignment.**Attach any code as an appendix if it contains a portion of your work.

1. Calculate the distance from the CTCP to the Black Knight Company and calculate how many Soldier hours it will take each day to keep the company supplied. This answer should validate the analysis conducted by the distribution section leader.

2. Create a formula to calculate the total number of Soldier Hours required each day to complete all routes. Use the formula to find the Soldier hours required based on the current location of the CTCP.

3. Find the total Soldier hours required per day including camouflage based on the location of the CTCP. This answer should be approixmately 90 hours.

4. Create a function, $SH(x, y)$, that takes the coordinates of the CTCP as an input and returns the number of Soldier hours per day as an output.

5. Calculate $\frac{\partial SH(x,y)}{dy}$ using a computer program and attach the code as an appendix to this assignment. (Hint: Mathematica Command “D”)

6. Using a numerical solver, find the optimal solution to place the CTCP to minimize the Soldier hours per day. (Hint: NSolve in Mathematica]

7. How many days would the CTCP need to be emplaced to make the move worth it if the process of moving the CTCP would take a total of 12 hours?

Instructor Solution

Part 1 - Individual Assignment

Consider that you are the Battalion S4 (Logistics Officer) for 3-66 AR in the 1st Brigade Combat Team, 1st Infantry Division. The current FLOT is along the 38 Easting and you are tasked with selecting a location for the Battalion CTCP. Background facts required for the project are listed in the background document. Refer to the posted scenario for additional details.

Submit the answers to the following questions on Canvas no later than **0740 17FEB26** IAW the DAAW. Write your answers in the provided boxes. As applicable, show your work for each problem. **You are not allowed to use generative AI to assist in completing this assignment.**Attach any code as an appendix if it contains a portion of your work.

1. Calculate the distance from the CTCP to the Black Knight Company and calculate how many Soldier hours it will take each day to keep the company supplied. This answer should validate the analysis conducted by the distribution section leader.

   Distance to Black Knight Company: $\sqrt{(39-48)^2+(17-13)^2} = \sqrt{97} = 9.84$

   $\sqrt{97}(km)/16(km/h) * 5(vehicles/convoy) * 2(Soldiers/vehicle) * 2(trips/convoy) * 2(convoys/day) = 24.62(hrs)$

2. Create a formula to calculate the total number of Soldier Hours required each day to complete all routes. Use the formula to find the Soldier hours required based on the current location of the CTCP. $5/2\sqrt{(39-x)^2+(11-y)^2} + 21/4\sqrt{(51-x)^2+(11-y)^2} + 7/4\sqrt{(39-x)^2+(14-y)^2} + 5/2\sqrt{(39-x)^2+(17-y)^2}$

   82.4471

3. Find the total Soldier hours required per day including camouflage based on the location of the CTCP. This answer should be approixmately 90 hours. 89.747

4. Create a function, $SH(x, y)$, that takes the coordinates of the CTCP as an input and returns the number of Soldier hours per day as an output. $SH(x, y) = 5/2\sqrt{(39-x)^2+(11-y)^2} + 21/4\sqrt{(51-x)^2+(11-y)^2} + 7/4\sqrt{(39-x)^2+(14-y)^2} + 5/2\sqrt{(39-x)^2+(17-y)^2} + 18.3 - 1.1(x-38)$

5. Calculate $\frac{\partial SH(x,y)}{dy}$ using a computer program and attach the code as an appendix to this assignment. (Hint: Mathematica Command "D") $-((5(11-y))/(2\sqrt{(39-x)^2+(11-y)^2})) - (21(11-y))/(4\sqrt{(51-x)^2+(11-y)^2}) - (7(14-y))/(4\sqrt{(39-x)^2+(14-y)^2}) - (5(17-y))/(2\sqrt{(39-x)^2+(17-y)^2})$

6. Using a numerical solver, find the optimal solution to place the CTCP to minimize the Soldier hours per day. (Hint: NSolve in Mathematica]

   NSolve[D[SH[x, y], x] == 0, D[SH[x, y], y] == 0, x, y, Reals] $x->45.4606, y->12.4326$

7. How many days would the CTCP need to be emplaced to make the move worth it if the process of moving the CTCP would take a total of 12 hours?

   Optimal SH/day= 88.1, Current SH/day= 89.7

   $89.7 - 88.1 = 1.6$hrs saved per day, $\frac{12}{1.6} = 7.5$ days

Point allocation : Question 4 and 5 worth 5 points. All other questions worth 4 points.

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OPTIMIZATION CAPSTONE: GROUP

Part 2 - Group Assignment

Consider that your team continues duty as the Battalion S4 of 3-66 AR. Based on your previous optimization efforts, you decided to place the CTCP at 11S NU 45460 12433. The Battalion Fire Support Officer (FSO) informs you that 1-5 Field Artillery Battalion has placed a circular Position Area for Artillery (PAA) of radius 2 km centered at 11S NU 45000 13000. You are not allowed to place the CTCP in the PAA, although you can drive vehicles through the PAA. Does this impact where we should place the CTCP? What is the new optimal location for the CTCP? Show this is the minimum using a technique you learned in class. Make a plot that helps the reader visualize the problem.

After successful artillery operations the FSO informs you that the PAA is being moved to a location outside of the current area of operations, but your S2 comes back to you with an update. The town of Tiefort City is a square city with corners 11S NU 43000 09000, 11S NU 43000 11000, 11S NU 45000 09000, 11S NU 45000 11000. The supreme leader of the enemy was born in Teifort City and he forbids any form of indirect fire on the town. Is it worthwhile to place the CTCP in Teifort City to avoid the cost of indirect fire? If so, where? Show that this is the minimum using a technique you learned in class. Make a plot that helps the reader visualize the problem.

Mathematica Hint: Look at the cadet starter code for assistance with the RegionFunction command.

Deliverables

Write a 3 to 5 page report, following the D/Math Writing Template that explains your rationale for selecting specific locations for the CTCP. You should address both operational situations in your report. The Methodology section must detail relevant simplifying assumptions. Your Results section should include recommendations for placement of the CTCP in both operational situation. The Discussion section should attempt to interpret your Results in the real, three-dimensional world. At a minimum your discussion section should address the following:

1. How would a real-world logistics situation differ from the scenario presented in this project and what are the risks associated with your analysis?
2. What additional details would you need to consider if you were going to optimize the placement of the CTCP to include altitude?
3. Interpret the impact of two additional assumptions that you made in the Methodology section.

Submit your report on Canvas NLT **01MAR26** IAW the DAAW. You may not use Artificial Intelligence to draft any portion of your report.

Bonus: Consider that the enemy is no longer a stationary threat but rather a force intent on creating havoc in your lines of communication (supply routes). In an appendix to your report describe how you would optimize the placement of the CTCP if the threat an enemy force was described by another mathematical function $T(x, y)$. At most set up a solution technique but do not solve.

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TAB N INTEGRATION CAPSTONE

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INTEGRATION CAPSTONE: SCENARIO

Background

During CP2, your battalion staff analyzed distance conditions across the Area of Operations (AO) and successfully identified the optimal location for the Combat Trains Command Post (CTCP). Using terrain constraints and operational modeling, the CTCP was positioned to maximize sustainment effectiveness while minimizing exposure to known threats.

Since establishing the CTCP, intelligence reporting indicates an increase in small autonomous drone reconnaissance flights operating near logistics nodes. These drones are assessed to be conducting surveillance in preparation for future indirect or precision attacks against sustainment operations.

The Battalion Commander now assesses:

"The CTCP location is sound, but we must understand where enemy drones are most likely to operate and how to best protect our sustainment hub."

Now serving as the Assistant Operations Officer (AS3), you are tasked with supporting the staff by providing a quantitative assessment of drone activity patterns and recommending placement of detection assets.

Intelligence Estimate

The Battalion S2 modeling cell developed a Joint Probability Density Function (JDF) describing the likelihood of drone presence across the AO relative to the newly established CTCP location.

During CP2, your staff transformed the operational terrain map into a simplified Cartesian coordinate system to support CTCP placement analysis. That coordinate transformation now serves as the baseline for this threat analysis. The CTCP location identified in CP2 has been translated to the origin $(0, 0)$, with positive x-axis oriented east, positive y-axis oriented north, and each unit representing 1 kilometer.

This coordinate framework allows the intelligence cell to express drone activity as a joint probability density function relative to the CTCP.

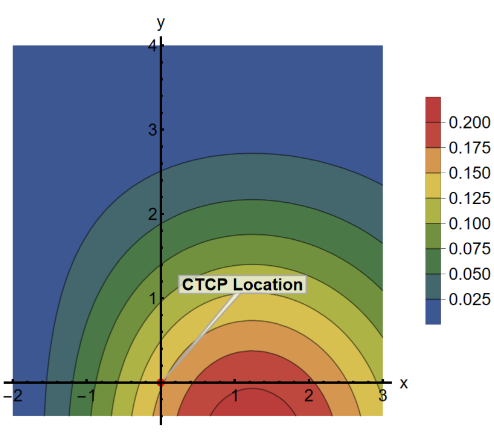

Figure 1: Contour Map for JDF

Drone activity is modeled by the following JDF:

$$f(x, y) = C(x + 2)(4 - y)e^{-(x^2+y^2)/8}$$

where the constant C is a normalization parameter for the JDF, and the operational region is given by:

$$R = \{(x, y)| \ -2 \leq x \leq 3, \ 0 \leq y \leq 4\}.$$

Figure 1 represents the contour map for the probability density function provided for the likelihood of drone presence across the AO during the next operational period.

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INTEGRATION CAPSTONE: CHECK ON UNDERSTANDING

Part 1 - Individual Assignment

Consider that you are the Assistant Operations Officer (AS3) for 3-66 AR in the 1st Brigade Combat Team, 1st Infantry Division. You are tasked with supporting the staff by providing a quantitative assessment of drone activity patterns and recommending placement of detection assets. Background facts required for the project are listed in the background document. Refer to the posted scenario for additional details.

Submit the answers to the following questions on Canvas no later than **0740 17MAR26** IAW the DAAW. Write your answers in the provided space. As applicable, show your work for each problem. **You are not allowed to use generative AI to assist in completing this assignment.** Attach any code as an appendix if it contains a portion of your work.

1. Before conducting any operational analysis, you must verify that the provided intelligence model represents a valid probability distribution. The Joint Probability Density Function (JDF) given by the Battalion S2 includes the constant C as a normalization parameter. **In the step, set up** and provide the double integral that enforces the properties required for a valid Joint Density Function.

2. Now, solve for the value of C to normalize and validate the JDF.

3. Briefly explain why determining C is operationally necessary before computing expected values or detection probabilities.

4. Using the given Joint Probability Density Function, compute the expected value for each variable and report the result as and ordered pair, $(E[X], E[Y])$, representing the expected location of drone activity relative to the CTCP.

5. Determine the location within the Area of Operations where the joint probability density function $f(x, y)$ attains its greatest value, and report the coordinates and the function value at this point. You may think of this as a multivariable optimization problem over a bounded region.

6. Briefly explain what the expected value location (from problem 4) represents operationally anb what the maximum-density location (from problem 5) represents and how the two are related or different.

7. Discuss why the expected value coordinates should not automatically be assumed to be the optimal location for placing a detection sensor.

8. The Battalion received counter-UAS detection sensors capable of detecting drones within a 1 km radius of their placement location. Detection occurs only within the boundaries of the Area of Operations (AO). Compute the probability that a drone is detected by a single sensor placed at selected locations within the AO. For each placement, calculate the total probability of drone presence within the 1 km detection region (restricted to the AO if necessary), and compare the resulting detection probabilities. At a minimum, evaluate the probability of detection when the sensor is positioned at:

    a. the expected value location found in problem 4,

    b. the maximum density location found in problem 5, and

    c. at least one additional location that you assess to be a plausible area of elevated drone activity based on your analysis of the distribution.

9. Briefly interpret how sensor placement influences detection effectiveness and what this suggests about protecting the CTCP.

Instructor Solution

Part 1 - Individual Assignment

Consider that you are the Assistant Operations Officer (AS3) for 3-66 AR in the 1st Brigade Combat Team, 1st Infantry Division. You are tasked with supporting the staff by providing a quantitative assessment of drone activity patterns and recommending placement of detection assets. Background facts required for the project are listed in the background document. Refer to the posted scenario for additional details.

Submit the answers to the following questions on Canvas no later than **0740 17MAR26** IAW the DAAW. Write your answers in the provided space. As applicable, show your work for each problem. **You are not allowed to use generative AI to assist in completing this assignment.** Attach any code as an appendix if it contains a portion of your work.

1. (4 points) Before conducting any operational analysis, you must verify that the provided intelligence model represents a valid probability distribution. The Joint Probability Density Function (JDF) given by the Battalion S2 includes the constant C as a normalization parameter. **In the step, set up** and provide the double integral that enforces the properties required for a valid Joint Density Function.

$$\boxed{\int_0^4 \int_{-2}^3 C(x+2)(4-y)e^{-(x^2+y^2)/8}\, dxdy = 1}$$

2. (4 points) Now, solve for the value of C to normalize and validate the JDF.

$$\boxed{C = 0.0183981}$$

3. (2 points) Briefly explain why determining C is operationally necessary before computing expected values or detection probabilities.

Some reference to probability between 0 and 1 and C acting as the scaling factor.

4. (4 points) Using the given Joint Probability Density Function, compute the expected value for each variable and report the result as and ordered pair, $(E[X], E[Y])$, representing the expected location of drone activity relative to the CTCP.

$$\boxed{(E[X], E[Y]) = (1.01634,\ 1.05204)}$$

5. (2 points) Determine the location within the Area of Operations where the joint probability density function $f(x, y)$ attains its greatest value, and report the coordinates and the function value at this point. You may think of this as a multivariable optimization problem over a bounded region.

$$\boxed{\text{Max density is } f(1.23538, 0) = 0.196747}$$

6. (2 points) Briefly explain what the expected value location (from problem 4) represents operationally and what the maximum-density location (from problem 5) represents and how the two are related or different.

   Some understanding that it is not the same point and explains why.

7. (2 points) Discuss why the expected value coordinates should not automatically be assumed to be the optimal location for placing a detection sensor.

   The density function is not constant or symmetric.

8. The Battalion received counter-UAS detection sensors capable of detecting drones within a 1 km radius of their placement location. Detection occurs only within the boundaries of the Area of Operations (AO). Compute the probability that a drone is detected by a single sensor placed at selected locations within the AO. For each placement, calculate the total probability of drone presence within the 1 km detection region (restricted to the AO if necessary), and compare the resulting detection probabilities. At a minimum, evaluate the probability of detection when the sensor is positioned at:

   a. (4 points) the expected value location found in problem 4,

   $$\boxed{\text{For } (E[X], E[Y]);\ P = 0.3762}$$

   b. (2 points) the maximum density location found in problem 5, and

   $$\boxed{\text{For } (1.23538, 0);\ P = 0.2569}$$

   c. (2 points) at least one additional location that you assess to be a plausible area of elevated drone activity based on your analysis of the distribution.

   Solution will vary but should be less than or equal to 0.401149.

9. (2 points) Briefly interpret how sensor placement influences detection effectiveness and what this suggests about protecting the CTCP.

   An interpretation of how density affects probability across different regions.

INTEGRATION CAPSTONE: GROUP

This is a GROUP assignment (two or three students). Your primary submission is an in-class presentation between 5 to 6 minutes in length with no more than 6 or 7 slides. See the "Deliverables" section of this project for the conditions under which you may use a seventh slide. You will also submit your Mathematica code. **Your instructor will terminate your presentation at 6 minutes and will not grade any material that was not presented. Please rehearse accordingly.** Presentations will happen on **Friday, March 27, 2026**.

Your presentation should cover the Intro, Methods, Results, and Discussion/Conclusions (notice no Abstract). Refer to Block Project 2's template for guidance on the content for each section. Your presentation will be graded on verbal and non-verbal presentation skills, timing, organization, visuals, and content. Use the equation editor in PowerPoint for all of your equations and refrain from copying and pasting snippets of your mathematica code in the presentation.

Authorized references include your textbook, *Mathematica*, and your partners. Any additional references used (such as internet sites or other cadets) must be cited correctly. For the use of generative Artificial Intelligence (e.g., ChatGPT), in addition to a citation in accordance with the DAAW, submit a link to the full transcript of your correspondence.

Part 2 - Group Assignment

1 Background

As previously reported, the Battalion has observed an increase in small unmanned aerial system (UAS) activity operating within the Area of Operations. Recent intelligence, surveillance, and reconnaissance (ISR) reports indicate drones are conducting pattern-of-life reconnaissance focused on sustainment routes and logistics nodes. While individual sightings remain sporadic, the S2 assesses that enemy drone behavior exhibits consistent spatial trends relative to the Combat Trains Command Post (CTCP) location.

The Commander's priority is protection of sustainment operations while maintaining freedom of maneuver. Due to limited counter-UAS detection assets, the staff must develop a quantitative assessment identifying where drone activity is most likely to occur and recommend positioning of detection systems to maximize early warning capability.

The S2 modeling cell has developed several candidate probabilistic threat models describing expected drone presence $g(x, y)$ across the Area of Operations. Each model is represented by a function $f_i(x, y)$ for $i = 1, 2, \cdots, 6$. To distribute the analytical workload, each group will be assigned one of these functions and will conduct all analysis using only the function assigned to their group.

Your assigned function represents the relative likelihood of drone activity, but it is not yet a valid joint probability density function because it has not been normalized. Before performing any probability calculations or interpreting the model operationally, your group must determine the normalization constant that ensures the function integrates to 1 over the AO.

Using your assigned function, complete the analysis required to support the Battalion Commander's decision regarding the placement of counter-UAS detection assets. Ensure that all calculations, visualizations, and interpretations in your briefing are based on your group's assigned model.

The Area of Operations is identical for all groups and is defined over the same coordinate domain;

the only component that changes your probability density for each group is the assigned function describing drone activity across the AO.

$$g(x,y) = \begin{cases} Cf_i(x,y) & \text{if } -2 \le x \le 3,\ 0 \le y \le 4 \\ 0 & \text{otherwise} \end{cases}$$

Where the constant C is the normalization parameter that you will need to compute for your JDF.

Group 1	$f_1(x,y) = 0.6\,e^{-\frac{(x-0.5)^2+(y-3)^2}{0.8}} + 0.4\,e^{-\frac{(x-2.2)^2+(y-1)^2}{0.6}}$
Group 2	$f_2(x,y) = e^{-\frac{(y-(0.8x+2))^2}{0.3}}\,e^{-\frac{(x-0.5)^2}{6}}$
Group 3	$f_3(x,y) = \left(\frac{x+2}{5}\right)^2\left(1-\frac{x+2}{5}\right)\left(\frac{y}{4}\right)\left(1-\frac{y}{4}\right)^3$
Group 4	$f_4(x,y) = (x+2)^2\,y\,e^{-\frac{x^2+y^2}{10}}$
Group 5	$f_5(x,y) = (x+2)(4-y)\,e^{-\frac{(x-1)^2+(y-2)^2}{6}}$
Group 6	$f_6(x,y) = ((0.7+\sin(1.2x)\cos(y))^2\,e^{-\frac{x^2+y^2}{12}}$

2 Final Briefing Requirement

Prepare and deliver a 6 minute, 6 or 7 slide presentation that discusses your analysis and recommendations to the Battalion Commander (aka, your instructor) and their staff (your family). Your presentation must be organized logically and include the following components:

Introduction: Briefly describe the operational problem, the purpose of your analysis, and how the joint probability model informs protection of the CTCP.

Methodology: Explain the analytical approach used to evaluate drone activity and detection effectiveness, including how expected location, maximum density, and detection probabilities were determined. Create your own and include a visual representation of the Joint Probability Density Function; such as a surface plot and corresponding contour map to help the audience understand the spatial distribution of drone activity and how the JDF computes detection probabilities. Do not copy and aste the figure used in the background document.

Results: Present your key quantitative findings, including the expected drone location, the maximum-density location, and the probability of detection for the sensor placements evaluated.

Interpretation: As members of the battalion staff, your responsibility is to provide a clear, quantitative assessment that informs the Commander's understanding of the threat and enables a confident, well-supported decision. The Battalion Commander has directed you to provide, at a minimum, the analysis of the following questions.

1. Where do we assess drone activity to be concentrated relative to the CTCP? What is the expected drone location?
2. Where should we position a single detection system to best protect the CTCP?
3. Would an additional detection asset significantly improve survivability of sustainment operations? Where should we position two different detection systems to best protect the CTCP?

Conclusion: Provide a concise recommendation for sensor placement (and additional assets, if applicable), supported by quantitative justification.

Your briefing should be concise, professional, and decision-focused. The Commander must be able to understand your recommendation and the reasoning behind it without reviewing your calculations.

Bonus: Rather than evaluating a limited set of candidate locations, determine the optimal placement of a single detection sensor by identifying the point (x_0, y_0) within the Area of Operations that maximizes the total probability of detection within the 1 km radius. Formulate and solve this as a continuous optimization problem over the AO, and report the optimal sensor location along with the corresponding maximum detection probability. Briefly explain the approach you used to search for the optimum and compare your result to the expected-value and maximum-density placements previously analyzed.

3 Deliverables

Submit your slides, code, and cover sheet for the presentation on Canvas NLT **Thursday, March 26, 2026**. You may use Artificial Intelligence to troubleshoot your coding but not to fully create or answer any portion of this exercise. Your presentation should thoroughly answer each paragraph and the questions posed in section 2 above. If your instructor requires it, you may also need to submit hard copies of your presentation and/or code. The presentation will cover the introduction to the problem, your methodology for solving the problem, the results of your methodology, and the conclusions/interpretations thereof.

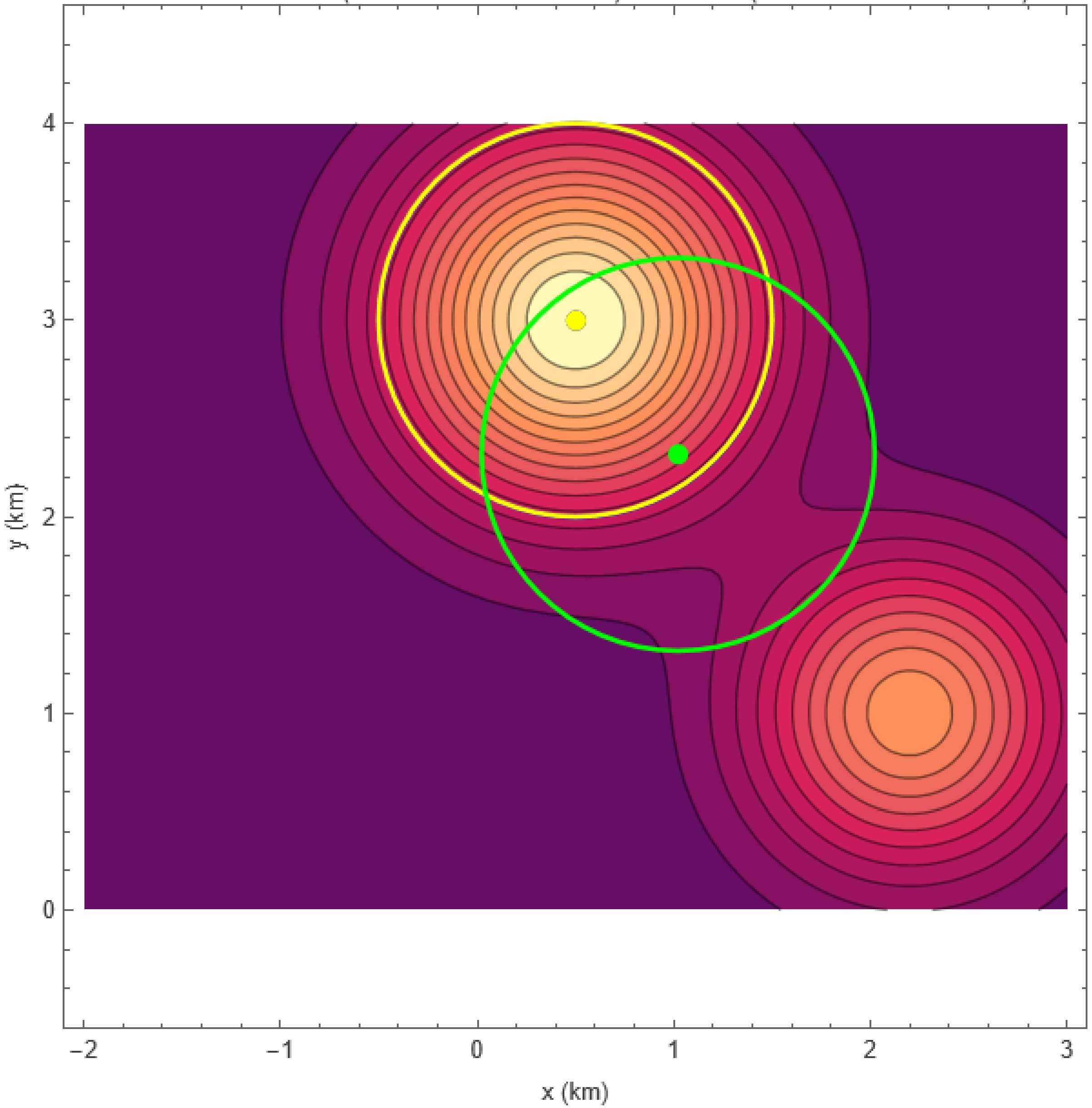
If you attempt the extra credit, you must include it during the presentation. If you do not brief the extra credit, you may not receive credit for your work there. You may use a seventh slide if you attempt the extra credit portion. This extra slide will contain only information relevant to the extra credit and no other required deliverables.

It is not recommended to take more than one slide of your six for the introduction, but the remainder is up to you. Any professional slide design is acceptable unless your instructor has specific guidance. The submission itself, as discussed in the introduction, consists of:

- USMA Cover Sheet signed by all group members.
- The presentation slides
- Mathematica Code
- Works cited and/or acknowledgment of assistance (as appropriate) per the DAAW.

Instructor Solution

$$f1(x, y) = 0.6 \exp\left(-\frac{(x-0.5)^2 + (y-3)^2}{0.8}\right) + 0.4 \exp\left(-\frac{(x-2.2)^2 + (y-1)^2}{0.6}\right)$$



Expected Drone Location: (Blue)

E[X] = 1.0215906432636846389

E[Y] = 2.3174566656446433428

Probability of Detection = 0.334308

, Max Density Point and Probability (Yellow)},

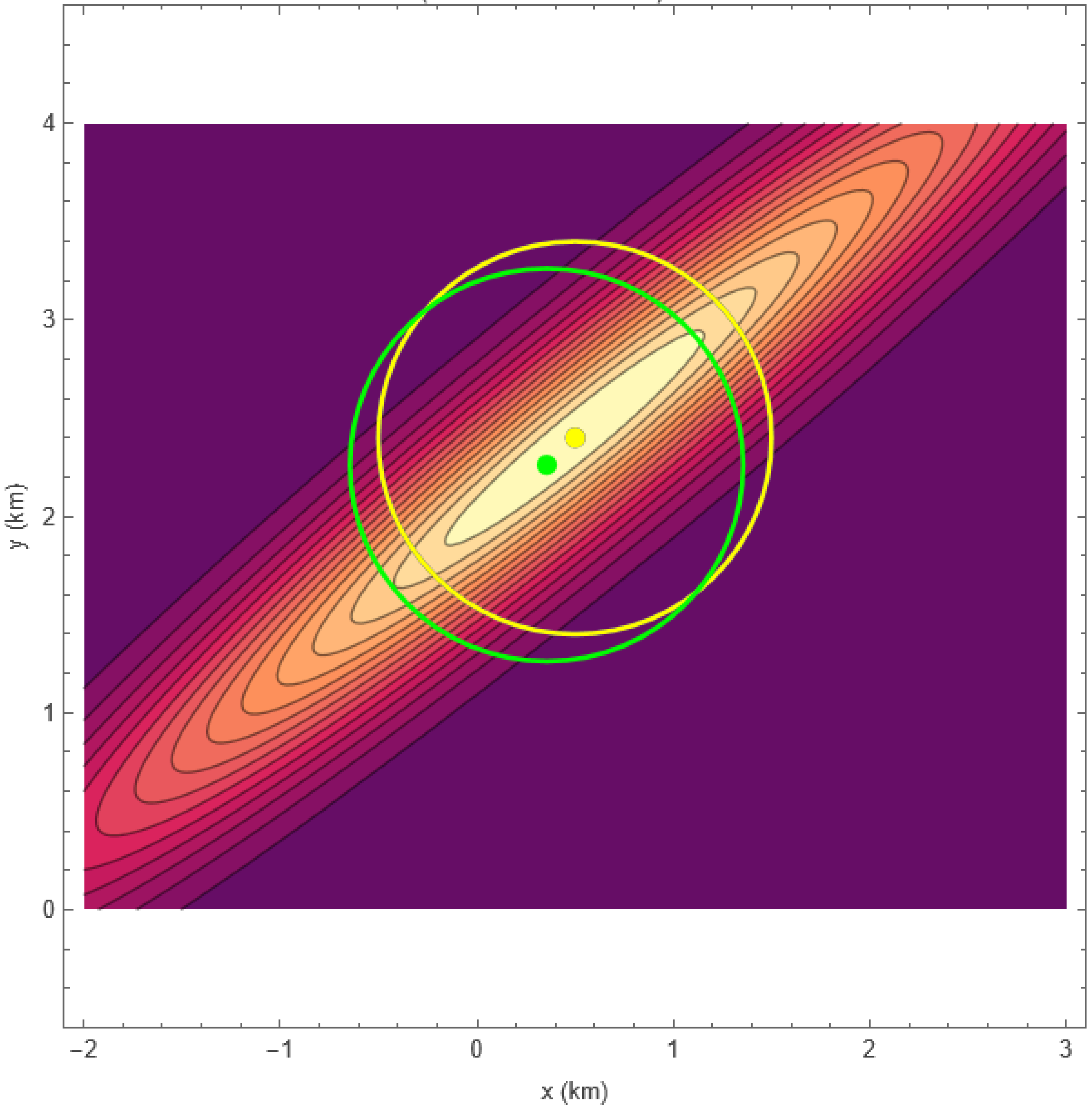
{0.513166}

{x1 → 0.500016, y → 2.99998}

Bonus: Optimal Probability Point (Green)

{0.513169, {x0 → 0.501939, y0 → 2.99772}}

$$f2(x, y) = \exp\left(-\frac{(y-(0.8x+2))^2}{0.3}\right)\exp\left(-\frac{1}{6}(x-0.5)^2\right)$$

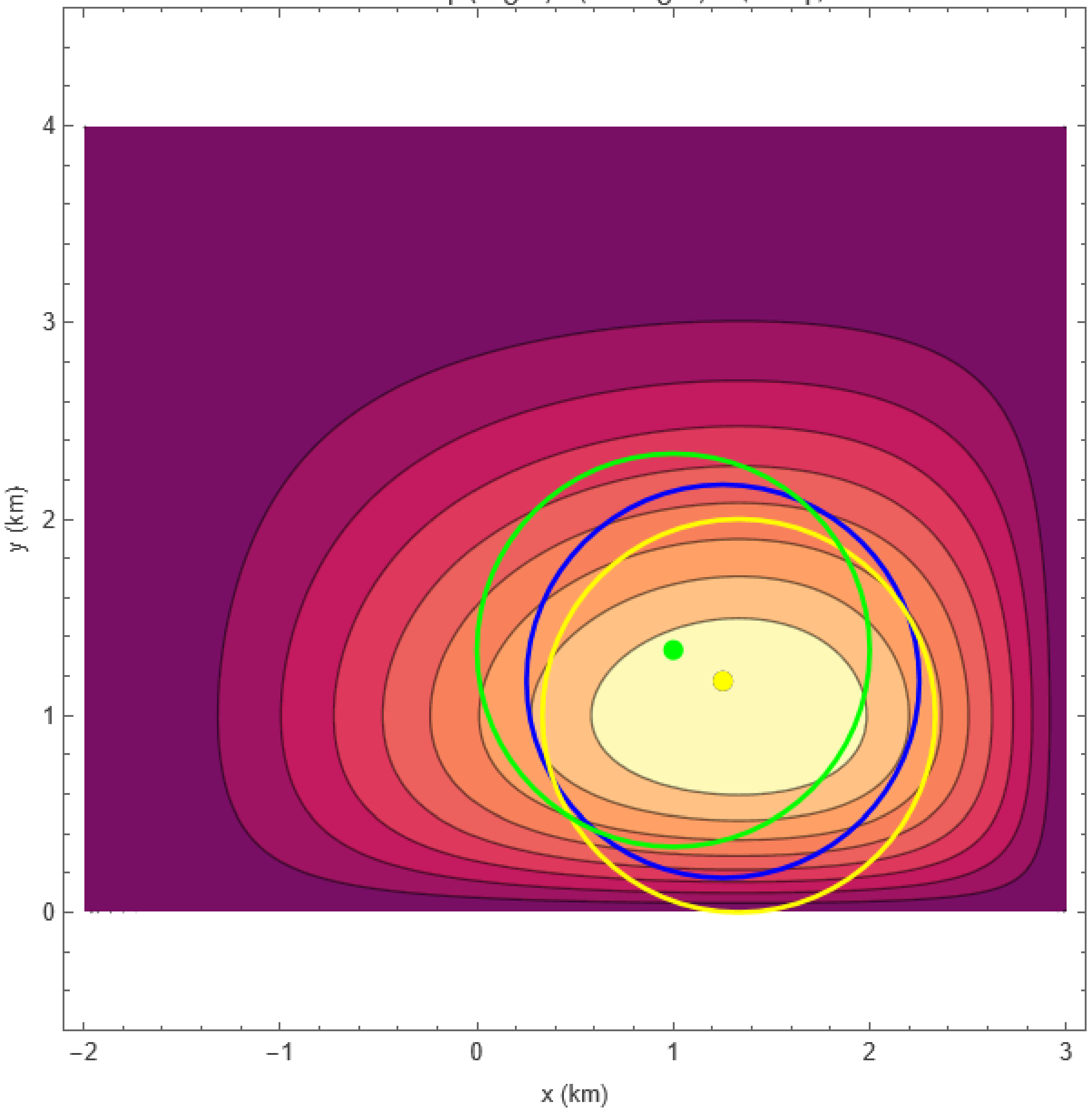


Expected Drone Location:(Blue)
E[X] = 0.35578909058987635665
E[Y] = 2.2620834859843920669
Probability of Detection = 0.417687

, Max Density Point and Probability (Yellow)},
{0.419307}
{x1 → 0.5, y → 2.4}

Bonus: Optimal Probability Point (Green)
{0.419307, {x0 → 0.5, y0 → 2.4}}

$$f3(x, y) = \frac{1}{4}\left(\frac{x+2}{5}\right)^2\left(1-\frac{x+2}{5}\right)y\left(1-\frac{y}{4}\right)^3$$

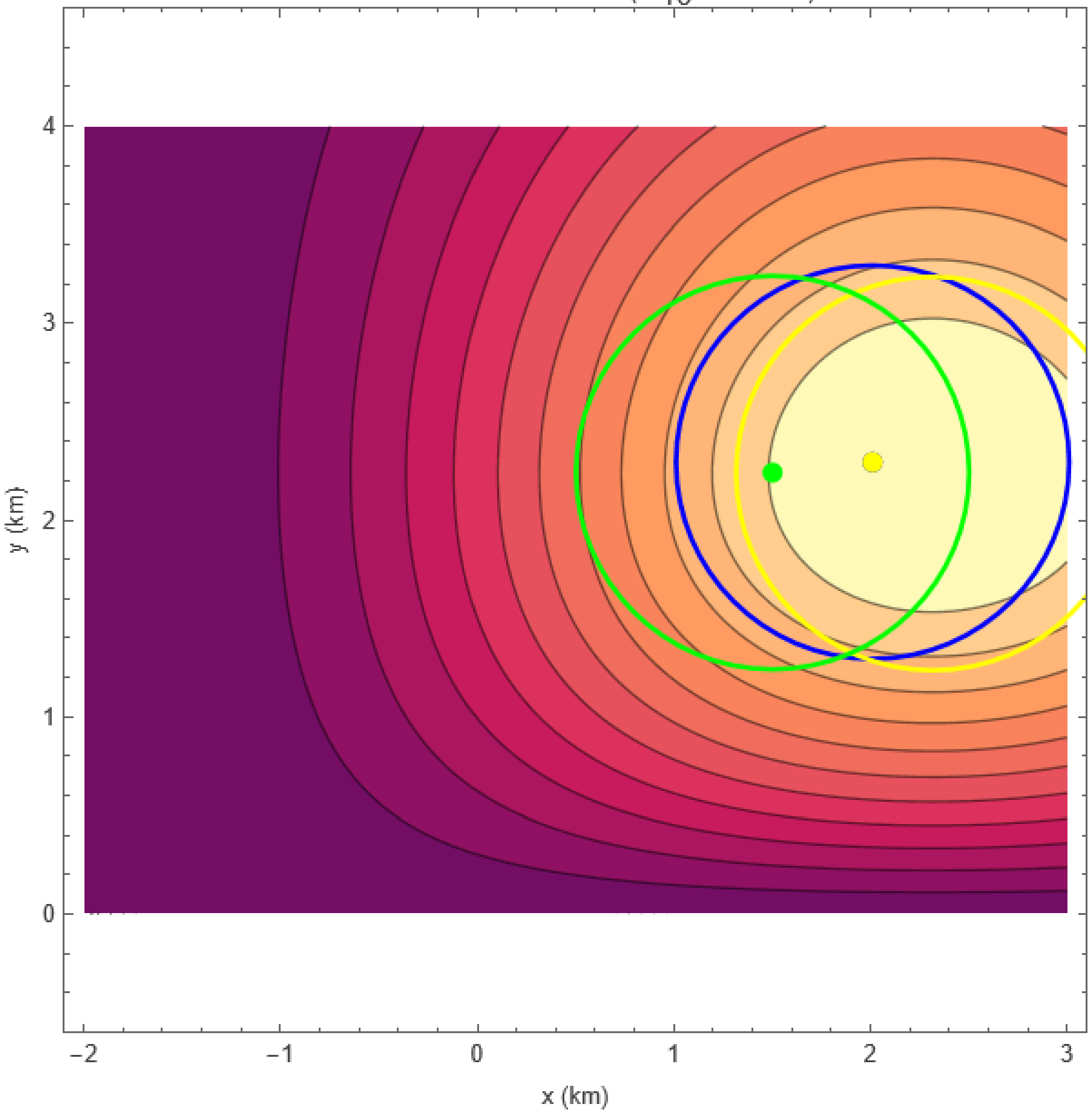


Expected Drone Location: (Blue)
E[X] = 1.0000000016747062573
E[Y] = 1.3333333463241361023
Probability of Detection = 0.449347

, Max Density Point and Probability (Yellow)},
{0.452896}
{x1 → 1.33333, y → 1.}

Bonus: Optimal Probability Point (Green)
{0.464309, {x0 → 1.25234, y0 → 1.17545}}

$$\text{f4}(x, y) = (x+2)^2\, y \exp\left(-\frac{1}{10}\left(x^2+y^2\right)\right)$$

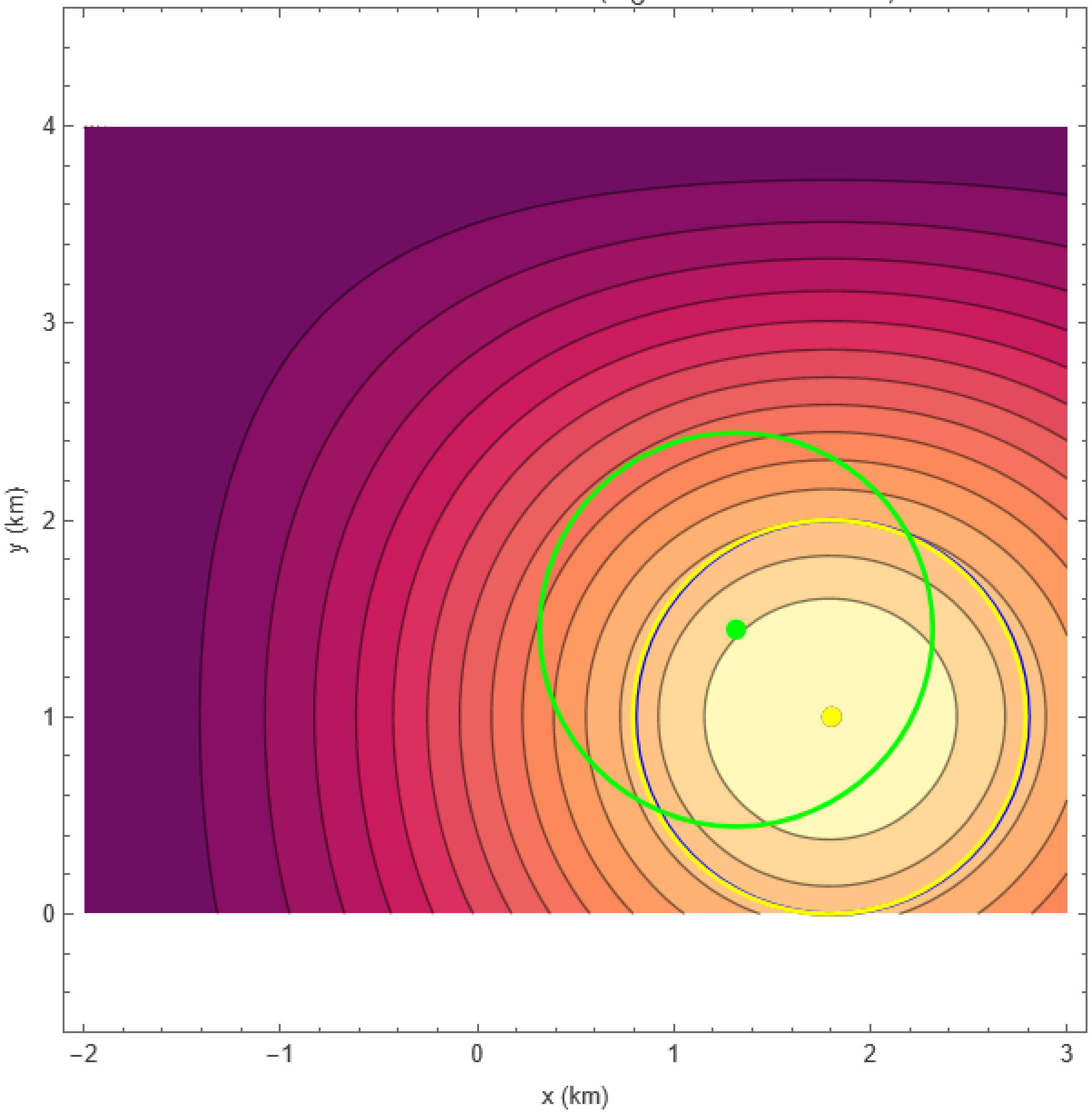


Expected Drone Location:(Blue)
E[X] = 1.5019038277919085453
E[Y] = 2.2409858521180252063
Probability of Detection = 0.330345

, Max Density Point and Probability (Yellow)},
{0.330696}
{x1 → 2.31662, y → 2.23607}

Bonus: Optimal Probability Point (Green)
{0.361164, {x0 → 2.01079, y0 → 2.29396}}

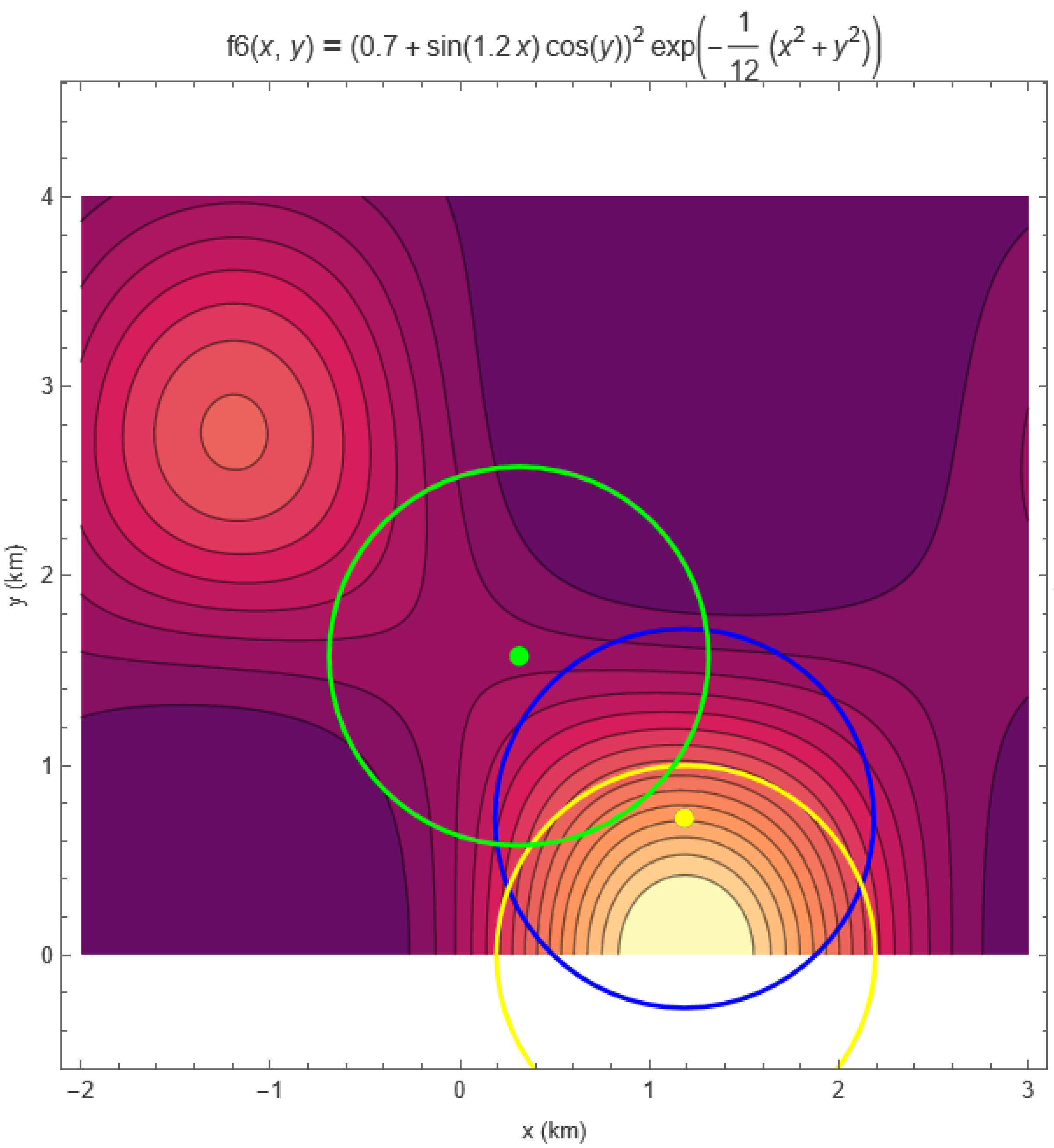
$$f5(x, y) = (x+2)(4-y)\exp\left(-\frac{1}{6}\left((x-1)^2+(y-2)^2\right)\right)$$




```
Expected Drone Location:(Blue)
E[X] = 1.3166353511642604540
E[Y] = 1.4437919484828103574
Probability of Detection = 0.344223

, Max Density Point and Probability (Yellow)}
{0.376162}
{x1 → 1.79129, y → 1.}

Bonus: Optimal Probability Point (Green)
{0.376171, {x0 → 1.803, y0 → 0.999885}}
```




```
Expected Drone Location:(Blue)
E[X] = 0.31216070544436457805
E[Y] = 1.5762307847074932827
Probability of Detection = 0.159288

, Max Density Point and Probability (Yellow)}
{0.309364}
{x1 → 1.19201, y → 0.0000171213}

Bonus: Optimal Probability Point (Green)
{0.414563, {x0 → 1.18515, y0 → 0.721232}}
```

TAB O VECTOR CALCULUS CAPSTONE

VECTOR CALCULUS CAPSTONE: SCENARIO

BC4 Project: Air Movement Under Pressure

Background

Following your analysis in BC3, the Combat Trains Command Post (CTCP) remains established at the origin $(0, 0)$ based on terrain, sustainment, and UAV threat optimization. However, the operational environment continues to evolve.

Consider that you are now the Assistant Operations Officer (AS3) for the 2-501st GSAB (General Support Aviation Battalion) that is supporting operations for 1st Brigade Combat Team, 1st Infantry Division. Enemy drone surveillance remains active, and now weather conditions are beginning to impact air mobility operations. Rotary-wing resupply missions must now account for environmental forces that affect fuel usage and flight efficiency.

The aviation battalion operations officer provides updated planning guidance:

"In the next few days, you will be planning to fly multiple resupply missions with our helicopters to support the established CTCPs in the AO, but winds have been unpredictable recently. Our flying budget is tight so you need to determine how much work the aircraft must perform and whether we can reduce it by adjusting our routes."

The helicopters will execute a resupply mission consisting of:

- Departure from a Brigade Support Area (BSA) at point $A = (15, 12)$
- Fly to a pickup zone (PZ) at point $B = (15, 0)$
- Delivery to the CTCP at point $C = (0, 0)$
- Return to the starting location A

The wind conditions in the area are modeled by the vector field:

$$\mathbf{F}(x, y) = \langle x^2/(y + 1), \quad y^2 \cos(y/\pi)\rangle$$

Ignore altitude when conducting your analysis (i.e. plan in a 2D world for these missions). All distances are measured in kilometers, so x and y represent position in km. The vector field $\mathbf{F}(x, y)$ is normalized, meaning its expression is dimensionless but represents the relative magnitude and direction of a force per unit mass (wind). Thus, line integrals of the form $\int_C \mathbf{F} \cdot d\mathbf{r}$ represent scaled work per unit mass along a path. Your analysis will determine how this field impacts the total work required to complete the mission.

Main Objective: Determine how environmental forces impact flight efficiency and whether route adjustments can reduce total work required for the mission.

VECTOR CALCULUS CAPSTONE: CHECK ON UNDERSTANDING

Part 1 - Individual Assignment

As the AS3 for 2-501st GSAB, you are tasked with supporting the staff by providing a quantitative assessment of winds affecting flight operations and attempting to mitigate costs in a resource constrained environment. Your analysis will aide the battalions ability to continue supporting the ground force. Refer to the posted scenario in the background document for additional details.

Submit the answers to the following questions on Canvas no later than **0740 27APR26** IAW the DAAW. Write your answers in the provided space. As applicable, show your work for each problem. **You are not allowed to use generative AI to assist in completing this assignment.** Attach any code as an appendix if it contains a portion of your work.

1. Compute: $\frac{\partial Q}{\partial x}$ and $\frac{\partial P}{\partial y}$

2. Based on your results, justify whether the vector field $\mathbf{F}(x, y)$ is conservative or non-conservative and explain at least two differences between a conservative and non-conservative vector field.

3. Provide a brief interpretation of what this vector field represents in the context of a helicopter's flight. How does the wind appear to affect the designated route?

4. Compute the work done by the vector field along a straight-line path from point A to point C.

5. Compute the work done by the vector field along a two-segment path from A to B to C.

6. Compare your results. Does the work depend on the path taken? Explain.

7. Combine your work above to evaluate the line integral around a closed triangular path: $\oint_C \mathbf{F} \cdot d\vec{r}$

8. Now let's use Green's Theorem to compute the same quantity: $\iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dA$

9. Compare the two methods and comment on which is more efficient and why.

Instructor Solution

Part 1 - Individual Assignment

As the AS3 for 2-501st GSAB, you are tasked with supporting the staff by providing a quantitative assessment of winds affecting flight operations and attempting to mitigate costs in a resource constrained environment. Your analysis will aide the battalions ability to continue supporting the ground force. Refer to the posted scenario in the background document for additional details.

Submit the answers to the following questions on Canvas no later than **0740 27APR26** IAW the DAAW. Write your answers in the provided space. As applicable, show your work for each problem. **You are not allowed to use generative AI to assist in completing this assignment.** Attach any code as an appendix if it contains a portion of your work.

Purpose: Assess your ability to conceptually understand and analyze paths or surfaces in vector fields with line integrals and Green's Theorem.

1. (5 points) Compute: $\frac{\partial Q}{\partial x}$ and $\frac{\partial P}{\partial y}$

$$\boxed{\frac{\partial Q}{\partial x} = 0, \quad \frac{\partial P}{\partial y} = -\frac{x^2}{(y+1)^2}}$$

2. (4 points) Based on your results, justify whether the vector field $\mathbf{F}(x, y)$ is conservative or non-conservative and explain at least two differences between a conservative and non-conservative vector field.

   Non-conservative field. The computed partials are not equal so the field does not satisfy the condition required for conservativeness. Comment on path independence, work, potential function, circulation...

3. (4 points) Provide a brief interpretation of what this vector field represents in the context of a helicopter's flight. How does the wind appear to affect the designated route?

   The vector field models wind conditions affecting helicopter motion during the mission. The wind field introduces variable resistance and assistance along the route, causing different levels of drift and effort depending on the segment traveled. The east-west component

$$P(x, y) = \frac{x^2}{y+1}$$

   indicates that wind strength increases with distance in the x-direction and becomes increasingly unstable as $y \to -1$, suggesting strong shear effects in lower regions. The north-south component

$$Q(x, y) = y^2 \cos\left(\frac{y}{\pi}\right)$$

   depends only on y, indicating oscillating vertical wind effects that vary with latitude.

4. (5 points) Compute the work done by the vector field along a straight-line path from point A to point C.

$$\boxed{W_{AC} = 307.163}$$

5. (4 points) Compute the work done by the vector field along a two-segment path from A to B to C.

$$\boxed{W_{AB} = 429.36, \quad W_{BC} = -1125}$$

$$\boxed{W_{ABC} = W_{AB} + W_{BC} = -695.64}$$

6. (4 points) Compare your results. Does the work depend on the path taken? Explain.

   Yes, the work depends on the path taken as $W_{AC} \neq W_{ABC}$. This confirms that the vector field is non-conservative, since work is path-dependent. Physically, this means the helicopter experiences different net wind effects depending on its route, even when starting and ending at the same locations.

7. (4 points) Combine your work above to evaluate the line integral around a closed triangular path:

   $\oint_C \mathbf{F} \cdot d\vec{r}$

$$\boxed{W_{Line} = W_{AB} + W_{BC} + (-W_{AC}) = -1002.8}$$

8. (5 points) Now let's use Green's Theorem to compute the same quantity:

   $\iint_D \frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\, dA$

$$\boxed{W_{Greens} = -1002.8}$$

9. (4 points) Compare the two methods and comment on which is more efficient and why.

   Both methods produce the same result. However, Green's Theorem is more efficient for closed curves because it converts the line integral into a double integral over the region D. Instead of computing multiple line segments individually, the problem becomes a single area-based computation, which is often simpler and more systematic.

VECTOR CALCULUS CAPSTONE: GROUP

Part 2 - Group Assignment / Presentation

This is a GROUP assignment (two or three students). Your primary submission is an in-class presentation between 5 to 6 minutes in length with a recommendation of 6 to 7 slides. See the "Deliverables" section of this project for the content that is required on your slides. You will also submit your Mathematica code. **Your instructor will terminate your presentation at 6 minutes and will not grade any material that was not presented. Please rehearse accordingly.** Presentations will happen on **Monday, May 4, 2026**.

Your presentation should cover the Intro, Methods, Results, and Discussion/Conclusions (notice no Abstract). Your presentation will be graded on verbal and non-verbal presentation skills, timing, organization, visuals, and content. Use the equation editor in PowerPoint for all of your equations and refrain from copying and pasting snippets of your mathematica code in the presentation **(export and publish instead if showing code enhances your presentation!)**.

Authorized references include your textbook, *Mathematica*, and your partners. Any additional references used (such as internet sites or other cadets) must be cited correctly. For the use of generative Artificial Intelligence (e.g., ChatGPT), submit a link to the full transcript of your correspondence in addition to a citation in accordance with the DAAW. You may use Artificial Intelligence to troubleshoot your coding but not to fully create presentations or answer any portion of this exercise.

1 Background

While conducting your initial analysis, the battalion staff continued refining their assessments as part of the planning process. Updated intelligence, surveillance, and weather reports indicate that atmospheric conditions will become more turbulent during the planned mission window. In response, the S2 developed a more advanced model of the operational environment using a spatially varying wind field represented by the vector field $\mathbf{F}(x, y)$. This model captures the combined effects of steady wind, terrain-induced flow over hills, and localized swirling regions that may either assist or resist aircraft movement.

The Battalion Commander's priority is to minimize the work experienced by each aircraft due to these wind conditions, thereby conserving fuel and extending operational capability. Given limited fuel resources, the staff must provide a quantitative analysis to determine which flight paths require the least work. Similar to your individual assessment, the work along a path C is modeled by the line integral.

To distribute the analytical workload, each group will be assigned a unique pickup zone (PZ) location within the same area of operations. **While all groups will work over the same coordinate domain, each will analyze a different route by defining points A as the BSA at $(15, 12)$, B as the designated PZ, and C as the CTCP fixed at $(0, 0)$. All calculations, visualizations, and conclusions should be based solely on your group's assigned location and corresponding flight paths.**

Your objective is to evaluate multiple possible routes between the given points, compute the work associated with each path, and determine which route minimizes the total work. Your final recommendation should identify the optimal flight path and justify your decision using both mathematical analysis and interpretation of the vector field.

2 Wind Field Model Over Terrain

The Area of Operations is identical for all groups and is defined over the same coordinate domain; the only component that changes is the intermediate pickup zone location each group is assigned.

Terrain Function:

$$h(x,y) = 84e^{-\frac{(x-5)^2+(y-8)^2}{20}} + 72e^{-\frac{(x-15)^2+(y-10)^2}{18}} + 42e^{-\frac{(x-20)^2+(y+2)^2}{16}} + 21e^{-\frac{(x-18)^2+(y-4)^2}{12}}$$

Terrain-Induced Flow:

$$\mathbf{F}_{\text{terrain}}(x,y) = -\nabla h(x,y)$$

Wind Component:

$$\mathbf{F}_{\text{wind}}(x,y) = \langle 10 + 1.5\cos(0.3y) + 0.75\sin(0.1xy),\ 4\sin(0.4x)\rangle$$

Localized Swirl of Air:

Let $\alpha = 12$.

$$\mathbf{F}_{\text{swirl}}(x,y) = \alpha\,\langle(-(y-2)),\ (x-11)\rangle \exp\left(-\frac{(x-11)^2+(y-2)^2}{11}\right)$$

Total Vector Field:

$$\mathbf{F}(x,y) = \mathbf{F}_{\text{wind}}(x,y) + \mathbf{F}_{\text{terrain}}(x,y) + \mathbf{F}_{\text{swirl}}(x,y)$$

Pickup Zone Locations

Group 1	$B_1 = (5,8)$
Group 2	$B_2 = (8,-7)$
Group 3	$B_3 = (18,4)$
Group 4	$B_4 = (20,-2)$
Group 5	$B_5 = (14,-5)$
Group 6	$B_6 = (12,-1)$

3 Final Briefing Requirement

Prepare and deliver a professional briefing that presents your analysis and recommendations to the Battalion Commander (your instructor). Your briefing should be clear, professional, and focused on decision-making. The Battalion Commander must be able to understand your recommendation and its justification without reviewing your detailed calculations.

Introduction: Briefly describe the operational scenario, including the updated wind field model and its impact on helicopter movement. Clearly state the purpose of your analysis.

Methodology: Explain the analytical tools used in your study. This should include how line integrals and/or Green's Theorem were used to compute work along flight paths, as well as how the vector field was visualized and interpreted. Include appropriate visualizations (vector field plots, stream plots, and route overlays) to support your analysis.

Results: Present your quantitative findings in three phases:

- **Phase 1 (Baseline Route Analysis):** Compute the total work along the original closed route and interpret whether the wind is overall assisting or opposing the helicopter.
- **Phase 2 (Route Optimization):** Propose and analyze **a modified route** that reduces total work by at least 5% (goal). Your new route may include curved or piecewise-defined paths. Clearly justify how your route leverages features of the vector field. Your new route must meet the following conditions:
  - Starts and ends at the BSA.
  - Visits your designated PZ and the CTCP.
- **Phase 3 (Comparison):** Compare the original and optimized routes, including total work and percent change.

Interpretation: As members of the battalion staff, provide a clear and actionable assessment for the Battalion Commander. At a minimum, address the following:

1. Does the wind field overall help or hinder movement along the original route?
2. How does your optimized route exploit the structure of the vector field?
3. What operational advantages are gained by adjusting the route (e.g., fuel savings, efficiency)?
4. Explain at least two assumptions or limitations present in your model.

Conclusion: Provide a concise, decision-focused recommendation that explains whether the route should be adjusted and justify your recommendation using your quantitative results.

Optional Extra Credit: Incorporate additional operational factors, such as drone risk from BC3, and develop a combined cost model that balances work and risk. Briefly describe how this impacts your recommended route.

4 Deliverables

Submit all content for the presentation on Canvas NLT **Sunday, May 3, 2026**. Your presentation should thoroughly answer each paragraph and the questions posed in section 3 above. If your instructor requires it, you may also need to submit hard copies of your presentation and/or code. The presentation will cover the introduction to the problem, your methodology for solving the problem, the results of your methodology, and the conclusions/interpretations thereof.

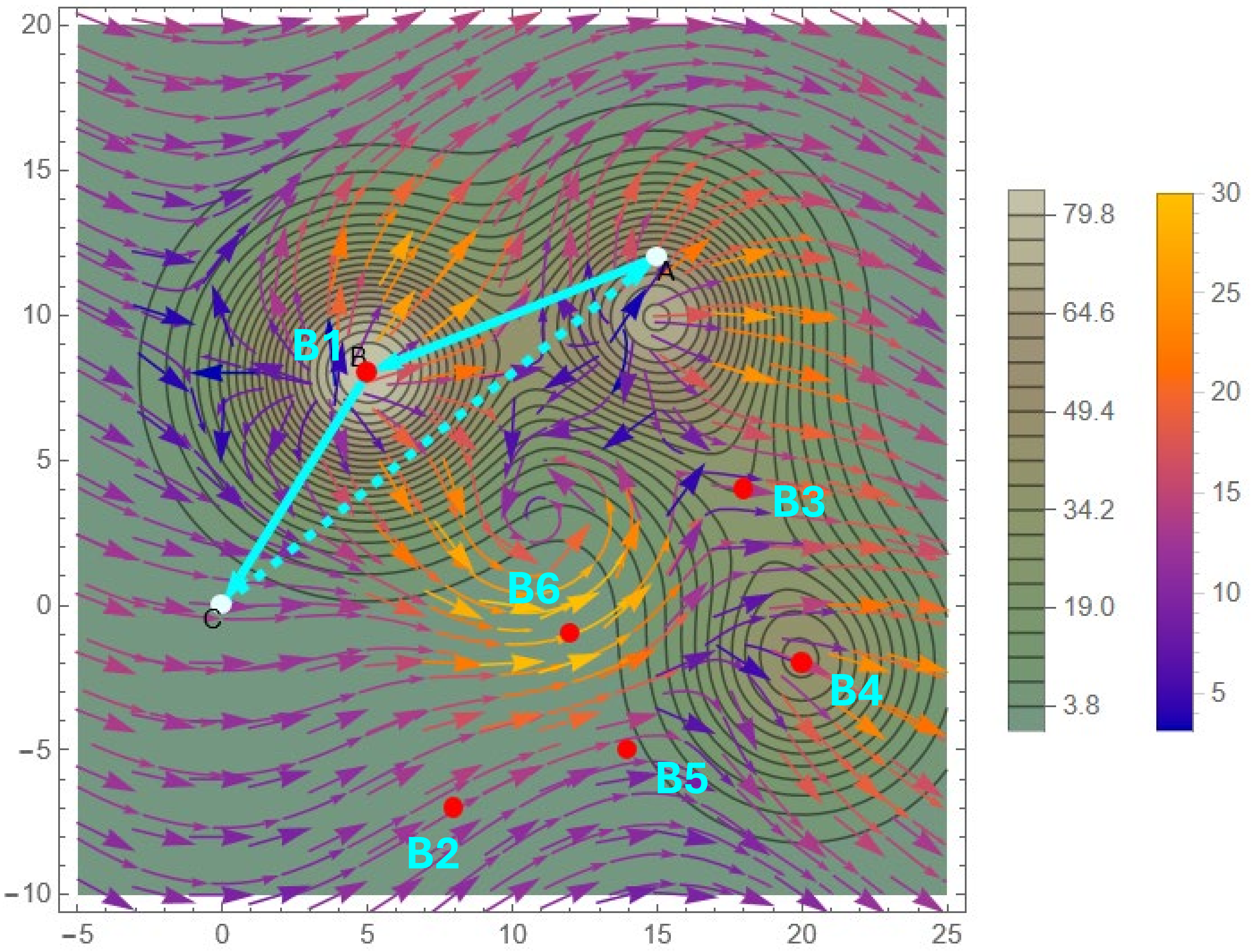
If you attempt the extra credit, you must include it during the presentation. If you do not brief the extra credit, you may not receive credit for your work. Use a dedicated slide to brief the extra credit portion. This slide will contain only information relevant to the extra credit and no other required deliverables. You will be allocated an additional minute to your briefing time to cover the additional material but must be declared before starting your presentation.

It is not recommended to take more than one slide for the introduction, but manage your presentation wisely. Any professional slide design is acceptable unless your instructor has specific guidance. The submission itself, as discussed in the introduction, consists of:

- USMA Cover Sheet signed by all group members.
- Presentation slides with appropriate visualizations of the vector field and all routes.
- *Mathematica* Code.
- Works cited and/or acknowledgment of assistance (as appropriate) per the DAAW.

Instructor Solution

Group 1: B = (5,8)



$$\int_0^1 \text{Ftotal}(15 - 15\,t,\ 12 - 12\,t).\{-15,\ -12\}\,dt$$

A→C -63.1779 **reverse for total work on route

$$\int_0^1 \text{Ftotal}(15 - 10\,t,\ 12 - 4\,t).\{-10,\ -4\}\,dt$$

A→B -103.984

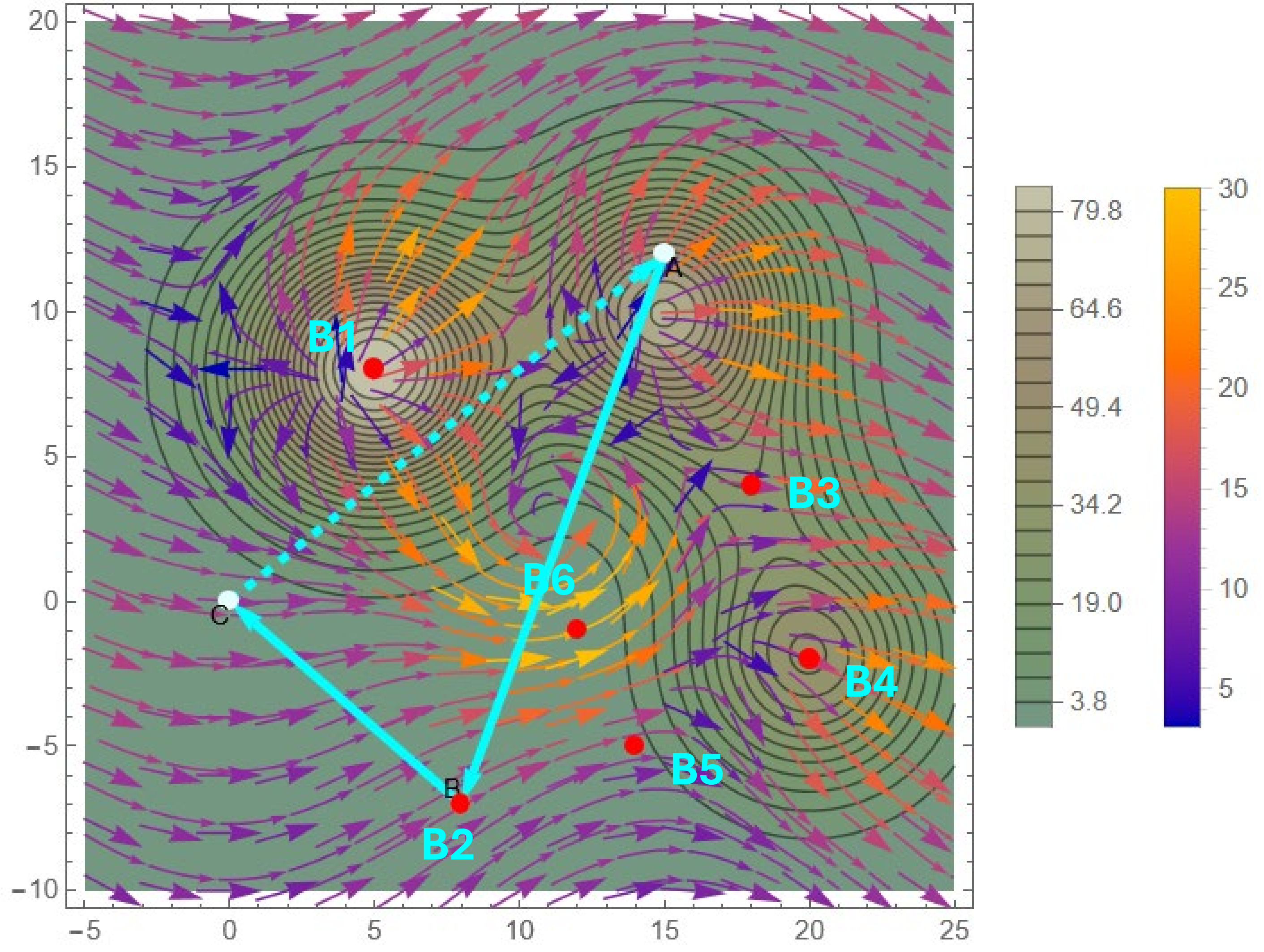
$$\int_0^1 \text{Ftotal}(5 - 5\,t,\ 8 - 8\,t).\{-5,\ -8\}\,dt$$

B→C 7.887

TOTAL: Out[137]= -32.9186

** Only CCW path so negative not needed on Green's Theorem

Group 2: B = (8,-7)



$$\int_0^1 \text{Ftotal}(15-15t,\ 12-12t).\{-15,\ -12\}\,dt$$

A→C -63.1779 **reverse for total work on route

$$\int_0^1 \text{Ftotal}(15-7t,\ 12-19t).\{-7,\ -19\}\,dt$$

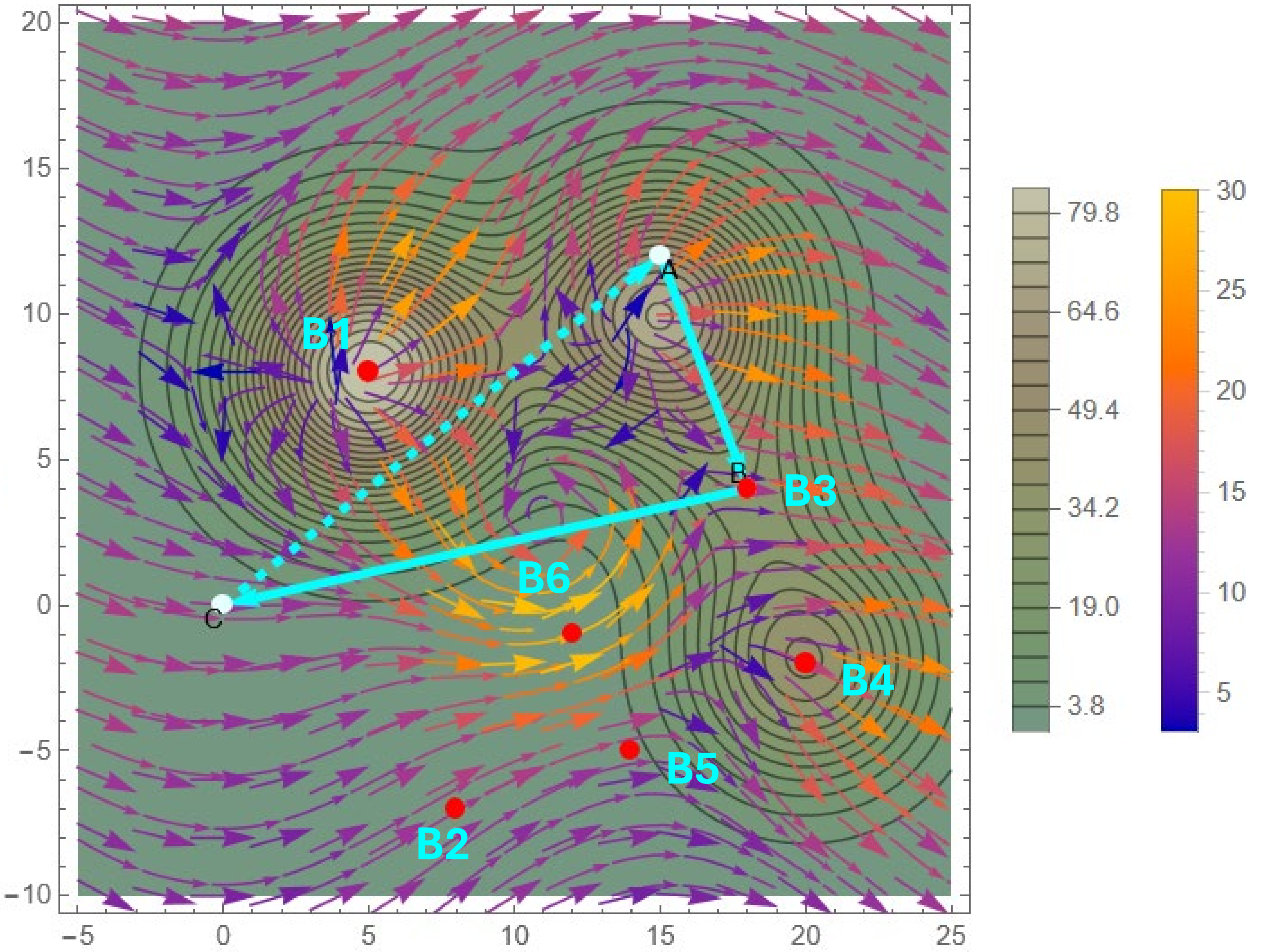
A→B 19.2779

$$\int_0^1 \text{Ftotal}(8-8t,\ 7t-7).\{-8,\ 7\}\,dt$$

B→C -67.769

TOTAL: Out[220]= 14.6868

Group 3: B = (18,4)



$$\int_0^1 \text{Ftotal}(15 - 15t,\ 12 - 12t).\{-15, -12\}\,dt$$

A→C -63.1779 **reverse for total work on route

$$\int_0^1 \text{Ftotal}(3t + 15,\ 12 - 8t).\{3, -8\}\,dt$$

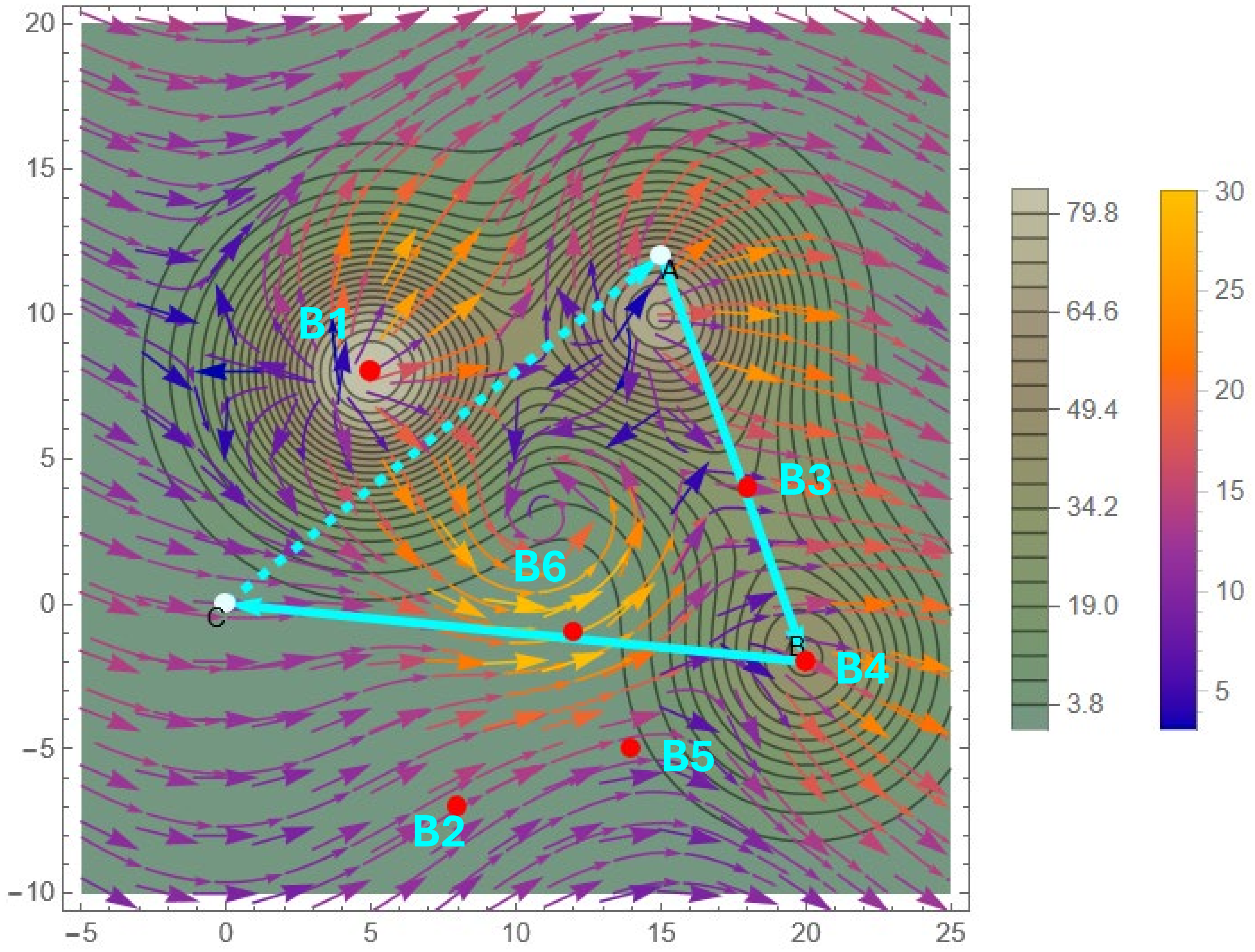
A→B 42.9897

$$\int_0^1 \text{Ftotal}(18 - 18t,\ 4 - 4t).\{-18, -4\}\,dt$$

B→C -144.94

TOTAL: Out[303]= -38.7726

Group 4: B = (20,-2)



$$\int_0^1 \text{Ftotal}(15 - 15t,\ 12 - 12t).\{-15, -12\}\, dt$$

A→C -63.1779 **reverse for total work on route

$$\int_0^1 \text{Ftotal}(5t + 15,\ 12 - 14t).\{5, -14\}\, dt$$

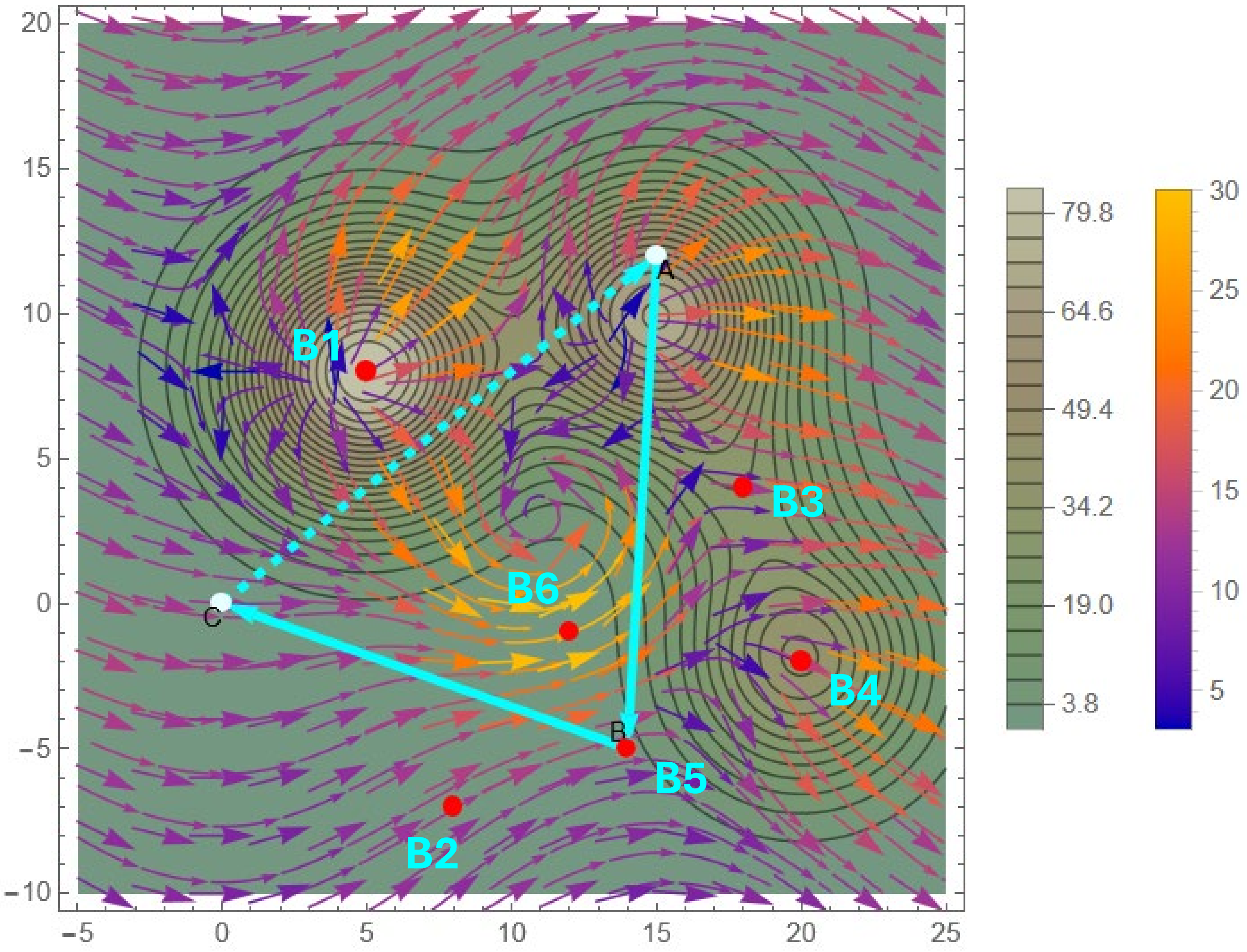
A→B 29.2635

$$\int_0^1 \text{Ftotal}(20 - 20t,\ 2t - 2).\{-20, 2\}\, dt$$

B→C -270.894

TOTAL: Out[386]= -178.453

Group 5: B = (14, -5)



$$\int_0^1 \mathrm{Ftotal}(15 - 15\,t,\ 12 - 12\,t).\{-15,\ -12\}\,dt$$

A→C -63.1779 **reverse for total work on route

$$\int_0^1 \mathrm{Ftotal}(15 - t,\ 12 - 17\,t).\{-1,\ -17\}\,dt$$

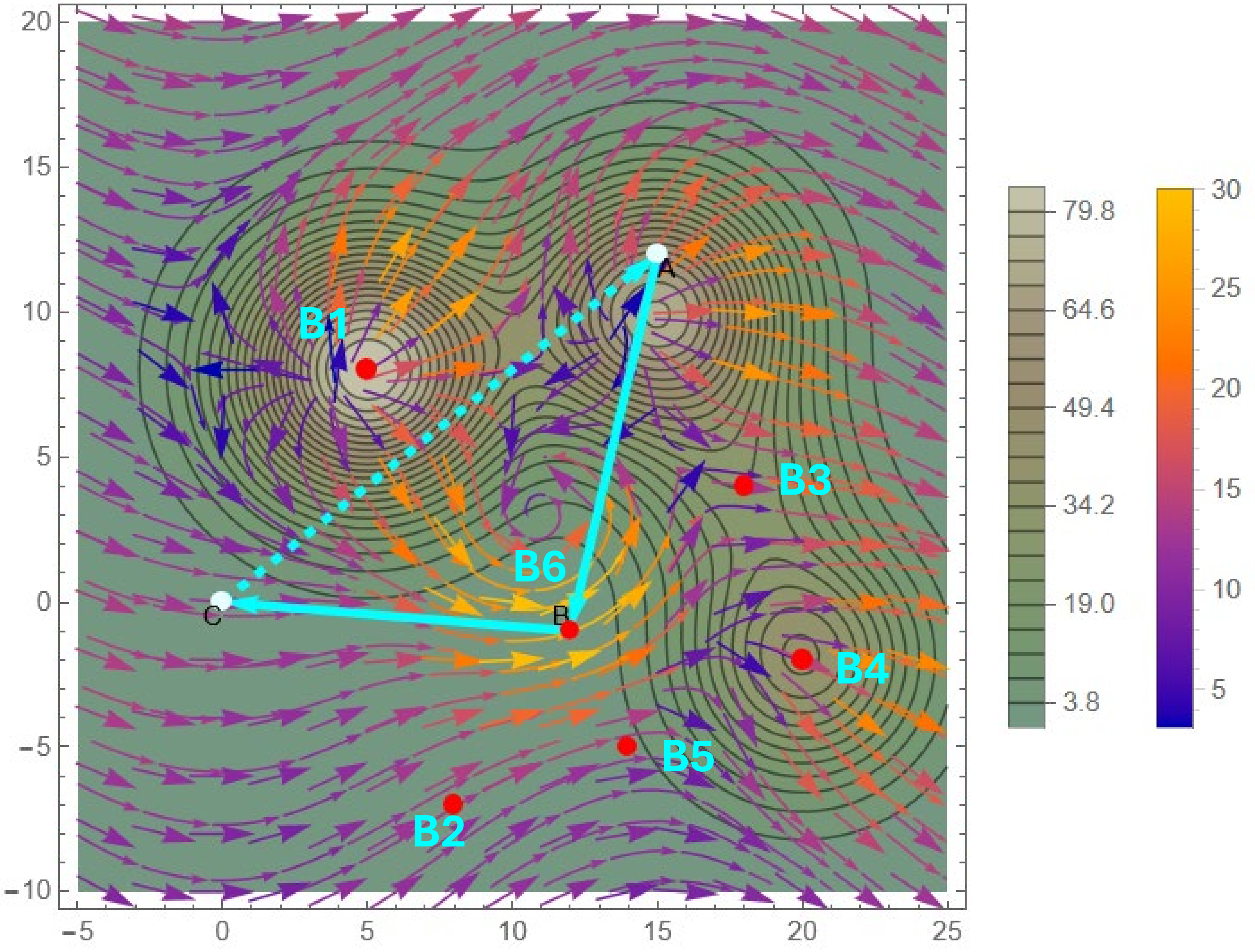
A→B -6.90046

$$\int_0^1 \mathrm{Ftotal}(14 - 14\,t,\ 5\,t - 5).\{-14,\ 5\}\,dt$$

B→C -172.558

TOTAL: Out[469]= -116.28

Group 6: B = (12, -1)



$$\int_0^1 \mathrm{Ftotal}(15 - 15t,\ 12 - 12t).\{-15, -12\}\, dt$$

A→C $\quad -63.1779$ **reverse for total work on route

$$\int_0^1 \mathrm{Ftotal}(15 - 3t,\ 12 - 13t).\{-3, -13\}\, dt$$

A→B $\quad -14.5715$

$$\int_0^1 \mathrm{Ftotal}(12 - 12t,\ t - 1).\{-12, 1\}\, dt$$

B→C $\quad -200.483$

TOTAL: Out[552]= -151.877